Entropic lattice Boltzmann models for thermal and compressible flows

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ENTROPIC LATTICE BOLTZMANN MODELS FOR THERMAL AND COMPRESSIBLE FLOWS

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To my family

Abstract

The lattice Boltzmann method (LBM) is an ever growing, powerful approach to computational fluid dynamics, based on a discrete velocity formulation of the Boltzmann equation. While success of LBM has expanded greatly over the past two decades to turbulence, multiphase flows, and thermal incompressible flows, all these applications are confined to the low Mach number domain of fluid dynamics. The situation is drastically different for compressible flows, where numerous attempts to derive a genuine lattice Boltzmann model for high Mach number flows have not achieved a comparable outcome. In this context, the main objective of this thesis is to develop a lattice Boltzmann model for simulation of compressible flows in the subsonic, transonic and supersonic regimes, for laminar and turbulent flows. Along this line, further models are developed for the simulation of thermal flows in the low Mach number regime and for conjugate heat transfer.

The compressible model is realized through three fundamental changes to the standard lattice Boltzmann scheme, namely the employment of admissible higher-order lattices, computation of exact entropic equilibrium, and use of the entropic over-relaxation scheme to improve numerical stability. In contrast to state-of-the-art fluid solvers, the algorithm employs Cartesian meshes, without any turbulence models, tuning parameters, or tracking of the shock front. However, the employment of higher-order lattices with large number of discrete velocities, especially in three dimensions, leads to, for some cases, a dramatic increase of computational cost when compared to the incompressible LBM. To this end, novel optimization strategies are also proposed, including the development of shifted lattices, and pruned lattices together with the accurate entropic guided equilibrium. In order to validate the compressible model, a number of two- and three-dimensional simulations of sub-, trans- and supersonic flows is presented, including shock dynamics, transonic and supersonic aerodynamics, turbulent flows and turbulence shock interaction.

Numerous heat transfer applications, however, do not involve high Mach number flows, thus allowing to neglect the compressibility effects and develop simplified models. To this end, two thermal lattice Boltzmann models are proposed in this thesis. The first is derived directly from the compressible model, simplifying both the lattice and the equilibrium distribution, and avoiding the employment of the second set of populations used to change the adiabatic exponent. Although this model is a simplified and lighter version of the compressible model, the computational cost may still be considerable. Hence, a second two-population model is also presented, and it is extended to the simulation of conjugate heat transfer problems. Also for these thermal models a thorough validation is exposed, for both two- and three-dimensional applications; this includes, among others, the thermal Couette flow, the simulation of convection around heated bodies and the simulation of the rotating and non-rotating turbulent Rayleigh-Bénard convection.

All models presented here are complemented with appropriate wall boundary conditions that are capable of handling arbitrary solid geometries and arbitrary lattice choices.

To conclude, by strongly incorporating the first and second laws of thermodynamics into the lattice Boltzmann method, this thesis presents one single algorithm capable of efficiently simulating a broad range of flow situations, starting from low Mach number flows and extending all the way to supersonic flows and conjugate heat transfer problems.

Zusammenfassung

Die Lattice-Boltzmann-Methode (LBM) ist eine leistungsstarker Ansatz zur numerischen Strömungssimulation, welcher auf einer diskreten Geschwindigkeitsformulierung der Boltzmann-Gleichung beruht. Während sich der Erfolg der LBM in den letzten zwei Jahrzehnten auf die Gebiete von porösen Medien, Mehrphasenströme und thermisch, inkompressible Strömungen ausgeweitet hat, beschränkten sich alle diese Anwendungen auf Strömungen im Regime niedriger Machzahlen. Zahlreiche Versuche ein Modell für kompressible Strömungen herzuleiten hatten keinen Bestand. Das Ziel dieser Arbeit ist es, ein lattice Boltzmann Modell für die Simulation von Unter-, Trans- und Überschallströmungen im laminaren sowie turbulenten Regime zu entwickeln. Im Zuge dessen, sind weitere neue Modelle zur Simulation von thermischen Unterschallströmungen entwickelt worden.

Das kompressibel Modell basiert auf drei fundamentalen Änderungen des Standard Lattice-Boltzmann Schemas, nämlich die Verwendung von zulässigen Gittern höherer Ordnung, des exakten entropischen Gleichgewichts sowie die entropische Relaxation. Im Gegensatz zu modernsten Strömungslösern beruht der Algorithmus auf kartesischen Gittern, ohne explizites Turbulenzmodell, empirischen Parametern oder die Bestimmung der Schockfront. Da die Verwendung von Gittern höherer Ordnung für die Simulation von sehr hohe Machzahlen zu einem untragbaren Rechenaufwand führt, werden Optimierungsstrategien wie beispielsweise die Verwendung von versetzten und reduzierten Gittern mit geführtem entropischen Gleichgewicht präsentiert. Für die Validierung des kompressiblen Modells wurden eine Reihe von zwei- und dreidimensionalen Simulationen im Unter-, Trans- und Überschallbereich durchgeführt, einschliesslich Schockdynamik, überschallaerodynamik, turbulente Strömungen sowie Turbulenz-Schock Interaktion.

Zusätzlich zum kompressiblen Strömungsmodell werden thermische Strömungsmodelle im Bereich niedriger Machzahl vorgestellt. Das erste Modell

wird direkt von dem kompressiblen Modell abgeleitet wobei Gitter und Gleichgewichtsverteilung simplifiziert werden und von der Verwendung eines zweiten Gitters zur Änderung des adiabatischen Exponenten abgesehen wird. Obwohl dieses Vorgehen eine Vereinfachung zum kompressiblen Modell darstellt, kann der Rechenaufwand beträchtlich sein. Aus diesem Grund wird ein Zwei-Populationen Modell entwickelt und zur Simulation von Wärmeübertragungsproblemen erweitert. Alle Modelle werden sowohl für simple zweidimensionale Probleme sowie für dreidimensionale turbulente Strömungen validiert. Ein besonderer Augenmerk wird auf die rotierende und nicht-rotierende Rayleigh-Bénard-Konvektion gelegt.

Diese Arbeit bietet, im Ramen der Lattice-Boltzmann-Methode, eine vollständige Methodik zu Simulation von Problemen in der Strömungsmechanik, bei denen Energieerhaltung berücksichtigt werden muss.

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Contents

Chapter 1

Introduction

Be it the next days weather forecast, understanding of supernova explosions, or designing next generation aircrafts, prediction of fluid flow phenomena is of paramount importance for human life. Along with its fascinating features and importance, description of fluid dynamics remains a challenge for mathematicians, physicists and engineers. In the continuum regime and in classical fluid dynamics, the Navier-Stokes (NS) equations are considered to provide the best description of fluid flows; however, analytical solutions to these equations are available only for overwhelmingly simplified problems, thus numerical simulations are required in order to tackle realistic situations. In the majority of the cases, in fact, the Reynolds number, defined as the ratio of inertial forces to viscous forces, is high and thus the flow becomes turbulent [1]. Simulations of turbulent flows face serious difficulties, due to the wide range of time and length scales involved, which inevitably make the problem computationally expensive even for the modern supercomputers. Further complications arise in the simulation of turbulent compressible flows [2, 3]; in this case, the Mach number, defined as the ratio between a characteristic velocity and the speed of sound, is high. When the Mach number exceeds the unity, the flow becomes supersonic and shock waves may appear. Particularly challenging is the interaction of shock waves with turbulence, which requires special treatment by conventional flow solvers. Indeed, higher-order discretization methods used for direct numerical simulations (DNS) of turbulent flows can cause Gibbs oscillations, leading to instabilities in the presence of shocks, whereas typical methods used to regularize shock computations demonstrate numerical dissipation which degrades accuracy of DNS in regions of high turbulence [2]. Therefore, hybrid

schemes combining higher-order methods with (W)ENO schemes through state-of-the-art shock sensors is the preferred approach for the simulation of high-speed compressible flows. The main drawback of these approaches, however, is the complexity of resulting numerical schemes, requiring flow-specific parameter tuning. Thus development of efficient, high-fidelity, robust DNS solvers which can treat seamlessly any type of compressible turbulent flow, in the sub-, trans- and supersonic regimes, is a topic of tremendous interest.
The lattice Boltzmann method (LBM) is a powerful, alternative approach to simulation of complex fluid dynamic problems [4, 5], including turbulent flows [6], surface and multiphase phenomena [7–12], micro-emulsions and soft-glassy materials [13], relativistic hydrodynamics [14], hemodynamics [15], and ab initio electronic structure calculations [16]. LBM is a recast of the continuum fluid mechanics in a form of simplified kinetic equations for the time evolution of populations $f_i(\boldsymbol{x}, t)$ of designer particles, with the basic rules of propagation on a regular space-filling lattice and collision at the lattice nodes $\boldsymbol{x}$, at time t. The discrete kinetic equations are designed in order to reproduce, in the hydrodynamic limit, the target equations of fluid dynamics, namely the Navier-Stokes equations. Foundation of this simplified mesoscopic discrete kinetic theory is that the macroscopic dynamic of the fluid is the collective behavior of particles at the microscopic level, and that the macroscopic dynamic is basically insensitive to the details of the underlying microscopic physics [17]. This way, the employment of complicated kinetic equations like the Boltzmann equation is avoided, as well as the simulation of each particle as in the case of molecular dynamic simulations [18].

1.1 Motivation

The lattice Boltzmann method brings along many of the advantages of molecular dynamics like a clear physical picture and easy implementation, and introduces some important features which may render it preferable with respect to conventional computational fluid dynamic (CFD) methods.
First, in the LBM the convection is exact and the non-linearity is local; this important features allow to stream particles along straight lines, transferring information from one lattice node to the other without any loss of information. The non-linearity resides in the collision process only and thus is local. These properties are in contrast to conventional CFD solvers which discretize directly the NS equations: in this case the momentum is transported along material lines, way more complicated trajectories than straight

lines, due to the fact that the convection term is non-linear and non-local at the same time [5].
Next, computation of the pressure, if required, is computed in LBM through the equation of state and is thus locally available from density (and temperature). Differently, in conventional incompressible flows solvers, iterations or relaxation are required, and information about the neighbor nodes need to be collected in order to solve the Poisson equation for the pressure.
Furthermore, implementation of boundary conditions is in principle straightforward, thus rendering LBM suitable for simulations with complex geometries. From the computational point of view, the nature of the basic LBM algorithm renders it suitable for parallel implementation and high performance computing (HPC). This is mainly due to the fact that at each time step every lattice node needs the information from nearby nodes only, for the advection step, while the collision operation is local.
Lastly, the kinetic nature of the lattice Boltzmann method offers the possibility to design implicit turbulence models in the under-resolved regimes, thus avoiding the employment of turbulence models which rely on tuning parameters or ad-hoc assumptions related to a particular flow.
Despite the many advantages, realizations of simulations of practical importance with high Reynolds number and high Mach with the standard lattice Boltzmann methods were restricted, on one side due to the inability to reach low viscosities, and on the other due to the insufficient Galilean invariance and accuracy of standard lattices. The first confined the range of applications to low Reynolds numbers without enlarging the grid, while the latter prevented the possibility of increasing the flow velocity to achieve transonic and supersonic conditions. Among a large number of suggestions [19–24], the entropic LBM (ELBM) for incompressible flows [25–29], disentangled the LBM from the low Reynolds regime by restoring the second law of thermodynamics (Boltzmann's $\mathcal{H}$-theorem). This basic physical requirement rendered the method nonlinearly stable numerically.

A robust extension of the LBM to thermal and compressible flows, together with a compliant $\mathcal{H}$-theorem, is a research topic of great interest. So far, a commonly accepted genuine lattice Boltzmann model for the simulation of the Fourier-Navier-Stokes equations, in particular for high subsonic, transonic and supersonic speeds, has not been yet established. In this thesis, entropic lattice Boltzmann models for the energy conserving Navier-Stokes equations are derived and validated. In particular, three main models are presented, two for the low Mach number limit and one for subsonic, transonic and supersonic regimes.

1.2 Outline of the thesis

The structure of the thesis is organized as follows.

- Chapter 2: An introduction to the kinetic theory of gases, together with the basics of the lattice Boltzmann method for incompressible flows, is presented. Details about the entropic and the entropic multi-relaxation lattice Boltzmann models and boundary conditions are then given. Finally, an overview over the existing lattice Boltzmann models for thermal and compressible flows is provided.

- Chapter 3: The entropic lattice Boltzmann model for the simulation of compressible flows in the subsonic, transonic and supersonic regimes is presented carefully. Details about the derivation of the higher-order admissible lattices, the exact entropic equilibrium, and the two-population model for variable adiabatic exponent are provided. The hydrodynamic limit is exposed in order to prove the correct recovery of the Fourier-Navier-Stokes equations. Finally, the boundary conditions for multi-speed lattices and the compressible model are detailed.

- Chapter 4: In order to reach high Mach number flow regimes, unduly large number of lattice speeds are required, as shown in Chapter 3. This is due to the drift of the LBM moments from the Maxwell-Boltzmann target moments when the temperature deviates from the reference and the velocity deviates from the reference velocity at rest ($u = 0$). In this Chapter, optimization strategies developed to increase the achievable Mach number and temperature range are described: the notion of reference frame at rest is challenged and the lattice kinetic theory in a co-moving Galilean reference frame is constructed. All the lattice velocities are shifted in a preferential direction; still, the weights associated to each velocity remain the same of the lattice at rest. The standard propagation-collision of the LBM is retained for integer shift velocities, and the new lattices enable a higher operating window in terms of Mach number.
  Furthermore, the employment of pruned lattices together with the entropic guided equilibrium allows to reduce the computational cost and the memory overhead of the model, while at the same time increasing the accuracy at high Mach numbers.

- Chapter 5: The entropic lattice Boltzmann method for compressible flows is validated against numerous two- and three-dimensional setups, in the transonic and supersonic regime and for shock dynamics. The first part of the Chapter is dedicated to the two-dimensional simulation of the transonic and supersonic flow around a NACA airfoil and the supersonic flow around a diamond airfoil. Next, the Busemann biplane is simulated in the sub-, trans- and supersonic regimes. The two-dimensional simulations conclude with the Schardin's problem followed by the simulation of one and two vortices interacting with a shock wave. In three-dimensions, the inviscid supersonic flows field around a double cone is presented, and is followed by the simulation of the transonic inviscid flow field around a finite wing. For flows including turbulence, the simulation of compressible isotropic decaying turbulence with shocklets is exposed, and the Chapter is concluded with the simulation of turbulence-shock interaction.

- Chapter 6: Two models for the simulation of thermal flows in the quasi-incompressible or mildly compressible limit are presented. The first model is based on multi-speed lattices and can be directly derived from the compressible model of Chapter 3; with this model, heat transport and dissipation as well as viscous heating can be simulated. Moreover, it may also be employed for acoustics in the low Mach number regime. The second model is based on a two-population approach, rendering it computationally simpler and less expensive than the multi-speed approach. Similar to the multi-speed thermal model, the two-population model can be employed for the simulation of heat transport and dissipation, including viscous dissipation effects, while it is not appropriate for problems involving finite speed of sound effects, like acoustics.

- Chapter 7: In this Chapter, the simulations employed to validate both the two-population and the multi-speed thermal models are presented. For the multi-speed model, two-dimensional problems have been simulated: these include the simulation of the thermal Couette flow in the laminar regime, a plane sound wave evolution at different Mach numbers and the simulation of two-dimensional heated circular and square cylinders. The two-population model, instead, has been employed in order to simulate three-dimensional turbulent Rayleigh-Bénard convection, with rotation of the frame of reference and without.

- Chapter 8: A conjugate heat transfer model employing the two-populations model of Chapter 6 is presented and validated. In particular, the different treatment between solid and fluid domains is specified, and the boundary conditions employed in order to treat the interface between solid and fluid are derived. In the second part of the chapter, the model is validated against many two-dimensional set-ups including solid-solid heat conduction and fluid-solid conjugate heat transfer. The chapter concludes with the simulation of three-dimensional conjugate heat transfer problem of a heated matrix of cubes immersed in a turbulent flow.

- Chapter 9: The algorithm for the block grid refinement in lattice Boltzmann methods is briefly reminded, and the rescaling of population in case of the entropic thermal and compressible models is derived. The grid refinement algorithm is validated with the simulation of the turbulent Rayleigh-Bénard convection and the turbulent flow field around a heated sphere in case of the thermal two-population model, while the simulation of the viscous supersonic flow field around a NACA0012 airfoil is employed to validate the rescaling of the populations in case of the compressible model.

Chapter 2

Lattice kinetic theory

2.1 Introduction

The lattice Boltzmann (LB) method (LBM) finds its roots in classical kinetic theory, and can be viewed as a discretization of the well known Boltzmann equation in combination with a few subtle modifications. This Chapter is thus introduced by providing an overview of the main features of the Boltzmann equation and the basic principles of statistical mechanics. The lattice Boltzmann method is then introduced, together with the entropic variants and the boundary conditions necessary for the simulation of real scenarios. The Chapter concludes by providing an overview over the existing models for the simulation of thermal and compressible flows with LBM.

2.2 The Boltzmann equation

The Boltzmann equation, or Boltzmann transport equation (BTE) was derived by Ludwig Boltzmann in 1872, and describes the statistical distribution and evolution of the particles in a moderately rarified gas. The dynamic of the Boltzmann equation may be used to predict the evolution of mass, momentum and energy of flows, or to derive the characteristic transport properties of a fluid, like viscosity, thermal conductivity, or even electrical conductivity for charged particles. The Boltzmann equation is an integro-differential equation for the time evolution of the one-particle distribution function $f(\boldsymbol{x}, \boldsymbol{c}, t)$, which is defined as the probability of finding a particle at position $\boldsymbol{x}$, with velocity vector $\boldsymbol{c}$ at time t, and is of the form

$$\partial_t f + c_\alpha \partial_\alpha f + G_\alpha \partial_{c_\alpha} f = \mathcal{Q}(f, f), \tag{2.1}$$

where G_α is the force acting on the particle per unit mass, and $\mathcal{Q}$ is the collision integral. The second term on the left hand side accounts for the free flight of the particles and the third for the external forces acting on the system, while on the right, the collision integral represents collision between two particles.
The collision mechanisms in Eq. (2.1) was derived by Boltzmann under the assumption of molecular chaos, or "Stosszahlansatz", which asserts the absence of correlation between two particles entering in binary collision. Under this assumption the collision term can be written as an integral over the particle velocities

$$\mathcal{Q} = \iiint \mathcal{W}(\boldsymbol{c}', \boldsymbol{c}_1' \mid \boldsymbol{c}, \boldsymbol{c}_1)\left[f(\boldsymbol{c}')f(\boldsymbol{c}_1') - f(\boldsymbol{c})f(\boldsymbol{c}_1)\right] d\boldsymbol{c}_1 d\boldsymbol{c}' d\boldsymbol{c}_1', \tag{2.2}$$

where $\mathcal{W}(\boldsymbol{c}', \boldsymbol{c}_1' \mid \boldsymbol{c}, \boldsymbol{c}_1)$ is the probability that, as a result of the binary collision, a pair of particles with initial velocity $\boldsymbol{c}$ and $\boldsymbol{c}_1$ changes its velocity to $\boldsymbol{c}'$ and $\boldsymbol{c}_1'$. The probability $\mathcal{W}(\boldsymbol{c}', \boldsymbol{c}_1' \mid \boldsymbol{c}, \boldsymbol{c}_1)$ is expressed depending on the details about intermolecular forces and underlying physics existing at the microscopic level.

For a detailed derivation of the Boltzmann equation and exhaustive discussions on the topic the reader is referred to References [30–34].

2.2.1 The collision operator

Much of the underlying physics of the Boltzmann equation, like macroscopic irreversibility or molecular details, is included in the collision term $\mathcal{Q}$. In a nutshell, this term suggests that the rate of change of free particles is only due to the loss of a pair of particles as they enter each other's sphere of influence, or as the gain of a pair of particles when they separate [31]. In the following, the main features of the collision integral are outlined.

2.2.1.1 Properties of the collision operator

The collision operator is subject to a number of important properties.

Locality As a consequence of the point particles approximation, the collision term is local in position space and non-local in particles velocity space. This is an important feature for computational purposes since in principle all information required to compute the collision operator are locally available.

Additive invariants The collision of particles, in the Boltzmann equation, is considered to be elastic. This has the consequence, that prior and after the collision, the number of particles density (mass) as well as the momentum vector and the total energy, are conserved. This can be expressed as

$$\int \left(\chi + \boldsymbol{\zeta} \cdot \boldsymbol{c} + \lambda \boldsymbol{c}^2\right) \mathcal{Q}(f, f) d\boldsymbol{c} = 0, \tag{2.3}$$

where χ, $\boldsymbol{\zeta}$ and λ are arbitrary constant scalars or vector respectively. This is an important feature of the Boltzmann equation, since it dictates that local changes in time of the hydrodynamic fields are a consequence of the free flight of particles only, i.e. they are merely due to a redistribution in space of the probability functions.

Zero point of the collision The condition of detailed balance [32], applies for equation

$$\mathcal{Q}(f, f) = 0, \tag{2.4}$$

and reads

$$f(\boldsymbol{c}_1) f(\boldsymbol{c}_2) = f(\boldsymbol{c}) f(\boldsymbol{c}'), \tag{2.5}$$

and it is valid for almost all velocities. Solution to Eq. (2.4) may be employed in perturbative schemes to solve the Boltzmann equation, as for example in the Chapman-Enskog method, as the lowest-order approximation to the distribution function. Condition Eq. (2.4) is satisfied by a class of functions of the form

$$f = \exp\left(\chi + \boldsymbol{\zeta} \cdot \boldsymbol{c} + \lambda \boldsymbol{c}^2\right). \tag{2.6}$$

In this class of functions, there exist a physically relevant solution, known as local Maxwellian distribution, parametrized by values of locally conserved quantities, like mass density $\rho(f)$, momentum $\rho \boldsymbol{u}(f)$ and total energy $\rho E(f) = (D/2)\rho(k_B/m)T(f) + 1/2\rho \boldsymbol{u}^2$:

$$f^{eq}(\boldsymbol{c},\rho,\boldsymbol{u},T)=\rho\left(\frac{m}{2\pi k_B T}\right)^{D/2}\exp\left(-\frac{m(\boldsymbol{c}-\boldsymbol{u})^2}{2k_B T}\right), \tag{2.7}$$

where D is the dimension of the physical space, m is the particles mass, T is the local temperature and k_B is the Boltzmann's constant.

Local entropy production inequality A further important property of the collision integral is Boltzmann's local entropy production inequality

$$\sigma(\boldsymbol{x},t)=-k_B\int \mathcal{Q}(f,f)\ln f d\boldsymbol{c}\geq 0, \tag{2.8}$$

for any distribution function f, assuming the integral exists. The equality holds if and only if f is the zero of the collision integral; in particular, it holds for the Maxwellian Eq. (2.7). Therefore, the collision integral describes the relaxation of the distribution function f toward the local maxwellian f^{eq}.

2.2.2 $\mathcal{H}$-theorem

The $\mathcal{H}$-theorem was introduced by Boltzmann in 1872 along with the Boltzmann equation, and it is one of the most outstanding contributions to statistical mechanics, providing a quantitative measure of the macroscopic irreversibility arising from the reversible microscopic behavior. Moreover, the Boltzmann equation extended the concept of entropy from equilibrium systems to non-equilibrium systems. Macroscopic irreversibility is measured by an increase in the entropy, which in turn is measured through the $\mathcal{H}$-function, defined as

$$\mathcal{H}(f)=\int f\ln f d\boldsymbol{c}. \tag{2.9}$$

The local $\mathcal{H}$-theorem states that the change in time of the $\mathcal{H}$-function, independent of spatial position, is proportional to the entropy production

$$\partial_t\mathcal{H}=-\frac{1}{k_B}\sigma. \tag{2.10}$$

For the case of space-dependent distribution functions, the local entropy density is defined as

$$S(\boldsymbol{x},t)=-k_B\mathcal{H}(\boldsymbol{x},t), \tag{2.11}$$

and obeys the entropy balance equation

$$\partial_t S(\boldsymbol{x},t) + \partial_\alpha J^s_\alpha(\boldsymbol{x},t) = \sigma(\boldsymbol{x},t), \tag{2.12}$$

where $J^s(\boldsymbol{x},t) = -k_B \int \boldsymbol{c} f(\boldsymbol{x},t) \ln f(\boldsymbol{x},t) d\boldsymbol{c}$ is the entropy flux. For a detailed discussion on the topic the reader is referred to Reference [32].

2.2.3 Models for the collision integral

The non-linear nature of the collision integral in the Boltzmann equation poses non-trivial mathematical complications; for practical purposes, this term requires simplifications or modeling techniques. Several approaches, known as *kinetic models*, have been proposed in the literature in order to model the collision term. The main purpose of these models is to retain the basic characteristics of the Boltzmann equation with the minimal loss of physical characteristics due to the simplifications. In the following, few of these models satisfying the criteria reported in previous subsections are reported.

2.2.3.1 Bhatnagar-Gross-Krook model

The most popular kinetic model for the collision, which does not violate the $\mathcal{H}$-theorem, is the Bhatnagar-Gross-Krook model (BGK) [35], which reads

$$\mathcal{Q}_{BGK} = -\frac{1}{\tau}\left[f - f^{eq}(\boldsymbol{c}, \rho(f), \boldsymbol{u}(f), T(f))\right]. \tag{2.13}$$

Here $\tau > 0$ is the relaxation time parameter, interpreted as a characteristic time for the relaxation toward local Maxwellian, which is of the same order of magnitude of the time between particles collisions [36]. The BGK collision model is still non-linear, but the linearity is of “lower dimension” as compared to the Boltzmann collision term. The BGK collision, in fact, is a non-linear function of the lower order moments of the distribution function f (zeroth, first and second order) only, while the original Boltzmann collision term is a function of the entire distribution function f (and then of its infinite number of moments). The main drawback of the BGK model is that the viscosity ν and the thermal conductivity α_{th} of the modeled gas cannot be adjusted independently, since both are related to the same relaxation time τ.

2.2.3.2 Fokker-Planck model

Another non-linear model for the collision integral is the Fokker-Planck model proposed in [37]:

$$\mathcal{Q}_{FP} = D_1 \left(\partial_{\boldsymbol{c}} \cdot \partial_{\boldsymbol{c}} f + \frac{1}{RT} \partial_{\boldsymbol{c}} \cdot (\boldsymbol{c} - \boldsymbol{u}) f \right). \tag{2.14}$$

This collision integral has the form of the self-consistent Fokker-Planck operator, describing diffusion in the velocity space. The diffusion coefficient $D_1 > 0$ may depend on the distribution function f. Also for the collision integral $\mathcal{Q}_{FP}$, the properties of the Boltzmann collision term and BGK model are preserved.

2.2.3.3 Generalized BGK models

There exist a class of kinetic models which generalized the BGK model to include quasi-equilibrium states, leading to a general form of the BGK type [38]. The quasi-equilibrium distribution function $f^*(\{\mathcal{M}(f)\})$ for the set of linear functionals $\{\mathcal{M}(f)\}$ is a distribution function which minimizes the $\mathcal{H}$-function under fixed values of $\{\mathcal{M}\}$. The generalized BGK collision integral can be written in the form

$$\mathcal{Q}_{QE} = -\frac{1}{\tau} \left[f - f^*(\{\mathcal{M}(f)\}) \right] + \mathcal{Q}_{Boltz} \left(f^*(\{\mathcal{M}(f)\}), f^*(\{\mathcal{M}(f)\}) \right), \tag{2.15}$$

where the second term on the right hand side, $\mathcal{Q}_{Boltz}$, denotes the value of the Boltzmann collision integral evaluated on the quasi-equilibrium manifold. This last term can be further simplified similar as in the BGK collision, and allows to build collision models with independent viscosity and thermal conductivity; these are employed to changed the Prandtl number in the compressible and thermal models exposed below. For further discussions about quasi-equilibrium and generalized BGK models the reader is referred to Reference [38].

2.2.4 Properties of the local equilibrium

The k^{th} order moment of the distribution function f can be written in general as:

$$\mathcal{M}_k(\boldsymbol{x}, t, f) = \int \underbrace{\boldsymbol{c}\boldsymbol{c}\ldots\boldsymbol{c}}_{k\text{-times}} f(\boldsymbol{x}, t, \boldsymbol{c}) d\boldsymbol{c}, \tag{2.16}$$

with k-factors of velocities $\boldsymbol{c}$. The lower order moments are the hydrodynamic fields, namely the particles mass density ρ, the particles momentum $\rho\boldsymbol{u}$ and the particles total energy ρE:

$$\int\{1,\boldsymbol{c},\boldsymbol{c}^2\}f(\boldsymbol{x},t,\boldsymbol{c})d\boldsymbol{c}=\{\rho(\boldsymbol{x},t),\rho\boldsymbol{u}(\boldsymbol{x},t),\rho E(\boldsymbol{x},t)\}=\{\rho,\rho\boldsymbol{u},\rho E\}, \tag{2.17}$$

and are often referred as the fields subject to conservation laws. The local equilibrium distribution, or local Maxwellian, defined by Eq. (2.7), has a number of important properties:

1. The first few moments of f^{eq}, i.e. the locally conserved quantities as mass density, momentum and total energy, are the same as its own parameter quantities $\rho(f)$, $\rho\boldsymbol{u}(f)$ and $\rho E(f)$:

$$\int\{1,\boldsymbol{c},\boldsymbol{c}^2\}f^{eq}(\boldsymbol{c},\rho,\boldsymbol{u},T)d\boldsymbol{c}=\{\rho(f),\rho\boldsymbol{u}(f),\rho E(f)\}. \tag{2.18}$$

2. When $f=f^{eq}$:

$$\mathcal{Q}(f^{eq},f^{eq})=0. \tag{2.19}$$

3. When $f=f^{eq}$:

$$\sigma=0. \tag{2.20}$$

4. The minimizer of the Boltzmann's $\mathcal{H}$-function subject to the local conservation laws (2.17) is the local Maxwellian (2.7).

2.2.5 Methods for reduced description

The problem of the reduced description arises when trying to find a set of closed equations for the hydrodynamic variables starting from the Boltzmann equation. Few non-trivial tasks are in order: from the choice of the proper macroscopic variables of interest, to the closure problem, i.e. the appearance of higher-order moments in the equations for the lower order moments. For the present case, a natural choice for the macroscopic variables of interest is for the conserved quantities like mass, momentum and energy. In order to provide a reduced description of the Boltzmann equation, several methods have been proposed in the past; among them, the Hilbert method, the Chapman-Enskog method, the Grad moments method, the method of invariant manifold and the quasi-equilibrium approximation are few of the most important.

2.2.5.1 The Hilbert method

A first approach to the problem of reduced description was proposed by David Hilbert in 1911 [39]. He introduced the notion of normal solutions,

$$f_H(\boldsymbol{c}, \rho(\boldsymbol{x}, t), \boldsymbol{u}(\boldsymbol{x}, t), T(\boldsymbol{x}, t)), \tag{2.21}$$

that is, solutions to the Boltzmann equation which depend on space and time only through the five hydrodynamic fields (2.17). The starting point for the method is the Boltzmann equation written in a singularly perturbed form

$$D_t f = \frac{1}{\epsilon}\mathcal{Q}(f, f), \tag{2.22}$$

where ϵ is a small parameter identified with the Knudsen number, the ratio between the mean free path of the molecules between collisions, and the characteristic scale variation of the hydrodynamic fields, and

$$D_t f \equiv \partial_t f + c_\alpha \partial_\alpha f. \tag{2.23}$$

In the Hilbert method, one seeks for the perturbative solution to the Hilbert expansion

$$f_H = \sum_{i=0}^{\infty} \epsilon^i f_H^{(i)}, \tag{2.24}$$

satisfying (2.22) order by order. Hilbert was able to demonstrate that this is formally possible; substituting (2.24) into (2.22), and matching various order in ϵ, the following sequence of integral equations is obtained:

$$\mathcal{Q}(f_H^{(0)}, f_H^{(0)}) = 0, \tag{2.25}$$

$$L_Q f_H^{(1)} = D_t f_H^{(0)}, \tag{2.26}$$

$$L_Q f_H^{(2)} = D_t f_H^{(1)} - 2\mathcal{Q}(f_H^{(0)}, f_H^{(1)}), \tag{2.27}$$

and so on for higher orders. Here L_Q is the linearized collision integral. From Eq. (2.25), it follow that $f_H^{(0)}$ is the local Maxwellian with undetermined parameters. Such parameters can be found by applying solvability condition to Eq. (2.26)

$$\int \{1, \boldsymbol{c}, \boldsymbol{c}^2\} D_t f_H^{(0)} d\boldsymbol{c} = 0, \tag{2.28}$$

resulting in the compressible Euler equations, providing then the parameters for the Maxwellian $f_H^{(0)}$. Hilbert was able to demonstrate that this procedure of constructing the normal solution can be carried out to arbitrary order n, where the distribution function $f_H^{(n)}$ is determined by the solvability condition at the next order. However, implementation of the Hilbert method requires solution to the macroscopic fields $\rho(\boldsymbol{x},t)$, $\boldsymbol{u}(\boldsymbol{x},t)$, and $T(\boldsymbol{x},t)$, obtained by solving a series of partial differential equations. Although possible theoretically, this a difficult task in practice. For more details concerning the Hilbert method the reader is refereed to Reference [32].

2.2.5.2 The Chapman-Enskog method

An alternative approach to reduced description was developed in 1917 by David Enskog [40] and independently by Sidney Chapman [36], and is known as the Chapman-Enskog method. The key innovation of their method was to seek an expansion of the time derivatives of the hydrodynamic variables rather than seeking the time-space dependencies of these function, as done in the Hilbert's method. In turn, this means that instead of looking for a perturbative solution of the Boltzmann equation in the neighborhood of local equilibrium, in this method, the equation of motion itself is expanded into slow and fast components.
In the Chapman-Enskog method one also starts with the singularly perturbed Boltzmann equation, and with the expansion

$$f_{CE} = \sum_{n=0}^{\infty} \epsilon^n f_{CE}^{(n)}. \tag{2.29}$$

However, the procedure of evaluation of the functions $f_{CE}^{(n)}$ differs from the Hilbert method. The first step is again based on the lowest order approximation

$$\mathcal{Q}(f_{CE}^{(0)}, f_{CE}^{(0)}) = 0, \tag{2.30}$$

which has the local Maxwellian as its solution. Next, the condition that the hydrodynamic variables as defined by the function f_{CE} coincide with the parameters of the local Maxwellian $f_{CE}^{(0)}$ requires

$$\begin{aligned}
&\int \{1, \boldsymbol{c}, \boldsymbol{c}^2\} f_{CE}^{(0)} d\boldsymbol{c} = \{\rho(\boldsymbol{x},t), \rho\boldsymbol{u}(\boldsymbol{x},t), \rho E(\boldsymbol{x},t)\}, \\
&\int \{1, \boldsymbol{c}, \boldsymbol{c}^2\} f_{CE}^{(k)} d\boldsymbol{c} = 0, \qquad k \geq 1.
\end{aligned} \tag{2.31}$$

The time derivative is then expanded as

$$\partial_t f_{CE} = \sum_{k=0}^{\infty} \sum_{r=0}^{k} \epsilon^k \partial_t^{(r)} f_{CE}^{(k-r)}, \tag{2.32}$$

where the meaning of this operator expansion is fixed by the solvability obtained by taking the moments of the kinetic equation. As an example, at first order one has:

$$L_Q f_{CE}^{(1)} = -\mathcal{Q}(f_{CE}^{(0)}, f_{CE}^{(0)}) + \partial_t^{(0)} f_{CE}^{(0)} + c_\alpha \partial_\alpha f_{CE}^{(0)}, \tag{2.33}$$

so that, by taking the first hydrodynamic moments of this equation, the time derivative operator $\partial_t^{(0)}$ is defined as

$$\partial_t^{(0)}\{\rho, \rho\boldsymbol{u}, \rho E\} \equiv -\int \left\{m, m\boldsymbol{c}, \frac{m\boldsymbol{c}^2}{2}\right\} c_\alpha \partial_\alpha f_{CE}^{(0)} d\boldsymbol{c}. \tag{2.34}$$

which correspond to the Euler equations.
The first correction $f_{CE}^{(1)}$ of the Chapman-Enskog method corresponds to dissipative hydrodynamic, where viscous momentum transfer, heat transfer and heat dissipation are in the Fourier-Navier-Stokes form. The Chapman-Enskog method was the first real success of the Boltzmann equation, recovering the macroscopic Fourier-Navier-Stokes equations from a microscopic description, and providing transport coefficients in terms of microscopic parameters.

However, higher-order corrections of the Chapman-Enskog method, as for example the Burnett or the super-Burnett description, face severe difficulties from both the theoretical as well as from the practical point of view [32, 33]. In particular, they result in unphysical instabilities of the equilibrium.

2.2.5.3 The Grad moment method

The idea of normal solutions employed previously by Hilbert and Chapman-Enskog, based on the time scales separation, was extended further by Grad, in 1949 [30]. His main idea, was based on the assumption that a finite number of lower order moments $\{\mathcal{M}'\}$, including the hydrodynamic moments and a subset of higher-order moments, evolves significantly slower than the rest of the higher-order moments $\{\mathcal{M}''\}$. It follows, that for times larger than a characteristic time t', the moments $\{\mathcal{M}''\}$ are uniquely defined by a combination of the lower order moments $\{\mathcal{M}'\}$, and that the distribution function is determined by the dynamics of the lower order moments $\{\mathcal{M}'\}$. This division of moments as fast and slow may be formulated as a finite order expansion of the distribution functions in terms of Hermite polynomials:

$$f_{Grad}(\{\mathcal{M}'\}, \boldsymbol{c}) = f^{eq}(\boldsymbol{c}, \rho, \boldsymbol{u}, \rho E)\left[1 + \sum_{i=0}^{k} \mathcal{A}^{(i)}(\{\mathcal{M}'\}) H^{(i)}(\boldsymbol{c} - \boldsymbol{u})\right], \tag{2.35}$$

where $H^{(i)}(\boldsymbol{c}-\boldsymbol{u})$ are various Hermite tensor polynomials, orthogonal with respect to the weight f^{eq}, while the coefficients of the expansion $\mathcal{A}^{(i)}(\{\mathcal{M}'\})$ are known functions of the slow moments $\{\mathcal{M}'\}$, and k is the highest order of $\{\mathcal{M}'\}$.

The evolution of the slower moments is obtained by substituting Eq. (2.35) into the Boltzmann equation and finding the moments of the resulting expression; this takes the name of Grad's moment equations. A well known example of this procedure is the Grad's thirteen moments approximation, where the set of slow moments $\{\mathcal{M}'\}$ includes the five conserved hydrodynamic moments, five components of the traceless part of the pressure tensor $\mathcal{P}_{\alpha\beta} = \int f[(c_\alpha - u_\alpha)(c_\beta - u_\beta) - (\boldsymbol{c}-\boldsymbol{u})^2/3\delta_{\alpha\beta}]d\boldsymbol{c}$, and the three components of the heat flux vector $q_\alpha = \int f[(c_\alpha - u_\alpha)(\boldsymbol{c}-\boldsymbol{u})^2/2]d\boldsymbol{c}$.
The validity of the decomposition of motions hypothesis cannot be evaluated a priori, so it is not surprising that the Grad's method failed to work in situations where it was believed to, primarily in phenomena with sharp time-space dependence such as strong shock waves. On the other hand, it succeeded in other situations, as for example in describing the transition between parabolic and hyperbolic propagation, and, in general, in situations slightly deviating from the classical Fourier-Navier-Stokes equations [41]. Finally, the Grad method played an important role in the foundation of phenomenological non-equilibrium thermodynamics based on a hyperbolic first-order equation, the so-called EIT (extended irreversible thermodynamic) [42, 43].

2.2.5.4 The method of invariant manifolds

A further generalization to the methods of Hilbert, Chapman-Enskog and Grad is the method of invariant manifold. The general objective of this method is to construct manifolds Ω such that the solution to the Boltzmann equation with any initial condition on the manifolds Ω remains on the same manifolds during time evolution. In order to avoid problems related to the positivity of the distribution functions and slowly converging series, the method relies on iterations rather than expansions.
In this method, for given values of macroscopic variables $\{\mathcal{M}'\}$, the microscopic distribution function f is not uniquely defined. The first step is to define a trial sub-manifold $\Omega \subset \Lambda^{(f)}$, where $\Lambda^{(f)}$ is the space of distribution functions, such that for the given $\{\mathcal{M}'\}$, the microscopic distribution function f is uniquely specified in Ω. For given values of $\{\mathcal{M}'\}$ the mapping $f^*(\{\mathcal{M}'\})$ depends only on $\mathcal{H}(f)$. In general, the trial manifold is not dynamically invariant.

The formal definition of the invariant manifold is the following: a manifold is called dynamically invariant if the vector field of the dynamic system is tangent to the manifold in every point, i.e., if Ω is a smooth manifold in the space of distribution function, and if f_Ω is an element of Ω, then Ω is invariant with respect to the dynamical system

$$\partial_t f = J(f), \quad \text{if } J(f_\Omega) \in T\Omega, \text{ for all } f_\Omega \in \Omega, \tag{2.36}$$

where $T\Omega$ is the tangent bundle of the manifold Ω. The invariance condition for a trial manifold Ω reads

$$P_\Omega(J(f_\Omega)) - J(f_\Omega) = 0, \tag{2.37}$$

where P_Ω is an arbitrary projector onto the tangent bundle of Ω. The invariance condition is considered as an equation which is solved iteratively, starting with an initial approximation Ω_0. A good example for an initial approximation Ω_0 is the manifold constructed from the quasi-equilibrium distributions [44].

2.2.5.5 The quasi-equilibrium approximation

The quasi-equilibrium approximation, also known as *entropic Grad method* or *maximum entropy method* or *generalized canonical ensemble*, is an important generalization of the Grad's method, see References [45, 46]. The quasi-equilibrium distribution function for a set of slow moments $\{\mathcal{M}'\}$ maximizes the entropy density S for fixed $\{\mathcal{M}'\}$. The quasi-equilibrium manifold $\Omega^*(\{\mathcal{M}'\})$ is the collection of quasi-equilibrium distribution functions for all admissible values of $\{\mathcal{M}'\}$. The quasi-equilibrium approximation is the simplest and extremely useful implementation of the hypothesis about a decomposition of motions in fast and slow modes. In the fast mode, the values of the slow macroscopic variables are frozen while all other variables equilibrate in a way consistent with the frozen value of macroscopic variables. In the slow mode, the macroscopic variables changes according to their evolution equation. A simple example of the quasi-equilibrium approximation is the generalized Gaussian function for $\{\mathcal{M}'\} = \{\rho, \boldsymbol{u}, \mathcal{P}\}$, with $\mathcal{P}$ the pressure tensor.

The quasi-equilibrium approximation provides (as anticipated) a good guess to the invariant manifold in many cases (see [47]) and, unlike the Grad or the Chapman-Enskog method gives a thermodynamic approximation, i.e. the distribution function is always positive and thus the approximate $\mathcal{H}$-function is a well defined convex functional, and the $\mathcal{H}$-theorem is valid for the approximation too. However, in general there is no guarantee that if one starts with a quasi-equilibrium distribution function as an initial condition, the distribution function will remain quasi-equilibrium distribution during the time evolution. This means, that the quasi-equilibrium distribution functions are not invariant.

2.2.6 Kinetic simulation algorithms

In order to solve the Boltzmann equation, numerical simulations are mostly required. The main challenge resides in the evaluation of the collision integral, which is a five dimensional integral in the case of the original Boltzmann equation. In order to circumvent this difficulty, two main approaches are followed: the first employs statistical evaluation of the collision term using the Monte-Carlo method, while the second is a deterministic method based on the solution of discrete model equations for the distribution function.

2.2.6.1 Direct simulation Monte-Carlo

The direct simulation Monte-Carlo (DSMC) is a statistical method for solving the full Boltzmann equation, developed in the in the 1960's by Graeme Bird [48]. This approach is based on a representation of the Boltzmann gas by a number of particles, whose dynamic is modeled by two processes: the free propagation, treated in a deterministic way, and the collision, treated in a statistical fashion. The modeling of collisions uses a random choice of two particles inside a cell in the space, and changes the velocity of these pairs in such a way as to comply with the conservation laws, and in accordance with the kernel of the Boltzmann collision integral. This method is widely employed in rarified gases simulations, see e.g. Reference [49].
The decoupling between advection and collision employed in the DSMC put a physical restriction on the time step used in the method. This restriction arises from the fact that the interval between two consequent collisions must be of the order of the collision interval. This put considerable computational effort in the calculation of the collision integral. In actual simulations, in order to reduce the computational effort, each simulated particle represents a large number of molecules; this has the drawback of introducing statistical errors. The computational cost of the method scales with N, the number of simulated particles, whereas statistical error is inversely proportional to the square root of N.
While this method is quite successful in describing the behavior of dilute gases in the limit of high Knudsen and Mach numbers, it faces enormous

problems in the simulation of turbulence and micro-flows in the limit of low Knudsen and Mach number, due to the large computational cost and slow convergence for this regime.

2.2.6.2 Discrete velocity models

Employment of the Boltzmann equation for hydrodynamic problems of engineering or research interest has been sought for long. However, a number of important simplifications needs to be incorporated in order to obtain an efficient solver. Such simplifications must retain the most important features of the Boltzmann equation while at the same time reducing the huge dimensionality of the problem.
Earlier discrete velocity models were successfully employed mostly in the description of shock waves profiles [50–52]. Only with the advent of the lattice-gas (LG) model [53], the idea of reproducing the Navier-Stokes hydrodynamic from overwhelmingly simplified discrete kinetic equations was considered. The method was consistent with the $\mathcal{H}$-theorem, thus was unconditionally stable, and the equilibria were non-polynomial, derived as maximizers of the Fermi-Dirac entropy. However, due to the lack of Galilean-invariance, computational complexity of the collision and too large statistical noise, the method could not become a useful tool for hydrodynamic simulations. The stepping stone to practical realizations was a pre-averaging of the LG model, via a Boltzmann formulation, leading to the lattice Boltzmann method [54, 55]. This way, the noise problem was solved, but the computational complexity was left over. However, this led to a number of sub-sequential improvements [55–60]. In the following, an overview of the lattice Boltzmann method for incompressible flows, together with its entropic variants and boundary conditions is presented.

2.3 Lattice Boltzmann method

In the lattice Boltzmann method, the continuous velocity space is discretized and only a minimum number of discrete velocities $\boldsymbol{v}_i$ is retained:

$$V_{n_Q} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_{n_Q}\}. \tag{2.38}$$

Here V_{n_Q} is the velocity-set composed of n_Q discrete velocity vectors, with a corresponding discrete population f_i each:

$$\boldsymbol{f} = \{f_1, f_2, \dots, f_{n_Q}\}. \tag{2.39}$$

The lattice Boltzmann equation for each discrete population f_i is found as follow. The starting point is the discretized (in terms of velocity space) Boltzmann equation for the populations f_i, with BGK collision

$$\partial_t f_i + v_{i\alpha}\partial_\alpha f_i = -\frac{1}{\tau}\left(f_i - f_i^{eq}\right), \tag{2.40}$$

which is integrated along characteristics $x_\alpha(s) = x_{0\alpha} + v_{i\alpha}\delta t s$ and $t(s) = t_0 + \delta t s$ for $s \in [0,1]$, where δt is the time step:

$$\begin{aligned} f_i(\boldsymbol{x}+\boldsymbol{v}_i\delta t, t+\delta t) - f_i(\boldsymbol{x},t) = -\frac{\delta t}{2\tau}\,&[(f_i(\boldsymbol{x},t) - f_i^{eq}(\boldsymbol{x},t) \\ &+(f_i(\boldsymbol{x}+\boldsymbol{v}_i\delta t, t+\delta t) - f_i^{eq}(\boldsymbol{x}+\boldsymbol{v}_i\delta t, t+\delta t))]\,. \end{aligned} \tag{2.41}$$

At this point the change of variable $f_i = af_i' + (1-a)f_i'^{eq}$ is performed and, by taking $a = \frac{2\tau}{2\tau+\delta t}$, the lattice Boltzmann BGK equation (LBGK) is recovered:

$$f_i(\boldsymbol{x}+\boldsymbol{v}_i\delta t, t+\delta t) - f_i(\boldsymbol{x},t) = \omega(f_i^{eq} - f_i), \tag{2.42}$$

where the left hand side is the shift operator representing advection of populations, and the right hand side is the collision, with $\omega = \frac{2\delta t}{2\tau+\delta t}$; the time step is often chosen as $\delta t = 1$.
At this point, what remains to be specified is the discrete velocity-set V_{n_Q} and the equilibrium populations f_i^{eq}. In the standard LBGK model for incompressible flows, the velocity-set is defined as the nodes of the third-order, D-dimensional Gauss-Hermite quadrature [61, 62], and each node is associated to the weight of the quadrature W_i. Each velocity-set is associated to a lattice designation of the form DdQn_Q, where d is the dimension. In Table 2.1, the lattices for the standard LBGK model, designed D2Q9 and D3Q27 for two- and three-dimensions respectively, and the associated discrete velocities and weights are reported.

In the standard formulation, equilibrium populations of Eq. (2.42) are written in a form of polynomials

$$f_i^{eq} = \rho W_i\left(1 + \frac{u_\alpha v_{i\alpha}}{T_0} + \frac{u_\alpha u_\beta\left(v_{i\alpha}v_{i\beta} - T_0\delta_{\alpha\beta}\right)}{2T_0^2}\right), \tag{2.43}$$

where $T_0 = c_s^2 = 1/3$ is the reference temperature, characteristic of the lattice, with c_s the characteristic speed of sound. The viscosity is related to the relaxation parameter by

V_i	W_i
D2Q9	
$\{(0,0)\}$	$4/9$
$\{(\pm 1,0),(0,\pm 1)\}$	$1/9$
$\{(\pm 1,\pm 1)\}$	$1/36$
D3Q27	
$\{(0,0,0)\}$	$8/27$
$\{(\pm 1,0,0),(0,\pm 1,0),(0,0,\pm 1)\}$	$2/27$
$\{(\pm 1,\pm 1,0),(\pm 1,0,\pm 1),(0,\pm 1,\pm 1)\}$	$1/54$
$\{(\pm 1,\pm 1,\pm 1)\}$	$1/216$

Table 2.1: Discrete velocities and corresponding weights for the incompressible LBM in two- and three-dimensions.

$$\nu = \left(\frac{1}{\omega} - \frac{1}{2}\right) T_0. \tag{2.44}$$

As anticipated in the Introduction, Chap. 1, the LBGK model as it is, suffers from instability problems in the case of under-resolved simulations. In order to remedy this problem, the entropic lattice Boltzmann model has been proposed.

2.3.1 Entropic lattice Boltzmann method

In the entropic lattice Boltzmann method (ELBM), the compliance with the $\mathcal{H}$-theorem, which was lost during the discretization process and construction of the equilibria, is re-established [25–28]. Two steps are followed in the ELBM in order to ensure the $\mathcal{H}$-theorem. The first is by defining consistently the local equilibrium f_i^{eq}; this is achieved by minimizing the discrete H-function

$$H = \sum_{i=1}^{n_Q} f_i \ln\left(\frac{f_i}{W_i}\right), \tag{2.45}$$

under constraint of conservation of mass and momentum (and total energy in the case of energy conserving models),

$$\rho = \sum_{i=1}^{n_Q} f_i, \quad \rho u_\alpha = \sum_{i=1}^{n_Q} v_{i\alpha} f_i, \tag{2.46}$$

leading to:

$$f_i^{eq} = \rho W_i \prod_{\alpha=1}^{D} \left(2 - \sqrt{1+3u_\alpha^2}\right) \left(\frac{2u_\alpha + \sqrt{1+3u_\alpha^2}}{(1-u_\alpha)}\right)^{v_{i\alpha}}, \tag{2.47}$$

for the lattice described in Tab. 2.1. The second step consists in ensuring the discrete time $\mathcal{H}$-theorem. This is done by first writing the lattice Boltzmann equation in an alternative way:

$$f_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) = f_i' = (1-\beta) f_i(\boldsymbol{x}, t) + \beta f_i^{mirr}(\boldsymbol{x}, t), \tag{2.48}$$

where f_i' is the post-collision state, β is the relaxation parameter related to viscosity, and $f_i^{mirr}(\boldsymbol{x}, t)$ is the mirror state, which in the case of the LBGK model reads

$$f_i^{mirr} = 2f_i^{eq} - f_i. \tag{2.49}$$

In ELBM, the mirror state is modified as

$$f_i^{mirr} = \alpha f_i^{eq} - (\alpha - 1) f_i, \tag{2.50}$$

where α is the entropic estimate, which is found by satisfying the entropic condition

$$H(f + \alpha(f^{eq} - f)) = H(f). \tag{2.51}$$

For more information about the entropic lattice Boltzmann method the reader is referred to References [63–70].

2.3.2 Entropic multi-relaxation lattice Boltzmann method

More recently, a new entropic variation of the lattice Boltzmann method, called entropic multi-relaxation lattice Boltzmann model, or KBC (Karlin-Bösch-Chikatamarla) model, has been introduced [29, 71, 72]. Differently from ELBM, this model is developed in the framework of the multi-relaxation-time (MRT) class of lattice Boltzmann models. This class of models, is based on the observation that, for the majority of lattice Boltzmann models, the discrete kinetic system has a higher dimensionality with respect to what

would be strictly needed to recover hydrodynamic equations. Those higher-order moments are then relaxed independently to increase stability without influencing the macroscopic dynamics. However, in MRT models, the question on how to relax the higher order moments arises; usually, relaxation times for higher-order moments are found by stability analysis or simply by tuning the parameters for a specific flow condition, thus, in general, MRT models are not parameter-free. In contrast to other MRT models, the KBC suggests that the rate at which those moments should be relaxed is determined by maximizing the entropy. Here, the main features of the model are remained.
In three-dimensions, the set of natural moments of populations f_i is given by

$$\rho M_{pqr} = \sum_{i=1}^{n_Q} v_{ix}^p v_{iy}^q v_{iz}^r f_i, \tag{2.52}$$

where, for the case of the D3Q27 lattice $p, q, r \in \{0, 1, 2\}$, and the first $D+1$ (or $D+2$ in case of energy conservation) moments are the conservation laws. This moment representation Eq. (2.52), spans a basis in which the populations can equivalently be expressed; following [29], populations f_i may be then decomposed into three main parts as

$$f_i = k_i + s_i + h_i, \tag{2.53}$$

where k_i is the kinematic part, which depends only on the locally conserved fields; s_i is the shear part, and depends on the stress tensor $\mathbf{\Pi} = \sum_{i=1}^{n_Q} \boldsymbol{v}_i \otimes \boldsymbol{v}_i f_i$ and optionally on other non-conserved fields; finally, h_i includes a linear combination of the remaining higher-order moments. The various moment representations are discussed in [71]. By employing populations decomposition (2.53), the mirror state (2.49) can be modified as follow

$$f_i^{mirr} = k_i + (2s_i^{eq} - s_i) + ((1-\gamma)h_i + \gamma h_i^{eq}), \tag{2.54}$$

where s_i^{eq} and h_i^{eq} are s_i and h_i evaluated at equilibrium. Here γ is a parameter which determines the relaxation of the higher-order moments; in the case of $\gamma_{LBGK} = 2$ the LBGK model is recovered, while for $\gamma = 1$ all higher-order moments are relaxed to equilibrium, as done for example in [22]. It should be noted, that for any choice of γ, the resulting LBM still recovers hydrodynamics with the same kinematic viscosity ν as for LBGK model (2.44). In KBC, it is required that the parameter γ maximizes the entropy (2.45) of the post-collision state f_i'; this leads to a non-linear equation for γ of the form

$$\sum_{i=1}^{n_Q} \Delta h_i \ln\left(1 + \frac{(1-\beta\gamma)\Delta h_i - (2\beta - 1)\Delta s_i}{f_i^{eq}}\right) = 0, \tag{2.55}$$

where $\Delta s_i = s_i - s_i^{eq}$ and $\Delta h_i = h_i - h_i^{eq}$. Note that, in this way, the defined *entropic stabilizer* γ appears not as a tunable parameter but rather it is computed on each lattice site at every time step from Eq. (2.55). Thus, the entropic stabilizer self-adapts to a value given by the local maximum entropy condition. By expanding Eq. (2.55) in Taylor series to the first order in Δs_i and Δh_i, and by defining the entropic scalar product

$$\langle X|Y\rangle = \sum_{i=1}^{n_Q} \frac{X_i Y_i}{f_i^{eq}}, \tag{2.56}$$

an estimate for the entropic stabilizer can be found:

$$\gamma = \frac{1}{\beta} - \left(1 - \frac{1}{\beta}\right) \frac{\langle \Delta s_i | \Delta h_i \rangle}{\langle \Delta h_i | \Delta h_i \rangle}. \tag{2.57}$$

It has been shown in [71] that the KBC models tend to the LBGK limit $\gamma_{LBGK} = 2$ in fully resolved simulations.

2.3.3 Boundary conditions

In order to simulate real flows configurations, a number of boundary conditions is needed to reproduce the presence of a wall, the inlet or the outlet of the domain and so on. Among the vast literature available on the topic concerning LBM [69, 73–81], one main type of boundary conditions is here considered for incompressible flows, due to the good results demonstrated and the compliance with the entropic formulation [82, 83]. Together, the domain boundary conditions for inflow and outflow are also shortly reminded.

2.3.3.1 Grad's boundary conditions

Grad's boundary conditions [79, 82, 83] are applied to those nodes for which some populations are missing or corrupted due to the presence of a wall. For each boundary node $\boldsymbol{x}_b$, the populations $f_i(\boldsymbol{x}_b, t)$ pointing from the solid to the fluid are missing and belong to the subset $\overline{D}$; they are unknown and need to be reconstructed as best as possible in order to complete the advection step. The point of departure for the reconstruction of the population is to find the so called "target values" for the hydrodynamic field, i.e. ρ_{tgt} and

$\boldsymbol{u}_{tgt}$ the target density and velocity respectively. For the target density, the general expression, valid also for moving boundaries, is

$$\rho_{tgt} = \rho_{bb} + \rho_s, \tag{2.58}$$

where

$$\rho_{bb} = \sum_{i\in\overline{D}} f_i^{bb} + \sum_{i\notin\overline{D}} f_i, \tag{2.59}$$

and

$$\rho_s = \sum_{i\in\overline{D}} 6W_i\rho_0\boldsymbol{v}_i \cdot \boldsymbol{u}_{w,i}, \tag{2.60}$$

where the first contribution ρ_{bb} denotes the no-flux density implied by the standard bounce-back boundary condition, for which the unknown distribution functions are replaced by the reflected populations as $f_i^{bb} = \tilde{f}_i$ with $\tilde{\boldsymbol{v}}_i = -\boldsymbol{v}_i$. The second contribution ρ_s is needed in case of moving boundaries, and accounts for the mass swept by the wall at a velocity $\boldsymbol{u}_{w,i}$; for more information see Reference [82].

The target velocity, on the other hand, is obtained by an averaged interpolation between the wall velocity $\boldsymbol{u}_{w,i} = \boldsymbol{u}_w(\boldsymbol{x}_{w,i}, t)$ at the intersection point $\boldsymbol{x}_{w,i}$ (between wall and missing population link), and the velocities $\boldsymbol{u}_{f,i} = \boldsymbol{u}_f(\boldsymbol{x}_{f,i}, t)$ at the neighbor fluid point $\boldsymbol{x}_{f,i} = \boldsymbol{x}_b + \boldsymbol{v}_i$ for $i \in \overline{D}$. For an averaged linear interpolation this leads

$$\boldsymbol{u}_{tgt} = \frac{1}{n_{\overline{D}}} \sum_{i\in\overline{D}} \frac{q_i\boldsymbol{u}_{f,i} + \boldsymbol{u}_{w,i}}{1+q_i}, \tag{2.61}$$

where $n_{\overline{D}}$ is the number of unknown populations, q_i is the distance from the wall along direction $\boldsymbol{v}_i$ normalized by the norm of $\boldsymbol{v}_i$, $q_i = \frac{\|\boldsymbol{x}_b - \boldsymbol{x}_{w,i}\|}{\|\boldsymbol{v}_i\|}$. This ensures correct wall velocity as well as it accounts for the momentum exerted from a moving object onto the fluid.

Finally, the approximation of the population is performed by including a contribution of the equilibrium part plus a non-equilibrium part evaluated at target conditions:

$$f_i^{miss} \simeq f_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}) + f_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, S_{\alpha\beta}), \tag{2.62}$$

where $f_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt})$ is the equilibrium Eq. (2.43) or Eq. (2.47) evaluated at target conditions, and $f_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, S_{\alpha\beta})$ is the non-equilibrium part also evaluated at target conditions, and employing the stress-tensor $S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha$. Evaluation of the gradient is performed using a semi-second order scheme. Approximation for the non-equilibrium part of the populations employs the Grad's approximation of the form

$$f_i^{(1)} = W_i \frac{P_{\alpha\beta}^{(1)}(v_{i\alpha}v_{i\beta} - T_0\delta_{\alpha\beta})}{2T_0^2}, \tag{2.63}$$

with $P_{\alpha\beta}^{(1)}$ the non-equilibrium pressure tensor

$$P_{\alpha\beta}^{(1)} = -\frac{1}{2\beta}\rho T_0 S_{\alpha\beta}, \tag{2.64}$$

derived from Chapman-Enskog analysis [82].

2.3.3.2 Domain boundary conditions

Usually, three types of domain boundary conditions are employed: inlet, outlet, and free-slip. In the following, each is shortly described.

Inlet For inlet boundary conditions, the density and velocity are usually prescribed, i.e. ρ_{tgt} and $\boldsymbol{u}_{tgt}$, thus all the populations of the nodes belonging to the inlet are simply set to equilibrium:

$$f_i = f_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}). \tag{2.65}$$

Outlet For outlet boundary conditions, density and velocity are usually not prescribed; for this case an often employed approach is to replace the missing population by their value at previous time step:

$$f_i^{miss} = f_i^{prev}. \tag{2.66}$$

Free-slip The free-slip condition is often used to mimic an open domain which should theoretically represent an "infinite" space. This is enforced by simply reflecting the populations with respect to the wall normal.

2.4 Energy conserving lattice Boltzmann models

Simulating thermal and compressible flows with the lattice Boltzmann method reduces, at least theoretically, to incorporate energy conservation in the collision process. This is however not a simple task; from one point of view, standard lattices suffer from limited Galilean invariance when the flow speed is increased after a certain threshold, while from the other point of view the number of lattice speeds of the standard model is simply to low to accommodate all the moments required to recover the Fourier-Navier-Stokes equations (this is imposed by the closure relation [84]). For this reason, energy conserving lattice Boltzmann models still remain an open issue, and a large number of propositions can be found in the literature, some more successful than the others. Before exposing the approaches employed in this work to ensure energy conservation in the lattice Boltzmann method, in the following, an overview over the main existing models on the topic is provided.

2.4.1 Correction terms models

One of the approaches employed to incorporate energy conservation in the standard LBM introduces correction terms [85, 86]; these are needed, as anticipated, due to the lack of sufficient isotropy of these standard lattices [87]. The general lattice Boltzmann equation for this model reads

$$f_i(\boldsymbol{x} + \boldsymbol{v}_i, t + 1) - f_i(\boldsymbol{x}, t) = \omega(f_i^{eq} - f_i) + \Psi_i + \Phi_i, \tag{2.67}$$

where the term Ψ_i is used to correct the spurious terms in the momentum equation, while the term Φ_i is used to correct those of the energy equation. The main drawback of this method is that it requires evaluation of gradients of velocity and temperature at each lattice site. While being suitable for thermal flows and also for moderately high Mach flows, such an approach of evaluating correction terms, similar to DNS solvers, requires highly non-dissipative schemes. This may increase the complexity of the code in the presence of walls and shock waves. Also, such an approach surrenders the main advantage of LB methods, namely locality of computations, which affects not just the computational efficiency and simplicity, but also in part the accuracy of the method.

2.4.2 Two-population models

A second way followed for the simulation of thermal flows with LBM is the two-population model, or double-distribution approach. In this method, the employment of correction terms is avoided by the employment of two sets of partially or fully coupled populations instead of just one. The first lattice is used for mass and momentum computations (recovering continuity and momentum equations), as in the case of isothermal LBM, while the other lattice is used for the temperature dynamics. To this end, the same models as for the isothermal LBM presented in Sec. 2.3 may be employed, while the second population is written in the form

$$g_i(\boldsymbol{x} + \boldsymbol{v}_i, t + 1) - g_i(\boldsymbol{x}, t) = Q_i^g, \tag{2.68}$$

where g_i are the populations designated for the energy equation, and Q_i^g is the corresponding collision term, which depends on the employed collision model. The first model proposed for the simulation of thermal flows with two populations was presented in [88]; in this case, the conservation law is

$$\rho E^{int} = \rho C_v T = \sum_{i=1}^{n_Q} g_i - \frac{1}{2} \sum_{i=1}^{n_Q} f_i r_i, \tag{2.69}$$

while the collision operator is of the form

$$Q_i^g = \omega_g (g^{eq} - g_i) + (\omega_g/2 - 1) f_i r_i, \tag{2.70}$$

where r_i is a non-local term containing the gradient of pressure and Jacobian of velocity, and g_i^{eq} is the appropriate equilibrium. The non-locality increases the complexity of the model, and, in particular, physical effects such as viscous heating do not emerge naturally and need to be introduced by forcing terms.

More recently, total energy was chosen instead of the internal energy as the conserved quantity on the second lattice [89, 90] to avoid non-locality of computations. Despite the small difference between the two models [89] and [90], in general, the conservation law and the collision term of the second lattice can be written as

$$2\rho E^{tot} = 2c_v T + \rho \boldsymbol{u}^2 = \sum_{i=1}^{n_Q} g_i, \tag{2.71}$$

$$Q_i^g = \omega_g (g^{eq} - g_i). \tag{2.72}$$

respectively. However, this approach still cannot be used for simulations of compressible flows at moderate Mach numbers or acoustics owing to the lack of sufficient isotropy of the first lattice. This drawback is partially eliminated in [91, 92] by introducing correction terms similar to Reference [87]; however the main disadvantages of the correction terms methods [85–87] continue to persist.

2.4.3 Multi-speed models

In the third approach, the introduction of correction terms and the use of a second set of populations is avoided by using lattices with more discrete velocities (multi-speed or higher-order lattices), required to adequately represent the moment system necessary to obtain the full thermal Fourier-Navier-Stokes equations. The lattice Boltzmann equation for these models is the same as for isothermal models (for fixed Prandtl number), Eq. (2.42), where only the set of velocities V_{n_Q} and the equilibrium are different. Early attempts in obtaining higher-order lattices were based on discretizing the Maxwell-Boltzmann distribution on the roots of Hermite polynomials [62]. While this route of higher-order Gauss-Hermite quadratures promised a systematic derivation of new complete Galilean-invariant LB models, since the roots of Hermite polynomials are irrational, the corresponding discrete velocities cannot be fitted into a regular space-filling lattice. Thus, one of the most important advantages of the LB methods, the exact space discretization of the advection step, is lost with the Gauss-Hermite quadrature-based off-lattice models. Regular space-filling lattices were obtained based on a quadrature representation of the equilibrium moments not necessarily Gauss-Hermite [93–98], where the nodes of the quadrature are chosen empirically in such a way as to reflect higher-order isotropy. However, none of these derivations could prove to construct an accurate numerical scheme without employing any special stabilization technique. Stabilization techniques like equilibration, for example, partially succeeded in the simulation of compressible flows as in [99]. An alternative way of deriving space-filling higher-order lattices was set by the development of the theory of admissible higher-order lattices [100, 101], which enabled systematic derivation of lattices of any size, being at the same time entropy supporting and guaranteeing the stability intrinsic to the lattice. First application of admissible higher-order lattices for turbulent flow simulations was demonstrated in [102].

In addition to the difficulties related in finding accurate and stable velocity-sets, this type of lattice require a more elaborate treatment of the boundary conditions.

2.4.4 Further compressible models

While all previous formulations were limited to low Mach numbers, some LB models succeeded in the simulation of shocks dynamics, in part by scarifying the turbulent characteristics of the flow field [103–108]. This was achieved by still employing multi-speed or two-population lattices as the previously described methods, but with the additional employment of some finite volumes or finite differences schemes for spatial discretization, and/or some Runge-Kutta schemes for the time integration, instead of the conventional exact advection of the standard LBM method. This brings along the effect of introducing additional uncontrolled numerical diffusion in the discretization scheme, and thus should be avoided for accurate simulations of turbulent flows.

Chapter 3

Entropic lattice Boltzmann method for compressible flows

3.1 Introduction

Predicting the behavior of compressible flows is crucial in many scientific and technological applications, ranging from magnetohydrodynamic and astrophysics, including interstellar shockwaves [109–111] and Solar corona magneto-convection [112, 113], to high-speed aerodynamics [114–119], turbomachinery [120–122], jet engines [123, 124], high-speed entry into planetary atmosphere [125, 126], abrasive blasting [127], and many others. Despite their widespread applications, predicting the time-space evolution of compressible flows remains a challenge, even more than for incompressible flows; incompressible and compressible flows, in fact, are stunningly different [128], with turbulence on one end and shock waves on the other. While the entire range is represented by the fundamental Fourier-Navier-Stokes equations, at least until moderate supersonic Mach numbers, the different physics of the two extremes (low and high Mach numbers) has led to the divergent paths that numerical simulations have long followed, each with different properties. The lack of a uniform approach which would bridge these two limits in a natural way is in fact well recognized [2, 3]. Standard discretization used for smooth flows can cause potentially dangerous Gibbs oscillations, leading to unstable schemes in the presence of shock waves, whereas typical methods used to regularize shock calculations exhibit excessive numerical viscosity [2]. Therefore, the main approaches to high-speed compressible flows are hybrid schemes, which typically combine higher-order methods with weighted essentially nonoscillatory schemes (WENO) through shock sensors, which determines which scheme should be used. A current alternative to the latter are nonlinear artificial viscosity methods, employing higher-order accurate schemes with additional diffusive terms introducing dissipation that adaptively adjust to the local regularity of the solution [2]. However, both approaches are characterized by complicated algorithms that may rely on tuning parameters dependent on the setup under consideration. This precludes the development of efficient, high-fidelity, robust direct numerical simulation solvers that can seamlessly treat any type of compressible turbulent flow (subsonic, transonic, and supersonic). Thus, high-fidelity reliable DNS of compressible or high Mach number flows still remain a challenge for the state-of-the art numerical methods.

On the other side, while the success of LB methods has expanded greatly over the past two decades to porous media, multiphase flows, and thermal incompressible flows, all these applications are confined to the low Mach number subdomain of fluid dynamics [4, 5]. The situation is drastically different for compressible flows, where numerous attempts to derive a genuine lattice Boltzmann model have failed to date (see Sec. 2.4.4). Earlier attempts were heavily reliant on tuning parameters or artificial dissipation arising from inexact propagation, or the introduction of correction terms and similar ad hoc techniques. With all this, lattice Boltzmann models for compressible flows drop short of a quantitative comparison to direct numerical simulations, unlike in the case of low Mach numbers.

In this Chapter, it is shown that the gap between the incompressible and compressible flow computations is covered with a single ELBM model [129, 130]. The model is capable of simulating a wide range of laminar and turbulent flows, from thermal and weakly compressible flows to transonic and supersonic flows and with the presence of arbitrarily complex objects. Three fundamental changes to the lattice Boltzmann scheme are introduced here for the development of such a model: the use of a properly chosen multispeed lattice, the accurate entropic equilibrium, and the entropic relaxation for the collision. In contrast to state-of-the-art fluid solvers, the algorithm employs Cartesian meshes without any turbulence model, tuning parameters, or tracking of the shock front.

The chapter is organized as follow: in Sec. 3.2 the lattice Boltzmann equation employed to simulate compressible flows with variable Prandtl number is exposed. Next, in Sec. 3.3 the construction of higher order lattices and an appropriate lattice choice for compressible simulations are presented. In Sec. 3.4 the accurate entropic equilibrium construction and related details are exposed, while in Sec. 3.5 the two-population model enabling variable adiabatic exponent is discussed. The thermo-hydrodynamic limit of the compressible LBM is derived in Sec. 3.6, the entropic estimate is treated in Sec. 3.7 and implementation details of the model are provided in Sec. 3.8. After a discussion on the wall boundary conditions for compressible LBM and, more in general, for multi-speed lattices, in Sec. 3.9, the chapter is closed by drawing the conclusions in Sec. 3.10.

3.2 Compressible lattice Boltzmann equation

Fundamental point of the compressible entropic lattice Boltzmann model is the conventional lattice Boltzmann equation, with the collision term modeled with the quasi-equilibrium model [38, 41, 131, 132]:

$$
\begin{aligned}
f_i(\boldsymbol{x}+\boldsymbol{v}_i, t+1) - f_i(\boldsymbol{x}, t) &= 2\beta_1(f_i^* - f_i) + 2\beta_2(f_i^{eq} - f_i^*) && (3.1)\\
&= 2\beta_1(f_i^{eq} - f_i) + 2(\beta_1 - \beta_2)\left[f_i^* - f_i^{eq}\right]. && (3.2)
\end{aligned}
$$

The left hand side (lhs) represents the standard streaming of populations and the right hand side (rhs) represents the collision. On the rhs of Eq. (3.1) two relaxation processes are represented: the first term is the relaxation to the quasi-equilibrium state f_i^*, while the second term is the subsequent relaxation to the equilibrium f_i^{eq}. The two independent relaxation parameters β_1 and β_2 are related to kinematic viscosity ν and thermal diffusivity α_{th}, depending on the Prandtl number Pr. Equation (3.1) may be rewritten in the form of Eq. (3.2), where the first term represents the LBGK collision, and the second one the deviation due to the quasi-equilibrium. It is evident that when $\beta_1 = \beta_2$ the collision term reduces to the LBGK collision form Eq. (2.42). The conservation laws, in the compressible model, additionally to the conservation of mass and momentum, need to satisfy also the total energy conservation, and reads

$$
\{\rho, \rho\boldsymbol{u}, 2\rho E^{tr}\} = \sum_{i=1}^{n_Q} \{1, \boldsymbol{v}_i, \boldsymbol{v}_i^2\} f_i(\rho, \boldsymbol{u}, T), \tag{3.3}
$$

with the translational energy defined as

$$2\rho E^{tr} = 2\rho C_v^{tr} T + \rho \boldsymbol{u}^2, \tag{3.4}$$

where $C_v^{tr} = D/2$ is the specific heat of an ideal monoatomic gas at constant volume in D-dimensions.
Following [131], the quasi-equilibrium populations f_i^* are required to satisfy the conservation of mass, momentum and total energy, as for the equilibrium populations,

$$\{\rho, \rho\boldsymbol{u}, 2\rho E^{tr}\} = \sum_{i=1}^{n_Q} \{1, \boldsymbol{v}_i, \boldsymbol{v}_i^2\} f_i^*, \tag{3.5}$$

and, depending on the Prandtl number, an additional quantity. If $\mathrm{Pr} < \mathrm{Pr}_{\mathrm{LBGK}}$, the additionally conserved quantity is the centered heat flux tensor

$$\overline{Q}_{\alpha\beta\gamma} = \sum_{i=1}^{n_Q} (v_{i\alpha} - u_\alpha)(v_{i\beta} - u_\beta)(v_{i\gamma} - u_\gamma) f_i \tag{3.6}$$

$$= \sum_{i=1}^{n_Q} (v_{i\alpha} - u_\alpha)(v_{i\beta} - u_\beta)(v_{i\gamma} - u_\gamma) f_i^*, \tag{3.7}$$

while if $\mathrm{Pr} > \mathrm{Pr}_{\mathrm{LBGK}}$, the additionally conserved quantity is the traceless centered pressure tensor

$$\overline{\Theta}_{\alpha\beta} = \sum_{i=1}^{n_Q} \left[(v_{i\alpha} - u_\alpha)(v_{i\beta} - u_\beta) - \frac{2}{D} \partial_\gamma u_\gamma \delta_{\alpha\beta} \right] f_i \tag{3.8}$$

$$= \sum_{i=1}^{n_Q} \left[(v_{i\alpha} - u_\alpha)(v_{i\beta} - u_\beta) - \frac{2}{D} \partial_\gamma u_\gamma \delta_{\alpha\beta} \right] f_i^*, \tag{3.9}$$

where $\mathrm{Pr}_{\mathrm{LBGK}}$ is the fixed Prandtl number in the case of $LBGK$.
Such a distinction between the cases of Prandtl number higher or lower than $\mathrm{Pr}_{\mathrm{LBGK}}$ is necessary to ensure compliance with the entropy principle for the collision (see [131] for more details). This way, by knowing the equilibrium distribution, the quasi-equilibrium populations can be written:

$$f_i^* = f_i^{eq} + W_i \frac{\left(\overline{Q}_{\alpha\beta\gamma} - Q_{\alpha\beta\gamma}^{eq}\right)(v_{i\alpha} v_{i\beta} v_{i\gamma} - 3 v_{i\gamma} T \delta_{\alpha\beta})}{6T^3}, \qquad \text{if } \mathrm{Pr} \leq 1, \tag{3.10}$$

$$f_i^* = f_i^{eq} + W_i \frac{\left(\overline{\Theta}_{\alpha\beta} - P_{\alpha\beta}^{eq}\right)(v_{i\alpha} v_{i\beta} - T \delta_{\alpha\beta})}{2T^2}, \qquad \text{if } \mathrm{Pr} > 1. \tag{3.11}$$

In the sequel, two key steps are employed to construct the compressible entropic lattice Boltzmann model. First, a proper choice of the discrete velocities $\boldsymbol{v}_i$ will be made in Sec. 3.3. Second, evaluation of the equilibrium f_i^{eq} for high Mach numbers will be described in Sec. 3.4.

3.3 Lattices for the compressible entropic lattice Boltzmann method

One of the basic features of the compressible ELBM is the employment of multi-speed, or higher-order, lattices. The derivation of multi-speed lattices is based, in this work, on the theory of admissible higher-order lattices presented in [100, 101] and on the later work [84]. Before presenting the higher-order lattices employed in the compressible model, an overview of the main steps in the derivation of these lattices is provided in the following.

From now on, the standard nomenclature DdQn^d for the lattices with $n_Q = n^d$ speeds in D-dimensions is considered. The hierarchy of admissible higher-order lattices is constructed, in higher dimensions, as the direct tensor product of D-copies of one-dimensional velocity-sets V_n, composed by n velocities, where n is considered to be odd only (lattices with even number discrete velocities can also be employed [101] but are not considered here due to simplicity). Discrete velocities $v_i \in V_n$ are chosen to be integer valued only, in order to fit into a lattice, i.e. $v_i = i,\ i \in \mathbb{Z}$. For the time being, the basic mirror symmetry of V_n is assumed; as a consequence, if $v_i \in V_n$ then also $v_{-i} \in V_n$, and thus rest particles with $v_0 = 0$ are always included. Operator $\langle \dots \rangle$ stands for summation over discrete velocity index i for the discrete case f_i, or integration over the continuos velocity space v in the continuous case f; thus $\rho = \langle f_i \rangle = \langle f \rangle$. Discrete velocities are chosen in such a way as to reproduce as much as possible the moments of the Maxwell-Boltzmann distribution function ($m/k_B = 1$)

$$f_v^M = \rho\sqrt{\frac{1}{2\pi T}}\exp\left(-\frac{(v-u)^2}{2T}\right), \tag{3.12}$$

which can be written, for order k, as

$$\rho M_k^M(u,T) = \left\langle v^k f_v^M \right\rangle. \tag{3.13}$$

Once the velocity-set is chosen, one of the most important information regarding the lattice, the reference temperature T_0, is first revealed. The

reference temperature is the temperature at which most of the possible (depending on the lattice size n) Maxwell moment relations are verified. Its derivation is done with the help of the closure relation and the matching condition [84], described in the following in two steps.
The first step consist in deriving the closure relation proper of the velocity-set V_n with n velocities; the closure relation consists of a linear relation between the n-th power of the discrete lattice velocities v_i^n, and the lower-order odd powers, starting with v_i^{n-2} and ending with v_i. Such a linear relation always exists, and reflects the fact that only n velocity polynomials, $1,\ v_i,\ \ldots,\ v_i^{n-1}$, are linearly independent. The closure relation is defined by writing

$$v_i^n = a_{n-2}v_i^{n-2} + a_{n-4}v_i^{n-4} + \ldots + a_1 v_i, \tag{3.14}$$

and then by substituting $(n-1)/2$ different non-zero values for the velocities v_i; finally the linear system of equations is solved for the coefficients $a_{n-2},\ \ldots,\ a_1$.
As indicatory examples, for the velocity set $V_3 = \{0, \pm 1\}$ (D1Q3), the closure relation is $v_i^3 = v_i$ (the cube of any velocity from the D1Q3 set is the velocity itself); for the velocity set $V_5 = \{0, \pm 1, \pm 3\}$ (D1Q5) it is $v_i^5 = 10v_i^3 - 9v_i$, and so on. The existence of the closure relation has a main important implication: the moment $\rho M_n = \langle f_i v_i^n \rangle$ cannot be assigned at one's will, and only the linear in u term of this moment at equilibrium can be made consistent with the corresponding Maxwell-Boltzmann value M_n^M. This is also the source of the Galilean invariance related to the standard lattice DdQ3^d, since the cubic term in the heat flux tensor (M_3^M), needed to recover the NS equations, is not correctly reproduced.

The second step in the derivation of the reference temperature is the matching condition. For example, for the D1Q3 lattice, the third-order moment $\rho M_3 = \left\langle f_i v_i^3 \right\rangle$ equals $M_3 = u$ for any population set. On the other hand, the Maxwell expression is $M_3^M = 3T_0 u + u^3$, which also contains the cubic term u^3, which cannot be made consistent with the previous expression. Only the linear term can be matched, $3T_0 u = u$, and only if the reference temperature is set to $T_0 = 1/3$. Similarly, for D1Q5, $M_5^M = 15T^2 u + O(u^3)$; thus, the matching condition for linear terms becomes $15T^2 - 30T_0 + 9 = 0$. The latter equation reveals two values of the reference temperature, $T_0 = 1 \pm 2/5$. This example also explains why the shortest D1Q5 lattice is $\{0, \pm 1, \pm 3\}$ and not $\{0, \pm 1, \pm 2\}$: for the latter, the closure relation is $v_i^5 = 5v_i^3 - 4v_i$, and T_0 is found as a solution of

$15T_0^2 - 15T_0 + 4 = 0$ which has no real-valued roots, and hence does not define any reference temperature [84]. This is also consistent with the findings in [100, 101]. In general, the closure relation to find the reference temperature is derived as follow [84]: the coefficients b_n of the linear in u term of the moment $M_n^M = b_n T^{(n-1)/2} u + O(u^3)$ (n odd) with $b_n = 1 \times 3 \times 5 \ldots \times n$ are found. Next the algebraic equation for the reference temperature,

$$b_n T^{(n-1)/2} - a_{n-2} b_{n-2} T^{(n-3)/2} - a_{n-4} b_{n-4} T^{(n-5)/2} - \ldots - a_1 = 0, \quad (3.15)$$

is solved. Positive roots (if they exist) define the reference temperature. If no positive roots are available, the corresponding lattice is ruled out of a further consideration. In Table 3.1, the higher-order lattices with $n =$ 3, 5, 7, 9, 11, together with the corresponding closure relations and reference temperatures are exposed.

n	V_n	Closure	T_0
3	$\{0, \pm 1\}$	$v_i^3 = v_i$	$1/3$
5	$\{0, \pm 1, \pm 3\}$	$v_i^5 = 10v_i^3 - 9v_i$	$1 \pm \sqrt{2/5}$
7	$\{0, \pm 1, \pm 2, \pm 3\}$	$v_i^7 = 14v_i^5 - 49v_i^3 + 36v_i$	0.697954
9	$\{0, \pm 1, \pm 2, \pm 3, \pm 5\}$	$v_i^9 = 39v_i^7 - 399v_i^5 + 1261v_i^3 - 900v_i$	0.756081, 2.175382
11	$\{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$	$v_i^{11} = 55v_i^9 - 1023v_i^7 + 7645v_i^5 - 21076v_i^3 + 14400v_i$	1.062794

Table 3.1: One-dimensional higher-order lattices with odd number of integer-valued velocities, $n =$ 3, 5, 7, 9, 11. Second column: lattice velocity set. Third column: closure relation, defining the reference temperature T_0 (fourth column) through the matching condition.

Once the lattice number of velocities has been chosen, based on accuracy considerations as outlined in Sec 3.4, and once the lattice links length has also been chosen, based on the previous considerations, the one-dimensional weights w_i of the corresponding lattice can be computed. The weights for symmetric set V_n, defined as the equilibrium populations at unity density $\rho = 1$ and zero velocity $u = 0$, are found by matching the moments of the Maxwellian Eq. (3.12) at $\rho = 1$ and $u = 0$, $f^M(0, T)$. This reduces to a linear $n \times n$ algebraic system of equations to be solved for the weights ($m = 0,\ 1,\ 2,\ \ldots,\ n-1$):

$$\langle w_i(T) v_i^m \rangle = \left\langle f^M(0,T) v^m \right\rangle . \tag{3.16}$$

Once the weights are computed, the temperature range associated to the lattice can be evaluated. The temperature range corresponds to the range in which, for a given temperature T, all the weights are positive. Together with the temperature range, the maximum to minimum temperature ratio can be evaluated; this quantity is important for simulations of practical application, since it relates the physical temperature ratio to the temperature ratio in lattice units. In Table 3.2, the temperature range and the ratio between maximum to minimum temperature are reported for the higher-order lattices with $n =\ 3,\ 5,\ 7,\ 9,\ 11$.

n	V_n	T_{min}	T_{max}	T_{max}/T_{min}
3	$\{0, \pm 1\}$	0.0	1.0	∞
5	$\{0, \pm 1, \pm 3\}$	1/3	3.0	9
7	$\{0, \pm 1, \pm 2, \pm 3\}$	$1 - \sqrt{2/5}$	$1 + \sqrt{2/5}$	4.44
9	$\{0, \pm 1, \pm 2, \pm 3, \pm 5\}$	0.69795	2.88131	4.13
11	$\{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$	0.75608	2.17538	2.88

Table 3.2: One-dimensional higher-order lattices with odd number of integer velocities, $n =\ 3,\ 5,\ 7,\ 9,\ 11$. Second column: minimal temperature T_{min}. Third column: maximal temperature T_{max}. Fourth column: temperature ratio T_{max}/T_{min}.

In higher dimensions, let's say $D = 3$, the weight W_i of each discrete velocity $\boldsymbol{v}_i$ in the natural Cartesian reference frame $\boldsymbol{v}_i = \{v_{ix}, v_{iy}, v_{iz}\}$, $i\alpha \in 0, \pm 1, \pm 2, \ldots$ is then the algebraic product of the corresponding one-dimensional weights

$$W_i = w_{ix} w_{iy} w_{iz} . \tag{3.17}$$

At this point, all the instruments to construct and employ higher-order lattices are provided. Before proceeding by detailing the equilibrium derivation, an example of multi-speed lattice is provided in the following in full detail.

3.3.1 The D2Q7^2 lattice for compressible flows

As will be demonstrated in Sec. 3.4, the minimum number of lattice velocities needed for the simulation of compressible flows with the present compressible ELBM, is $n = 7$ in one-dimension. Its higher-dimensions counterparts (the D2Q7^2 or the D3Q7^3 lattices), then, can be employed for the simulation of transonic or supersonic flows. Thereafter, the D2Q7^2 lattice is thus discussed in details, since it provides an exhaustive overview of the main features of all the higher-order lattices that can be employed for the simulation of compressible flows in two- or three-dimensions.

The base of the D2Q7^2 lattice is the velocity-set $V_7 = \{0, \pm 1, \pm 2, \pm 3\}$, so that the entire discrete velocity-set composed by velocities $\boldsymbol{v}_i$ is the tensor product of the one-dimensional velocity-set, i.e.

$$\boldsymbol{v}_i = \{v_{ix}, v_{iy}\}, \quad i\alpha \in 0, \pm 1, \pm 2, \pm 3. \tag{3.18}$$

The D2Q7^2 lattice and corresponding velocities are exposed in Fig. 3.1.

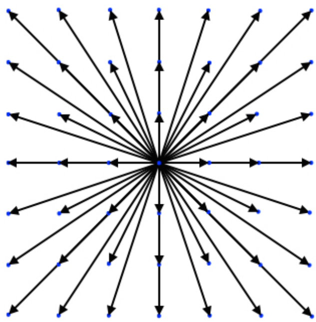


Figure 3.1: Visualization of the D2Q7^2 lattice.

Next, as anticipated by Table 3.1, the closure relation for the one-dimensional velocity-set V_7 is

$$v_i^7 = 14v_i^5 - 49v_i^3 + 36v_i, \tag{3.19}$$

implying that starting from the seventh-order moment of the DdQ7^d lattices, the higher-order moments are no more independents. From the coefficients of the closure relation (3.19), it is now possible to find the matching condition for the reference temperature:

$$105T^3 - 210T^2 + 147T - 36 = 0. \tag{3.20}$$

Only one real solution to Eq. (3.20) exists and is $T_0 = 0.697954$ as anticipated in Tab. 3.1.
Next, the one-dimensional weights for the velocity set V_7 can be derived from the system of equations (3.16), and reads:

$$w_0 = \frac{(36 - 49T + 42T^2 - 15T^3)}{36}, \qquad w_{\pm 1} = \frac{T(12 - 13T + 5T^2)}{16},$$
$$w_{\pm 2} = \frac{T(-3 + 10T - 5T^2)}{40}, \qquad w_{\pm 3} = \frac{T(4 - 15T + 15T^2)}{720}. \quad (3.21)$$

The positivity condition imposed on each weight provides the maximum and minimum temperature allowed by the lattice, together with the maximum to minimum temperature ratio, provided in Tab. 3.2.
Again, the two-dimensional weights W_i of the lattice D2Q7^2, corresponding to velocity $\boldsymbol{v}_i$, are derived as the algebraic product of the corresponding one-dimensional weights:

$$W_i = w_{ix} w_{iy}. \quad (3.22)$$

3.4 Entropic accurate equilibrium

The equilibrium computation is the second key element of the compressible entropic lattice Boltzmann model. For this reason, many details about equilibrium construction, computational cost, and comparison with other types of equilibria are exposed. Moreover, the accuracy that can be reached with the present approach, in terms of Mach number and temperature, is examined. The section is closed with further details about the equilibrium by presenting related interesting properties, in particular compared to other types of equilibria.

3.4.1 Equilibrium construction

Three mainstream approaches were followed in the literature to derive the equilibrium for the lattice Boltzmann equation. The first and most popular is the top-down approach [5], essentially based on writing the equilibrium in the form of a polynomial and then tuning the coefficients of the polynomials to recover the (Fourier-)Navier-Stokes equations [93, 133] as the limit of the LBM equation. Alternatively, the equilibrium was derived from the continuous Maxwell-Boltzmann distribution by expanding (in a Taylor series or Hermite basis) to the desired order [62, 134]. The last and most recent approach for the derivation of the equilibrium is based on entropic construction, where one first discretizes the continuous entropy function Eq. (2.9) and then computes the equilibrium as the point of minimum discrete entropy [135]. This contrasts with the second approach that first computes the equilibrium of the continuous H-function (resulting in the Maxwell-Boltzmann distribution) and then discretizes it to attain the discrete equilibrium. This discretization obviously violates the entropy maximum condition that was valid for the Boltzmann equation [136]. Hence, the entropic approach is the only possibility of having a valid equilibrium that is defined as the point of maximum entropy, thus bringing back thermodynamic consistency and stability to the LBM scheme. It is worth noting that most expressions for equilibrium in the LB literature are polynomial in nature. Also, in the case of the entropic lattice Boltzmann method, the solution of the minimization of the discrete H-function, although nonlinear, is written approximately as a polynomial in velocity (small perturbations around zero flow velocity). Polynomial equilibria suffer from two limitations, namely, limited positivity (the f_i^{eq} become negative at high Ma numbers) and inaccuracies at large values of flow velocities. Loss of positivity renders the physical nature of the populations void and is considered as one of the sources of numerical instabilities in the scheme [137]. Moreover, a polynomial that is obtained as an expanded solution to the entropic minimization problem cannot be used at large values of flow velocities, due to exponential divergence from the desired solution arising from the finite nature of the polynomials. Hence it is clear that any polynomial solution to the equilibrium cannot be used for high-Mach-number compressible flows. Thus, it is of quintessential importance that an accurate method of evaluation of equilibrium is found for any possible extension of LB methods to high-velocity flows.

Here the entropic definition of equilibrium is adopted, and the derivation is conducted by minimizing the discrete entropy function Eq. (2.45), with $W_i = W_i(T)$ the temperature-dependent weights described in Sec. 3.3. Minimization of H is carried out under the constraints of local conservation of mass ρ, momentum $\rho\boldsymbol{u}$, and translational energy $2\rho E^{tr}$ Equations. (3.3), and is solved with the method of the Lagrange Multipliers (LM), leading to:

$$\delta H + \delta\left[\chi\rho + \zeta_\alpha(\rho u_\alpha) + \lambda(2\rho E^{tr})\right] = 0, \tag{3.23}$$

where $\chi(\boldsymbol{u}, T)$, $\zeta(\boldsymbol{u}, T)$ and $\lambda(\boldsymbol{u}, T)$ are Lagrange multipliers corresponding to conservation of mass, momentum and translational energy, respectively. The formal solution to the minimization problem reads

$$f_i^{eq} = \rho W_i \exp\left[\chi + \zeta_\alpha v_{i\alpha} + \lambda v_i^2\right]. \tag{3.24}$$

The closed form of f_i^{eq} is computed by applying the conservation laws (3.3) to (3.24). This results in $D+2$ equations for $D+2$ unknown LMs. In the case of the standard lattice ($V_1 = \{0, \pm 1\}$) and isothermal flows, this leads to a quadratic system of equations, in terms of Lagrange multipliers, that was earlier solved analytically in [135]. However, for larger lattices, say, the D1Q5 and D1Q7 lattices (and the respective lattices in higher dimensions), and with energy conservation, this leads to a system of $D+2$ equations of the order $9D+3$ (for the largest velocity being $v_3 = 3$), thus ruling out any possibility of computing them with an exact analytical solution. The main problem here reduces then in finding an appealing way of computing the exact entropic equilibrium for any given velocity and temperature for any lattice. The first application of entropy supporting higher-order lattices like in [102] for isothermal flows and in [138] for thermal flows employed an expanded version around zero velocity of the LM included in the equilibrium (3.24). This version of the entropic equilibrium, however, suffers the same problem of positivity as for the conventional polynomial when the velocity is increased.
Here, a direct numerical evaluation of LMs using the rapidly converging Newton-Raphson method is suggested. The simulations shows that such an approach converges to an accurate solution (with error of order 10^{-8} on the LMs) with approximatively five iterations.
At a first glance, such a computation of the equilibrium at every node and every time-step could appear extremely computationally demanding. In the following, the numerical computation of the equilibrium and its influence on the overall computational time of the algorithm are analyzed.

3.4.2 Computational cost of numerical equilibrium

In order to estimate the performance of the accurate numerical equilibrium compared to a standard polynomial equilibrium evaluation, in the following, the computational time required to run a constant homogenous advection flow at a Mach $\mathrm{Ma_a} = 0.05$ is reported. Considered are the accurate numerical equilibrium Eq. (3.24), and a fourth order approximation of the equilibrium in polynomial form. The fourth order approximation is theoretically applicable only if n_Q is "large enough" i.e. if the lattice velocity space can accommodate all higher order moments required for recovering the compressible flow hydrodynamic equations. The fourth order approximation is given by

$$f_i^{eq} = \rho W_i \left(1 + \frac{u_\alpha v_{i\alpha}}{T} + \frac{u_\alpha u_\beta X_{i\alpha\beta}}{2T^2} + \frac{u_\alpha u_\beta u_\gamma Y_{i\alpha\beta\gamma}}{6T^3} + \frac{u_\alpha u_\beta u_\gamma u_\mu Z_{i\alpha\beta\gamma\mu}}{24T^4}\right), \tag{3.25}$$

with

$$X_{i\alpha\beta} = v_{i\alpha} v_{i\beta} - T\delta_{\alpha\beta}, \tag{3.26}$$

$$Y_{i\alpha\beta\gamma} = X_{i\alpha\beta} v_{i\gamma} - v_{i\alpha} T\delta_{\beta\gamma} - v_{i\beta} T\delta_{\alpha\gamma}, \tag{3.27}$$

$$Z_{i\alpha\beta\gamma\mu} = Y_{i\alpha\beta\gamma} v_{i\mu} - X_{i\alpha\beta}\delta_{\gamma\mu} T - X_{i\alpha\gamma}\delta_{\beta\mu} T - X_{i\beta\gamma}\delta_{\alpha\mu} T. \tag{3.28}$$

One should notice, that this approximation can be employed for the simulation of energy conserving flows, but only for moderately low Mach number flows, due to truncation errors and non-positivity of the populations, as discussed in Sec. 3.4.4.
In Table 3.3, the computational time measurements are reported normalized by the fastest evaluation, namely the polynomial evaluation on the DdQ3^d lattice, for the two different equilibrium evaluations, for the three different lattices DdQ3^d, DdQ5^d and DdQ7^d, and in two- and three-dimensions. Firstly, interesting to notice is that the computational time required by the

	DdQ3^d	DdQ5^d	DdQ7^d
$2D$			
Polynomial	1	4.12	8.13
Numerical	2.02	4.83	8.74
$3D$			
Polynomial	1	8.91	25.66
Numerical	1.58	10.83	29.77

Table 3.3: Computational time for the LBGK algorithm with evaluation of the equilibrium using a polynomial and numerical method, for three different lattices DdQ3^d, DdQ5^d and DdQ7^d, in two- and three-dimensions. Computational time is normalized by the fastest implementation, i.e. with polynomial form and on the DdQ3^d lattice.

numerical equilibrium on every lattice is not more than the double the computational time of the polynomial equilibrium. Once the number of speeds in a lattice is increased, the difference between the numerical equilibrium and the polynomial form decreases, in both two- and three-dimensions. This is explained by the fact that numerical equilibrium first computes $(D+2)$ Lagrange Multipliers, which are then employed to construct the equilibrium distributions, while for the polynomial form n_Q, fourth-order polynomials need to be computed. Table 3.3 clearly shows, also, that the computational cost of the simulation depends mainly on the number of discrete lattice velocities, while the type of equilibrium plays a secondary role. This analysis shows clearly that the employment of the accurate form of the equilibrium is not heavily penalizing the performance of the simulation with respect to a polynomial form of the equilibrium.

A general comment on the computational cost of the present model is in order here. As just seen, the computational cost of the present compressible model is one to two orders of magnitude larger (in 2 and 3 dimensions respectively) than its incompressible counterpart, the DdQ3^d. However, for compressible (or high Mach number) simulations the time step is in general one to two orders of magnitude higher than incompressible simulations, since the time step is directly proportional to the characteristic Mach number of the simulation. This implies that no significant increase of computational cost may be observed in present simulations with respect to incompressible LBM simulation (in terms of time to solution or large scale turn-over time). Further it must be pointed out that incompressible ELBM simulations were shown to be one to two order of magnitude faster than DNS simulations using state-of-the-art techniques, for both periodic flows [102] and flow in complex geometries [72].

Interesting to analyze, is also the influence of computational load with variation of the flow field (ρ, $\boldsymbol{u}$ and T) for the numerical evaluation of the equilibrium. In fact, since the Newton-Raphson solver for the equilibrium employs previous time step LM as initial guess, it is expected that a high time variability of the flow would influence the performance. The influence of the time variability of the flow field is measured by running simulations of compressible decaying isotropic turbulence similar to the simulations presented in [129], for different initial turbulent Mach numbers $\mathrm{Ma_t}$. The initial turbulent Mach number is considered here as the quantitative measure of the time variability of the flow. In Table 3.4 the influence of the time-variability of the flow is shown (in terms of $\mathrm{Ma_t}$) by measuring the computational time of the various decaying turbulence simulations. The lattice employed for the simulations is the D3Q7^3. As one can notice, the computational time for the simulation increases only slightly with increasing turbulent Mach number. In fact, the computational time is increased at maximum of 7%, and when the turbulent Mach number is quite high (the local Mach number may reach values above $\mathrm{Ma_{loc}} \simeq 2.0$). This test also includes the uneven load

Ma_t	0	0.1	0.2	0.3	0.4	0.5	0.6
t/t_0	1.000	1.047	1.023	1.059	1.062	1.070	1.067

Table 3.4: Computational time dependence on turbulent Mach number $\mathrm{Ma_t}$.

balancing (for HPC or parallel simulations) experienced in the simulation due to varying local Mach numbers, i.e. the fact that in some processors the flow field varies much less or much more than in other processors.

3.4.3 Equilibrium accuracy

In this Section, the numerical equilibrium accuracy is analyzed in terms of deviation of the moments computed by the numerical equilibrium with respect to their corresponding Maxwell-Boltzmann moments:

$$\epsilon_m = \frac{M_m^{eq} - M_m^M}{M_m^M}. \tag{3.29}$$

In Figure. 3.2, the five (D2Q5^2) and seven (D2Q7^2) speed lattices are compared for the errors in evaluation of the equilibrium heat flux $q_x^{eq} = \sum_{i=1}^{n_Q} v_{ix} v_i^2 f_i^{eq}$ and equilibrium contracted fourth order moment $R_{xx}^{eq} = \sum_{i=1}^{n_Q} v_{ix} v_{ix} v_i^2 f_i^{eq}$ when compared to their Maxwell-Boltzmann counterpart at various Mach numbers. Two types of Mach number are here defined: $\mathrm{Ma_t}$ stands for turbulent Mach number and $\mathrm{Ma_a}$ is the advection Mach number. The equilibrium is computed for a Mach number $\mathrm{Ma_a}$ in the x-direction plus a Mach number $\mathrm{Ma_t}$ superimposed in both the x- and y-directions. In the same plot, for completeness, the errors of the eleven (D2Q11^2) velocities lattice are also shown. Important to notice is that the aforementioned moments, in particular the q_x^{eq} moment, need to be recovered as accurately as possible by the kinetic model, in order to recover the correct thermo-hydrodynamic limit. One can notice that the errors in the moments are small until an advection Mach number of about $\mathrm{Ma_a} \leq 1.2$ for the lattice with a base velocity number of $n_Q = 7^d$, while for the lattice with $n_Q = 5^d$ the errors grow rapidly already in the subsonic regime. This imply that in order to perform supersonic simulations the minimum number of lattice velocities is $n_Q = 7^d$ in d-dimensions. For this reason, from now on, $n_Q = 7^d$ is considered as the minimum number of lattice velocities in order to perform compressible flow simulations. Moreover, it is important to observe

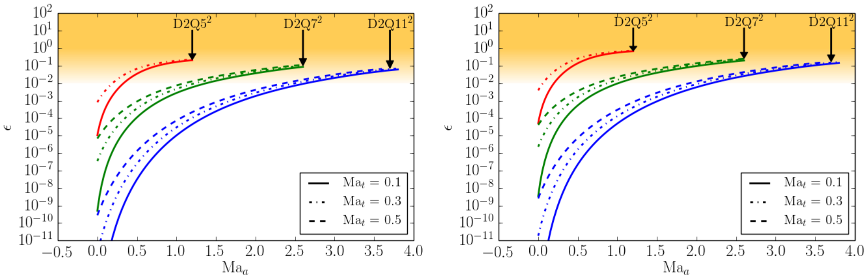


Figure 3.2: Deviation of the contracted, x-component of third order moment q_x^{eq} (left) and, xx-component of fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann counterpart values, as a function of turbulent and advection Mach number $\mathrm{Ma_t}$ and $\mathrm{Ma_a}$ respectively, for three different lattice sizes: D2Q5^2, D2Q7^2 and D2Q11^2.

that if one needs to increase the Mach number, with the entropic lattice Boltzmann model it is just required to increase the number of lattice velocities. For example, from Fig. 3.2 it is possible to observe that with the eleven speeds lattice it is possible to simulate flows where the Mach number can reach values until $\mathrm{Ma} \lesssim 2.5$, thus further validating the hierarchy of lattices proposed in [84, 100, 101].

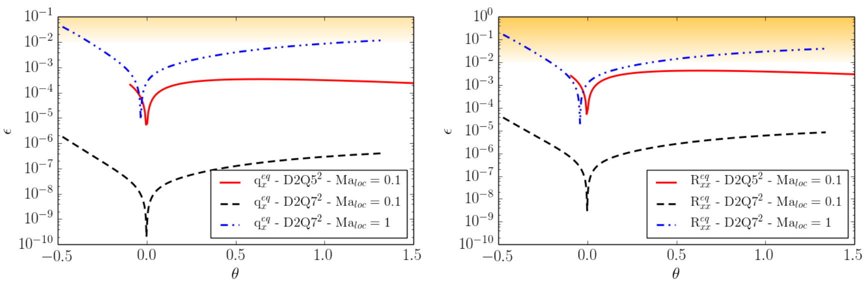


Figure 3.3: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of the reduced temperature θ for two different lattice sizes: D2Q5^2 at local Mach $\mathrm{Ma_{loc}} = 0.1$ and D2Q7^2 at local Mach $\mathrm{Ma_{loc}} = 0.1$ and $\mathrm{Ma_{loc}} = 1$.

In Fig. 3.3, the same errors as for Fig. 3.2 are plotted as a function of the reduced temperature $\theta = \frac{T - T_0}{T_0}$, and for two different lattices D2Q5^2 and

D2Q7^2 at different Mach numbers. Errors are shown within the reduced temperature range appropriate for the lattice as discussed in Sec. 3.3. First of all, one can notice, similarly as before, that by increasing the number of velocities in a lattice, the errors decrease by order of magnitudes, and the same is valid even when moving away from reference temperature. Moreover, by increasing the Mach number, the errors when deviating from reference temperature are amplified. However, considering that typical medium strength shock experience a reduced temperature jump of $\Delta\theta \simeq 0.3$, it is safe to state that even when increasing the Mach number the deviations from the equilibrium are small enough to perform supersonic flow simulations with shocks on the DdQ7^d lattice.

3.4.4 Equilibrium positivity

In order to conclude this section on accurate entropic equilibrium, a remark on an interesting property of the numerical equilibrium employed in this work is in order. For the lattice DdQ7^d discussed in Sec. 3.3, a quantitative impression of the difference between evaluating the equilibrium exactly with numerical procedure and evaluating it with polynomials is provided. In Figure 3.4, populations f_0^{eq}, f_1^{eq}, f_{-2}^{eq} and f_3^{eq} are plotted as a function of the Mach number Ma for three different ways of evaluation, for the one-dimensional case D1Q7: a third and a fourth order Grad's approximation (in terms of $\boldsymbol{u}$) of equilibrium (3.24), and the accurate numerical equilibrium (Newton-Raphson evaluation).

It is clear that as the Mach number is increased, especially after the Mach number reach the sonic value, the polynomial approximations of the populations start to diverge from the exact solution to the minimization problem, leading to an accumulation of errors. Moreover, an increase of the degree of approximation, like from a third order in velocity to a fourth order in velocity, has only a minor influence in the convergence to the correct solution. Important to notice is also, that after a certain value of the Mach number the populations evaluated with a polynomial approximation become negative, violating the positivity constraints imposed by the construction of the equilibrium, with the possibility then of leading to an unstable scheme. From Figure 3.4 is then possible to conclude that an accurate numerical evaluation of the equilibrium is not just a more precise way of evaluating the equilibrium distribution function, but it is also an unavoidable solution for compressible lattice Boltzmann simulations.

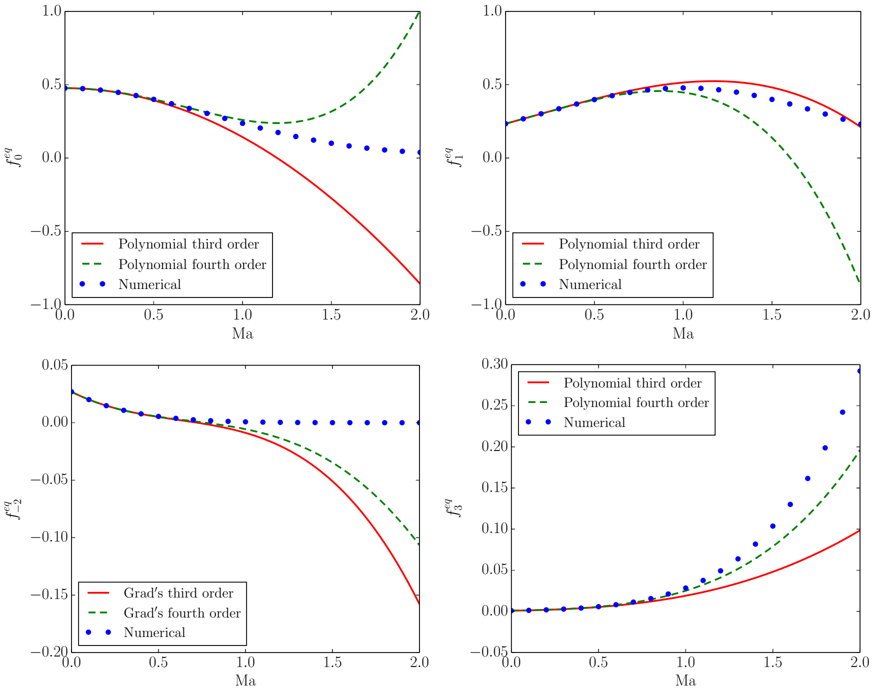


Figure 3.4: Equilibrium population f_0^{eq}, f_1^{eq}, f_{-2}^{eq} and f_3^{eq} evaluated as a function of Mach number with three different methods: third order and fourth order polynomial approximation in velocity, and exact numerical evaluation.

3.5 Two-population model for variable adiabatic exponent

The compressible ELBM as described until now imposes an adiabatic coefficient given by $\gamma_{tr} = \frac{D+2}{D}$, owing to the fact that f_i^{eq} was matched to the Maxwell-Boltzmann distribution for a monatomic gas. To change arbitrarily the adiabatic exponent γ_{ad} proper to real gases, some changes in the ELBM scheme described until now are in order.
Previous approaches employed to make variable adiabatic exponent γ_{ad}, such as in [103] or [107], are limited to models with a prescribed polynomial form of the equilibrium. Since the present way of computing equilibrium is exact and needs numerical evaluation, an alternative way is employed, fol-

lowing the original idea of [139] subsequently extended for the LBM in [140]. In this model, another set of populations g_i is introduced, which carry the energy related to the internal degrees of freedom of molecules (rotational and vibrational) and thus enabling a variable γ_{ad}. In this case, the kinetic equations are written for both f- and g-populations:

$$f_i(\boldsymbol{x}+\boldsymbol{v}_i, t+1) - f_i(\boldsymbol{x}, t) = 2\beta_1\left(f_i^{eq} - f_i\right) + 2\left(\beta_1 - \beta_2\right)[f_i^* - f_i^{eq}], \quad (3.30)$$

$$g_i(\boldsymbol{x}+\boldsymbol{v}_i, t+1) - g_i(\boldsymbol{x}, t) = 2\beta_1\left(g_i^{eq} - g_i\right) + 2\left(\beta_1 - \beta_2\right)[g_i^* - g_i^{eq}], \quad (3.31)$$

where f_i^{eq} is the same as Eq. (3.24), while the equilibrium for the second set of population can be directly set as

$$g_i^{eq} = (2C_v - D)T f_i^{eq}. \quad (3.32)$$

This way, the conservation laws for the present model become

$$\rho = \sum_{i=1}^{n_Q} f_i, \quad \rho\boldsymbol{u} = \sum_{i=1}^{n_Q} f_i \boldsymbol{v}_i, \quad 2\rho E^{tot} = \sum_{i=1}^{n_Q}\left[f_i \boldsymbol{v}_i^2 + g_i\right], \quad (3.33)$$

where

$$2\rho E^{tot} = 2C_v \rho T + \rho \boldsymbol{u}^2, \quad (3.34)$$

is the total energy for a polyatomic gas with heat capacity at constant volume $C_v = 1/(\gamma_{ad} - 1)$. The quasi-equilibrium g_i^* is defined consistently to f_i^* employing a Grad's approximation

$$g_i^* = g_i^{eq} + W_i \frac{(\overline{\boldsymbol{q}}^g - \boldsymbol{q}^{g,eq}) \cdot \boldsymbol{v}_i}{T}, \qquad \text{if Pr} \le 1, \quad (3.35)$$

$$g_i^* = g_i^{eq} + W_i\left[K - \rho(2C_v - D)T\right], \qquad \text{if Pr} > 1, \quad (3.36)$$

where

$$\overline{\boldsymbol{q}}^{\boldsymbol{g}} = \sum_{i=1}^{n_Q} (\boldsymbol{v}_i - \boldsymbol{u}) g_i \quad (3.37)$$

$$= \sum_{i=1}^{n_Q} (\boldsymbol{v}_i - \boldsymbol{u}) g_i^*, \quad (3.38)$$

is the contracted and centered heat flux tensor associated with the internal degrees of freedom, and

$$K = \sum_{i=1}^{n_Q} g_i \tag{3.39}$$

$$= \sum_{i=1}^{n_Q} g_i^*. \tag{3.40}$$

In order to test thermodynamic consistency of the two-population model for variable adiabatic exponent, the speed of sound $c_s(T)$ is here measured as a function of the temperature for two different adiabatic exponents, i.e. for the one corresponding of a monoatomic gas, $\gamma_{tr} = (D+2)/D$ with $D = 2$ and for $\gamma_{ad} = 1.4$, characteristic of air. In Figure 3.5, the speed of sound for the two different adiabatic coefficients is compared to the theory of ideal gases, $c_s(T) = \sqrt{\gamma_{ad} T}$. The comparison clearly shows that the two-

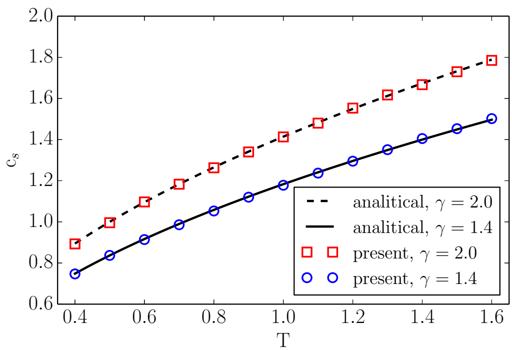


Figure 3.5: Speed of sound comparison as a function of the temperature for the two-population model and the theory, for $\gamma_{tr} = 2$ and $\gamma_{ad} = 1.4$.

population model reproduces the correct adiabatic exponent over the entire range of temperature admissible by the DdQ7^d lattice.

3.6 Thermo-hydrodynamic limit

In the following, the thermo-hydrodynamic limit of Eqs. (3.30) and (3.31) together with the conservation laws (3.33), for the case of a D-dimensional gas, is derived following [130]. The thermo-hydrodynamic limit is derived using the Chapman-Enskog method, under the assumption the second, third and fourth order Maxwell-Boltzmann moments are recovered by the equilibrium populations, as shown in Sec. 3.3. The starting point is the expansion of the shift operator of Eqs. (3.30) and (3.31) in a Taylor series till second order

$$\left[\delta t\left(\partial_t+\partial_\mu v_{i\mu}\right)+\frac{\delta t^2}{2}\left(\partial_t+\partial_\mu v_{i\mu}\right)\left(\partial_t+\partial_\nu v_{i\nu}\right)\right] f_i \\ =2\beta_1\left(f_i^{eq}-f_i\right)+2\left(\beta_1-\beta_2\right)\left[f_i^*-f_i^{eq}\right], \tag{3.41}$$

$$\left[\delta t\left(\partial_t+\partial_\mu v_{i\mu}\right)+\frac{\delta t^2}{2}\left(\partial_t+\partial_\mu v_{i\mu}\right)\left(\partial_t+\partial_\nu v_{i\nu}\right)\right] g_i \\ =2\beta_1\left(g_i^{eq}-g_i\right)+2\left(\beta_1-\beta_2\right)\left[g_i^*-g_i^{eq}\right], \tag{3.42}$$

and by introducing a characteristic time scale of the flow, Θ, and a reduced time $t'=t/\Theta$. Similarly, reduced velocities and reduced coordinates are introduced, $v_i'=v_i/c$ and $x'=x/(c\Theta)$, where $c=1$, and Eqs. (3.41), (3.42) are rewritten in terms of the reduced variables t', x', and primes are dropped to simplify notation. After introduction of the smallness parameter $\epsilon=\delta t/\Theta$, Eqs. (3.41) and (3.42) can be written as

$$\left[\epsilon\left(\partial_t+\partial_\mu v_{i\mu}\right)+\frac{\epsilon^2}{2}\left(\partial_t+\partial_\mu v_{i\mu}\right)\left(\partial_t+\partial_\nu v_{i\nu}\right)\right] f_i \\ =2\beta_1\left(f_i^{eq}-f_i\right)+2\left(\beta_1-\beta_2\right)\left[f_i^*-f_i^{eq}\right], \tag{3.43}$$

$$\left[\epsilon\left(\partial_t+\partial_\mu v_{i\mu}\right)+\frac{\epsilon^2}{2}\left(\partial_t+\partial_\mu v_{i\mu}\right)\left(\partial_t+\partial_\nu v_{i\nu}\right)\right] g_i \\ =2\beta_1\left(g_i^{eq}-g_i\right)+2\left(\beta_1-\beta_2\right)\left[g_i^*-g_i^{eq}\right]. \tag{3.44}$$

A multi-scale expansion of the time derivative operator, of the populations, and of the quasi-equilibrium f- and g-populations, can be performed to second order as,

$$\epsilon\partial_t = \epsilon\partial_t^{(1)} + \epsilon^2\partial_t^{(2)} + \ldots \tag{3.45}$$

$$f_i = f_i^{(0)} + \epsilon f_i^{(1)} + \epsilon^2 f_i^{(2)} + \ldots \tag{3.46}$$

$$g_i = g_i^{(0)} + \epsilon g_i^{(1)} + \epsilon^2 g_i^{(2)} + \ldots \tag{3.47}$$

$$f_i^* = f_i^{*(0)} + \epsilon f_i^{*(1)} + \epsilon^2 f_i^{*(2)} + \ldots \tag{3.48}$$

$$g_i^* = g_i^{*(0)} + \epsilon g_i^{*(1)} + \epsilon^2 g_i^{*(2)} + \ldots \tag{3.49}$$

Inserting Eqs. (3.45), (3.46) and (3.48) into Eq. (3.43), and Eqs. (3.45), (3.47) and (3.49) into Eq. (3.44), analysis to order $\epsilon^{(0)}$, $\epsilon^{(1)}$ and $\epsilon^{(2)}$ is performed.

Zeroth order terms Terms to the order $\epsilon^{(0)}$ lead to,

$$f_i^{(0)} = f_i^{*(0)} + \frac{\beta_2}{\beta_1}\left(f_i^{eq} - f_i^{*(0)}\right), \tag{3.50}$$

$$g_i^{(0)} = g_i^{*(0)} + \frac{\beta_2}{\beta_1}\left(g_i^{eq} - g_i^{*(0)}\right). \tag{3.51}$$

Since the quasi-equilibrium population f_i^* and g_i^* are constructed to satisfy

$$f_i^{*(0)} = f_i^{eq}, \tag{3.52}$$

and

$$g_i^{*(0)} = g_i^{eq}, \tag{3.53}$$

it is found that the leading term in the expansions (3.46) and (3.47) is the local equilibrium,

$$f_i^{(0)} = f_i^{eq}, \qquad g_i^{(0)} = g_i^{eq}. \tag{3.54}$$

Local conservation laws imply:

$$\sum_{i=1}^{n_Q}\{1, v_{i\alpha}\}f_i = \sum_{i=1}^{n_Q}\{1, v_{i\alpha}\}f_i^{eq}, \tag{3.55}$$

for mass and momentum conservation, and

$$\sum_{i=1}^{n_Q}\left[v_i^2 f_i + g_i\right] = \sum_{i=1}^{n_Q}\left[v_i^2 f_i^{eq} + g_i^{eq}\right], \tag{3.56}$$

for the total energy conservation, which, combined with Eqs. (3.46), (3.47) and (3.54) lead to the solvability conditions

$$\sum_{i=1}^{n_Q}\{1, v_{i\alpha}\}f_i^{(1)} = \sum_{i=1}^{n_Q}\{1, v_{i\alpha}\}f_i^{(2)} = ... = 0, \tag{3.57}$$

and

$$\sum_{i=1}^{n_Q}\left[v_i^2 f_i^{(1)} + g_i^{(1)}\right] = \sum_{i=1}^{n_Q}\left[v_i^2 f_i^{(2)} + g_i^{(2)}\right] = ... = 0. \tag{3.58}$$

First order terms Collecting terms to the order $\epsilon^{(1)}$, the expression for the first-order f- and g-populations is obtained:

$$f_i^{(1)} = f_i^{*(1)}\left(1 - \frac{\beta_2}{\beta_1}\right) - \frac{1}{2\beta_1}\left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) f_i^{(0)}, \tag{3.59}$$

$$g_i^{(1)} = g_i^{*(1)} \left(1 - \frac{\beta_2}{\beta_1}\right) - \frac{1}{2\beta_1} \left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) g_i^{(0)}. \tag{3.60}$$

Applying solvability conditions (3.57) to Eq. (3.59), and using the conditions

$$\sum_{i=0}^{n_Q} \{1, v_{i\alpha}\} f_i^{*(1)} = 0, \tag{3.61}$$

obtained from the definition of the quasi-equilibrium populations, the thermo-hydrodynamic equations of mass and momentum are recovered to first order:

$$\partial_t^{(1)} \rho = -\partial_\alpha \left(\rho u_\alpha\right), \tag{3.62}$$

$$\partial_t^{(1)} u_\alpha = -u_\beta \partial_\beta u_\alpha - \frac{1}{\rho} \partial_\alpha \left(\rho T\right). \tag{3.63}$$

The term ρT can be identified as $p = \rho T$, which is the equation of state for ideal gases, with p being the pressure. In order to obtain first-order equations for the total energy, the solvability condition (3.58) is applied to the sum of equations (3.59) and (3.60), while the definition of the quasi-equilibrium is used to achieve

$$\sum_{i=1}^{n_Q} \left[v_i^2 f_i^{*(1)} + g_i^{*(1)}\right] = 0, \tag{3.64}$$

so that:

$$\partial_t^{(1)} T = -u_\alpha \partial_\alpha T - \frac{1}{\rho C_v} \rho T \left(\partial_\alpha u_\alpha\right). \tag{3.65}$$

The sofar derived equations composed by (3.62), (3.63) and (3.65) are the compressible Euler equations.

Second order terms Collection of second order terms $\epsilon^{(2)}$in Eqs. (3.43) and (3.44) leads to

$$\left[\partial_t^{(2)} + \frac{1}{2} \left(\partial_t^{(1)} \partial_t^{(1)} + \partial_\mu \partial_\nu v_{i\mu} v_{i\nu} + 2\partial_t^{(1)} \partial_\mu v_{i\mu}\right)\right] f_i^{(0)}$$
$$= -2\beta_1 f_i^{(2)} - 2\left(\beta_2 - \beta_1\right) f_i^{*(2)} - \left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) f_i^{(1)}, \tag{3.66}$$

$$\left[\partial_t^{(2)} + \frac{1}{2}\left(\partial_t^{(1)}\partial_t^{(1)} + \partial_\mu\partial_\nu v_{i\mu}v_{i\nu} + 2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right] g_i^{(0)}$$
$$= -2\beta_1 g_i^{(2)} - 2\left(\beta_2 - \beta_1\right) g_i^{*(2)} - \left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) g_i^{(1)}. \tag{3.67}$$

Applying Eq. (3.59) into (3.66) and Eq. (3.60) into (3.67) and considering that $f_i^{*(2)} = g_i^{*(2)} = 0$ by construction of the quasi-equilibrium, Eqs. (3.66) and (3.67) become

$$\left[\partial_t^{(2)} - \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\left(\partial_t^{(1)}\partial_t^{(1)} + \partial_\mu\partial_\nu v_{i\mu}v_{i\nu} + 2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right] f_i^{(0)}$$
$$= -2\beta_1 f_i^{(2)} - \left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) f_i^{*(1)}\left(1 - \frac{\beta_2}{\beta_1}\right), \tag{3.68}$$

$$\left[\partial_t^{(2)} - \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\left(\partial_t^{(1)}\partial_t^{(1)} + \partial_\mu\partial_\nu v_{i\mu}v_{i\nu} + 2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right] g_i^{(0)}$$
$$= -2\beta_1 g_i^{(2)} - \left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) g_i^{*(1)}\left(1 - \frac{\beta_2}{\beta_1}\right). \tag{3.69}$$

Applying solvability condition for the mass, the second order contribution to the continuity equation vanishes,

$$\partial_t^{(2)}\rho = 0. \tag{3.70}$$

Applying solvability condition for the momentum to (3.68) and solvability condition for the total energy to (3.68)+(3.69), leads to

$$\partial_t^{(2)}\left(\rho u_\alpha\right) = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\partial_\beta\left(\partial_t^{(1)} P_{\alpha\beta}^{eq} + \partial_\gamma Q_{\alpha\beta\gamma}^{eq}\right) - \left(1 - \frac{\beta_2}{\beta_1}\right)\partial_\beta P_{\alpha\beta}^{*(1)}, \tag{3.71}$$

$$\partial_t^{(2)}\left(2\rho E^{tot}\right) = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\partial_\alpha\left(\partial_t^{(1)} q_\alpha^{eq} + \partial_\beta R_{\alpha\beta}^{eq}\right) - \left(1 - \frac{\beta_2}{\beta_1}\right)\partial_\alpha q_\alpha^{*(1)}. \tag{3.72}$$

At this point, a distinction between the cases where $\mathrm{Pr} \leq 1$ and $\mathrm{Pr} > 1$ has to be made, as a consequence of the different quasi-equilibria employed in Eq. (3.2) in order to satisfy the $\mathcal{H}$-theorem.

Case Pr ≤ 1 As anticipated, in the case where Pr ≤ 1, the additional quasi-conserved quantity by the quasi-equilibrium population are the centered heat transfer tensor, $\overline{Q}_{\alpha\beta\gamma}$, for f-populations,

$$f^* = f^* \left(\rho, u_\alpha, T, \overline{Q}_{\alpha\beta\gamma}\right), \tag{3.73}$$

while for g-populations is the contracted centered heat flux related to the rotational and vibrational degrees of freedom:

$$g^* = g^* \left(\rho, u_\alpha, T, \overline{q}^g_\alpha\right). \tag{3.74}$$

This additional conservation guarantees the following form for the first order quasi-equilibrium heat flux and the first order quasi-equilibrium pressure tensor:

$$q^{*(1)}_\alpha = q^{*(1),f}_\alpha + q^{*(1),g}_\alpha = q^{(1)}_\alpha - 2u_\gamma P^{(1)}_{\alpha\gamma}, \tag{3.75}$$

$$P^{*(1)}_{\alpha\beta} = 0, \tag{3.76}$$

where $q^{*(1),f}_\alpha$ and $q^{*(1),g}_\alpha$ are the contributions from the f- and g- populations respectively to the contracted first order heat flux tensor.
By projecting Eqs. (3.59) and (3.60) on the desired moment space, it is possible to obtain $q^{(1)}_\alpha$ as

$$q^{(1)}_\alpha = q^{*(1)}_\alpha \left(1 - \frac{\beta_2}{\beta_1}\right) - \frac{1}{2\beta_1}\left(\partial^{(1)}_t q^{eq}_\alpha + \partial_\beta R^{eq}_{\alpha\beta}\right), \tag{3.77}$$

and, by using (3.76), the first order pressure tensor $P^{(1)}_{\alpha\beta}$,

$$P^{(1)}_{\alpha\beta} = -\frac{1}{2\beta_1}\left(\partial^{(1)}_t P^{eq}_{\alpha\beta} + \partial_\gamma Q^{eq}_{\alpha\beta\gamma}\right). \tag{3.78}$$

By inserting (3.78) in (3.75) and (3.75) in (3.77), an expression for $q^{*(1)}_\alpha$ is obtained:

$$q^{*(1)}_\alpha = -\frac{1}{2\beta_2}\left(\partial^{(1)}_t q^{eq}_\alpha + \partial_\beta R^{eq}_{\alpha\beta}\right) - \frac{2u_\beta}{2\beta_2}\left(\partial^{(1)}_t P^{eq}_{\alpha\beta} + \partial_\gamma Q^{eq}_{\alpha\beta\gamma}\right). \tag{3.79}$$

Computing explicitly the right-hand-side of Eq. (3.79) leads:

$$q^{*(1)}_\alpha = -\frac{1}{2\beta_2} 2\left(C_v + 1\right)\rho T \partial_\alpha T, \tag{3.80}$$

where $C_p = C_v + 1$ is the heat capacity at constant pressure. At this point, computation of second order time derivative of momentum and energy is straight forward: using (3.76) the second order momentum equation is obtained:

$$\partial_t^{(2)} u_\alpha = -\frac{1}{\rho}\partial_\beta \Pi_{\alpha\beta}, \tag{3.81}$$

where $\Pi_{\alpha\beta}$ is the stress tensor,

$$\Pi_{\alpha\beta} = -\mu\left(S_{\alpha\beta} - \frac{2}{D}\partial_\gamma u_\gamma \delta_{\alpha\beta}\right) + \xi \partial_\gamma u_\gamma \delta_{\alpha\beta}, \tag{3.82}$$

with

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha, \tag{3.83}$$

the strain rate tensor,

$$\mu = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, \tag{3.84}$$

the dynamic viscosity, and

$$\xi = \left(\frac{1}{C_v} - \frac{2}{D}\right)\mu, \tag{3.85}$$

the bulk viscosity, characteristic of any kinetic scheme with internal degrees of freedom, $C_v \neq D/2$. Left hand-side of Eq. (3.72) is transformed using second-order time derivatives of mass and momentum, Eqs. (3.70) and (3.81):

$$\partial_t^{(2)}\left(\rho E^{tot}\right) = \rho C_v \partial_t^{(2)} T - u_\alpha \partial_\beta \Pi_{\alpha\beta}. \tag{3.86}$$

The right-hand side of Eq. (3.72) is transformed by computing explicitly first-order time derivative of q_α^{eq} and spatial derivative of $R_{\alpha\beta}^{eq}$; by making use of Eqs. (3.62), (3.63) and (3.65) one obtains:

$$\partial_t^{(1)} q_\alpha^{eq} + \partial_\beta R_{\alpha\beta}^{eq} = 2\left(C_v + 1\right)\rho T \partial_\alpha T + 2\rho T u_\beta \left(S_{\alpha\beta} - \frac{1}{C_v}\left(\partial_\gamma u_\gamma\right)\delta_{\alpha\beta}\right). \tag{3.87}$$

By substitution of (3.87) and (3.86) in (3.72) the second order derivative of the temperature is obtained:

$$\partial_t^{(2)} T = \frac{1}{\rho C_v}\partial_\alpha\left(\kappa \partial_\alpha T\right) - \frac{1}{\rho C_v}\left(\partial_\alpha u_\beta\right)\Pi_{\alpha\beta}, \tag{3.88}$$

where κ is the thermal conductivity given by,

$$\kappa = C_p\left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T. \tag{3.89}$$

Case Pr > 1 For the case where Pr > 1, the additional quasi-conserved quantity by the quasi-equilibrium population is the traceless pressure tensor, $\overline{\Theta}_{\alpha\beta}$, for f-populations,

$$f^* = f^* \left(\rho, u_\alpha, T, \overline{\Theta}_{\alpha\beta}\right), \tag{3.90}$$

while in the case of g-population the following holds:

$$g^* = g^* \left(\rho, u_\alpha, T, K\right). \tag{3.91}$$

This additional conservation guarantees the following form for the first order quasi-equilibrium heat flux and the first order quasi-equilibrium pressure tensor:

$$q_\alpha^{*(1)} = 2u_\gamma P_{\alpha\gamma}^{(1)}, \tag{3.92}$$

$$P_{\alpha\beta}^{*(1)} = P_{\alpha\beta}^{(1)}. \tag{3.93}$$

By proceeding similarly as for the case Pr ≤ 1 the same equations are recovered, but with exchanged relaxation parameters for the viscosity and thermal diffusivity respectively:

$$\mu = \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, \tag{3.94}$$

$$\kappa = C_p \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T. \tag{3.95}$$

3.6.0.1 Thermo-Hydrodynamic Equations

Summing up contributions of first and second order, the Fourier-Navier-Stokes equations for compressible flows are recovered as follows:

$$\partial_t \rho + \partial_\alpha \left(\rho u_\alpha\right) = 0, \tag{3.96}$$

$$\partial_t u_\alpha + u_\beta \partial_\beta u_\alpha = -\frac{1}{\rho}\partial_\alpha \left(\rho T\right) - \frac{1}{\rho}\partial_\beta \Pi_{\alpha\beta}, \tag{3.97}$$

$$\partial_t T + u_\alpha \partial_\alpha T = -\frac{1}{\rho C_v}\rho T \left(\partial_\alpha u_\alpha\right) - \frac{1}{\rho C_v}\left(\partial_\alpha u_\beta\right)\Pi_{\alpha\beta} + \frac{1}{\rho C_v}\partial_\alpha \left(\kappa \partial_\alpha T\right), \tag{3.98}$$

where, as before, the stress tensor is defined as,

$$\Pi_{\alpha\beta} = -\mu\left(S_{\alpha\beta} - \frac{2}{D}\partial_\gamma u_\gamma \delta_{\alpha\beta}\right) + \xi \partial_\gamma u_\gamma \delta_{\alpha\beta},$$

with the strain rate tensor,

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha.$$

Expressions for dynamic viscosity, bulk viscosity, thermal conductivity and adiabatic exponent are given by,

$$\mu = \begin{cases} \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} \le 1, \\ \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} > 1, \end{cases} \qquad \xi = \left(\frac{1}{C_v} - \frac{2}{D}\right)\mu,$$

$$\kappa = \begin{cases} C_p\left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} \le 1, \\ C_p\left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} > 1, \end{cases} \qquad \gamma_{ad} = \frac{C_p}{C_v},$$

and the expression for the Prandtl number becomes:

$$\mathrm{Pr} = \begin{cases} \frac{(1-\beta_1)\beta_2}{(1-\beta_2)\beta_1}, & \text{if } \mathrm{Pr} \le 1, \\ \frac{(1-\beta_2)\beta_1}{(1-\beta_1)\beta_2}, & \text{if } \mathrm{Pr} > 1. \end{cases}$$

3.7 Entropic estimate

Since the LBGK models allow limited stability for high Reynolds number flows, the kinetic equations (3.30) and (3.31) are here extended to their entropic variant based on [25]:

$$f_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) - f_i(\boldsymbol{x}, t) = \alpha\beta_1 \left(f_i^* - f_i\right) + 2\beta_2 \left(f_i^{eq} - f_i^*\right), \tag{3.99}$$

$$g_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) - g_i(\boldsymbol{x}, t) = \alpha\beta_1 \left(g_i^* - g_i\right) + 2\beta_2 \left(g_i^{eq} - g_i^*\right). \tag{3.100}$$

In the above equations, equilibrium populations f_i^{eq} and g_i^{eq} are the same as in Sec. 3.4 and 3.5, while the relaxation parameter related to the viscosity is replaced by $\alpha\beta_{1/2}$, where α is the maximal overrelaxation parameter, which is the positive root of the entropy condition:

$$H(f + \alpha(f^{eq} - f)) = H(f), \tag{3.101}$$

where H is the entropy function (2.45). This formulation is based on the assumption that the entropic estimate (3.101) serves to stabilize the flow only through viscosity, without affecting the thermal conductivity.
The entropic relaxation parameter α was originally conceived for stabilization of high Reynolds number flow simulations to handle large velocity gradients [136]. In that case, fluctuations of α around $\alpha = 2$ due to the entropic estimate acts as a built-in sub-grid model. The entropic relaxation parameter α then played a crucial role in stabilizing multiphase flow simulations with large density gradients [11].
It will be shown in Chap. 5 that entropic estimate helps to stabilize the compressible flow simulations where large gradients of velocity, density and temperature are present. The stabilizing effect of the entropic estimate is triggered automatically and is of importance for two reasons: in the turbulent regions the entropic estimate acts as a built in sub-grid model as it would be for incompressible flows, and in the presence of shocks, it naturally helps avoiding the Gibbs oscillations which would otherwise corrupt the flow field. It is remarkable that the entire scheme described thus far is free of any tuning parameter and the required stabilization of the flows occurs automatically by simply respecting second law of thermodynamic, without the need to identify the regions that would destabilize the simulations.

3.7.1 Implementation details

In order to solve Eq. (3.101), the first step of the implementation consists in checking the allowable values for α guaranteeing that the discrete H-function (2.45) can be evaluated, i.e:

$$f_i + \alpha\,(f_i^{eq} - f_i) > 0. \tag{3.102}$$

Assuming that populations f_i are always positives, the positivity condition (3.102) translates to the values for minimum and maximum entropic estimate as:

$$\begin{cases} \alpha_{min} = \max\{\delta_i, \quad i \in 1, 2, ..., n_Q, \qquad if \quad \delta_i < 0\}, \\ \alpha_{max} = \min\{\delta_i, \quad i \in 1, 2, ..., n_Q, \qquad if \quad \delta_i > 0\}, \end{cases} \tag{3.103}$$

where $\delta_i = \frac{f_i}{f_i - f_i^{eq}}$.
In the case $\alpha_{max} < \alpha_{LBGK}$ $(= 2)$ the value $\alpha = \alpha_{max}$ is set directly, otherwise, Eq. (3.101) is solved with the Newton-Raphson method for α.

3.8 Realization of the compressible ELBM

The equations and expressions needed for the implementation of the compressible entropic lattice Boltzmann model are here summarized. The two kinetic lattice Boltzmann equations including quasi-equilibrium model read:

$$f_i(\boldsymbol{x}+\boldsymbol{v}_i,t+1)-f_i(\boldsymbol{x},t)=\alpha\beta_1\left(f_i^*-f_i\right)+2\beta_2\left(f_i^{eq}-f_i^*\right), \tag{3.104}$$

$$g_i(\boldsymbol{x}+\boldsymbol{v}_i,t+1)-g_i(\boldsymbol{x},t)=\alpha\beta_1\left(g_i^*-g_i\right)+2\beta_2\left(g_i^{eq}-g_i^*\right), \tag{3.105}$$

where the f_i^{eq} equilibrium is computed by solving numerically the equations

$$\{\rho,\rho\boldsymbol{u},D\rho T+\rho\boldsymbol{u}^2\}=\sum_{i=1}^{n_Q}\{1,\boldsymbol{v}_i,v_i^2\}f_i^{eq}(\rho,\boldsymbol{u},T), \tag{3.106}$$

for χ, ζ_α and λ in

$$f_i^{eq}=\rho W_i\exp\left(\chi+\zeta_\alpha v_{i\alpha}+\lambda v_i^2\right), \tag{3.107}$$

while the g_i^{eq} equilibrium is consequently given by:

$$g_i^{eq}=(2C_v-D)Tf_i^{eq}. \tag{3.108}$$

Quasi-equilibrium f-populations can be written in the compact form as:

$$f_i^*=f_i^{eq}+W_i\frac{\left(\overline{Q}_{\alpha\beta\gamma}-Q^{eq}_{\alpha\beta\gamma}\right)\left(v_{i\alpha}v_{i\beta}v_{i\gamma}-3v_{i\gamma}T\delta_{\alpha\beta}\right)}{6T^3},\qquad \text{if Pr}\le 1, \tag{3.109}$$

$$f_i^*=f_i^{eq}+W_i\frac{\left(\overline{\Theta}_{\alpha\beta}-P^{eq}_{\alpha\beta}\right)\left(v_{i\alpha}v_{i\beta}-T\delta_{\alpha\beta}\right)}{2T^2},\qquad \text{if Pr}> 1, \tag{3.110}$$

where

$$\overline{Q}_{\alpha\beta\gamma}=\sum_{i=1}^{n_Q}(v_{i\alpha}-u_\alpha)(v_{i\beta}-u_\beta)(v_{i\gamma}-u_\gamma)f_i. \tag{3.111}$$

and

$$\overline{\Theta}_{\alpha\beta}=\sum_{i=1}^{n_Q}\left[(v_{i\alpha}-u_\alpha)(v_{i\beta}-u_\beta)-\frac{2}{D}\partial_\gamma u_\gamma\delta_{\alpha\beta}\right]f_i. \tag{3.112}$$

Quasi-equilibrium g-populations reads:

$$g_i^* = g_i^{eq} + W_i \frac{(\overline{\boldsymbol{q}}^{\boldsymbol{g}} - \boldsymbol{q}^{eq}) \cdot \boldsymbol{v}_i}{T}, \qquad \text{if } \Pr \leq 1, \tag{3.113}$$

$$g_i^* = g_i^{eq} + W_i \left[K - \rho(2C_v - D)T\right], \qquad \text{if } \Pr > 1, \tag{3.114}$$

where

$$\overline{\boldsymbol{q}}^{\boldsymbol{g}} = \sum_{i=1}^{n_Q} (\boldsymbol{v}_i - \boldsymbol{u}) g_i, \tag{3.115}$$

$$K = \sum_{i=1}^{n_Q} g_i \tag{3.116}$$

and $W_i(T)$ being the weights computed according to (3.17). The velocity set is chosen according to the Mach number requirements; for transonic and supersonic flows the DdQ7^d lattice can be used.
The relaxation parameters β_1 and β_2 are computed by the corresponding dynamic viscosity and thermal conductivity:

$$\beta_1 = \begin{cases} \frac{1}{\frac{2\mu}{\rho T}+1}, & \text{if } \Pr \leq 1, \\ \frac{1}{\frac{2\kappa}{\rho(C_v+1)T}+1}, & \text{if } \Pr > 1, \end{cases}$$

$$\beta_2 = \begin{cases} \frac{1}{\frac{2\kappa}{\rho(C_v+1)T}+1}, & \text{if } \Pr \leq 1, \\ \frac{1}{\frac{2\mu}{\rho T}+1}, & \text{if } \Pr > 1, \end{cases}$$

and the heat capacity C_v is derived from the desired adiabatic exponent γ_{ad} from:

$$C_v = \frac{1}{\gamma_{ad} - 1}. \tag{3.117}$$

Finally, the compressible ELBM algorithm is summarized in three main steps:

1. **Propagation step:** populations associated with discrete velocities $\boldsymbol{v}_i$ are moved to the corresponding adjacent links:

$$f_i(\boldsymbol{x}, t) \rightarrow f_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1), \tag{3.118}$$

$$g_i(\boldsymbol{x}, t) \rightarrow g_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1), \tag{3.119}$$

which results in a new set of populations, $f_i^{'}(\boldsymbol{x}, t)$ and $g_i^{'}(\boldsymbol{x}, t)$.

2. **Boundary conditions:** in the presence of walls or domain boundary conditions, the missing populations are replaced according to the description given below in Sec. 3.9.

3. **Collision step:** in this last step, populations $f_i^{'}(\boldsymbol{x},t)$ and $g_i^{'}(\boldsymbol{x},t)$ are updated by the rule:

$$f_i^{'}(\boldsymbol{x},t) \leftarrow f_i^{'}(\boldsymbol{x},t) + \alpha\beta_{1/2}\,(f_i^{*} - f_i^{\prime}) + 2\beta_{2/1}\,(f_i^{eq} - f_i^{*})\,, \tag{3.120}$$

$$g_i^{'}(\boldsymbol{x},t) \leftarrow g_i^{'}(\boldsymbol{x},t) + \alpha\beta_{1/2}\,(g_i^{*} - g_i^{\prime}) + 2\beta_{2/1}\,(g_i^{eq} - g_i^{*})\,, \tag{3.121}$$

where the two relaxation parameters, equilibrium populations and quasi-equilibrium populations have been previously described.

3.9 Boundary conditions

The boundary conditions employed in the present compressible ELBM model are an extension to those presented in Sec. 2.3.3 for incompressible flows [82, 83]. In general, population $f_i(\boldsymbol{x}_b,t)$ (and $g_i(\boldsymbol{x}_b,t)$) associated with a boundary node residing into the fluid domain $\boldsymbol{x}_b$ (black and gray in Figure 3.6) may belong to two different subsets, namely the subset $\overline{D}$ denoting velocities pointing from the solid into the fluid domain and the subset of velocities pointing from the fluid into the solid or fluid domain. Populations belonging to $\overline{D}$, $f_i(\boldsymbol{x}_b,t), g_i(\boldsymbol{x}_b,t) \in \overline{D}$ are missing and need to be approximated in order to complete the advection step. Due to the employment of a multi-speed lattice, nodes are distinguished between those nodes belonging to the boundary and which are the nearest to the wall, $\boldsymbol{x}_{b,n}$ (black in Figure 3.6) and the nodes belonging to the boundary set but which are only next to nearest, $\boldsymbol{x}_{b,f}$ (gray in Figure 3.6). Formally, they are distinguished by the criterium:

$$\text{if} \qquad \max\left(\frac{\|\boldsymbol{r}_i\|}{\|\boldsymbol{v}_i^2\|}\right) \leq 1 \text{ and } \|\boldsymbol{v}_i^2\| \leq D, \text{for } i \in \overline{D} \quad \rightarrow \quad \boldsymbol{x}_{b,n} \tag{3.122}$$

$$\text{else if} \qquad \max\left(\frac{\|\boldsymbol{r}_i\|}{\|\boldsymbol{v}_i^2\|}\right) \leq 1 \text{ and } \|\boldsymbol{v}_i^2\| \nleq D, \text{for } i \in \overline{D} \quad \rightarrow \quad \boldsymbol{x}_{b,f} \tag{3.123}$$

where $\boldsymbol{r}_i = \boldsymbol{x}_b - \boldsymbol{x}_{w,i}$ and $\boldsymbol{x}_{w,i}$ is the intersection point of lattice link $\boldsymbol{v}_i$ with the wall.

First, it should be noted, that only missing populations are replaced, for both $\boldsymbol{x}_{b,n}$ and $\boldsymbol{x}_{b,f}$ nodes. At this point, one needs to formulate an approximation for the missing populations $f_i(\boldsymbol{x}_b,t)$ and $g_i(\boldsymbol{x}_b,t)$ at boundary node

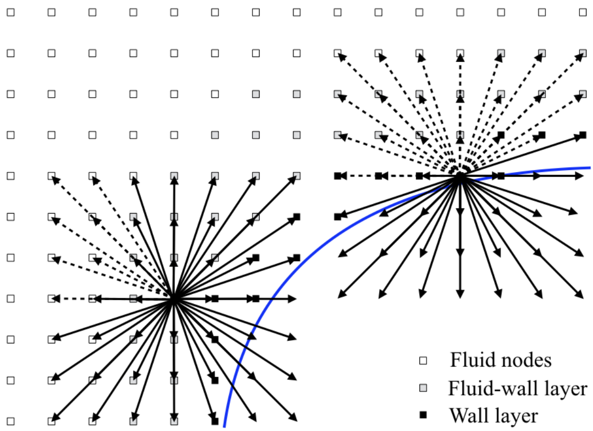


Figure 3.6: Scheme representing boundary nodes (black, $\boldsymbol{x}_{b,n}$ and gray $\boldsymbol{x}_{b,f}$) for an arbitrary wall, and the corresponding populations, represented in dashed arrow when missing.

$\boldsymbol{x}_b$; here, the missing populations are computed by a chosen equilibrium part and a non-equilibrium part, approximated by first-order Chapman-Enskog solution in the populations, leading to:

$$f_i^{\text{miss}} = f_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}) + f_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}, \nabla\boldsymbol{u}_{tgt}, \nabla T_{tgt}), \quad (3.124)$$

$$g_i^{\text{miss}} = g_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}) + g_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}, \nabla\boldsymbol{u}_{tgt}, \nabla T_{tgt}), \quad (3.125)$$

where ρ_{tgt}, $\boldsymbol{u}_{tgt}$, T_{tgt}, $\nabla\boldsymbol{u}_{tgt}$ and ∇T_{tgt} are the target density, velocity, temperature, velocity Jacobian and temperature gradient respectively and need to be specified. Before proceeding by specifying target values, the approximation for the first order part of both f- and g-populations needs to be provided. For f-populations, a Grad's approximation of the form

$$f_i^{(1)} = W_i \left\{ P_{\alpha\beta}^{(1)} \frac{(v_{i\alpha}v_{i\beta} - T\delta_{\alpha\beta})}{2T^2} + Q_{\alpha\beta\gamma}^{(1)} \frac{(v_{i\alpha}v_{i\beta}v_{i\gamma} - 3v_{i\gamma}T\delta_{\alpha\beta})}{6T^3} \right\}, \quad (3.126)$$

is used, where $P^{(1)}_{\alpha\beta}$ and $Q^{(1)}_{\alpha\beta\gamma}$ are the first order pressure tensor and heat flux tensor. First order pressure and heat flux tensors are derived from Champan-Enskog analysis, thus by projecting Eq. (3.59) into the second and third order moment space:

$$P^{(1)}_{\alpha\beta} = -\frac{1}{2\beta_1}\rho T\left(S_{\alpha\beta} - \frac{1}{C_v}\partial_\gamma u_\gamma \delta_{\alpha\beta}\right), \tag{3.127}$$

where $S_{\alpha\beta}$ is given by Eq. (6.79), and

$$\begin{aligned} Q^{(1)}_{\alpha\beta\gamma} &= -\tfrac{1}{2\beta_2}\rho T\left[\partial_\alpha T\delta_{\beta\gamma} + \partial_\beta T\delta_{\alpha\gamma} + \partial_\gamma T\delta_{\alpha\beta}\right] \\ &\quad + u_\alpha P^{(1)}_{\beta\gamma} + u_\beta P^{(1)}_{\alpha\gamma} + u_\gamma P^{(1)}_{\alpha\beta}. \end{aligned} \tag{3.128}$$

For g-populations, a Grad's approximation of the form

$$g^{(1)}_i = W_i\left\{K^{(1)} + q^{g(1)}_\alpha \frac{v_{i\alpha}}{T} + R^{g(1)}_{\alpha\beta}\frac{(v_{i\alpha}v_{i\beta} - T\delta_{\alpha\beta})}{2T^2}\right\}, \tag{3.129}$$

is used, where

$$K^{(1)} = \sum_{i=1}^{n_Q} g^{(1)}_i, \tag{3.130}$$

is the first order contribution of the zeroth order moment of the g-populations, $q^{g(1)}_\alpha$ is the first order contribution to the contracted heat flux associated to g-populations, and $R^{g(1)}_{\alpha\beta}$ is the first order contribution to the contracted fourth order moment of g-populations. All these moments can be derived analytically by Chapman-Enskog method, thus by projecting Eq. (3.60) to the desired order moment:

$$K^{(1)} = -\frac{1}{2\beta_1}\rho T(2C_v - D)\left(\frac{1}{C_v}\partial_\gamma u_\gamma\right), \tag{3.131}$$

$$q^{g(1)}_\alpha = -\frac{1}{2\beta_2}\rho T(2C_v - D)\partial_\alpha T + K^{(1)}u_\alpha, \tag{3.132}$$

$$R^{g(1)}_{\alpha\beta} = -\frac{1}{2\beta_1}\rho T(2C_v - D)\Big(TS_{\alpha\beta} + u_\alpha\partial_\beta T + u_\beta\partial_\alpha T \tag{3.133}$$

$$- \frac{1}{C_v}\partial_\gamma u_\gamma(2T\delta_{\alpha\beta} + u_\alpha u_\beta)\Big). \tag{3.134}$$

At this point, only target values need to be specified. The target values need to be chosen based on which type of boundary conditions one wants to simulate: inlet, no-slip or free-slip (which includes adiabatic wall treatment for the temperature) boundary condition. For the case of the outlet boundary condition an alternative approach is employed, i.e. by replacing all populations belonging to the outlet by previous time-step populations. Before proceeding with the description of target quantities, it is important to notice that only target quantities belonging to $\boldsymbol{x}_{b,n}$ boundary nodes need to be specified, while for the boundary nodes $\boldsymbol{x}_{b,f}$ previous time step local quantities are employed. Temperature gradients and velocity Jacobian are instead evaluated by a semi-second order centered scheme, based on previous time step quantities. For the case of the inlet boundary condition, target quantities need not to be computed since they are prescribed by the set-up, so they are chosen corresponding to the prescribed conditions at the inlet.

3.9.1 Target values for no-slip boundary conditions

Only target values ρ_{tgt}, $\boldsymbol{u}_{tgt}$, T_{tgt} for nodes $\boldsymbol{x}_{b,n}$ remains to be specified. For the case of no-slip (wall) boundary conditions the target density ρ_{tgt} correspond to the density obtained after applying bounce-back rule to the f-populations:

$$\rho_{tgt} = \sum_{i \in \overline{D}} f_i^{bb} + \sum_{i \notin \overline{D}} f_i, \tag{3.135}$$

where the missing populations f_i^{bb} are replaced by the reflected populations as $f_i^{bb} = \tilde{f}_i$ with $\tilde{\boldsymbol{v}}_i = -\boldsymbol{v}_i$. The target velocity $\boldsymbol{u}_{tgt}$ at a given boundary node $\boldsymbol{x}_{b,n}$ is obtained by an interpolation scheme involving the wall velocity $\boldsymbol{u}_{w,i} = \boldsymbol{u}(x_{w,i}, t)$ (which usually is 0) at the intersection point $\boldsymbol{x}_{w,i}$ and the velocities $\boldsymbol{u}_{f,i} = \boldsymbol{u}(\boldsymbol{x}_{f,i}, t)$ at the adjacent fluid nodes $\boldsymbol{x}_{f,i} = \boldsymbol{x}_b + \boldsymbol{v}_i$ for $i \in \overline{D}$. For an averaged linear interpolation, this yields:

$$\boldsymbol{u}_{tgt} = \frac{1}{n_{\overline{D}}} \sum_{i \in \overline{D}} \frac{(\|\boldsymbol{r}\|_i / \|\boldsymbol{v}\|_i)\, \boldsymbol{u}_{f,i} + \boldsymbol{u}_{w,i}}{(\|\boldsymbol{r}\|_i / \|\boldsymbol{v}\|_i) + 1}, \tag{3.136}$$

where $n_{\overline{D}}$ is the number of unknown populations satisfying: $\|\boldsymbol{v}_i^2\| \leq D,\ i \in \overline{D}$. Similarly, the same procedure is applied to the temperature:

$$T_{tgt} = \frac{1}{n_{\overline{D}}} \sum_{i \in \overline{D}} \frac{(\|\boldsymbol{r}\|_i / \|\boldsymbol{v}\|_i)\, T_{f,i} + T_{w,i}}{(\|\boldsymbol{r}\|_i / \|\boldsymbol{v}\|_i) + 1}. \tag{3.137}$$

The derivatives, for a generic field Φ are evaluated with a semi-second order scheme for unstructured grid:

$$\frac{\partial \Phi}{\partial x} = \frac{\Phi_{j+1} - \Phi_w}{1 + \|\boldsymbol{r}\|_i}, \tag{3.138}$$

or

$$\frac{\partial \Phi}{\partial x} = \frac{\Phi_w - \Phi_{j-1}}{1 + \|\boldsymbol{r}\|_i}, \tag{3.139}$$

depending on the position of the wall with respect to the boundary node, and where Φ_w is the quantity Φ imposed at the wall (for example $\boldsymbol{u}_w$ or T_w).

3.9.2 Target values for free-slip and adiabatic wall boundary conditions

Target conditions for free-slip and adiabatic wall boundary conditions can be constructed by starting from the no-flux condition for density and temperature:

$$\left.\frac{\partial \rho}{\partial n}\right|_{\partial\Omega} = 0, \tag{3.140}$$

$$\left.\frac{\partial T}{\partial n}\right|_{\partial\Omega} = 0, \tag{3.141}$$

where n is the coordinate normal to the boundary surface $\partial\Omega$, as depicted in Fig. 3.7. For the velocity field, the same condition is imposed for the

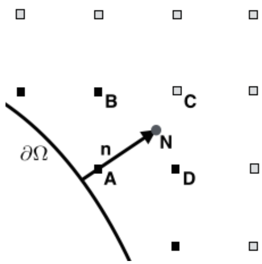


Figure 3.7: Scheme representing boundary nodes (black, $\boldsymbol{x}_{b,n}$ and gray $\boldsymbol{x}_{b,f}$) and normal $\boldsymbol{n}$ of the wall. Point N is employed to extrapolate flow-field values in the direction normal to the wall. A, B, C and D are the point employed in the bilinear interpolation to compute N.

component of velocity normal to the boundary surface $\partial\Omega$, $\boldsymbol{u}_\perp$:

$$\left.\frac{\partial \boldsymbol{u}_\perp}{\partial n}\right|_{\partial\Omega} = 0. \tag{3.142}$$

On the other side, the tangent component of velocity, $\boldsymbol{u}_\parallel$, is the projection on the surface $\partial\Omega$ of the velocity field $\boldsymbol{u}(\boldsymbol{x}_b)$ at the boundary node:

$$\boldsymbol{u}_\parallel = \boldsymbol{u} - (\boldsymbol{u} \cdot \boldsymbol{n})\boldsymbol{n}, \tag{3.143}$$

where $\boldsymbol{n}$ is the vector normal with respect to surface $\partial\Omega$. In order to satisfy the no-flux conditions, values of density, temperature and velocity are evaluated in the normal direction, by bilinear interpolation, as depicted in Fig. 3.7. Given the normal direction, and then its end-point N considering a constant distance from A of $\|\boldsymbol{n}\| = \sqrt{2}$ in $2D$ or $\|\boldsymbol{n}\| = \sqrt{3}$ in $3D$, the neighbors points A, B, C and D are identified. Next a bilinear interpolation (in $2D$), for a given field Φ, is performed according to:

$$\begin{aligned}\Phi_N = \left(\left(D_x - A_x\right)\left(B_y - A_y\right)\right)^{-1} \qquad & \left(\Phi_A \left(D_x - N_x\right)\left(B_y - N_y\right)\right. \\ & \Phi_B \left(D_x - N_x\right)\left(N_y - A_y\right) \\ & \Phi_C \left(N_x - A_x\right)\left(N_y - A_y\right) \\ & \left.\Phi_D \left(N_x - A_x\right)\left(B_y - N_y\right)\right).\end{aligned} \tag{3.144}$$

Here Φ_N is the field to be found at point N, and Φ_A, Φ_B, Φ_C and Φ_D are the known fields at respective points and where A_x, A_y, etc. are the x and y coordinate respectively for point A and the same for the other points. Similar procedure for fields at point N, Φ_N, can be performed in three dimensions by trilinear interpolation.
Once Φ_N is found, a first order approximation for the derivative in the direction of the normal is assumed

$$\frac{\partial \Phi}{\partial n} \simeq \frac{\Phi_N - \Phi_A}{\|N - A\|} = 0. \tag{3.145}$$

This allows to approximate the values of the fields at node A

$$\Phi_A = \Phi_N. \tag{3.146}$$

which correspond to the target values $\Phi_{tgt} = \Phi_A$.

3.10 Conclusion

An extension of the LBM presented here allows, with one single algorithm, to simulate a broad range of applications ranging from low Mach number and thermal flows to transonic and supersonic flow regimes, also in the presence of arbitrarily complex obstacles. This is in sharp contrast to conventional solvers which require identification of different flow regimes and employing the appropriate numerical approach for it.
The model is based on three key elements: higher-order entropy supporting lattices, entropic accurate equilibrium, and entropic time stepping, thus retaining the simplicity of the isothermal LBM. In particular, the novel entropic accurate evaluation of equilibrium is argued as a corner stone, and perhaps unavoidable requirement for simulation of compressible flows using the lattice Boltzmann method. Moreover, detailed boundary conditions were presented in order to simulate the presence of solid walls. These boundaries are validated in Chap. 5 through a number of standard benchmarks in two and three dimensions.
The simplicity and robustness of the present approach are highlighted by the fact that the construction (adherence to the second law of thermodynamics) and boundary conditions presented here are independent of the choice of the lattice (standard and multispeed lattices) and simulation dimensions.

Chapter 4

Optimization strategies for the compressible lattice Boltzmann method

4.1 Introduction

As evident from the previous Chapter, one of the main drawbacks of the compressible model is the increased number of populations which are needed to guarantee accuracy of the pertinent equilibrium moments when the Mach number is increased. In order to simulate mildly supersonic flows, at least $n_Q = 7^3$ velocities are needed, and this number is doubled when the adiabatic coefficient γ_{ad} is chosen differently from that of monoatomic gases. When such a large number of velocities is employed, the computational cost is increased considerably, as shown in Table 3.3. It grows even further when the Mach number needs to be increased, thus by employing at least $n_Q = 11^3$ number of velocities in three-dimensions. It is clear that the computational cost may become easily prohibitive.

In this Chapter, optimization strategies for the reduction of the computational cost and for increasing the Mach number are proposed. Together, modifications of the lattice which can be employed to extend the temperature range of a lattice are also proposed.
The first aspect is developed in Sec. 4.2, where shifted lattices are presented.

With this kind of lattices, the operating domain in terms of Mach number is shifted, and centered at a higher shift velocity. This allows to greatly increase the Mach number achievable by a given lattice. In Sec. 4.3, the temperature range of a given lattice is increased by modifying the lattice link sizes, without increasing the number of populations. Finally, in Sec. 4.4, the computational costs related to large lattices for the simulation of high Mach number flows is reduced by pruning the original lattices. This, however, would increase dramatically the inaccuracies in the pertinent equilibrium moments; the remedy for that is the employment of the entropic guided equilibrium strategy, where the correct recovery of the pertinent equilibrium moments is enforced.

4.2 Lattice kinetic theory in a co-moving Galilean reference frame

As outlined in the introduction, the accuracy for a given lattice, in terms of the deviation from the continuous Maxwell-Boltzmann moments, is guaranteed only for a limited deviation of the local Mach number from its reference value $\mathrm{Ma}_0 = 0$. This issue is here addressed by retaining the same number of populations. Straightforward would be, in fact, to increase the number of lattice discrete velocities: when passing from the lattice with $n_q = 7^d$ to the lattice with $n_q = 11^d$, for example, between an advection Mach number of 1 and 2 the reduction of errors is one to two-orders of magnitude (see Fig. 3.2). At the same time, however, the computational cost would become prohibitive, as also demonstrated by Table 3.3, since it increases at least linearly with the number of populations; thus, simulations with the D2Q11^2 or D3Q11^3 lattice may be easily unfeasible.
In the following, a novel idea about how to increase the limited Mach number, or better, on how to shift the operating Mach number, is provided. To that end the lattice DdQ7^d is here considered, since it is the lattice considered in this work for compressible flows. The first step is the proof that the LB method stays invariant in any Galilean reference frame [141, 142]. The proof is based on a simple observation about the Galilean invariance of the lattice equilibrium distribution. Based on that, shifted lattices are introduced, which, nonetheless, are realized with the standard propagation-collision algorithm, on the same grids and with the same boundary conditions as in the familiar symmetric case.

It is well known, that deviations of the flow velocity u (in one dimension) from the reference value $u = 0$ in the lattice Boltzmann method are followed by an increase of the errors in the equilibrium moments from the corresponding Maxwell-Boltzmann value. This was also outlined in Sec. 3.4.3, where the accuracy in reproducing Maxwell-Boltzmann moments with higher-order lattices, and by increasing the Mach number, was investigated. Clearly, the origin of these errors is the choice of the preferred reference frame $U = 0$ at the outset.

4.2.1 Galilean invariance of weights

Here, the notion of a reference frame at rest is challenged. The equilibrium is rather defined in a frame moving, say, with the velocity U in the x-direction. This is achieved by considering a *shifted* discrete velocity-set V_n' with the velocities

$$v_i' = v_i + U, \;\; v_i \in V_n, \tag{4.1}$$

and finding the shifted weights $w_i(U,T)$ by matching the moments of the Maxwellian $f_v^{\mathrm{M}}(U,T)$ computed at velocity U and temperature T:

$$f_v^{\mathrm{M}}(U,T) = \sqrt{\frac{1}{2\pi T}} \exp\left\{-\frac{(v-U)^2}{2T}\right\}. \tag{4.2}$$

For symmetric lattices the weights are found by solving the following system of equations ($m = 0, ..., n-1$)

$$\langle w_i(0,T) v_i^m \rangle = \langle f_v^{\mathrm{M}}(0,T) v^m \rangle, \tag{4.3}$$

where the operator $\langle ... \rangle$ stands for summation over discrete velocity index i for the discrete case (left) or integration over the continuous velocity v (right). Instead, for shifted weights $w_i(U,T)$ one has:

$$\langle w_i(U,T)[v_i + U]^m \rangle = \langle f_v^{\mathrm{M}}(U,T) v^m \rangle. \tag{4.4}$$

Key observation of [141] is that the weights $w_i(U,T)$ are Galilean invariant: for any set V_n and for any shift U the relation

$$w_i(U,T) = w_i(0,T), \tag{4.5}$$

is valid. The proof of (4.5) follows immediately from Galilean invariance of the moments of the Maxwellian: using binomial theorem and Eq. (4.3), for

$m = 0, \ldots, n-1$, it follows

$$\begin{aligned}\langle f_v^{\mathrm{M}}(U,T)v^m\rangle &= \sum_{k=0}^{m} \begin{pmatrix} m \\ k \end{pmatrix} U^k \langle f_v^{\mathrm{M}}(0,T)v^{m-k}\rangle \\ &= \sum_{k=0}^{m} \begin{pmatrix} m \\ k \end{pmatrix} U^k \langle w_i(0,T)v_i^{m-k}\rangle \\ &= \langle w_i(0,T)[v_i+U]^m\rangle.\end{aligned}$$

Hence, Eq. (4.4) is equivalent to

$$\langle [w_i(U,T) - w_i(0,T)][v_i+U]^m\rangle = 0. \tag{4.6}$$

Since the shift U is arbitrary, the result (4.5) follows.

4.2.2 Shifted lattices

Galilean invariance of the weights Eq. (4.5) has a number of immediate important implications. First, the construction of the discrete velocity set in higher dimensions remains the same as in Sec. 3.3. As an example, in two dimensions, the shift U in the x-direction corresponds to the tensor product $V'_{n_x} \otimes V_{n_y}$. Second, the symmetric velocity sets generate space-filling lattices. This is crucial for the realization of the time-marching LB scheme Eq. (3.1). Now, if the shift U is itself integer, the corresponding shifted lattice is also space filling. For example, with $U = 1$, the shifted set V_7 becomes $V'_7 = \{-2, -1, 0, 1, 2, 3, 4\}$. The shifted lattice, with the discrete velocities $V'_{7x} \otimes V_{7y}$ is shown in Fig. 4.1. Next, the weights corresponding to the velocities of the shifted lattice are the same as for the symmetric case: $W_i(U,T) = w_{ix}(0,T)w_{iy}(0,T)$. Moreover, since the weights do not change under a transform to a co-moving reference frame, the entropy function is Galilean invariant, and is given by Eq. (2.45). Consequently, the equilibrium populations are form invariant. They are found by the same minimization techniques as in the familiar symmetric case, that is, $H \to min$, subject to the constraints $\rho = \langle f_i\rangle$, $\rho\boldsymbol{u} = \langle f_i \boldsymbol{v}'_i\rangle$, and $\rho DT + \rho\boldsymbol{u}^2 = \left\langle f_i \boldsymbol{v}'^2_i\right\rangle$, where v'_i are shifted velocities.

In summary, the surprising feature of these shifted, non-symmetric lattices is that simulations in the co-moving reference frame require no change in the LB algorithm and can be performed using the same grids as in the symmetric case. All that is needed is to replace the velocities $\boldsymbol{v}_i$ in the propagation part of the LB equation (3.1) with the velocities $\boldsymbol{v}'_i$ of the shifted lattice described above.

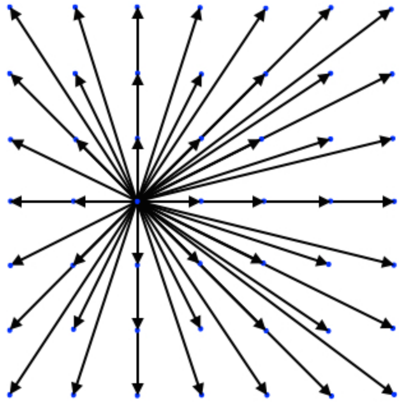


Figure 4.1: Visualization of the D2Q7^2 lattice with shift $U = 1$ in the x direction.

4.2.3 Errors for shifted lattices

The errors associated with the equilibrium computed on shifted lattices are shown in Fig. 4.2, for the same quantities as presented in Fig. 3.2; namely, the errors in evaluation of the equilibrium heat flux $q_x^{eq} = \sum_{i=1}^{n_Q} v_{ix} v_i^2 f_i^{eq}$ and contracted fourth order equilibrium moment $R_{xx}^{eq} = \sum_{i=1}^{n_Q} v_{ix} v_{ix} v_i^2 f_i^{eq}$ when compared to their Maxwell-Boltzmann counterpart. Presented are the errors for the lattice with no shift, and with a shift $U = 1$ and $U = 2$. The errors in Fig. 3.2 are plotted now as a function of the u_x velocity, in order to show the effect of the lattice shift. When the velocity matches the shift velocity the errors in the moments become zero. One can notice how the operating domain, in terms of Mach number and errors in the moments, shifts together with the shift velocity of the lattice. It is thus evident that the operating domain can be chosen in a preferential way in order to match with the requirements in terms of Mach number. This is particularly appealing for the simulation of compressible flows: the operation window of the co-moving lattice Boltzmann is centered around $u = U$, in contrast to the symmetric case which operates around $u = 0$. Clearly, the errors also show that, while in principle any shift velocity can be chosen, for practical simulations it is not possible. In the presence of walls in fact, where e.g. no-slip boundary condition is employed in the rest frame of reference, the Mach number is locally very low, and even negative. As a consequence, for the DdQ7^d lattice and a shift $U \geq 2$, the errors of the shifted lattice become

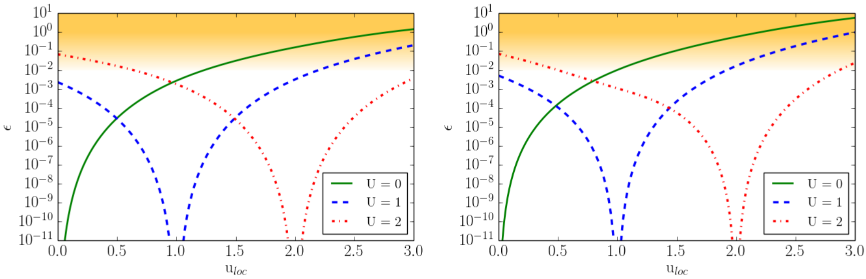


Figure 4.2: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component of the contracted fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of local u_x velocity, for non-shifted lattice and for lattices with shift $U = 1$ and $U = 2$.

too large.

In order to understand the effectiveness of the shifted lattice, test simulations of a vortex advected at different Mach numbers are performed. After a certain Mach number, the non-shifted lattice is expected to introduce some numerical errors, due to the lack of Galilean invariance in the moments. As noted in Figure 3.2, due to these numerical errors the vortex is no longer advected correctly and starts to deform. The same is expected also for the shifted lattice, but around a higher Mach number (a unity shift in errors). This is demonstrated in Fig. 4.3 for a characteristic vortex Mach number of $\mathrm{Ma}_v = 0.5$ and different advection Mach numbers $\mathrm{Ma}_\mathrm{a} = \{0.3, 0.8, 1.3, 1.8, 2.3\}$, and by means of pressure contours. On the very left, the initial conditions are represented for both shifted (bottom), and non-shifted (top) lattices. As one can notice, at a relatively low advection Mach number, both non- and shifted lattices propagate the vortex correctly. The same happens until an advective Mach number of around $\mathrm{Ma}_v = 0.8$, which results in a maximum local Mach number of $\mathrm{Ma}_\mathrm{loc,max} \simeq 1.3$. By further increasing the velocity to $\mathrm{Ma}_\mathrm{a} = 1.3$ ($\mathrm{Ma}_\mathrm{loc,max} = 1.8$) one can notice that the vortex for a non-shifted lattice starts to deform, while for the shifted lattice this does not happen. For an advection Mach number $\mathrm{Ma}_\mathrm{a} = 1.8$ ($\mathrm{Ma}_\mathrm{loc,max} = 2.3$), the vortex for the non-shifted lattice is completely deformed while for the shifted lattice it remain unaffected. Finally, for an advection Mach number of around $\mathrm{Ma}_\mathrm{a} = 2.3$ ($\mathrm{Ma}_\mathrm{loc,max} = 2.8$), the equilibrium computation for the non-shifted lattice does not admit a solution, while for the shifted-lattice the vortex starts to deform similarly to symmetric lattice at a $\mathrm{Ma}_\mathrm{v} = 1.3$, clearly demonstrating

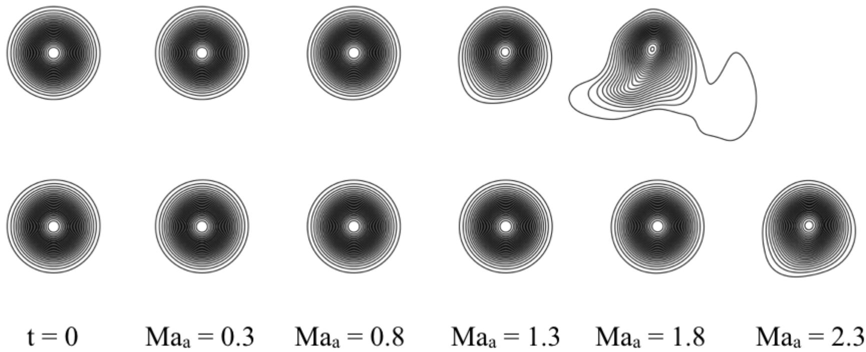


Figure 4.3: Vortex advection comparison at different advection Mach number. Top row: symmetric lattice. Bottom row: shifted lattice with $U = 1$.

an increase in the operating range of a lattice.

Second order convergence has been checked for lattices without and with shifts $U = 1$ and $U = 2$, by simulating the Green-Taylor vortex with a mean advection speed of $U = 0$, $U = 1$, $U = 2$, respectively. The result of the grid convergence study is presented in Fig. 4.4 and shows no difference between the results of a non- and shifted lattices. Plotted is the L_2 norm of the error, which is defined as:

$$L_2 = \sqrt{\frac{\sum_i \left(u_{i,LBM} - u_{i,ref}\right)^2}{\sum_i u_{i,ref}^2}}, \tag{4.7}$$

where $u_{i,LBM}$ is the velocity at sample i with the present simulations, and $u_{i,ref}$ is the velocity computed from the analytical solution.

The limited operating domain for a given multi-speed lattice, in terms of Mach number, has been here extended by modifying the lattice without increasing the number of lattice speeds. This was achieved by shifting the lattice speeds by one or more unities in a preferential direction. It was found, that the weights corresponding to the new class of shifted lattices are in fact, the same weights of the corresponding non-shifted lattice. Also thanks to the recently introduced exact entropic equilibrium, which is computed independently from the lattice velocity set, an increase in the Mach number by a unity or more can be easily achieved just by skewing the lattice velocity-set in a preferential direction, still by employing the conventional collision-propagation algorithm of the standard lattice Boltzmann method.

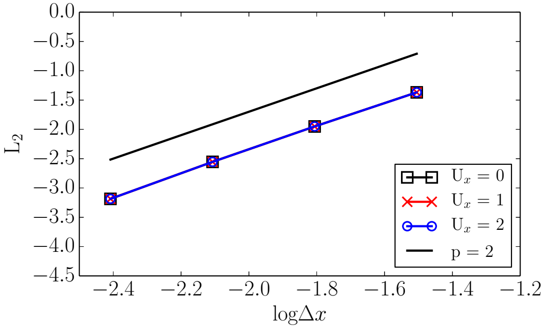


Figure 4.4: Grid convergence study for the Green-Taylor vortex. Results are shown for non-shifted lattice and lattices with shift of $U = 1$ and $U = 2$.

4.3 Lattices with increased temperature range

The range of temperatures supported by the lattice is increased based on the following observation. By considering the velocity set $V_7 = \{0, \pm 1, \pm 2, \pm r\}$, where for $r = 3$ the standard lattice employed for compressible flows, as in Sec. 3.3, is recovered, it is noted that by increasing the longest lattice link to $r = 4$ the temperature range increases significantly, while the lattice still belongs to the same hierarchy of the admissible higher order lattices. Table 4.1 reports the minimum T_{min} and maximum temperatures T_{max}, reference temperature T_0, reduced temperatures θ_{min} and θ_{max}, and the temperature ratio T_{max}/T_{min} for the two lattices with $r = 3$ and $r = 4$.

V_7	T_{min}	T_{max}	T_0	θ_{min}	θ_{max}	$\frac{T_{max}}{T_{min}}$
$\{0, \dots, \pm 3\}$	0.36754	1.63246	0.69795	−0.4734	1.3389	4.4416
$\{0, \dots, \pm 4\}$	0.34969	2.71682	1.91103	−0.8170	0.4217	7.7692

Table 4.1: Temperature range comparison for the original DdQ7^d lattice and its expanded version for increased temperature range.

The temperature ratio T_{max}/T_{min} is the most important parameter to characterize the temperature range. An increase of the temperature range, however, comes at the expense of increased errors in the q^{eq}_{α} and $R^{eq}_{\alpha\beta}$ moments when the Mach number is increased, as shown in Fig. 4.5.

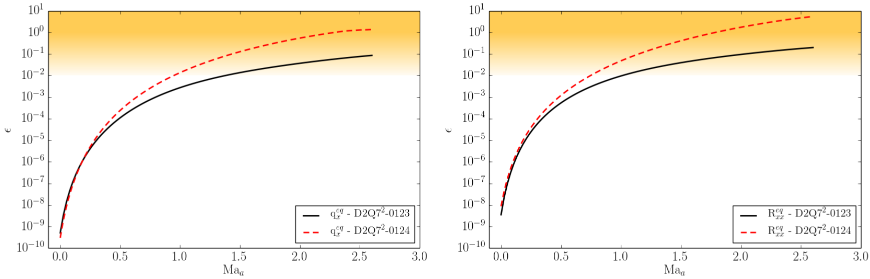


Figure 4.5: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann counterpart values, as a function of Mach number, for two different lattices: the D2Q7^2-0123 and the D2Q7^2-0124.

In Figure 4.6 the same errors are shown as a function of the temperature. At the considered Mach number of $\mathrm{Ma}_{\mathrm{loc}} = 0.1$ the errors stay small over the entire temperature range for both the lattices.

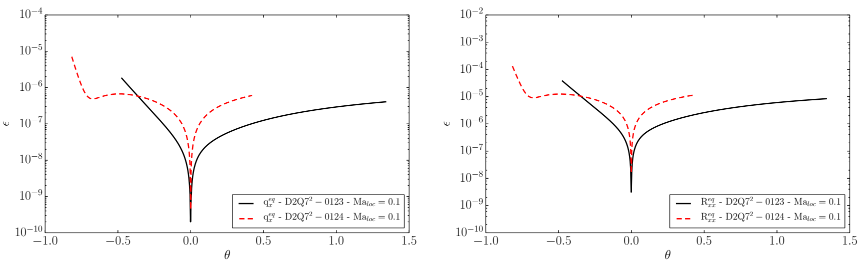


Figure 4.6: Deviation of the contracted, x-component, third order moment q_x^{eq} (left) and xx-component, fourth-order moment R_{xx}^{eq} (right), from the Maxwell-Boltzmann values as a function of the reduced temperature θ, at a local Mach number of $\mathrm{Ma}_{loc} = 0.1$, and for two different lattices: the D2Q5^2-0123 and the D2Q7^2-0124.

In conclusion, since the errors of the D2Q7^2-0124 lattice being higher than for the D2Q7^2-0123 lattice, the latter should be the preferred choice. However, it may happen that for certain simulations the temperature range that needs to be achieved is greater than the maximum allowed by the standard DdQ7^d lattice, while the Mach number in the simulation does not reach supersonic values. In this particular case the D2Q7^2-0124 (or D3Q7^3-0124) lattice could become particularly useful.

4.4 Lattice pruning and guided equilibrium

The question on optimizing the compressible model in terms of computational cost and memory overhead is still unanswered. The only way to drastically reduce the memory overhead is by reducing the number of populations. This way, also the computational cost would decrease. To this end, pruning of the lattice, which consists in omitting certain lattice velocities based on their energy shell, may be an option. However, the accuracy of the lattice, in the present case the DdQ7^d, is already limited in the mildly supersonic regime if no shifting technique is employed. What it is suggested here, is to slightly modify the approach to construct the equilibrium, i.e. by employing a guided equilibrium which enforces accuracy on the required equilibrium moments. These two techniques, lattice pruning and guided equilibrium are briefly described in the following. The combination of the two allows to decrease by one order of magnitude the memory overhead, by three to four times the computational cost and at the same time to increase the accuracy at high Mach numbers.

4.4.1 Lattice pruning

Lattice pruning has already been employed in the past to reduced the number of velocities of a given lattice: as an example, the D3Q15 or the D3Q19 lattices are prunes of the D3Q3^3 lattice. Prunes of higher-order lattices have also been proposed [84, 101, 102], recovering the D3Q41 lattice from the D3Q5^3. Other reduced velocity-set lattices have also been proposed in two- and three-dimensions, without employing pruning techniques, but by empirically proposing the velocity-set and then by finding the corresponding weights [97, 143].
Here, the pruning procedure is further extended to the lattice employed for the simulation of compressible flows, the D3Q7^3. Elimination of discrete velocities from a given lattice becomes simple when symmetry considerations are accounted. If one wants to discard an arbitrary discrete velocity (i, j, k), in order to maintain symmetry along the x-direction, and then to conserve the x-momentum, the discrete velocity $(-i, j, k)$ needs also to be discarded. Similar considerations applies in the y- and z-directions. At this point, since also $(i, -j, k)$ have been discarded, x- and z- reflections

$(-i,-j,k)$ and $(i,-j,-k)$ need to be discarded. This procedure is further applied until, at the end, all discrete velocities $(\pm i,\pm j,\pm k)$ are discarded. Equivalently, populations belonging to the same energy shell $\epsilon = i^2+j^2+k^2$ and with the same velocity shell $\eta = |i|+|j|+|k|$ can be discarded. For the D3Q7^3 lattice, the number of energy shells is 19, while the number of velocity shells is 20. These are given by (4.8).

$$
\begin{array}{llll}
\epsilon = 0, & \eta = 0 & : (0,0,0) & \ldots 1 \\
\epsilon = 1, & \eta = 1 & : (\pm 1,0,0) & \ldots 6 \\
\epsilon = 2, & \eta = 2 & : (\pm 1,\pm 1,0) & \ldots 12 \\
\epsilon = 3, & \eta = 3 & : (\pm 1,\pm 1,\pm 1) & \ldots 8 \\
\epsilon = 4, & \eta = 2 & : (\pm 2,0,0) & \ldots 6 \\
\epsilon = 5, & \eta = 3 & : (\pm 2,\pm 1,0) & \ldots 24 \\
\epsilon = 6, & \eta = 4 & : (\pm 2,\pm 1,\pm 1) & \ldots 24 \\
\epsilon = 8, & \eta = 4 & : (\pm 2,\pm 2,0) & \ldots 12 \\
\epsilon = 9, & \eta = 3 & : (\pm 3,0,0) & \ldots 6 \\
\epsilon = 9, & \eta = 5 & : (\pm 2,\pm 2,\pm 1) & \ldots 24 \\
\epsilon = 10, & \eta = 4 & : (\pm 3,\pm 1,0) & \ldots 24 \\
\epsilon = 11, & \eta = 5 & : (\pm 3,\pm 1,\pm 1) & \ldots 24 \\
\epsilon = 12, & \eta = 6 & : (\pm 2,\pm 2,\pm 2) & \ldots 8 \\
\epsilon = 13, & \eta = 5 & : (\pm 3,\pm 2,0) & \ldots 24 \\
\epsilon = 14, & \eta = 6 & : (\pm 3,\pm 2,\pm 1) & \ldots 48 \\
\epsilon = 17, & \eta = 7 & : (\pm 3,\pm 2,\pm 2) & \ldots 24 \\
\epsilon = 18, & \eta = 6 & : (\pm 3,\pm 3,0) & \ldots 12 \\
\epsilon = 19, & \eta = 7 & : (\pm 3,\pm 3,\pm 1) & \ldots 24 \\
\epsilon = 22, & \eta = 8 & : (\pm 3,\pm 3,\pm 2) & \ldots 24 \\
\epsilon = 27, & \eta = 9 & : (\pm 3,\pm 3,\pm 3) & \ldots 8
\end{array}
\tag{4.8}
$$

A vast number of combinations can be chosen in order to discard velocity and energy shells. Among others, the lattice composed by the velocity and energy shells

$$\begin{array}{llll}
\epsilon = 0, & \eta = 0 & : (0,0,0) & \dots 1 \\
\epsilon = 1, & \eta = 1 & : (\pm 1,0,0) & \dots 6 \\
\epsilon = 3, & \eta = 3 & : (\pm 1,\pm 1,\pm 1) & \dots 8 \\
\epsilon = 4, & \eta = 2 & : (\pm 2,0,0) & \dots 6 \\
\epsilon = 8, & \eta = 4 & : (\pm 2,\pm 2,0) & \dots 12 \\
\epsilon = 9, & \eta = 3 & : (\pm 3,0,0) & \dots 6
\end{array} \tag{4.9}$$

is here considered as an example, with a total number of discrete velocities $n_Q = 39$. The D3Q39 lattice is visualized in Fig. 4.7.

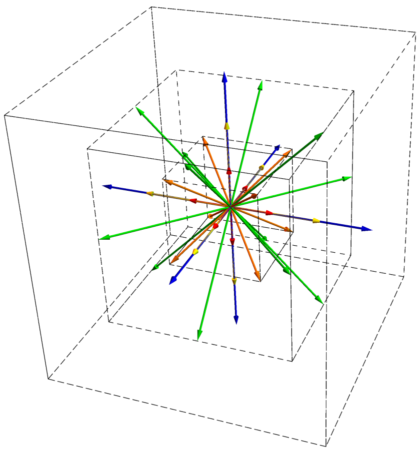


Figure 4.7: Visualization of the D3Q39 lattice. Red: $\epsilon = 1$. Orange: $\epsilon = 3$. Yellow: $\epsilon = 4$. Green: $\epsilon = 8$. Blue: $\epsilon = 9$.

This number of discrete velocities allows to decrease by at least one order of magnitude the memory overhead of the computations, and, in the case the numerical equilibrium of Sec. 3.4 would be employed, to also gain one order of magnitude in the computational cost, as shown in Sec. 3.4.2. However, as pointed out in Sec. 3.4.3, the already limited accuracy of the DdQ7^d lattice would degrade even further, rendering impossible the simulation of compressible flows. A major change is thus required in order to enforce the correct recovery of the pertinent equilibrium moments. To this end, a modified version of the so-called "entropic guided equilibrium" is employed, and presented in the next section.

4.4.2 Entropic guided equilibrium

In order to remove the aforementioned errors in the pertinent equilibrium moments, the method of guided equilibrium [63, 85] is here employed. As for the equilibrium derivation of Sec. 3.4.1, an entropy function, which will be defined later, is minimized under constraints of local conservation laws of mass, momentum and energy, and, additionally, under the conditions some pertinent moments are of the Maxwell-Boltzmann form. In the present case, the conservations and enforced moments are similar to those of the Grad's thirteen moments method, i.e. the conservation laws without the total energy, the pressure tensor

$$P^{eq}_{\alpha\beta} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} f^{eq}_i = \rho T \delta_{\alpha\beta} + \rho u_\alpha u_\beta, \tag{4.10}$$

and the contracted heat flux tensor

$$q^{eq}_\alpha = \sum_{i=1}^{n_Q} v_{i\alpha} v_i^2 f^{eq}_i = \left(2\rho E^{tr} + 2\rho T\right) u_\alpha. \tag{4.11}$$

The total energy is not included in the conservations since it is automatically recovered by the inclusion of the pressure tensor. Here, an entropy function of the form

$$H(f) = \sum_{i=1}^{n_Q} f_i \ln f_i, \tag{4.12}$$

is considered. Differently from the H function Eq. 2.45, the weights are not included here because of two reasons: the first is that the inclusion of the weights imply the equilibrium distribution to be valid only within a certain temperature range, because of positivity constraints. The second is that inclusion of the weights is not needed in the present construction, since equilibrium moments are directly enforced, and do not need to be recovered only by enforcing conservation laws. The minimization problem is expressed in terms of Lagrange multipliers, and results in

$$\delta H + \delta \left[\chi\rho + \zeta_\alpha(\rho u_\alpha) + \pi_{\alpha\beta} P^{eq}_{\alpha\beta} + \lambda_\alpha q^{eq}_\alpha\right] = 0, \tag{4.13}$$

where $\chi(\rho, \boldsymbol{u}, T)$, $\zeta_\alpha(\rho, \boldsymbol{u}, T)$, $\pi_{\alpha\beta}(\rho, \boldsymbol{u}, T)$, and $\lambda_\alpha(\rho, \boldsymbol{u}, T)$ are Lagrange multipliers corresponding to conservation of mass, momentum, pressure tensor and contracted heat flux tensor respectively. The formal solution is

$$f^{eq}_i = \rho \exp\left(\chi + \zeta_\alpha v_{i\alpha} + \pi_{\alpha\beta} v_{i\alpha} v_{i\beta} + \lambda_\alpha v_{i\alpha} v_i^2\right). \tag{4.14}$$

As for the accurate entropic equilibrium, the closed form of f_i^{eq} (4.14) is computed by inserting (4.14) in the expression for the respective conservation laws and equilibrium moments. This results in 8 equations for 8 unknown LMs in two-dimensions, and 13 equations for 13 unknown LMs in three-dimensions. Also in this case, direct numerical evaluation of LMs using the rapidly converging Newton-Raphson method is employed. Clearly, inclusion of more constraints in the minimization problem leads to higher computational cost of the equilibrium. Anyway this is largely compensated by the reduction in the computational cost related to the lattice, and, in regard to simulations with large number of CPUs or with grid refinement, the advantage is even larger. By measuring the computational time of simulations, it has been found that the gain is of the order of 3 to 4 times with respect to the original compressible model in three dimensions.
In order to show the accuracy of combining the pruned D3Q39 lattice with the guided entropic equilibrium, in Fig. 4.8 advection of vortices at different Mach numbers is presented for the original compressible model of Chap. 3 and the present approach. The employed vortex Mach number is $\mathrm{Ma_v} = 0.2$. The vortices are visualized by means of pressure contours. The behavior of

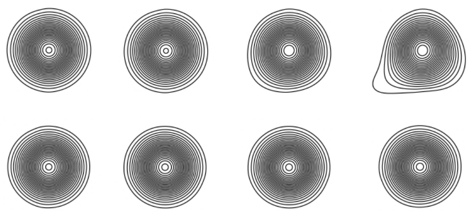


Figure 4.8: Vortex advection comparison at different advection Mach number. Top row: D3Q343 lattice with entropic equilibrium. Bottom row: D3Q39 lattice with guided entropic equilibrium.

the vortex in Fig. 4.8 is similar to the one of Fig. 4.3, where the symmetric lattice was compared to the shifted ($U = 1$) lattice. For relatively low Mach numbers, from $\mathrm{Ma_a} = 0.6$ to $\mathrm{Ma_a} = 0.9$, both models behave the same, with the vortex which is advected correctly, as shown by the pressure isocontours. Starting from a Mach number of around $\mathrm{Ma_a} = 1.2$, the vortex for the case of the original compressible model starts to slightly deform; this is even more visible for the higher Mach number $\mathrm{Ma_a} = 1.5$. Differently, the vortex in the case of the present model is not deformed, and it is advected correctly until a Mach number of around $\mathrm{Ma_a} = 1.5$. However, for the present modified formulation of the compressible model, simulations of higher Mach numbers,

above $\mathrm{Ma_a} = 1.5$, leads to unstable simulations (without the employment of shifted lattices), at least when the entropic estimate is not employed.
To conclude, the improvement of the present model with respect to the original compressible entropic lattice Boltzmann method of Chapter 3 are evident: the reduction of the computational cost by a factor $3-4$, together with the much lower memory overhead, and the higher accuracy at higher Mach numbers, render the present optimization very attractive, allowing to tackle more complex and computationally demanding simulations. Such a search for an improved high performance lattice Boltzmann model for compressible flows is left for future work. This example however, shows the proof-of-concept for further improvements.

4.5 Conclusion

Optimization strategies for the compressible lattice Boltzmann method have been discussed. First, shifted lattices for a shift of the operation domain, in therms of Mach number, has been presented, allowing to reach higher Mach numbers in the simulations maintaining high accuracy in the lower order equilibrium moments. This is very effective in those flows where the mean flow has a preferential direction. For example, in the simulation of supersonic flows around obstacles, or simulations with stationary shocks and mean cross-flow like the simulation of shock-vortex interaction or shock-turbulence interaction. Further sample simulations using shifted lattices are presented in the next chapter.
Next, modifications in the lattice discrete velocity size showed to be a moderately effective way of modifying the temperature range of the lattice, while at the same time retaining admissibility by entropic considerations.
Finally, reduction of the lattice size by pruning the original lattice, together with the employment of entropic guided equilibrium, proved to be an effective way of reducing computational cost and memory overhead, while at the same time improving accuracy and temperature range.

Chapter 5

Compressible flows simulations

5.1 Introduction

Numerical simulations are required in order to solve the Fourier-Navier-Stokes equations for practical and complex applications, which include turbulence, complex geometries, complex shock dynamics and turbulence-shock interaction in the case of supersonic flows. Unlike low Mach number turbulence, where high-order numerical schemes have met with significant success, compressible flows with and without shocks do not have a universal numerical scheme that can capture the shock and the smooth part of the flow simultaneously, due to conflicting grid and numerical requirements; the former requires high dissipation to stabilize the simulations while the latter requires low dissipation for accuracy of results [2]. In fact, hybrid schemes are a necessity for flows with discontinuities; however it is believed that the key role for the success of hybrid methods is played by shock sensors [2, 3]. Also, DNS simulations of compressible flows require small simulation time step and large grids. Thus success in this regime was quite limited [2].
For the smooth turbulent part of the flow, in order to minimize the numerical dissipation, high order explicit and implicit central schemes were devised by [144]. Such methods were widely used in the literature, especially for wave propagation problems in which nonlinearities are weak [145]. However, for high-Reynolds number fluid turbulence such a discretization

leads to numerical instabilities, owing to the accumulation of aliasing errors resulting from discrete evaluation of the non-linear convective terms, or due to the failure to discretely preserve quadratic invariants associated with the conservation equations [2]. For high-Reynolds number turbulence, alternative discretization techniques capable of ensuring stability in the inviscid limit was then proposed. One of many is the upwinding approach, commonly used by the gas-dynamic community, and based on the idea that Euler equations propagates along characteristics, and then that a stable numerical method should also propagate its information in the same characteristics direction [146]. Extension to higher order accuracy was achieved by [147] and was often used for shock-free compressible turbulence. Stability of higher-order central schemes was also achieved by filtering, which promise to cure non-linear instabilities while retaining high-order accuracy [144, 148, 149]. Some other methods attempted to design nonlinearly stable numerical schemes by replicating the energy-preservation properties of the governing equations in the discrete sense, see [150–153] to cite a few. Finally, stabilization of numerical schemes was also achieved by enforcing, at the discrete level, the conservation properties associated with entropy [151, 154–156].
All the previously cited methods suffer from spurious Gibbs oscillations and numerical instabilities near shock jumps in case of transonic and supersonic simulations [2]. Two strategies are followed in the literature to avoid these instabilities: shock-fitting approaches treat shock waves as genuine discontinuities, with dynamics governed by their own algebraic equations, using Rankine-Hugoniot relations as a set of nonlinear boundary conditions to relate the states on the two sides of the discontinuity [157]; this kind of approach guarantees usually higher accuracy but is restricted to very simple shock topology [2]. The other strategy is the shock-capturing approach and it is the most widely diffused in the computation of flows with shocks. In this case, the same discretization scheme is used in all points, and achieve regularization of the shock through additional numerical dissipation, inhibiting the onset of Gibbs oscillations. Abundant shock-capturing schemes satisfying the total variation diminishing (TVD) conditions can be found in the literature and comprehensive review is given by [158]. However, such schemes are only first order accurate [159] and then suffer in general from loss of accuracy. Major success in higher-order shock-capturing schemes was first reached by the family of essentially nonoscillatory (ENO) schemes [159–162] and then by the class of weighted essentially nonoscillatory (WENO) schemes [163–168]. Despite the excellent performance of WENO schemes in the shocked regions of the flow, their extension to smooth flows becomes very expensive due to the large number of floating operations required for the evaluation of the smoothness measurements. Moreover, spectral-like

resolution can never be achieved [2].
Due to the contrasting nature of higher-order schemes and shock-capturing schemes, in order to simulate situations where both turbulence (smooth flow) and shocks are present, hybrid schemes or artificial viscosity schemes are required. The idea behind hybrid schemes is to have a baseline spectral-like scheme with shock-capturing capability through local replacement with a classical shock-capturing method [169–171]. While the computational expense of hybrid schemes can be reduced until 50%, the key ingredient of such methods are shock sensors [172–175], which are, however, heavily reliant on tuning parameters [2]. With the employment of artificial viscosity schemes, additional viscosity is introduced in the shocked regions of the flow [176–179]. Also in this case, introduction of viscosity is modeled and relies on tuning parameters.

As an alternative to all the above numerical schemes for the discretization of the Fourier-Navier-Stokes equations, the compressible entropic lattice Boltzmann model presented in Chap. 3 is employed, in the following, to simulate a series of compressible flow set-ups, from laminar to turbulent flows, with and without shocks, in the subsonic, transonic and supersonic regime, and in two- and three-dimensions.
First, the simulation of the two-dimensional supersonic flow field around a diamond shaped airfoil is presented in Sec. 5.2, and is followed by the transonic and supersonic flow field around a two-dimensional NACA0012 airfoil, in Sec. 5.3, for viscous and inviscid flows. Next, the Busemann biplane is simulated over a wide range of Mach numbers, in Sec. 5.4, while the Schardin's problem is simulated and compared to experiments and other simulations in Sec. 5.5. Moreover, the interaction of one and two vortices with a shock wave is studied in Sec. 5.6, while the two-dimensional simulations conclude with the simulation of the Richtmyer-Meshkov instability, in Sec 5.7. The first presented three-dimensional simulation is that of a double cone immersed in supersonic inviscid flow field, in Sec. 5.8. Next, simulation of the transonic inviscid flow field around the Onera M6 wing is presented in Sec. 5.9. Turbulent compressible simulations include the isotropic compressible decaying turbulence, in Sec. 5.10, followed by the turbulence-shock interaction in Sec. 5.11.

5.2 Supersonic diamond airfoil

The supersonic, inviscid flow field around a two-dimensional diamond-shaped airfoil is here simulated [142]. The inflow Mach number is set to $\mathrm{Ma}_\infty = 2.2$, while the Reynolds number is set to $\mathrm{Re} = CU_\infty/\nu = 10^6$, with the chord of the airfoil resolved with $C = 500$ lattice points, and U_∞ the inlet velocity; the angle of attack is set to $\mathcal{A} = 3°$; the angle of the solid geometry is set to $2\theta = 6°$; finally the adiabatic exponent is fixed to $\gamma_{ad} = 1.4$. Due to the high Mach reached in the simulation a shifted lattice with $U = 1$ is employed. Free-slip boundary conditions are set at the wall of the airfoil. A snapshot of the flow field is presented in Fig. 5.1. Typical oblique shocks are formed starting from the leading and trailing edges, while expansions waves are formed at the rear edges of the airfoil. The simulation results

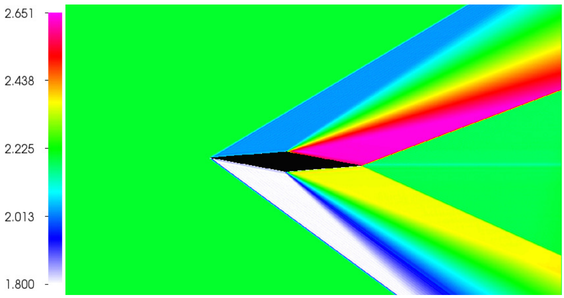


Figure 5.1: Mach number distribution around the diamond shaped airfoil immersed in a supersonic flow field at $\mathrm{Ma}_\infty = 2.2$ and $\mathrm{Re} = 10^6$.

are compared with the analytical expression [180] for shocks angles, Mach and pressure distributions around the airfoil. In Fig. 5.2, the various zones which are compared, are shown together with the corresponding angles definition. The angles of the oblique shocks with respect to the surface of the

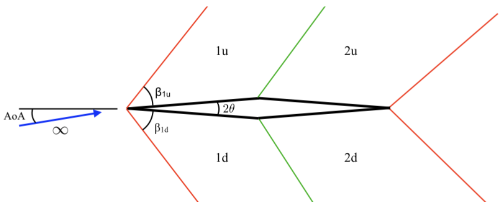


Figure 5.2: Scheme of the various zones and angles compared with analytical solutions.

airfoil depend on the upstream Mach number Ma_∞, the angle of attack $\mathcal{A}$ and the angle of the solid geometry 2θ, and are given by the non-linear equations:

$$\tan(\theta - \mathcal{A}) = 2\cot\beta_{1u}\frac{\mathrm{Ma}_\infty^2 \sin^2\beta_{1u} - 1}{\mathrm{Ma}_\infty^2\left(\gamma_{ad} + \cos 2\beta_{1u}\right) + 2}, \tag{5.1}$$

$$\tan(\theta + \mathcal{A}) = 2\cot\beta_{1d}\frac{\mathrm{Ma}_\infty^2 \sin^2\beta_{1d} - 1}{\mathrm{Ma}_\infty^2\left(\gamma_{ad} + \cos 2\beta_{1d}\right) + 2}, \tag{5.2}$$

for the upper and lower oblique shocks respectively. Once the angles of the shock are found, it is possible to express the pressure and the Mach number after the oblique shocks:

$$\frac{p_{1u/d}}{p_\infty} = 1 + \frac{2\gamma_{ad}}{\gamma_{ad}+1}\left(\mathrm{Ma}_\infty^2 \sin^2\beta_{1u/d} - 1\right), \tag{5.3}$$

$$\mathrm{Ma}_{1u} = \frac{1}{\sin^2\left(\beta_{1u} - \theta + \mathcal{A}\right)}\frac{\left(1 + \frac{\gamma_{ad}+1}{2}\mathrm{Ma}_\infty^2\sin^2\beta_{1u}\right)}{\gamma_{ad}\mathrm{Ma}_\infty^2\sin^2\beta_{1u} - \frac{\gamma_{ad}-1}{2}}, \tag{5.4}$$

$$\mathrm{Ma}_{1d} = \frac{1}{\sin^2\left(\beta_{1d} - \theta - \mathcal{A}\right)}\frac{\left(1 + \frac{\gamma_{ad}+1}{2}\mathrm{Ma}_\infty^2\sin^2\beta_{1d}\right)}{\gamma_{ad}\mathrm{Ma}_\infty^2\sin^2\beta_{1d} - \frac{\gamma_{ad}-1}{2}}. \tag{5.5}$$

In order to find the post-expansion-fan distributions ($2u$ and $2d$), one needs first to evaluate the Prandtl-Meyer function $\nu(\mathrm{Ma})$ for the pre-expansion Mach number:

$$\begin{aligned}\nu(\mathrm{Ma}_{1u/d}) = {} & \sqrt{\frac{\gamma_{ad}+1}{\gamma_{ad}-1}}\arctan\sqrt{\frac{\gamma_{ad}-1}{\gamma_{ad}+1}\left(\mathrm{Ma}_{1u/d}^2 - 1\right)} \\ & - \arctan\sqrt{\mathrm{Ma}_{1u/d}^2 - 1}.\end{aligned} \tag{5.6}$$

Next, the Mach number after the expansion can be evaluated by solving the same equation for the Mach number $\mathrm{Ma}_{2u/d}$, but for a modified Prandtl-Meyer value:

$$\begin{aligned}\nu(\mathrm{Ma}_{1u/d}) + 2\theta = {} & \sqrt{\frac{\gamma_{ad}+1}{\gamma_{ad}-1}}\arctan\sqrt{\frac{\gamma_{ad}-1}{\gamma_{ad}+1}\left(\mathrm{Ma}_{2u/d}^2 - 1\right)} \\ & - \arctan\sqrt{\mathrm{Ma}_{2u/d}^2 - 1}.\end{aligned} \tag{5.7}$$

Pressure is derived by the pressure upstream of the expansion fan and the Mach number upstream and downstream of the expansion fan:

$$\frac{p_{2u/d}}{p_{1u/d}} = \left(\frac{1 + \frac{\gamma_{ad}-1}{2}\mathrm{Ma}_{1u/d}^2}{1 + \frac{\gamma_{ad}-1}{2}\mathrm{Ma}_{2u/d}^2}\right)^{\frac{\gamma_{ad}}{\gamma_{ad}-1}}. \tag{5.8}$$

In Table 5.1, the results obtained with the ELBM solver are compared for pressure distribution, Mach number distributions and oblique shock angles, together with analytical solution employing thin airfoil approximation [180].

		P_1/P_∞	P_2/P_∞	Ma_1	Ma_2	β
Up	Analytical [-]	1.1944	0.5616	2.0860	2.5693	29.4033
	Present [-]	1.1951	0.5615	2.0859	2.5702	29.6203
	Error [%]	0.059	0.018	0.048	0.035	0.738
Down	Analytical [-]	1.6717	0.8334	1.8609	2.3077	34.7915
	Present [-]	1.6708	0.8334	1.8581	2.3074	34.7091
	Error [%]	0.054	0.018	0.015	0.013	0.237

Table 5.1: Comparison between analytical quantities and ELBM results for the problem of a diamond airfoil at high Mach. Compared are the pressure ratio P_*/P_∞ and the Mach number Ma_* for the first half and the second half of the airfoil, denoted 1 and 2 respectively, and the leading edge oblique shock angle β; every quantity for both the upper and lower surface.

The agreement between theory and the simulation with the ELBM solver is excellent demonstrating the accuracy of the compressible model and the shifted lattice.

5.3 Transonic and supersonic NACA0012 airfoil

A series of simulations is here reported for different types of flows around a NACA0012 airfoil, from transonic to supersonic, and for viscous and inviscid conditions.

5.3.1 Transonic inviscid NACA0012 airfoil

The first two simulations of the NACA0012 airfoil consider transonic conditions and inviscid flow, reproduced by imposing free-slip boundary conditions at the wall together with a high Reynolds number. Two different situations were performed: the first with an angle of attack of the airfoil of

$\mathcal{A} = 0°$ and a Mach number of $\mathrm{Ma}_\infty = 0.82$ and the other with an angle of attack of $\mathcal{A} = 2°$ and an inflow Mach number of $\mathrm{Ma}_\infty = 0.75$ [130]. For both cases the Reynolds number was set to $\mathrm{Re} = 10^6$, the chord was resolved with $C = 200$ points and no shift were employed for the lattice, i.e. $U = 0$. In Figure 5.3 the distribution of Mach number around the airfoil for the case with an angle of attack $\mathcal{A} = 0°$, on the left, and for the case with $\mathcal{A} = 2°$, on the right, are represented.

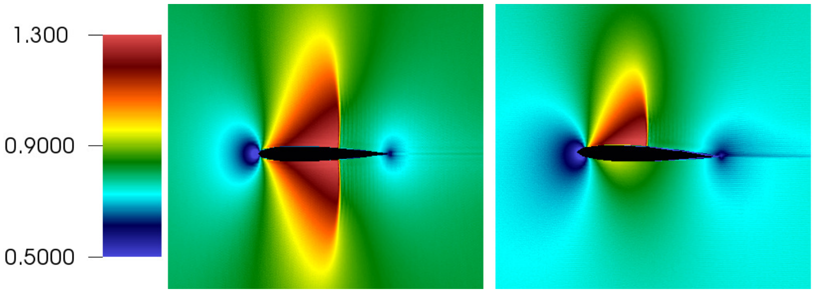


Figure 5.3: Distribution of Mach number around the NACA0012 airfoil with $\mathrm{Ma}_\infty = 0.82$ and $\mathcal{A} = 0°$ (left), and $\mathrm{Ma}_\infty = 0.75$ and $\mathcal{A} = 2°$ (right).

The plot shows the typical shock wave formation for both cases: on both sides of the airfoil for $\mathrm{Ma}_\infty = 0.82$ and $\mathcal{A} = 0°$ and on the upper surface for the case $\mathrm{Ma}_\infty = 0.75$ and $\mathcal{A} = 2°$. This is due to the progressively increasing velocity on the airfoil surface, which gradually becomes supersonic. The pressure coefficient $c_p = (P - P_\infty)/(0.5\rho U_\infty^2)$ distribution for both cases has been compared with experimental measurements, the results of Euler equations solver, and potential equation solver, and is shown in Fig. 5.4.

The ELBM solver compares very well with the Euler equations solver, while the shock is more steep if compared to the experimental measurements. This is due to the fact that in the present simulation free-slip boundary conditions were employed, while in the experiment the boundary layer plays a fundamental role in shaping the shock near the wall, diffusing the pressure. The pre- and post-shock conditions are however captured correctly. Comparison of the ELBM result with the potential solver solution is also provided, and it can be noted that both have similar behavior, apart from the fact that the ELBM solver computes the post shock pressure correctly, while the potential solver does not.
For the case with an angle of attack $\mathcal{A} = 2°$, the pressure coefficient on the non-shocked (lower) surface and computed with the ELBM solver compare perfectly with the measurements, while the Euler and potential solvers are

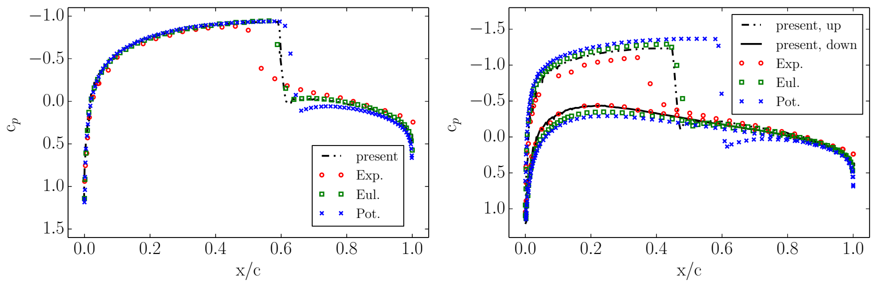


Figure 5.4: Pressure coefficient comparison for the transonic flow field around a NACA0012 airfoil for two different Mach and angle of attack, at Reynolds $\mathrm{Re} = 10^6$. Left: $\mathrm{Ma}_\infty = 0.82$ and $\mathcal{A} = 0°$. Right: $\mathrm{Ma}_\infty = 0.75$ and $\mathcal{A} = 2°$. Compared are the ELBM simulations with experimental measurements [181], Euler solver and transonic potential equation solver simulations [182].

slightly incorrect. When the upper shocked surface is analyzed, one can see that the ELBM solution is close to the Euler solution. The deviation from the experiments is due to the reason explained for the case with zero angle of attack, and gets amplified when the angle of attack is increased.

5.3.2 Transonic viscous NACA0012 airfoil

Viscous and boundary layer effects are included by imposing no-slip and adiabatic boundary condition on the surface of the airfoil. The freestream Mach number was set to $\mathrm{Ma}_\infty = 0.85$, the Reynolds number to $\mathrm{Re} = 10^4$ and the angle of attack to $\mathcal{A} = 0°$ [130]. In order to capture the boundary layer correctly, the chord was resolved with $C = 800$ points. On the left, Figure 5.5 shows the distribution of Mach number over the airfoil.

In this configuration, a complex flow field forms on the airfoil surface and downstream: due to the shear developing from the boundary layer, vortex shedding is initiated downstream the airfoil, with sound waves departing from it and accumulating at the shock front developed above (or below for the lower airfoil surface) the boundary layer. Further upstream an other shock forms: both shocks travel downstream and upstream unsteadily influenced by the vortex shedding.
Together, in Figure 5.5 the distribution of entropic estimate α is plotted for another time-step, on the right snapshot. One can notice how it follow the

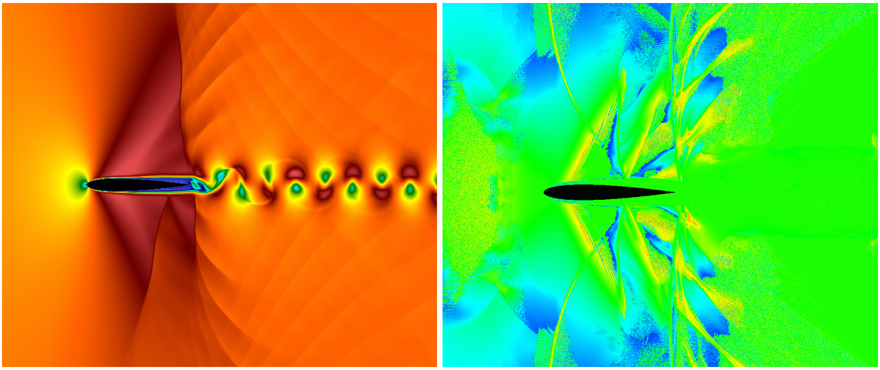


Figure 5.5: Mach number (left) and entropic estimate (right) distributions around the NACA0012 airfoil with $\mathrm{Ma}_\infty = 0.85$, $\mathrm{Re} = 10000$ and $\mathcal{A} = 0°$, shown at two different time-steps.

flow field patterns (compared to the Mach number distribution), in particular in the location of the shocks and of the waves emitted by the downstream vortex shedding, helping to stabilize the flow field. In fact, if the entropic estimate was set constant to $\alpha = 2$ (LBGK model), the simulation would become unstable. In order to compare quantitatively the solution obtained with ELBM, in Figure 5.6 the pressure coefficient distribution on the airfoil surface is plotted at normalized time $t = 13.8$.

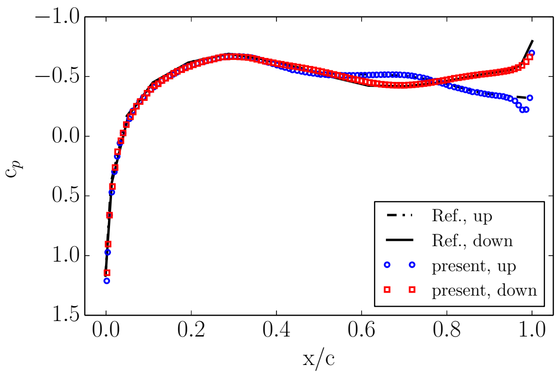


Figure 5.6: Pressure coefficient comparison for the viscous transonic flow field around a NACA0012 airfoil with $\mathrm{Ma}_\infty = 0.85$, $\mathrm{Re} = 10000$ and $\mathcal{A} = 0°$. Compared is the ELBM simulation with reference [183].

The comparison with the simulations of [183] shows once again the excellent results obtained with ELBM method.

5.3.3 Supersonic inviscid NACA0012 airfoil

As a next simulation, the two-dimensional flow around an airfoil immersed in a supersonic flow at Mach number $\mathrm{Ma}_\infty = 1.4$, with a Reynolds number $\mathrm{R}e = 3 \cdot 10^6$ and a grid resolution of $C = 200$ for the chord is considered [129]. To compare the simulation results to the solution of Euler equations, free-slip boundary conditions were employed on the airfoil surface. The expected stationary bow shock is clearly visible in the Mach distribution presented in Fig. 5.7, in the lower part. For this setup, compared is the pressure coefficient upstream and on the airfoil surface to the Euler solver [184] in Fig. 5.8. The ELBM result is in excellent agreement with the supersonic flow solver. Significance of the entropic estimate is demonstrated

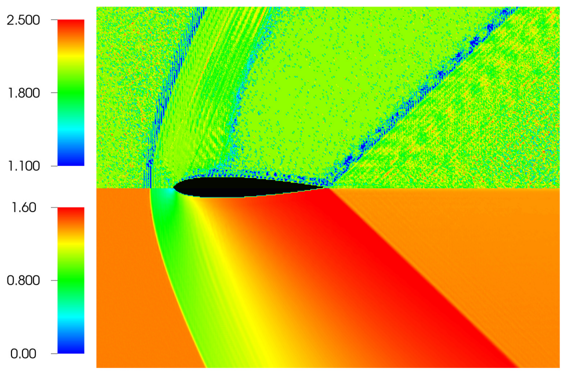


Figure 5.7: Flow around a NACA0012 airfoil immersed in a supersonic flow field at $\mathrm{Ma}_\infty = 1.4$ and $\mathrm{R}e = 3 \cdot 10^6$ based on chord length and inflow speed. Bottom panel: Mach number distribution; top panel: distribution of entropic estimate α. Steady state was reached after 5000 lattice steps.

in the upper part of Fig. 5.7, by showing the distribution of α. Note that α deviates significantly from the near-equilibrium value $\alpha = 2$ [25] in the regions near the shock waves, thus damping the Gibbs oscillations. This is demonstrated by the inset in Fig. 5.8 with a zoom into the inner structure of the bow shock: only small amplitude oscillations appear and they are confined in the near shock region. Also, the shock front is resolved with just five to seven grid points. This provides evidence that the fully explicit and purely local ELBM, free of any tuning parameter, is able to stabilize the shock automatically and on demand.

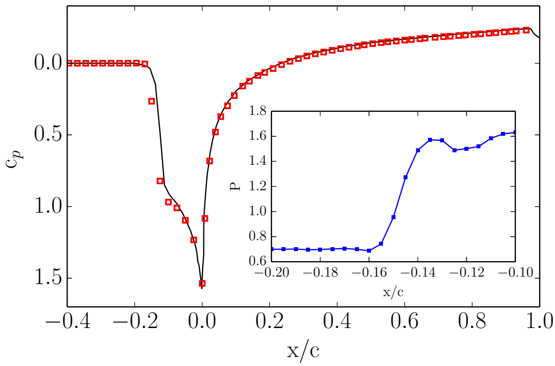


Figure 5.8: Pressure coefficient c_p upstream and on the NACA0012 airfoil surface. Symbol: ELBM. Line: supersonic flow solver [184]. Space reduced by chord length C. Inset: zoom of pressure distribution through the shock.

5.3.4 Supersonic viscous NACA0012 airfoil

Further validation of the model for high Mach flow simulations is carried out by simulating the viscous supersonic flow field around a NACA0012 airfoil with a free stream Mach number of $\mathrm{Ma}_\infty = 1.5$, Reynolds number $\mathrm{Re} = 10^4$ and zero angle of attack $\mathcal{A} = 0°$ [142]. The chord was resolved with a uniform grid of $C = 800$ points and a shifted lattice with $U = 1$ was employed. Figure 5.9 shows the temperature distribution around the airfoil at a normalized time $t/(C/U_\infty) = 232.3$. One can observe the typical bow shock forming in front of the airfoil for supersonic conditions, as well as the the oblique shocks starting as a form of a lambda-shock from the trailing edge. Moreover, due to the shear layer developing from the trailing edge of the airfoil, vortex shedding is initiated downstream. In order to validate the simulation, in Fig. 5.10 the pressure coefficient along the upstream, airfoil surface and downstream direction is plotted, and it is compared with reference [185]. The plot demonstrates the capability of the model and shifted lattices to correctly compute the pressure distribution. It is clear that introduction of stationary walls is possible also for shifted lattices which assume a mean flow.

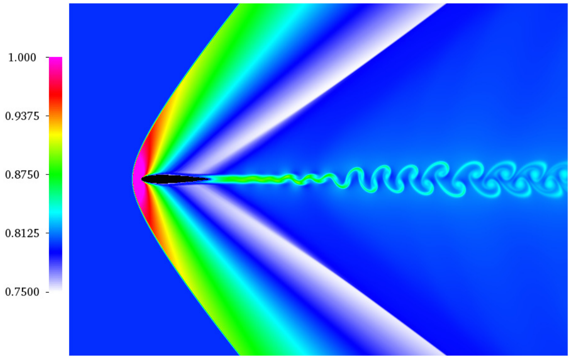


Figure 5.9: Snapshot of temperature around the NACA0012 airfoil with a free stream Mach of $\mathrm{Ma}_\infty = 1.5$, a Reynolds number $\mathrm{Re} = 10^4$ and an angle of attack $\mathcal{A} = 0°$.

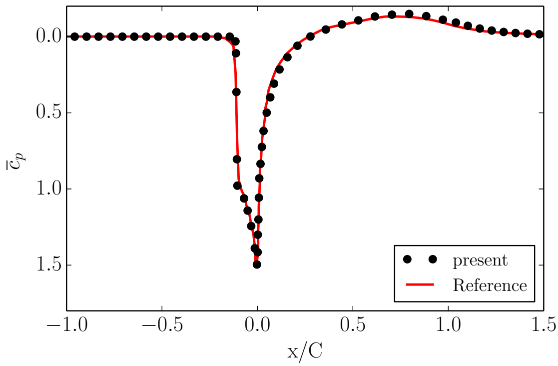


Figure 5.10: Pressure coefficient in front of the airfoil, on the airfoil surface and behind the airfoil for the simulation of the NACA0012 airfoil at free stream Mach of $\mathrm{Ma}_\infty = 1.5$, a Reynolds number $\mathrm{Re} = 10^4$ and an angle of attack $\mathcal{A} = 0°$.

5.4 Busemann biplane

In order to validate the model and the shifted lattices over a wide range of Mach numbers, the simulations of the Busemann biplane [186] are here shown for its two-dimensional configuration [142]. The Busemann biplane is a biplane composed of two half-diamond airfoils (as shown in the insets in Fig. 5.11), and it was originally conceived in order to reduce wave-drag for supersonic flight. It is well know, that conventional airfoils experience a dramatic increase of the drag coefficient c_d near to sonic conditions, and then, with increasing Mach number, the drag coefficient starts to progressively decrease again. The same happens in the case of the Busemann biplane, where

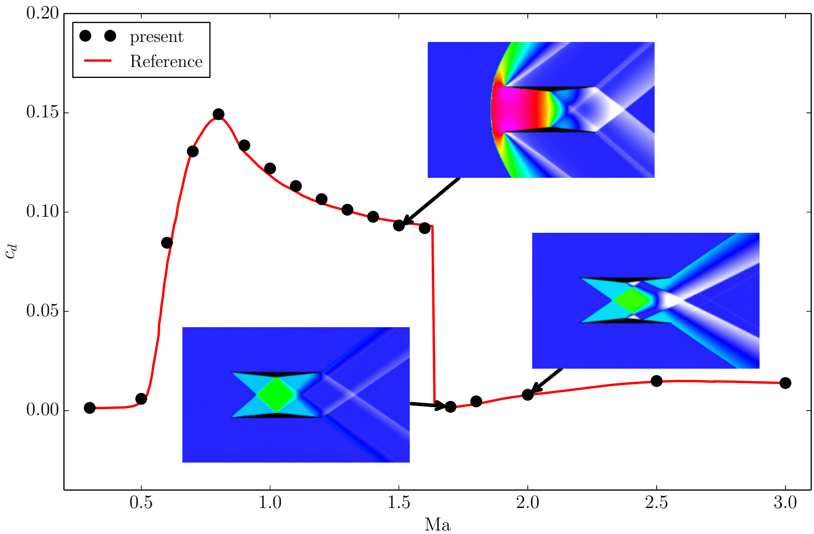


Figure 5.11: Drag coefficient c_d as a function of the free stream Mach number for the Busemann biplane simulations. Insets represent a snapshot of the pressure distribution around the biplane for three different Mach numbers: $\mathrm{Ma}_\infty = 1.5$, top; $\mathrm{Ma}_\infty = 1.7$, bottom left; $\mathrm{Ma}_\infty = 2.0$, bottom right. The pressure is shown between $\mathrm{P}_{\mathrm{min}} = 0.5$, white, and $\mathrm{P}_{\mathrm{max}} = 1.9$, pink, in lattice units.

at its simplest configuration, the drag coefficient increases dramatically near to sonic conditions, and, similar to conventional diamond airfoils, its drag coefficient starts to decrease. However, for the Busemann biplane, after a critical Mach number, the drag coefficient drops suddenly, well below that of the normal diamond shaped airfoil. This is reflected in Fig. 5.11, where the drag coefficient computed with the present ELBM model is compared to the simulations reported in [186].
Simulations of the Busemann biplane are performed in its simplest configuration, with two symmetric half diamond airfoils with zero angle of attack. The chord of the airfoil was resolved with $C = 400$ grid points. The range of simulated Mach numbers varies between $\mathrm{Ma}_{\mathrm{min}} = 0.3$ to $\mathrm{Ma}_{\mathrm{max}} = 3.0$. Along with the increasing Mach number, different lattices have been em-

ployed: from a Mach number of $\mathrm{Ma}_{\min} = 0.3$ to $\mathrm{Ma} = 1.3$ the conventional symmetric lattice D2Q49 ($U = 0$) was employed. From a Mach number of $\mathrm{Ma} = 1.4$ to $\mathrm{Ma} = 2.0$ the lattice with a shift of $U = 1$ was employed. Finally, for the two highest Mach numbers a lattice with a shift of $U = 2$ in the x-direction was used. The results of the ELBM model agree well with the simulations reported in [186]. In particular, the sensitive drag crises happening between a Mach number of $\mathrm{Ma} = 1.6$ and $\mathrm{Ma} = 1.7$ is well captured. This condition of drag crises, corresponds to free-stream Mach number at which the oblique shock waves developing from the leading edges reattach to the mid-point of the opposite airfoil, thus reflecting nearly onto its own trailing edge.
In Figure 5.11, together with the plot of the drag coefficient as a function of the free-stream Mach number, the pressure distribution around the biplane is shown at three representative Mach numbers: before, near and after drag crises. At $\mathrm{Ma} = 1.5$, before the drag crises, a detached bow shock is formed in front of the airfoils, which is responsible for a dramatic rise of the pressure which in turn increases wave drag of the biplane. When the Mach is increased sufficiently, the bow shock approaches gradually the leading edge of the airfoils, and finally it gradually transforms in two attached oblique shocks at leading edges of the airfoils. At this point drag dramatically decreases. The minimum drag coefficient is obtained at a Mach number of around $\mathrm{Ma} = 1.7$, which is represented by the lower left inset. As anticipated above, for this case the oblique shocks of the leading edges attain an optimal configuration in terms of drag coefficient. This is due to a combination of the geometric angle of the oblique shock wave and the geometry of the airfoil; at this Mach number, the oblique shock matches the middle edge of the opposite half-diamond airfoil. This has two effects: an influence on the expansion wave formed there, and a reflection of the oblique shock which reaches exactly the trailing edge of the proper airfoil [186]. This perfect matching induces the minimum drag coefficient reported in Fig. 5.11. When the Mach number is further increased, e.g. like in the case represented by the lower right inset, $\mathrm{Ma} = 2.0$, the oblique shock configuration does not match no more with the geometry of the biplane, and the drag coefficient starts to increase again.
The computation and validation against other simulations of a sensitive quantity such as the drag coefficient, clearly demonstrates the capability of the model to accurately perform simulations over a wide range of Mach numbers. Also, it is demonstrated here the typical use of the shift velocity: as the mean Mach number of the flow increases, the shift velocity is increased thus minimizing the errors arising from the non-optimal use of the lattice frame velocity.

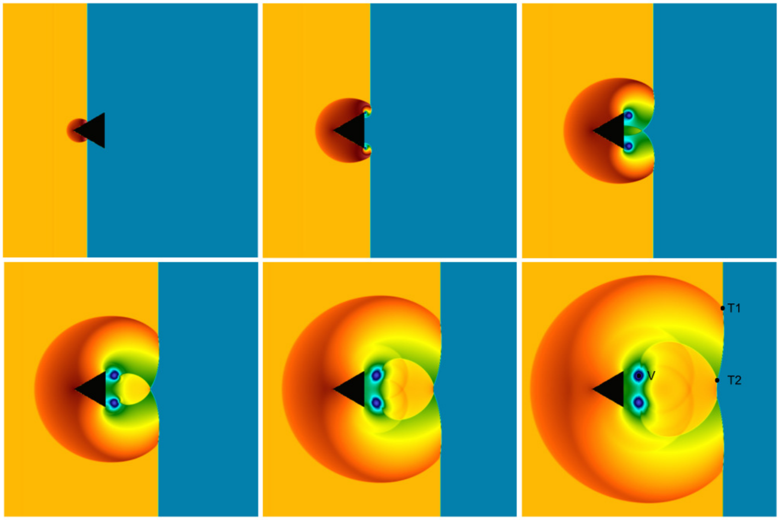


Figure 5.12: Evolution of the pressure for the Schardin's problem. Time evolves from left to right and from top to bottom.

5.5 Schardin's problem

The simulation of the Schardin's problem is here presented [130]. In this set-up a planar shock-wave impinges on a triangular wedge, reflecting and refracting, thus creating complex shock-shock and shock-vortex interactions [187, 188]. A typical evolution of the flow field for such a problem is shown in Fig. 5.12 by plotting the pressure distribution for a shock wave traveling at $\mathrm{Ma} = 1.34$ and $\mathrm{Re} = 2000$ based on the wedge length, resolved with $L = 300$ points.

At the first time instance, the shock wave has just collided with the wedge: part of the shock wave is reflected back, while the remaining part, which is free to travel, continues in the same direction. At this point a triple point is also created on the upper and lower wedge edges. At the second time instance, the traveling shock wave has already passed the wedge trailing edge, where it creates two vortices at the two corners. Part of the shock wave is thus deviated and travels along the wedge trailing edge, from both sides toward the center. In the third snapshot, the traveling shock wave interacts with its mirrored counterpart, and refracts. At this point, a second triple point is created at the refraction location. The next snapshots shows the evolution of the two triple points and of the vortex.

In Figure 5.13 the evolutions of the position of the triple point T1, the

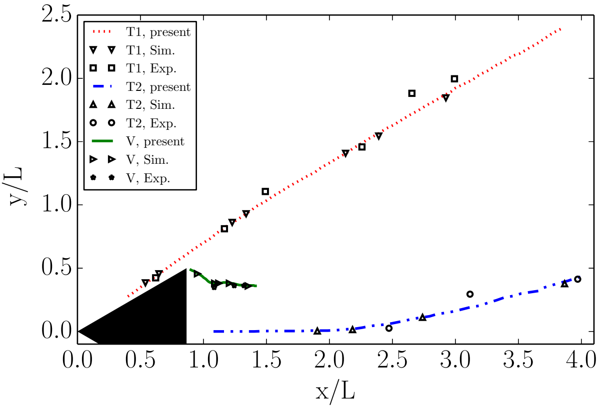


Figure 5.13: Evolution of the position of the triple point T1, the triple point T2 and the vortex V.

triple point T2 and the vortex center V are represented and compared to experimental and simulations results.

5.6 Vortex-shock interaction

In the following, the simulation of the well studied problem of vortex-shock interaction is presented; the D2Q7^2 lattice with a shift of $U = 1$ was employed. The present solutions [142] are compared to the DNS simulations of [189]. A two-dimensional vortex characterized by a vortex Mach number $\mathrm{Ma_v}$, is advected at a Mach number $\mathrm{Ma_a}$ and subsequently interacts with a stationary shock-wave. The advection Mach number corresponds to the relative Mach number of the shock with respect to the upstream velocity, $\mathrm{Ma_s} = \mathrm{Ma_a}$; the shock is kept at rest. The initial flow field of the vortex is given by:

$$u_\theta(r) = \sqrt{\gamma_{ad} T}\mathrm{Ma_v} r \exp\left((1 - r^2)/2\right), \tag{5.9}$$

and

$$u_r = 0. \tag{5.10}$$

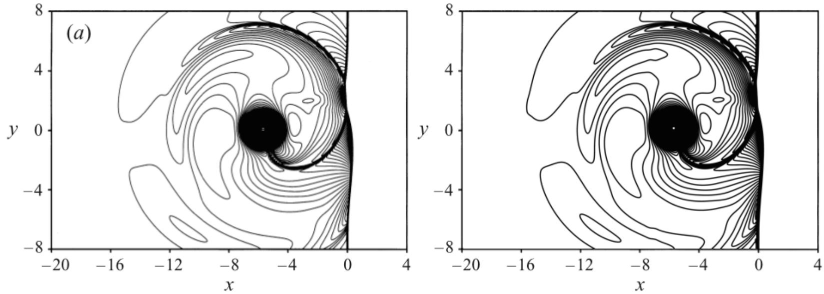


Figure 5.14: Snapshot of density for both DNS (left) and ELBM simulations (right) for the case $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.5$ and $\mathrm{Re} = 400$. Contour levels are from $\rho_{min} = 0.92$ to $\rho_{max} = 1.55$ with an increment of $\Delta\rho = 0.0053$.

Above, u_θ and u_r are the tangential and radial velocity of the vortex respectively, and r is the reduced radius $r = r'/R$, with $R = 60$ the characteristic radius. Pressure and density distributions are given by:

$$p(r) = \frac{1}{\gamma_{ad}}\left[1 - \frac{\gamma_{ad}-1}{2}\mathrm{Ma_v^2}\exp\left(1-r^2\right)\right]^{\gamma_{ad}/(\gamma_{ad}-1)}, \tag{5.11}$$

$$\rho(r) = \left[1 - \frac{\gamma_{ad}-1}{2}\mathrm{Ma_v^2}\exp\left(1-r^2\right)\right]^{1/(\gamma_{ad}-1)}. \tag{5.12}$$

The vortex flow field is superimposed on a constant flow with Ma_a upstream of the shock front. For further details on the numerical set-up the reader is referred to [189]. Three different configurations has been simulated: $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.5$ and $\mathrm{Re} = 400$; $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.25$ and $\mathrm{Re} = 800$; $\mathrm{Ma_a} = 1.05$, $\mathrm{Ma_v} = 0.25$ and $\mathrm{Re} = 400$. The Reynolds number was defined as $\mathrm{Re} = \frac{a_\infty R}{\nu}$, with the speed of sound $a_\infty = \sqrt{\gamma_{ad} T}$.

In Figure 5.14, the isocontours of density ρ are compared to the results reported in [189] for the case $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.5$ and $\mathrm{Re} = 400$ at a time $t' = t/(R/a_\infty) = 8$. From density contours, one can notice the deformed shock, at $x = 0$, the deformed vortex, at $x \simeq -5$ and $y \simeq 0$, and the reflected shocks developing from the original planar shock, one of which is still connected to the vortex. The density contours show in general a good comparison between DNS and ELBM simulations.
In Figure 5.15 the radial and tangential sound pressure distributions are compared respectively. The sound pressure is defined as $\Delta p = (p - p_s)/p_s$, where p_s is the pressure behind the shock wave. The radial sound pressure

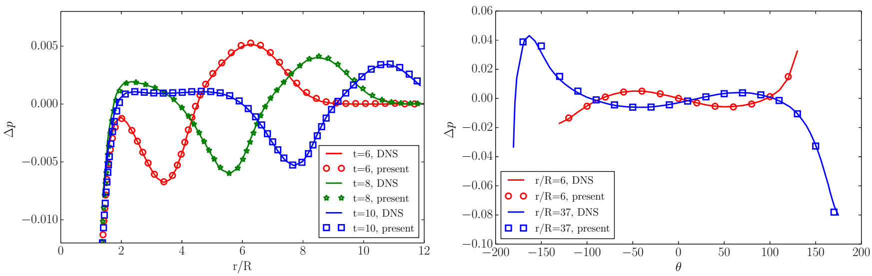


Figure 5.15: Comparison of the sound pressure distribution Δp in the radial and tangential directions for the case $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.25$ and $\mathrm{Re} = 800$. Radial sound pressure distribution Δp comparison (left) measured at an angle $\theta = -45°$ with respect to x-axis. Times: $t' = 6$, $t' = 8$, $t' = 10$. Tangential sound pressure distribution Δp comparison (right) measured at two different radius $r = 6$ and $r = 3.7$. Time: $t' = 6$.

distribution was measured in cylindrical coordinates with origin at the vortex center, at an angle $\theta = -45°$ with respect to the x-axis and at different times $t' = 6$, $t' = 8$, $t' = 10$. The tangential sound pressure distributions were measured in the tangential coordinate at two different radii $r = 6$ and $r = 37$, at time $t' = 6$. From the radial distribution one can notice how both the sound precursor (upper sound pressure peak) and the second sound propagate radially from the vortex center with time. Moreover the peak sound Δp_m corresponding to the maximum peak pressure decays with time, for both precursor and second sound. From tangential distribution it is possible to observe the quadrupolar nature of the sound generated by the vortex-shock interaction [189]: one can see in fact, that the second sound, measured at $r = 3.7$ is of opposite sign with respect to the precursor, at $r = 6$.
In Figure 5.16, the peak sound pressure Δp_m of the precursor is plotted, measured at $\theta = -45°$ at different times, and at the corresponding radial position. It can be remarked, that the amplitude of the generated sound increases by increasing both shock and vortex Mach numbers Ma_s, Ma_v. To finalize the results on vortex-shock interaction, in Fig. 5.17 is shown the simulation of a pair of vortices interacting with a shockwave with the same parameters as the simulation presented in Fig. 5.14: $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.5$ and $\mathrm{Re} = 400$. Compared is the sound pressure of the ELBM simulation and the DNS of [189]. Also in this case the comparison shows good agreement between the DNS and the ELBM method.

One should notice the following point on the non-dimensional time-

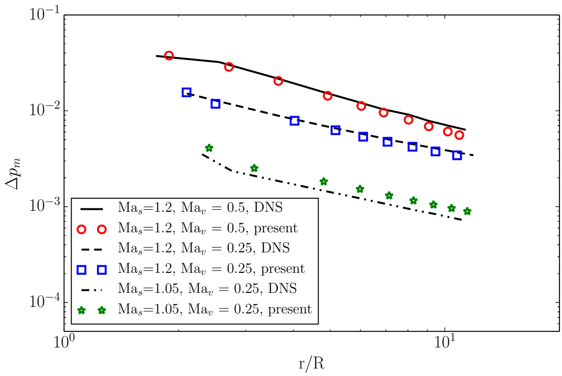


Figure 5.16: Comparison of decay of peak sound pressure.

step of the present simulations compared to DNS simulations. The non-dimensional time step for the present ELBM is $\delta t \cdot a_\infty/R = 1.67 \cdot 10^{-2}$ while for the reference DNS it is $\Delta t \cdot a_\infty/R = 1.75 \cdot 10^{-4}$; this shows one of the advantages of the ELBM method over the DNS, requiring two order of magnitude less time steps. Strong coupling of lattice Boltzmann time step with the advection velocity or Mach number is a disadvantage for LBM for low Mach number applications; however this becomes an advantage for compressible flows.

5.7 Richtmyer-Meshkov instability

The last two-dimensional simulation has the aim of validating the increased temperature range lattice with the simulation of Richtmyer-Meshkov instability (RMI) [190]. In order to reproduce the experiment of [191] in fact, a minimum temperature ratio of $T_{max}/T_{min} \simeq 4.5$ has to be allowed by the lattice; therefore the standard D2Q7^2 lattice cannot be used and the increased temperature range lattice needs to be employed.
The RMI problem consists of a shockwave, characterized by a shock Mach number $\mathrm{Ma_s}$, passing through a stratified fluid with a density interface which is of sinusoidal shape with an initial amplitude a_o, and characterized by the pre-shock Atwood number $\mathrm{A} = (\rho_2 - \rho_1)/(\rho_1 + \rho_2)$. Here ρ_1 and ρ_2 are the density upstream and downstream of the interface. After the shock front passes the interface, its amplitude grows and the RMI develops. The experimental conditions of [191] have been adopted in the simulation: the

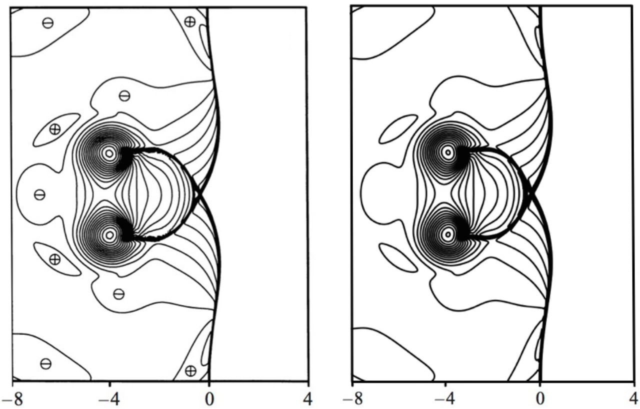


Figure 5.17: Snapshot of sound pressure for both DNS, on the left ($\oplus$ and $\ominus$ indicate positive and negative sound pressure respectively), and ELBM simulations, on the right, for the case $\mathrm{Ma_a} = 1.2$, $\mathrm{Ma_v} = 0.5$ and $\mathrm{Re} = 400$. Contour levels are from $\Delta p_{min} = -0.459$ to $\Delta p_{max} = 0.301$ with an increment of $\Delta p = 0.027$.

incident shock strength was $\mathrm{Ma_s} = 1.21$, while the pre-shock Atwood number was $A = (\rho_2 - \rho_1)/(\rho_1 + \rho_2) = 0.601$. The fluid was characterized by a mean adiabatic exponent of $\gamma_{ad} = 1.276$. The initial interface amplitude is $ka_0 = 0.21$ where k is the wavenumber of the perturbation $k = 2\pi/\lambda$. For further details the reader is referred to [191]. In the simulation, $\lambda = 1878$ grid points in the transversal direction with respect to the shock motion were adopted.

In Figure 5.18, eight snapshots of the evolution of the density are reported. Time evolves from left to right and from top to bottom. The normalized time τ is defined as $\tau = (t - t_0)ku_0$, where u_0 is the speed of the flow behind the density interface after the shock wave passes through it, and t_0 is the time at which the shock passes through the interface.

The initial conditions are represented by $\tau = -0.03$, where the shock wave is on the left of the sinusoidal density interface. After the passage of the shock, as shown for $\tau = 0.09$, part of the shock travels in the same direction, while the other part is refracted back upstream (see left interface). During the passage of the shock, vorticity is deposed near the density interface, thus initiating the growth of the amplitude of the sinusoidal interface, $\tau = 1.13 - 2.29$. After the initial growth, the instability presents the typical formation of mushrooms-like form, $\tau = 3.44 - 6.92$.

The simulation has been validated by measuring the non-dimensional growth

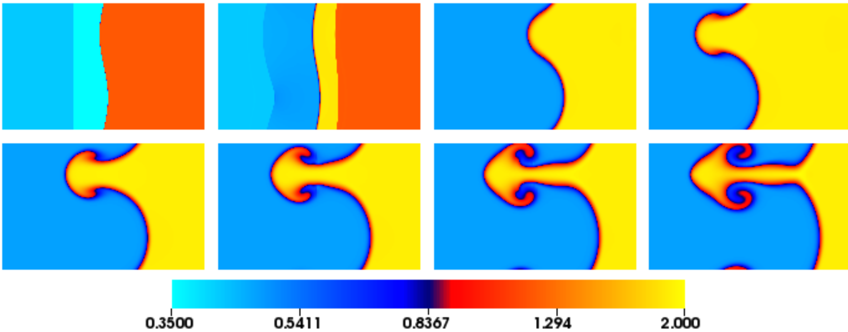


Figure 5.18: Snapshots of the evolution of the density for the RMI. Time instants from left to right and from top to bottom; first row: $\tau = -0.03$, $\tau = 0.09$, $\tau = 1.13$, $\tau = 2.29$. Second row: $\tau = 3.44$, $\tau = 4.60$, $\tau = 5.76$, $\tau = 6.92$.

rate of the instability $a(\tau)$ and comparing it with the experimental measurements of [191] and the WENO simulations of [192]. The results are reported in Fig. 5.19, which shows the good matching of the ELBM simulation with the experimental results. It should be noted that the WENO results have been non-dimensionalized with the experimental parameters ($u_0 = 628 cm/s$).

5.8 Supersonic double cone

As a first three-dimensional simulation, the results obtained by the three dimensional simulation of the supersonic flow field around a double cone geometry are shown [130]. The Mach number was set to $\mathrm{Ma}_\infty = 1.3$, the Reynolds number to $\mathrm{Re} = 10^6$ and with zero angle of attack. Free-slip boundary conditions were imposed on the cone surface in order to compare with analytical computations based on inviscid equations. In Figure 5.20 an overview of the flow field around the double cone is provided: the slice shows distribution of pressure coefficient while the orange colored isosurface of density depicts the conical shock forming in front of the cones. In order to validate the simulation, in Tab. 5.2 the angle of the cone formed by the axisymmetric oblique shock, the mean pressure coefficient and the mean Mach number on the first cone surface are compared with the analytical computations of [194].

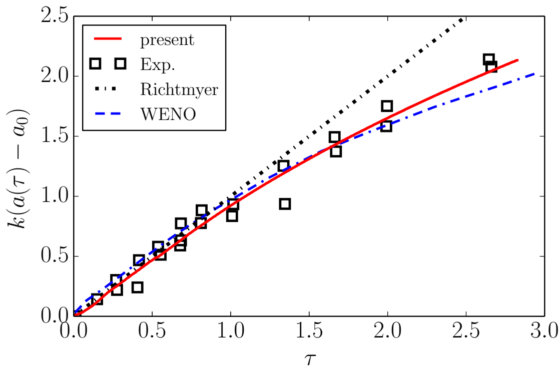


Figure 5.19: Normalized amplitude growth rate for the Richtmyer-Meshkov instability simulation as a function of normalized time. Plotted are the amplitude evolution for the ELBM, the experiment [191], the WENO simulation [192] and the original analytical prediction of Richtmyer [193] for the initial growth.

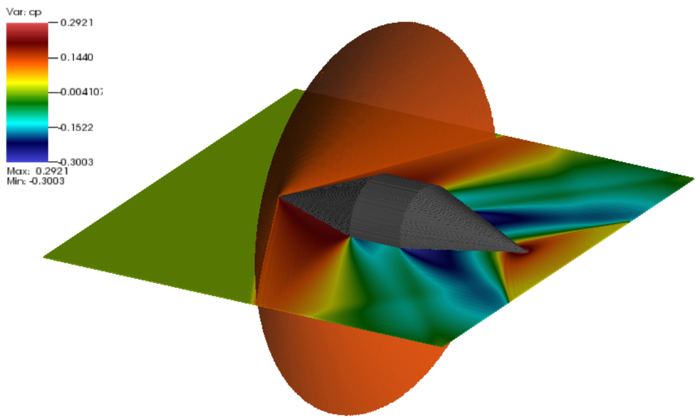


Figure 5.20: Slice of the pressure coefficient distribution together with isosurface of density in order to visualize the conical shockwave forming in front of the double cone.

Also for the three-dimensional case the comparison is satisfactory, in particular considering the low resolution employed for the simulation.

	θ_s	$\mathrm{Ma_w}$	c_{p_w}
Analytical	58.0°	0.970	0.219
Present	58.7°	0.959	0.220
ϵ [%]	1.2	1.1	0.6

Table 5.2: Errors with respect to analytical computations [194] of the semi-angle of the conical shock θ_s, the mean Mach number on the surface of the first cone $\mathrm{Ma_w}$, and on the mean pressure coefficient on the surface of the first cone c_{p_w}. The first cone is resolved with a length of $C = 150$ lattice points.

5.9 Onera M6 wing

In the following, the results of the simulation of the transonic, inviscid flow field around the Onera M6 wing are reported. This kind of swept wing of intermediate aspect ratio is a standard test example for numerical methods for transonic 3D flows, also thanks to the formation of a significant vortex above the wing tip, which interacts with the lambda-shock forming due transonic conditions.

The inflow Mach number was set to $\mathrm{Ma}_\infty = 0.839$, while the Reynolds number was $\mathrm{Re} = 11.7 \cdot 10^6$, the angle of attack $\mathcal{A} = 3.06°$ and the mean chord was resolved with $C_m = 108.9$ lattice points. For more details about wing geometry and set-up the reader is referred to [195].

In Figure 5.21, an overview of the flow field is provided by showing streamlines colored by vorticity, and by the isosurface of sonic conditions, in turquoise. The flow field presents the usual characteristics typical of the transonic flow around a three dimensional, finite wing: at the tip of the wing the characteristic tip vortex is formed, shown by the helical behavior of the streamlines together with the increase in the vorticity (from green to red). Moreover, on the upper part of the wing, a lambda-shock forms, and is shown by the sonic surface, which divides the domain into supersonic and subsonic regions. The shock, furthermore, continue toward the tip of the wing and reaches also the lower part: this is consistent with the results reported in the literature [195, 196]. The shock formation is also revealed by the sectional streamwise plots of pressure coefficient c_p, provided in Fig. 5.23, where ELBM results are compared to experiments and Euler solver simulations [195]. In particular, pressure coefficient profiles along spanwise direction reveal the formation of the lambda-shock on the upper wing surface: the upstream leg of the lambda-shock, located near the

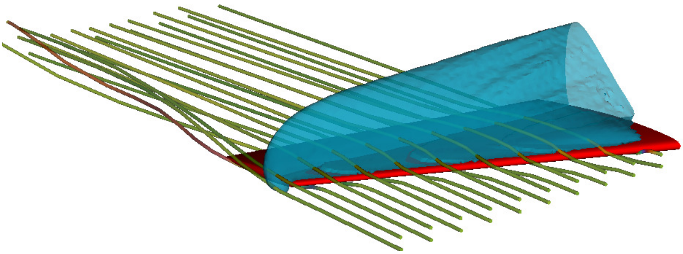


Figure 5.21: Snapshot of the flow field around the Onera M6 wing in a transonic, inviscid flow at $\mathrm{Ma}_\infty = 0.839$, $\mathrm{Re} = 11.7 \cdot 10^6$ and $\mathcal{A} = 3.06°$. Shown are the streamlines colored by vorticity, and the isosurface of sonic conditions, in turquoise.

leading edge, is the foot of a supersonic-supersonic 3D shock, whereas the downstream leg of the lambda-shock is a 3D shock terminating the local supersonic region (see c_p distributions at $y/b = 0.2$ and $y/b = 0.65$). Near the tip of the wing the two shocks merge into a single one due to the presence of a strong tip vortex influencing the flow on the upper side of the wing (see the c_p distribution at $y/b = 0.95$).
Even more insight into the characteristic of the flow field is provided by the sectional spanwise plots of c_p (fore, mid and aft chord), in Fig. 5.23, where the ELBM solution is compared again with the experimental results and the simulation of an Euler solver. Inboard, the pressure varies only slowly with the distance from the root, evidencing the two-dimensional character of the flow field, until it approaches the tip of the wing. There, the most intriguing feature of the three-dimensional flow are present: at section $x/C = 0.27$ a small amount of negative lift is produced near the tip, as evidenced by the present simulations and references (the c_p of the upper surface is smaller than for the lower surface); at the next two section this contribution returns again positive. This further confirms the presence of the rotational flow visualized by streamlines and vorticity.

5.10 Compressible decaying isotropic turbulence

To further validate the capability of the present model as a predictive method for subsonic and supersonic turbulent flows in three-simensions, simulations of compressible decaying turbulence following direct numerical

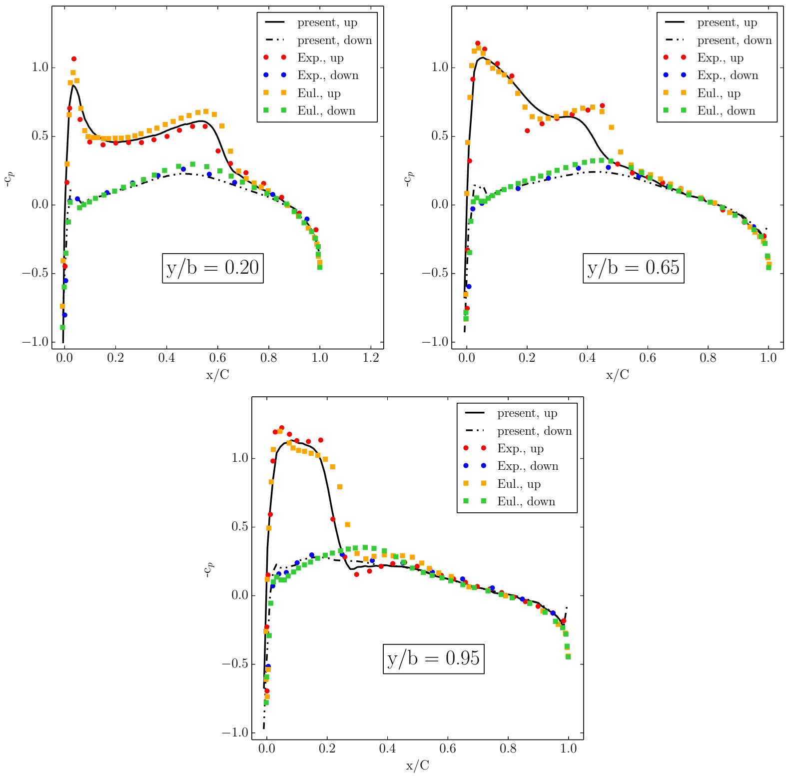


Figure 5.22: Pressure coefficient c_p on the Onera M6 wing, shown at three different wing sections in the streamwise direction, at $y/b = 0.2$, $y/b = 0.65$, and $y/b = 0.95$.

simulations of [197] are here reported. Three cases were considered at three different initial turbulent Mach number: a low Mach number case with $\mathrm{Ma_t} = 0.1$ and a Taylor Reynolds number $\mathrm{Re}_\lambda = 72$, representative of the compressible but subsonic regime; a medium initial turbulent Mach number $\mathrm{Ma_t} = 0.3$ with Taylor Reynolds number $\mathrm{Re}_\lambda = 72$, representative of compressible and near to sonic regime, and high initial turbulent Mach number $\mathrm{Ma_t} = 0.488$ with initial Taylor Reynolds number $\mathrm{Re}_\lambda = 175$, representative of the transonic/supersonic regime. Simulations were performed on uniform cubic grid of $L = 394$ points and the initial turbulent velocity field was generated by random Fourier modes according to the energy spectrum of the form $E(k) = Ak^4 \exp(-2k^2/k_0^2)$, and uniform pressure and density similar

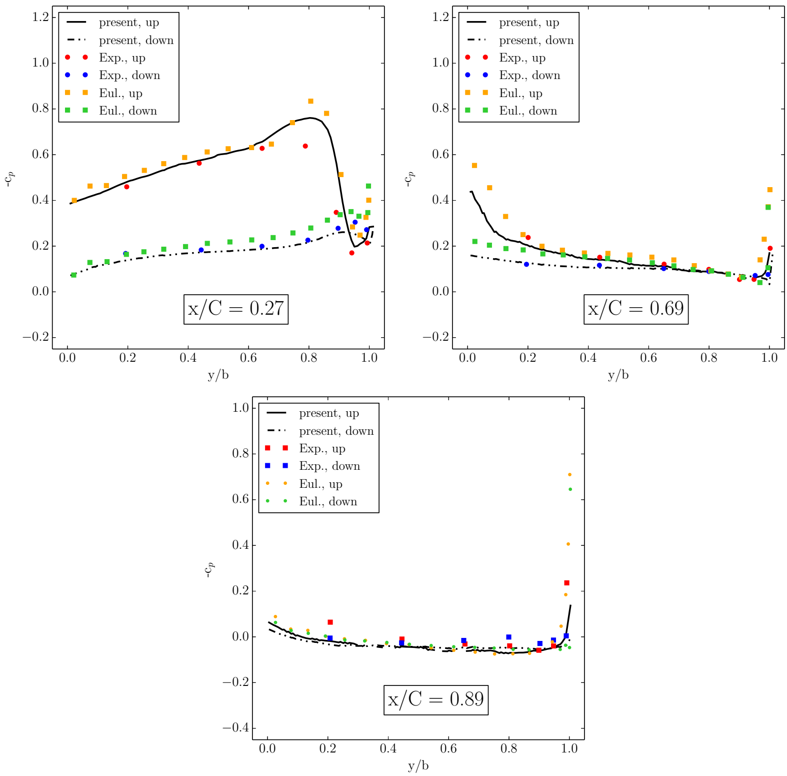


Figure 5.23: Pressure coefficient c_p on the Onera M6 wing, shown at three different sectional spanwise positions, at $x/C = 0.27$, $x/C = 0.69$, and $x/C = 0.89$.

to [197].

As a first validation, the Taylor Reynolds number decay is reported in Fig. 5.24 together with the results of the DNS [197], for the cases $\mathrm{Ma_t} = 0.3$ and $\mathrm{Ma_t} = 0.488$, as a function of the reduced time t/τ where $\tau = u'/L_I$ is the large-eddy turnover time at $t = 0$. It is possible to notice how the decays of Taylor Reynolds number follow correctly those of the DNS simulations.

Next, in Figure 5.25 the decay of turbulent kinetic energy is plotted for cases $\mathrm{Ma_t} = 0.1$ and $\mathrm{Ma_t} = 0.488$. For both cases the ELBM simulations reproduce correctly the decay in turbulent kinetic energy. Additionally, in the same Fig. 5.25, the probability density function (PDF) of the local Mach number $\mathrm{Ma_{loc}}$ is provided for time $t/\tau = 1.56$ and case $\mathrm{Ma_t} = 0.488$. The

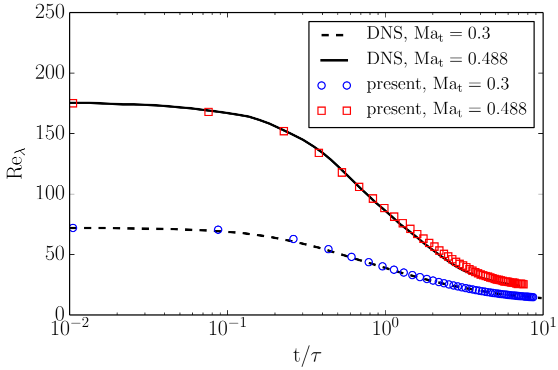


Figure 5.24: Time evolution of Re_λ for cases $Ma_t = 0.3$ and $Ma_t = 0.488$. DNS results from [197].

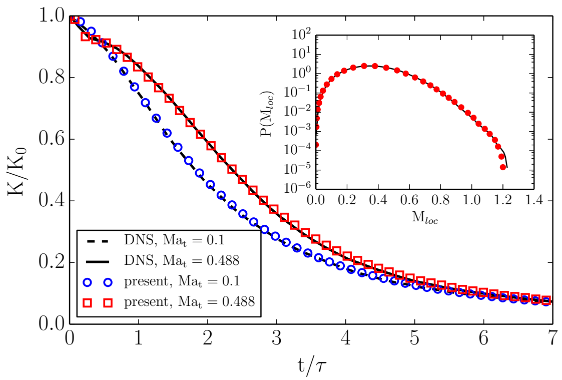


Figure 5.25: Time evolution of reduced turbulent kinetic energy K/K_0 for cases $Ma_t = 0.3$ and $Ma_t = 0.488$. Inset: Pdf of local Mach number at $t/\tau = 1.56$ for case $Ma_t = 0.488$. Line: DNS [197]. Symbol: present.

PDF shows, that for this particular case, where the initial turbulent Mach number is sufficiently high, rare events of local supersonic flow may happen in the domain, owing to the large fluctuations of velocity in the random decaying flow field. In this regions, the flow field may compress and the appearance of "shocklets" [197] should be noticeable.

In Figure 5.26, the compressed zones in the domain are visualized by a volume rendering of the velocity divergence. In particular, high compression domains, the shocklets, are seen in red, whereas, in contrast, small deviation of positive divergence are visualized in turquoise. Next, in Figure 5.27, the

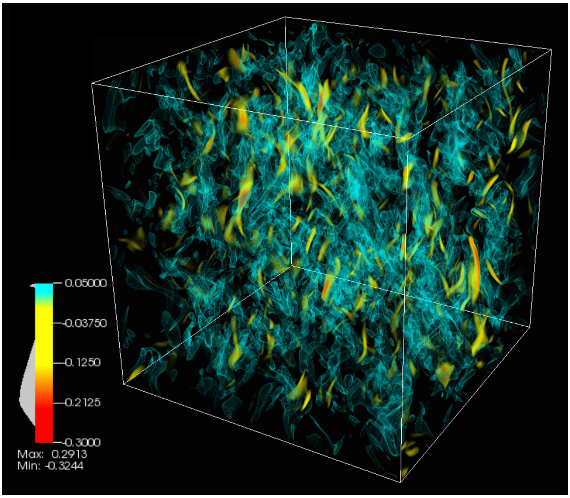


Figure 5.26: Volume rendering of velocity divergence $\nabla \cdot \boldsymbol{u} < 0$ for decaying compressible turbulence simulation at $t/\tau = 1.56$ for the case $\mathrm{Ma_t} = 0.488$. High compression domains (shocklets) are seen here in red.

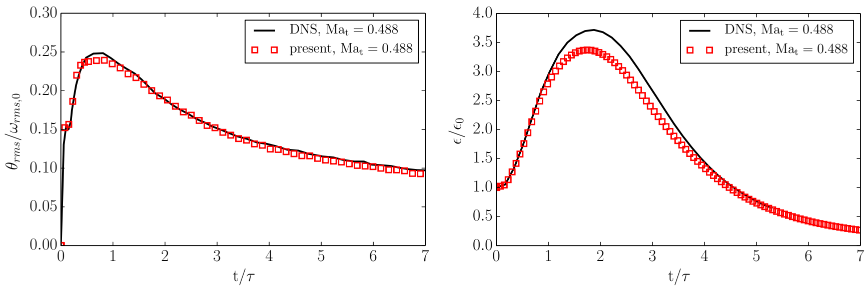


Figure 5.27: Time evolution of rms divergence θ_{rms} normalized by initial rms vorticity $\omega_{\mathrm{rms},0}$ (left), and of turbulent kinetic energy dissipation rate ϵ (right), for the case $\mathrm{Ma_t} = 0.488$.

time evolution of rms divergence θ_{rms} normalized by initial rms vorticity $\omega_{\mathrm{rms},0}$ and of turbulent kinetic energy dissipation rate ϵ are plotted for the case $\mathrm{Ma_t} = 0.488$. First, important to notice, is that the rms divergence shows a sharp increase at the beginning of the simulation, owing to the homogeneous initial condition of pressure and density. Other types of initial conditions could have been used following [197], leading to a smoother evolution of the rms velocity divergence. Second, in general, the evolution is quite well captured by the ELBM model, although an underestimation of the peak values for both rms velocity divergence and energy dissipation may be observed. One of the reasons is that these values have been evaluated by a second order accurate finite difference scheme; the employment of higher order schemes may provide better results, as suggested by the good match of all other quantities which does not require the evaluation of gradients, like the turbulent kinetic energy.

As a last validation concerning global statistical quantities, the time evo-

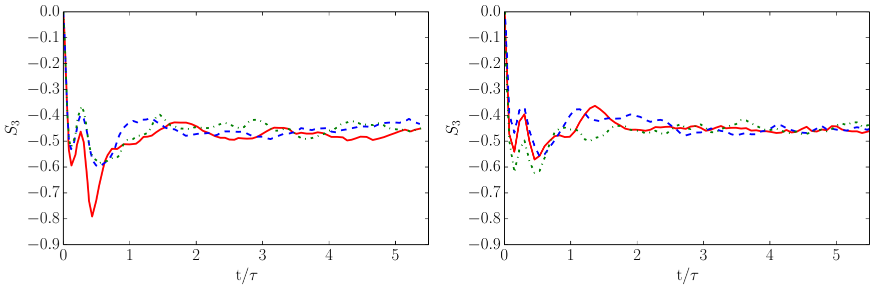


Figure 5.28: Time evolution of skewness S_3 for both DNS (left), and ELBM (right), for all directions x, y and z.

lution of skewness S_3 is shown in Fig. 5.28, on the left for the DNS and on the right for the present simulations, in all x, y and z directions. The time evolution of skewness shows a sharp decrease for both DNS and ELBM, followed by a relaxation to a nearly constant value of S_3 in the interval $\{-0.5, -0.4\}$ typical of isotropic decaying turbulence for $t/\tau > 1$.

To conclude the overview over the results on compressible isotropic decaying turbulence, in Fig. 5.29 the evolution of energy spectra is shown for the case $\mathrm{Ma_t} = 0.488$. DNS results and ELBM results has been plotted separately to ease the visualization. In general, the spectra at various times show good agreement between DNS and ELBM simulations. At time $t = 0$, shown in dashed lines, most of the energy is contained in the large scales. Comparison of the two curves for time $t = 0$ indicate that kinetic energy distribution at various scales of the initial condition is correctly initialized for the ELBM simulation. As time evolves one can notice a progressive transfer of energy from the large scales to the smaller scales, as typical for decaying isotropic turbulence. Moreover, the smallest scale progressively decreases starting from the initial time $t = 0$ and for later times.

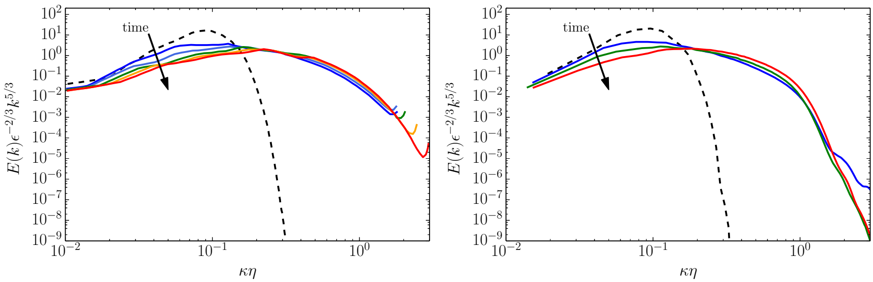


Figure 5.29: Compensated energy spectrum at various times for case $\mathrm{Ma_t} = 0.488$, for DNS (left) and ELBM (right). The dotted curve is the spectrum at time $t = 0$.

5.11 Turbulence-shock interaction

The interaction of turbulence with shocks is a fundamental phenomenon happening in a wide range of fluid mechanics fields: from supernova explosions to internal confinement fusion, supersonic flight and propulsion, and shock wave lithotripsy [198]. In many of these applications, shock-turbulence interaction includes additional complexities like real gas effects, multiple species, non-uniform mean flow, or streamline curvature [199]. The most fundamental problem, where these additional complexities are avoided, is the interaction of isotropic turbulence passing through a nominally normal shock-wave in an ideal gas. The problem of isotropic turbulence shock interaction (TSI) has been extensively studied analytically [200–203], experimentally [204–207] and by direct numerical simulations [169, 198, 208–212]. Here, preliminary results of the simulation of TSI with the compressible entropic lattice Boltzmann model are presented, for two of the cases reported in [198]: the turbulent Mach number of the isotropic turbulence is $\mathrm{Ma_t} = 0.22$, while two shock Mach number has been considered, i.e. $\mathrm{Ma_t} = 1.28$ and $\mathrm{Ma_s} = 1.5$. These two cases correspond to two different regimes: the wrinkled shock regime and the broken shock regime.
Some difficulties arises when simulating turbulence-shock interaction with the LBM. One of these arises when setting the correct inflow conditions: isotropic turbulent inflow conditions are generated by previously running a simulation of decaying isotropic turbulence, as suggested in [198] and performed in Sec. 5.10, starting with a Taylor Reynolds number $\mathrm{Re}_\lambda = 140$ and a turbulent Mach number $\mathrm{Ma_t} = 0.33$, and run for about three eddy turnover times $t/\tau = 3$. At this point, the generated flow field is used

as a database, which, together with Taylor's hypothesis [213], is used to specify the time-dependent inflow turbulence. Difficulties arises due to the employment of a multi-speed lattice, since unknown populations away from the nearest boundary points have not a clear flow field definition. In this preliminary simulations, the Grad's boundary conditions of Sec. 3.9 are employed, and the flow field obtained by the Taylor's hypothesis is employed for every boundary node. Improvements to this approach may be obtained by following [198] for the generation of the inlet flow field. Another difficulty is associated to the non-stationarity of the shock when interacting with the turbulence. In order to collect statistics in fact, stationarity of the shock needs to be guaranteed. This requires changes to the imposed pressure at the outflow. However, with the present boundary conditions, where populations are simply replaced by previous time-step populations, the pressure cannot be controlled. Once the pressure at the outflow boundary condition can be controlled, the technique to control the shock position of [198] may be employed.
In the present simulations, the computational domain is $2L \times L \times L$, with $L = 384$, the shock is positioned at about $k_0 x \simeq 8$, with $k_0 = 8\pi/L$ and initial conditions for mean streamwise velocity, temperature and density are derived by Rankine-Hugoniot conditions for shock without turbulence. A lattice shift velocity $U = 1$ in the x-direction has been employed.

The mean profiles of density and velocity are shown in Fig. 5.30 for the two cases $\mathrm{Ma_t} = 1.28$ and $\mathrm{Ma_s} = 1.5$.

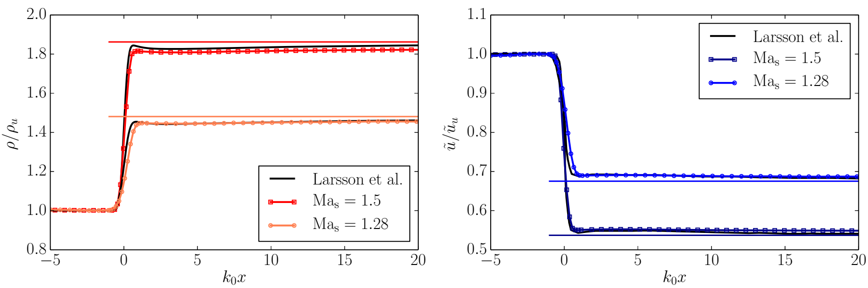


Figure 5.30: Mean density (left) and streamwise velocity (right) profiles along streamwise direction, normalized by the pre-shock states. Thin horizontal lines denoting the laminar post-shock states. Markers are clustered in the mean shock position in order to ease the visualization.

For both ELBM and DNS simulations, the jumps at the shock are consistently smaller than the Rankine-Hugoniot relations (laminar jump), also shown in the plots. Moreover, the deviations from the laminar jumps increase for larger values of $\mathrm{Ma_t}$. This is reproduced by the ELBM simulations, although the matching with DNS is not perfect. One reason may be the resolution, which in the ELBM simulations is lower in the region near the shock. Moreover, a slight movement of the shock was observed in the present simulations, which may be responsible for the flattening of the shock profile, in particular for the case $\mathrm{Ma_s} = 1.28$.

In Figure 5.31, the the streamwise and transverse Reynolds stresses are shown for cases $\mathrm{Ma_s} = 1.28$ and $\mathrm{Ma_s} = 1.5$ along the streamwise direction. The streamwise stress $\widetilde{u_1''u_1''}$ has a large peak at the mean streamwise location of the shock, and than evolves non-monotonically. This non-monotone evolution can be attributed to the transfer of energy from acoustical to vortical modes behind the shock [198]. Moreover, a higher shock Mach number leads to a larger increase of the streamwise stress peak after the shock. The transverse stress $\widetilde{u_2''u_2''}$ component increases just after the shock position, and then decreases monotonically. Also for this case, a higher shock Mach number leads to a larger stress intensity after the shock.

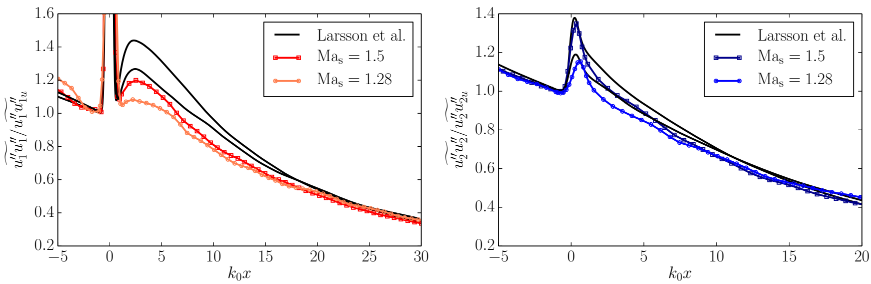


Figure 5.31: Streamwise (left) and transverse (right) stresses profiles along streamwise direction, normalized by the pre-shock states. Markers are clustered in the mean shock position in order to ease the visualization.

Although the trends are qualitatively captured by the ELBM model, a quantitative comparison cannot be provided by the present simulations yet, most likely due to the resolution, which is not sufficient for this regime of Reynolds and Mach numbers.

The vorticity variances for both simulations cases are plotted in Fig. 5.32 for streamwise and transverse components.

The transverse vorticity is amplified directly at the shock and then decays. Differently, the streamwise vorticity is initially unaffected by the shock,

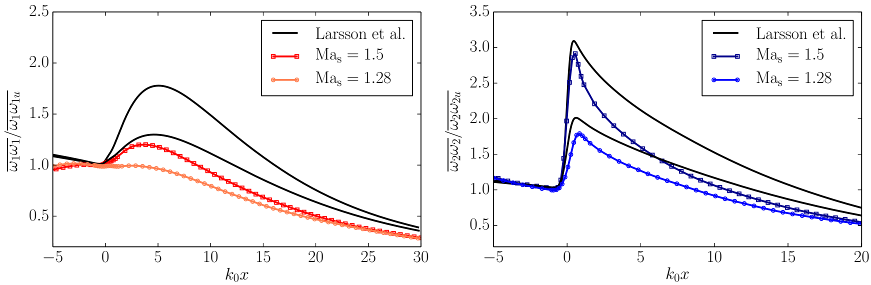


Figure 5.32: Streamwise (left) and transverse (right) vorticity variances profiles along streamwise direction, normalized by the pre-shock states. Markers are clustered in the mean shock position in order to ease the visualization.

and then increases until it equilibrates with the transverse components. The effect of increasing the shock Mach number is simply to increase the magnitude of both components after passing the shock. As for the stresses, the ELBM model is able to predict the qualitative evolution of the trends for different Mach numbers, but, at present resolution, is unable to quantitatively reproduce all the profiles of the DNS simulations.

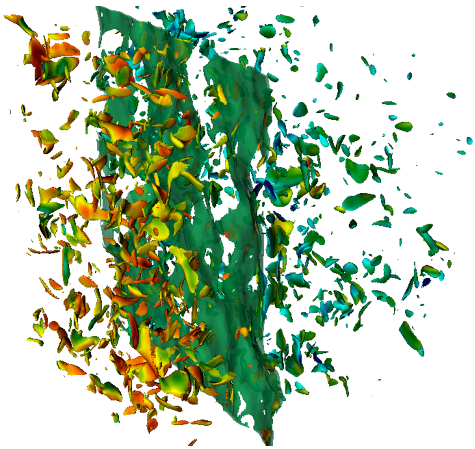


Figure 5.33: Flow field overview for the case $\mathrm{Ma_s} = 1.28$, corresponding to the broken shock regime. Shown is the vorticity isosurface colored by velocity magnitude, showing turbulent structures, and large negative divergence, showing the shock surface.

An overview of the flow field for the two different cases representing the two different regimes is provided in Figs. 5.33 and 5.34.

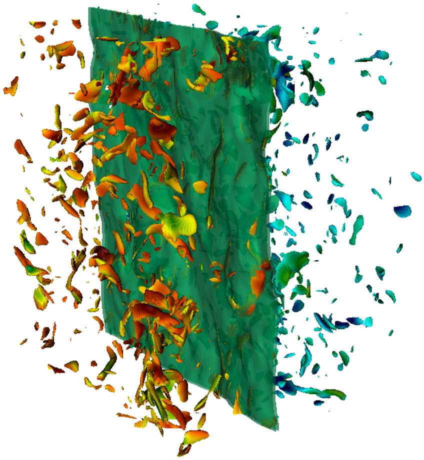


Figure 5.34: Flow field overview for the case $\mathrm{Ma_s} = 1.5$, corresponding to the wrinkled shock regime. Shown is the vorticity isosurface colored by velocity magnitude, showing turbulent structures, and large negative divergence, showing the shock surface.

Interesting to notice is the surface of the shock for the two different cases: for the lower shock Mach number cases in fact, the turbulence shock interaction is in the broken shock regime, which brings to the formation of "holes" in the shock surface, due to the stretching effect of the turbulence. The same does not happen for the case with $\mathrm{Ma_s} = 1.5$, because the turbulence is not strong enough with respect to the strength of the shock to brake it. For this case, even if the shock surface is not broken, it is no more planar and it is distorted by the turbulence. The holes in the shock, for the broken shock condition, have the main effect of smoothing the hydrodynamic quantities through it. This is exemplarily shown in Fig. 5.35, where instantaneous density profiles are plotted for three different cases: the averaged density profile, the profile representative of a "strong interaction" and the profile representative of a "weak interaction" [198].

The plot shows how in the weak interaction the transition of the density through the mean shock position is more or less smooth. Oppositely, in the strong interaction the density experience a sudden jump at the mean shock position, followed by an expansion just after the shock. This is in qualitative agreement with the DNS results [198].

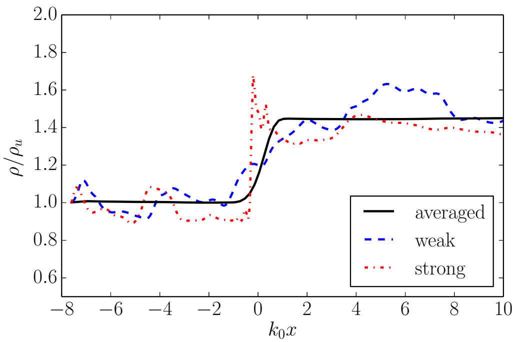


Figure 5.35: Instantaneous profiles of density along streamwise direction. Shown are weak and strong interaction together with the mean density profile.

To conclude this section about isotropic turbulence-shock interaction simulations with the ELBM, the following points arise. The present compressible model is able to provide the qualitative features of turbulence-shock interaction, in terms of Mach number and flow field characteristics, even for under-resolved conditions. However, further work should be done towards more resolved simulations, both by including grid refinement and by increasing the performance of the model in terms of computational cost. Perhaps, the lack of sufficient grid nodes to resolve the shock front results in the increased numerical dissipation that leads to a quicker decay of turbulence after the shock front. Further studies with increased grid sizes and precise estimation of dissipation at the shock front are left for the future work.

5.12 Conclusion and outlook

The compressible entropic lattice Boltzmann model has been validated by numerous two- and three-dimensional set-ups, from pure shock dynamics, as the supersonic flow field around different airfoils, or the Schardin's problem, to the interaction of vortices and turbulence with shocks, in the viscous trans- and supersonic NACA airfoil, the shock-vortex interaction and the turbulence-shock interaction. The majority of the simulations have been validated quantitatively, while for the more complex case like the turbulence shock interaction only qualitative agreement could be reached for most of the quantities, in particular due to the computational cost of the present model for three-dimensional simulations, which restricted the grid resolution. Despite these difficulties in the most complex case, in general, the present model and simulations have provided, for the first time, a way of simulating compressible flows by a genuine lattice Boltzmann approach, and over a wide range of different laminar and turbulent sub-, trans- and supersonic flows in a quantitative fashion. It must be emphasized that the results do not employ any turbulence model, tuning parameter or tracking of the shock front, and retains, overall, the main characteristic simplicity of the lattice Boltzmann method.

Further work may be pursued in this direction: the model needs to be validated for more complex compressible flows configurations, such as the boundary-layer-shock interaction, the turbulent viscous flow field around three-dimensional, real airfoils at high Reynolds number, and other practical applications. In order to perform such simulations, block grid refinement needs to be imperatively included efficiently in the simulations, while at the same time the model requires improvements, in particular from the point of view of the computational cost and memory overhead. In this sense, the present compressible model and the presented simulations represent a solid starting point never tackled by any lattice Boltzmann method before.

Chapter 6

Entropic lattice Boltzmann method for thermal flows

6.1 Introduction

Thermal flows are that kind of flows where temperature changes are important, while compressibility effects become small, owing to the low Mach number limit. A vast number of situations exists, where thermal flows are important for bot scientific and technical applications; among others, some examples are geophysical convection, including convection in the core of a planet, oceanic circulation or atmospheric dynamics [214–216], or the heat transfer in heat exchangers and chips [217, 218]. The physical phenomena that needs to be retained in the simulation of thermal flows are the advection and diffusion of temperature, the viscous heating due to dissipation, the compression work in the case where compressibility starts to play a role, and finally some additional heat sources like radiation or chemical reactions.

Over the years, a large variety of suggestions have been presented for the simulation of thermal flows with lattice Boltzmann method, each one with some degree of complexity and success in the field, see the overview provided in Sec. 2.4. In this Chapter, two thermal models, one based on the multi-speed approach Sec. 2.4.3, and the other based on the two-population approach Sec. 2.4.2, are discussed, each one with its advantages and disadvantages.

The chapter is organized as follow: in Sec. 6.2 the multi-speed entropic lattice Boltzmann model for thermal flows is presented. Next, in Sec. 6.3 the two-population entropic lattice Boltzmann model for thermal flows is detailed, including the hydrodynamic limit, its entropic multi-relaxation time formulation and the boundary conditions. Finally, in Sec. 6.4 the conclusions and outlook over the two models are drawn.

6.2 Multi-speed entropic lattice Boltzmann model

Despite from a temporal point of view, the multi-speed entropic lattice Boltzmann method for thermal flows [138] has been developed prior to the compressible entropic LBM [129], in this work it is presented as a particular case of the latter. The main principles of the two models are in fact the same, but, in the thermal model, the lattice and the equilibrium are simpler, and the second population may be avoided, since the adiabatic exponent does not play in general any role. For these reasons, only a brief overview over the model is provided in the following.

The Lattice Boltzmann equation employed for thermal simulations, including quasi-equilibrium model for the collision, reads

$$f_i(\boldsymbol{x}+\boldsymbol{v}_i,t+1)-f_i(\boldsymbol{x},t)=2\beta_1\left(f_i^{*}-f_i\right)+2\beta_2\left(f_i^{eq}-f_i^{*}\right), \tag{6.1}$$

where the equilibrium and quasi-equilibrium populations, f_i^{eq} and f_i^{*} respectively, can be written in compact form, until third order in velocity, as

$$\begin{aligned}F_i=W_i\Big(&\rho+\frac{\rho u_\alpha v_{i\alpha}}{T}+\frac{\left(M_{\alpha\beta}-\rho T\delta_{\alpha\beta}\right)\left(v_{i\alpha}v_{i\beta}-T\delta_{\alpha\beta}\right)}{2T^2}\\&+\frac{\left(M_{\alpha\beta\gamma}-\rho T\left(u_\alpha\delta_{\beta\gamma}+u_\beta\delta_{\alpha\gamma}+u_\gamma\delta_{\alpha\beta}\right)\right)\left(v_{i\alpha}v_{i\beta}v_{i\gamma}-3Tv_{i\gamma}\delta_{\alpha\beta}\right)}{6T^3}\Big).\end{aligned} \tag{6.2}$$

Here, the temperature dependent weights $W_i=W_i(T)$ depends on the lattice, and the moments $M_{\alpha\beta}$ and $M_{\alpha\beta\gamma}$ needs to be chosen depending on the equilibrium or quasi-equilibrium. For equilibrium populations, the moments correspond to the equilibrium pressure tensor and equilibrium energy flux tensor respectively

$$\begin{aligned} M_{\alpha\beta} &= P^{eq}_{\alpha\beta} = \rho T \delta_{\alpha\beta} + \rho u_\alpha u_\beta, \\ M_{\alpha\beta\gamma} &= Q^{eq}_{\alpha\beta\gamma} = \rho T \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}\right) + \rho u_\alpha u_\beta u_\gamma. \end{aligned} \tag{6.3}$$

For quasi-equilibrium populations, the second and third order moments are defined based on the Prandtl number. For $Pr \leq Pr_{BGK} = 1$:

$$M_{\alpha\beta} = P^{*}_{\alpha\beta} = P^{eq}_{\alpha\beta}, \tag{6.4}$$

$$\begin{aligned} M_{\alpha\beta\gamma} = Q^{*}_{\alpha\beta\gamma} = Q_{\alpha\beta\gamma} &- u_\alpha \left(P_{\beta\gamma} - P^{eq}_{\beta\gamma}\right) \\ &- u_\beta \left(P_{\alpha\gamma} - P^{eq}_{\alpha\gamma}\right) \\ &- u_\gamma \left(P_{\alpha\beta} - P^{eq}_{\alpha\beta}\right). \end{aligned} \tag{6.5}$$

For $Pr > Pr_{BGK} = 1$:

$$M_{\alpha\beta} = P^{*}_{\alpha\beta} = P_{\alpha\beta}, \tag{6.6}$$

$$\begin{aligned} M_{\alpha\beta\gamma} = Q^{*}_{\alpha\beta\gamma} = Q^{eq}_{\alpha\beta\gamma} &+ u_\alpha \left(P_{\beta\gamma} - P^{eq}_{\beta\gamma}\right) \\ &+ u_\beta \left(P_{\alpha\gamma} - P^{eq}_{\alpha\gamma}\right) \\ &+ u_\gamma \left(P_{\alpha\beta} - P^{eq}_{\alpha\beta}\right), \end{aligned} \tag{6.7}$$

where the second and third order tensors can be computed directly from populations:

$$P_{\alpha\beta} = \sum_{i=1}^{n_Q} f_i v_{i\alpha} v_{i\beta}, \tag{6.8}$$

$$Q_{\alpha\beta\gamma} = \sum_{i=1}^{n_Q} f_i v_{i\alpha} v_{i\beta} v_{i\gamma}. \tag{6.9}$$

The relaxation parameters β_1 and β_2 are computed by the corresponding viscosity and thermal conductivity:

$$\mu = \begin{cases} \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} \leq 1, \\ \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} > 1, \end{cases}$$

and

$$\kappa = \begin{cases} C_p \left(\frac{1}{2\beta_2} - \frac{1}{2}\right) \rho T, & \text{if } \mathrm{Pr} \leq 1, \\ C_p \left(\frac{1}{2\beta_1} - \frac{1}{2}\right) \rho T, & \text{if } \mathrm{Pr} > 1, \end{cases}$$

where the heat capacity at constant pressure is fixed in this model to $C_p = \frac{D}{2} + 1$.
The lattice employed by the model is based on the one-dimensional velocity set $V_5 = \{0, \pm 1, \pm 3\}$, which in two-dimensions becomes the D2Q25-ZOT lattice [138], with the following velocity set:

$$\begin{aligned} C_0 \quad &= \{(0,0)\}, \\ C_1 \quad &= \{(0,\pm 1),(\pm 1,0)\}, \\ C_2 \quad &= \{(\pm 1,\pm 1)\}, \\ C_3 \quad &= \{(0,\pm 3),(\pm 3,0)\}, \\ C_4 \quad &= \{(\pm 1,\pm 3),(\pm 3,\pm 1)\}, \\ C_5 \quad &= \{(\pm 3,\pm 3)\}, \end{aligned}$$

and with weights

$$\begin{aligned} W_{(0,0)} &= \frac{1}{81}(T(3T-10)+9)^2, & W_{(0,\pm 1)} &= \frac{T}{48}(3-T)(T(3T-10)+9), \\ W_{(\pm 1,\pm 1)} &= \frac{9T^2}{256}(T-3)^2, & W_{(0,\pm 3)} &= \frac{T}{1296}(3T-1)(9+T(3T-10)), \\ W_{(\pm 1,\pm 3)} &= \frac{T^2}{768}(T-3)(1-3T), & W_{(\pm 3,\pm 3)} &= \frac{T^2}{20736}(1-3T)^2. \end{aligned} \tag{6.10}$$

The hydrodynamic limit of the model, in the low-Mach number approximation, recovers the same equations as for the compressible model Sec. 6.3.4.1,

$$\partial_t \rho + \partial_\alpha (\rho u_\alpha) = 0, \tag{6.11}$$

$$\partial_t u_\alpha + u_\beta \partial_\beta u_\alpha = -\frac{1}{\rho}\partial_\alpha (\rho T) - \frac{1}{\rho}\partial_\beta \Pi_{\alpha\beta}, \tag{6.12}$$

$$\partial_t T + u_\alpha \partial_\alpha T = -\frac{1}{\rho C_v}\rho T (\partial_\alpha u_\alpha) - \frac{1}{\rho C_v}(\partial_\alpha u_\beta)\Pi_{\alpha\beta} + \frac{1}{\rho C_v}\partial_\alpha (\kappa \partial_\alpha T), \tag{6.13}$$

where the stress tensor is now defined as,

$$\Pi_{\alpha\beta} = -\mu \left(S_{\alpha\beta} - \frac{2}{D}\partial_\gamma u_\gamma \delta_{\alpha\beta} \right),$$

with the strain rate tensor,

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha.$$

Expressions for dynamic viscosity, and thermal conductivity are given by,

$$\mu = \begin{cases} \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} \leq 1, \\ \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} > 1, \end{cases} \tag{6.14}$$

$$\kappa = \begin{cases} C_p\left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} \leq 1, \\ C_p\left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T, & \text{if } \mathrm{Pr} > 1, \end{cases} \tag{6.15}$$

and the expression for the Prandtl number becomes:

$$\mathrm{Pr} = \begin{cases} \frac{(1-\beta_1)\beta_2}{(1-\beta_2)\beta_1}, & \text{if } \mathrm{Pr} \leq 1, \\ \frac{(1-\beta_2)\beta_1}{(1-\beta_1)\beta_2}, & \text{if } \mathrm{Pr} > 1, \end{cases} \tag{6.16}$$

where the heat capacities and adiabatic exponent are fixed by the dimensionality of the problem:

$$\begin{aligned} C_v &= \frac{D}{2}, \\ C_p &= \frac{D}{2} + 1, \\ \gamma_{ad} &= \frac{C_p}{C_v} = \frac{D+2}{D}. \end{aligned} \tag{6.17}$$

The boundary conditions are also the same as presented in Sec. 3.9, without the employment of the second populations and with the fixed heat capacities (6.17).

The thermo-hydrodynamic limit shows that the Fourier-Navier-Stokes equations are recovered correctly, in particular when the Mach number is low. This allows to simulate any phenomena related to thermal fluid dynamics in the low Mach number limit: convection-diffusion of temperature, viscous heating, and acoustics. This is demonstrated by a number of two dimensional simulations in Chap. 7.
However, due to the large number of lattice velocities, in particular in three-dimensions, with $n_Q = 125$, the efficiency of the lattice Boltzmann method is somehow lost. For this reason, a more efficient model, based on the two-population approach, is presented in the next Section 6.3.

6.3 Two-population entropic lattice Boltzmann model

In the two-population entropic lattice Boltzmann model for thermal flows [90, 219], in order to include the temperature dynamics into the isothermal LBM framework, the evolution equations for the thermo-hydronamic fields are split into two different lattices, one responsible for solving the mass and momentum equations and the other for solving the energy equation. The conservation laws of the thermal LBM are defined as follows:

$$\rho = \sum_{i=1}^{n_Q} f_i, \tag{6.18}$$

$$\rho u_\alpha = \sum_{i=1}^{n_Q} f_i v_{i\alpha}, \tag{6.19}$$

$$2\rho E = 2\rho C_v T + \rho u^2 = \sum_{i=1}^{n_Q} g_i. \tag{6.20}$$

The vector f_i represents populations of the lattice responsible for the mass and momentum conservation equations, while vector g_i represents the populations of the lattice responsible for the energy conservation equation. Although, in principle, the number of velocities for representing the f and g kinetics needs not to be the same, this is assumed here without any restriction. The heat capacity at constant volume C_v is imposed by the continuos Maxwell-Boltzmann distribution for a D-dimensional monoatomic gas:

$$C_v = \frac{D}{2}. \tag{6.21}$$

6.3.1 Equilibrium moments

Without loss of generality, both f- and g-lattices are here equivalent. Specifically, both distributions are discretized with standard lattices, which are D-dimensional tensor products of the fundamental velocity set $V_3 = \{0, \pm 1\}$. This results in the D2Q9 and D3Q27 lattices. These models are subject to the closure relation $v_i^3 = v_i$, which restricts them to incompressible flows simulations.

The employment of standard lattices imposes the following expressions for the pressure and heat flux tensors:

$$P^{eq}_{\alpha\beta} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} f_i^{eq} = \rho T_0 \delta_{\alpha\beta} + \rho u_\alpha u_\beta, \tag{6.22}$$

$$Q^{eq}_{\alpha\beta\gamma} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} v_{i\gamma} f_i^{eq} = \rho T_0 \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta} \right), \tag{6.23}$$

where $T_0 = 1/3$ is the reference temperature of the lattice. The moments G^{eq}_{lmn} of the energy distribution function g_i are then made consistent with the fixed reference temperature T_0 arising by the employment of the incompressible model for mass and momentum equations. These moments are derived by employing the fixed-temperature operator $\tilde{O}_\alpha$ [90]

$$\tilde{O}_\alpha = T_0 \frac{\partial}{\partial u_\alpha} + u_\alpha, \tag{6.24}$$

resulting in

$$G^{eq}_{lmn} = \left[\tilde{O}_x \right]^l \left[\tilde{O}_y \right]^m \left[\tilde{O}_z \right]^n (2\rho E), \tag{6.25}$$

for $D = 3$ dimensions. Applying the fixed temperature operator to the total energy definition in order to generate higher-order moments, and considering the closure relation $v_i^3 = v_i$, the equilibrium moments of the g-populations results in

$$\begin{aligned} q^{eq}_\alpha &= \sum_{i=1}^{n_Q} v_{i\alpha} g_i^{eq} \\ &= \left(2\rho E + 2\rho T_0 \right) u_\alpha, \end{aligned} \tag{6.26}$$

$$\begin{aligned} R^{eq}_{\alpha\beta} &= \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} g_i^{eq} \\ &= 2\rho E \left(T_0 \delta_{\alpha\beta} + u_\alpha u_\beta \right) + 2\rho T_0 \left(T_0 \delta_{\alpha\beta} + 2 u_\alpha u_\beta \right), \end{aligned} \tag{6.27}$$

$$\begin{aligned} S^{eq}_{\alpha\beta\gamma} &= \sum_{i=0}^{n_Q} v_{i\alpha} v_{i\beta} v_{i\gamma} g_i^{eq} \\ &= 2\rho T_0 \left(E + 2T_0 \right) \left(u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta} \right), \end{aligned} \tag{6.28}$$

where, in particular, the higher-order moments pertinent to the analysis of the hydrodynamic limit are the equilibrium energy flux q^{eq}_α and the once-contracted fourth-order moment $R^{eq}_{\alpha\beta}$, whereas the contracted fifth-order moment $S^{eq}_{\alpha\beta\gamma}$ is only needed for the derivation of the Grad's boundary conditions for the present model.

6.3.2 Kinetic equations

The evolution equations of both f- and g-populations are a modified version of the lattice Bhatnagar-Gross-Krook (LBGK) kinetic equation, the quasi-equilibrium model [90, 131], which enable variable Prandtl number similarly as in the case of the compressible model Sec. 3.5. The kinetic equations are:

$$f_i\left(\boldsymbol{x}+\boldsymbol{v}_i,t+1\right)-f_i\left(\boldsymbol{x},t\right)=2\beta_1\left(f_i^{eq}-f_i\right)+2\left(\beta_1-\beta_2\right)\left[f_i^{*}-f_i^{eq}\right], \tag{6.29}$$

$$g_i\left(\boldsymbol{x}+\boldsymbol{v}_i,t+1\right)-g_i\left(\boldsymbol{x},t\right)=2\beta_1\left(g_i^{eq}-g_i\right)+2\left(\beta_1-\beta_2\right)\left[g_i^{*}-g_i^{eq}\right], \tag{6.30}$$

where $\beta_1,\beta_2\in[0,1]$ are the relaxation parameters related to viscosity and thermal conductivity, f_i^{eq} and g_i^{eq} are the populations at equilibrium and f_i^* and g_i^* are the quasi-equilibrium populations. The choice of quasi-equilibrium states depends on the Prandtl number. In the case in which $\mathrm{Pr}\le 1$, $f_i^*=f_i^{eq}$, while the quasi-equilibrium g_i^* populations are designed to conserve, additionally to the total energy, the centered third-order heat-flux tensor, leading to [131, 219]:

$$\sum_{i=1}^{n_Q} g_i^*=2\rho E, \tag{6.31}$$

$$\sum_{i=1}^{n_Q} v_{i\alpha}g_i^*=q_\alpha-2u_\beta\left(P_{\alpha\beta}-P_{\alpha\beta}^{eq}\right), \tag{6.32}$$

$$\sum_{i=1}^{n_Q} v_{i\alpha}v_{i\beta}g_i^*=R_{\alpha\beta}^{eq}. \tag{6.33}$$

In the case in which $\mathrm{Pr}>1$, the f_i^* populations need to conserve, additionally to the mass and momentum, the pressure tensor:

$$P_{\alpha\beta}^*=P_{\alpha\beta}, \tag{6.34}$$

with the consequence, that the g_i^* populations have to be of the form

$$\sum_{i=1}^{n_Q} g_i^*=2\rho E, \tag{6.35}$$

$$\sum_{i=1}^{n_Q} v_{i\alpha}g_i^*=q_\alpha^{eq}+2u_\beta\left(P_{\alpha\beta}-P_{\alpha\beta}^{eq}\right), \tag{6.36}$$

$$\sum_{i=1}^{n_Q} v_{i\alpha}v_{i\beta}g_i^*=R_{\alpha\beta}^{eq}, \tag{6.37}$$

in order to be consistent with the theoretical third order quasi-equilibrium heat flux tensor on the f-lattice, as will be proved in the hydrodynamic limit Sec. 6.3.4.

6.3.3 Equilibrium and quasi-equilibrium construction

In the isothermal entropic lattice Boltzmann method, equilibrium populations are obtained by minimizing the discrete entropy function H Eq. (2.45), under local conservation laws (6.18) and (6.19). In the low Mach number limit, the equilibrium obtained as the minimizer of H can be expanded and written in a polynomial form until a certain order. In this case the Grad's distribution form [79] can be employed as a second order approximation to the entropic equilibrium. The same form of the Grad's distribution can be employed also to construct equilibria and quasi-equilibria of the g-populations by simply choosing the appropriate equilibrium and quasi-equilibrium moments. The generic form for all equilibrium and quasi-equilibrium populations can thus be written as:

$$F_i = W_i \left(M_0 + \frac{M_\alpha v_{i\alpha}}{T_0} + \frac{(M_{\alpha\beta} - M_0 T_0 \delta_{\alpha\beta})\,(v_{i\alpha} v_{i\beta} - T_0 \delta_{\alpha\beta})}{2T_0^2} \right). \quad (6.38)$$

F_i is the sought population set, M_0, M_α and $M_{\alpha\beta}$ are its moments of 0th, 1st and 2nd order. The weights employed in the present work, which are function of the chosen lattice, are the same as for the incompressible model of Sec. 2.3, as well as the corresponding velocity vectors.
The moments to be plugged in the Grad's distribution depend on the role of the sought populations; accordingly, in order to build the equilibrium distributions, the 0th, 1st and 2nd order equilibrium moments have to be employed. The same apply for the quasi-equilibrium populations also, where respective moments need to be used. All the moments required to perform the collision step are summarized in Table 6.1.

6.3.4 Thermo-hydrodynamic limit

In the following, the thermo-hydrodynamic limit of Eqs. (6.29) and (6.30) together with the conservation laws (6.18)-(6.20), for the case of a D-dimensional gas, is derived following [90, 130] and using the Chapman-Enskog method.

F_i		M_0	M_α	$M_{\alpha\beta}$
f_i^{eq}		ρ	ρu_α	$P_{\alpha\beta}^{eq}$
g_i^{eq}		$2\rho E$	q_α^{eq}	$R_{\alpha\beta}^{eq}$
$f_i^*,$	$\mathrm{Pr} \le 1$	ρ	ρu_α	P^{eq}
$f_i^*,$	$\mathrm{Pr} > 1$	ρ	ρu_α	$P_{\alpha\beta}$
$g_i^*,$	$\mathrm{Pr} \le 1$	$2\rho E$	$q_\alpha - 2u_\beta\left(P_{\alpha\beta} - P_{\alpha\beta}^{eq}\right)$	$R_{\alpha\beta}^{eq}$
$g_i^*,$	$\mathrm{Pr} > 1$	$2\rho E$	$q_\alpha^{eq} + 2u_\beta\left(P_{\alpha\beta} - P_{\alpha\beta}^{eq}\right)$	$R_{\alpha\beta}^{eq}$

Table 6.1: Moments for equilibrium and quasi-equilibrium construction.

The starting point is the expansion of the shift operator of Eqs. (6.29) and (6.30) in a Taylor series till second order

$$\left[\delta t\left(\partial_t + \partial_\mu v_{i\mu}\right) + \frac{\delta t^2}{2}\left(\partial_t + \partial_\mu v_{i\mu}\right)\left(\partial_t + \partial_\nu v_{i\nu}\right)\right] f_i \\ = 2\beta_1\left(f_i^{eq} - f_i\right) + 2\left(\beta_1 - \beta_2\right)[f_i^* - f_i^{eq}], \tag{6.39}$$

$$\left[\delta t\left(\partial_t + \partial_\mu v_{i\mu}\right) + \frac{\delta t^2}{2}\left(\partial_t + \partial_\mu v_{i\mu}\right)\left(\partial_t + \partial_\nu v_{i\nu}\right)\right] g_i \\ = 2\beta_1\left(g_i^{eq} - g_i\right) + 2\left(\beta_1 - \beta_2\right)[g_i^* - g_i^{eq}], \tag{6.40}$$

and by introducing a characteristic time scale of the flow, Θ, and a reduced time $t' = t/\Theta$. Similarly, reduced velocities and reduced coordinates are introduced, $v_i' = v_i/c$ and $x' = x/(c\Theta)$, where $c = 1$, and Eqs. (6.39), (6.40) are rewritten in terms of the reduced variables t', x', where primes are dropped to simplify notation. After introduction of the smallness parameter $\epsilon = \delta t/\Theta$, Eqs. (6.39) and (6.40) can be written as

$$\left[\epsilon\left(\partial_t + \partial_\mu v_{i\mu}\right) + \frac{\epsilon^2}{2}\left(\partial_t + \partial_\mu v_{i\mu}\right)\left(\partial_t + \partial_\nu v_{i\nu}\right)\right] f_i \\ = 2\beta_1\left(f_i^{eq} - f_i\right) + 2\left(\beta_1 - \beta_2\right)[f_i^* - f_i^{eq}], \tag{6.41}$$

$$\left[\epsilon\left(\partial_t + \partial_\mu v_{i\mu}\right) + \frac{\epsilon^2}{2}\left(\partial_t + \partial_\mu v_{i\mu}\right)\left(\partial_t + \partial_\nu v_{i\nu}\right)\right] g_i \\ = 2\beta_1\left(g_i^{eq} - g_i\right) + 2\left(\beta_1 - \beta_2\right)[g_i^* - g_i^{eq}]. \tag{6.42}$$

A multi-scale expansion of the time derivative operator, of the populations, and of the quasi-equilibrium f- and g-populations, can be performed to second order as,

$$\begin{aligned}
\epsilon\partial_t &= \epsilon\partial_t^{(1)} &&+ \epsilon^2\partial_t^{(2)} &&+ \ldots && && (6.43)\\
f_i &= f_i^{(0)} &&+ \epsilon f_i^{(1)} &&+ \epsilon^2 f_i^{(2)} &&+ \ldots && (6.44)\\
g_i &= g_i^{(0)} &&+ \epsilon g_i^{(1)} &&+ \epsilon^2 g_i^{(2)} &&+ \ldots && (6.45)\\
f_i^* &= f_i^{*(0)} &&+ \epsilon f_i^{*(1)} &&+ \epsilon^2 f_i^{*(2)} &&+ \ldots && (6.46)\\
g_i^* &= g_i^{*(0)} &&+ \epsilon g_i^{*(1)} &&+ \epsilon^2 g_i^{*(2)} &&+ \ldots && (6.47)
\end{aligned}$$

Inserting Eqs. (6.43), (6.44) and (6.46) into Eq. (6.41), and Eqs. (6.43), (6.45) and (6.47) into Eq. (6.42), analysis to order $\epsilon^{(0)}$, $\epsilon^{(1)}$ and $\epsilon^{(2)}$ is performed.

Zeroth order terms Terms to the order $\epsilon^{(0)}$ lead to,

$$f_i^{(0)} = f_i^{*(0)} + \frac{\beta_2}{\beta_1}\left(f_i^{eq} - f_i^{*(0)}\right), \tag{6.48}$$

$$g_i^{(0)} = g_i^{*(0)} + \frac{\beta_2}{\beta_1}\left(g_i^{eq} - g_i^{*(0)}\right). \tag{6.49}$$

Since the quasi-equilibrium population $f_i^{*(0)}$ and $g_i^{*(0)}$ are constructed to satisfy

$$f_i^{*(0)} = f_i^{eq}, \tag{6.50}$$

and

$$g_i^{*(0)} = g_i^{eq}, \tag{6.51}$$

it is found that the leading term in the expansions (6.44) and (6.45) is the local equilibrium,

$$f_i^{(0)} = f_i^{eq}, \qquad g_i^{(0)} = g_i^{eq}. \tag{6.52}$$

Local conservation laws imply:

$$\sum_{i=1}^{n_Q}\{1, v_{i\alpha}\} f_i = \sum_{i=1}^{n_Q}\{1, v_{i\alpha}\} f_i^{eq}, \tag{6.53}$$

for mass and momentum conservation, and

$$\sum_{i=1}^{n_Q} g_i = \sum_{i=1}^{n_Q} g_i^{eq}, \tag{6.54}$$

for the total energy conservation, which, combined with Eqs. (6.44), (6.45) and (6.52) lead to the solvability conditions

$$\sum_{i=1}^{n_Q}\{1, v_{i\alpha}\} f_i^{(1)} = \sum_{i=1}^{n_Q}\{1, v_{i\alpha}\} f_i^{(2)} = ... = 0, \tag{6.55}$$

and

$$\sum_{i=1}^{n_Q} g_i^{(1)} = \sum_{i=1}^{n_Q} g_i^{(2)} = ... = 0. \tag{6.56}$$

First order terms Collecting terms to the order $\epsilon^{(1)}$, the expressions for the first-order f- and g-populations are obtained:

$$f_i^{(1)} = f_i^{*(1)}\left(1 - \frac{\beta_2}{\beta_1}\right) - \frac{1}{2\beta_1}\left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) f_i^{(0)}, \tag{6.57}$$

$$g_i^{(1)} = g_i^{*(1)}\left(1 - \frac{\beta_2}{\beta_1}\right) - \frac{1}{2\beta_1}\left(\partial_t^{(1)} + \partial_\mu v_{i\mu}\right) g_i^{(0)}. \tag{6.58}$$

Applying solvability conditions (6.55) to Eq. (6.57), and using the conditions

$$\sum_{i=0}^{n_Q}\{1, v_{i\alpha}\} f_i^{*(1)} = 0, \tag{6.59}$$

obtained from the definition of quasi-equilibrium populations, the thermo-hydrodynamic equations of mass and momentum are recovered to first order:

$$\partial_t^{(1)}\rho = -\partial_\alpha\left(\rho u_\alpha\right), \tag{6.60}$$

$$\partial_t^{(1)} u_\alpha = -u_\beta \partial_\beta u_\alpha - \frac{1}{\rho}\partial_\alpha\left(\rho T_0\right). \tag{6.61}$$

The term ρT_0 can be identified as $p = \rho T_0$, which is the equation of state for ideal gases in the incompressible limit, where the temperature is fixed, with p being the pressure. In order to obtain first-order equations for the total energy, the solvability condition (6.56) is applied to Eq. (6.58), while the definition of the quasi-equilibrium on the g-populations is used to achieve

$$\sum_{i=1}^{n_Q} g_i^{*(1)} = 0, \tag{6.62}$$

so that:

$$\partial_t^{(1)} T = -u_\alpha \partial_\alpha T - \frac{2}{\rho D}\rho T_0\left(\partial_\alpha u_\alpha\right). \tag{6.63}$$

The sofar derived equations composed by (6.60), (6.61) and (6.63) are the incompressible Euler equations.

Second order terms Collection of second order terms $\epsilon^{(2)}$in Eqs. (6.41) and (6.42) leads to

$$\left[\partial_t^{(2)}+\frac{1}{2}\left(\partial_t^{(1)}\partial_t^{(1)}\;+\;\partial_\mu\partial_\nu v_{i\mu}v_{i\nu}+2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right]f_i^{(0)}$$
$$=-2\beta_1 f_i^{(2)}-2\left(\beta_2-\beta_1\right)f_i^{*(2)}-\left(\partial_t^{(1)}+\partial_\mu v_{i\mu}\right)f_i^{(1)}, \tag{6.64}$$

$$\left[\partial_t^{(2)}+\frac{1}{2}\left(\partial_t^{(1)}\partial_t^{(1)}\;+\;\partial_\mu\partial_\nu v_{i\mu}v_{i\nu}+2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right]g_i^{(0)}$$
$$=-2\beta_1 g_i^{(2)}-2\left(\beta_2-\beta_1\right)g_i^{*(2)}-\left(\partial_t^{(1)}+\partial_\mu v_{i\mu}\right)g_i^{(1)}. \tag{6.65}$$

Applying Eq. (6.57) into (6.64) and Eq. (6.58) into (6.65) and considering that $f_i^{*,(2)}=g_i^{*,(2)}=0$ by construction of the quasi-equilibrium, Eqs. (6.64) and (6.65) become

$$\left[\partial_t^{(2)}-\left(\frac{1}{2\beta_1}-\frac{1}{2}\right)\left(\partial_t^{(1)}\partial_t^{(1)}\;+\;\partial_\mu\partial_\nu v_{i\mu}v_{i\nu}+2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right]f_i^{(0)}$$
$$=-2\beta_1 f_i^{(2)}-\left(\partial_t^{(1)}+\partial_\mu v_{i\mu}\right)f_i^{*(1)}\left(1-\frac{\beta_2}{\beta_1}\right), \tag{6.66}$$

$$\left[\partial_t^{(2)}-\left(\frac{1}{2\beta_1}-\frac{1}{2}\right)\left(\partial_t^{(1)}\partial_t^{(1)}\;+\;\partial_\mu\partial_\nu v_{i\mu}v_{i\nu}+2\partial_t^{(1)}\partial_\mu v_{i\mu}\right)\right]g_i^{(0)}$$
$$=-2\beta_1 g_i^{(2)}-\left(\partial_t^{(1)}+\partial_\mu v_{i\mu}\right)g_i^{*(1)}\left(1-\frac{\beta_2}{\beta_1}\right). \tag{6.67}$$

Applying solvability condition for the mass, the second order contribution to the continuity equation vanishes,

$$\partial_t^{(2)}\rho=0. \tag{6.68}$$

Applying solvability condition for the momentum to (6.66) and solvability condition for the total energy to (6.67), leads to

$$\partial_t^{(2)}\left(\rho u_\alpha\right)=\left(\frac{1}{2\beta_1}-\frac{1}{2}\right)\partial_\beta\left(\partial_t^{(1)}P_{\alpha\beta}^{eq}+\partial_\gamma Q_{\alpha\beta\gamma}^{eq}\right)-\left(1-\frac{\beta_2}{\beta_1}\right)\partial_\beta P_{\alpha\beta}^{*(1)}, \tag{6.69}$$

$$\partial_t^{(2)}\left(2\rho E\right)=\left(\frac{1}{2\beta_1}-\frac{1}{2}\right)\partial_\alpha\left(\partial_t^{(1)}q_\alpha^{eq}+\partial_\beta R_{\alpha\beta}^{eq}\right)-\left(1-\frac{\beta_2}{\beta_1}\right)\partial_\alpha q_\alpha^{*(1)}. \tag{6.70}$$

At this point, a distinction between the cases where $\mathrm{Pr}\leq 1$ and $\mathrm{Pr}>1$ has to be made, as a consequence of the different quasi-equilibria employed in order to satisfy the $\mathcal{H}$-theorem.

Case $\mathrm{Pr}\leq 1$ As anticipated, in the case where $\mathrm{Pr}\leq 1$, the additional quasi-conserved quantity by the quasi-equilibrium leads to the following form for the first order quasi-equilibrium pressure tensor and the first order quasi-equilibrium heat flux:

$$P_{\alpha\beta}^{*(1)}=0, \tag{6.71}$$

$$q_\alpha^{*(1)}=q_\alpha^{(1)}-2u_\gamma P_{\alpha\gamma}^{(1)}. \tag{6.72}$$

By projecting Eqs. (6.57) and (6.58) on the desired moment space, it is possible to obtain $q_\alpha^{(1)}$ as

$$q_\alpha^{(1)}=q_\alpha^{*(1)}\left(1-\frac{\beta_2}{\beta_1}\right)-\frac{1}{2\beta_1}\left(\partial_t^{(1)}q_\alpha^{eq}+\partial_\beta R_{\alpha\beta}^{eq}\right), \tag{6.73}$$

and, by using (6.71), the first order pressure tensor $P_{\alpha\beta}^{(1)}$ as

$$P_{\alpha\beta}^{(1)}=-\frac{1}{2\beta_1}\left(\partial_t^{(1)}P_{\alpha\beta}^{eq}+\partial_\gamma Q_{\alpha\beta\gamma}^{eq}\right). \tag{6.74}$$

By inserting (6.74) in (6.72) and neglecting velocity contributions of order $O(u^3)$ pertinent to the low Mach number limit, and inserting (6.72) in (6.73), an expression for $q_\alpha^{*(1)}$ is obtained:

$$q_\alpha^{*(1)}=-\frac{1}{2\beta_2}\left(\partial_t^{(1)}q_\alpha^{eq}+\partial_\beta R_{\alpha\beta}^{eq}\right)-\frac{2u_\beta}{2\beta_2}\left(\partial_t^{(1)}P_{\alpha\beta}^{eq}+\partial_\gamma Q_{\alpha\beta\gamma}^{eq}\right). \tag{6.75}$$

Computing explicitly the right-hand-side of Eq. (6.75) leads:

$$q_\alpha^{*(1)}=-\frac{1}{2\beta_2}D\rho T_0\partial_\alpha T, \tag{6.76}$$

where $C_v=C_p=\frac{D}{2}$ relates the dimension to the heat capacities at constant volume and constant pressure, which for the thermal two-population model are the same. At this point, computation of second order time derivatives of momentum and energy is straight-forward: using (6.71) the second order momentum equation is obtained:

$$\partial_t^{(2)} u_\alpha = -\frac{1}{\rho}\partial_\beta \Pi_{\alpha\beta}, \tag{6.77}$$

where $\Pi_{\alpha\beta}$ is the stress tensor of the form,

$$\Pi_{\alpha\beta} = -\mu S_{\alpha\beta}, \tag{6.78}$$

with

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha, \tag{6.79}$$

the strain rate tensor, and

$$\mu = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T_0, \tag{6.80}$$

the dynamic viscosity. Left hand-side of Eq. (6.70) is transformed using second-order time derivatives of mass and momentum, Eqs. (6.68) and (6.77):

$$\partial_t^{(2)}(\rho E) = \rho\frac{D}{2}\partial_t^{(2)} T - u_\alpha \partial_\beta \Pi_{\alpha\beta}. \tag{6.81}$$

The right-hand side of Eq. (6.70) is transformed by computing explicitly first-order time derivative of q_α^{eq} and spatial derivative of $R_{\alpha\beta}^{eq}$; by making use of Eqs. (6.60), (6.61) and (6.63) one obtains:

$$\partial_t^{(1)} q_\alpha^{eq} + \partial_\beta R_{\alpha\beta}^{eq} = D\rho T_0 \partial_\alpha T + 2\rho T_0 u_\beta S_{\alpha\beta}. \tag{6.82}$$

By substitution of (6.82) and (6.81) in (6.70) the second order derivative of the temperature is obtained:

$$\partial_t^{(2)} T = \frac{2}{\rho D}\partial_\alpha\left(\kappa \partial_\alpha T\right) - \frac{2}{\rho D}\left(\partial_\alpha u_\beta\right)\Pi_{\alpha\beta}, \tag{6.83}$$

where κ is the thermal conductivity given by,

$$\kappa = \frac{D}{2}\left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T_0. \tag{6.84}$$

Case Pr > 1 For the case where Pr > 1, the additional quasi-conserved quantity by the quasi-equilibrium population is the traceless pressure tensor, $\overline{\Theta}_{\alpha\beta}$, for f-populations,

$$f^* = f^*\left(\rho, u_\alpha, T, \overline{\Theta}_{\alpha\beta}\right), \tag{6.85}$$

This additional conservation guarantees the following form of the first order quasi-equilibrium pressure tensor and, by consistency with the f-lattice, for the first order quasi-equilibrium heat flux:

$$P^{*(1)}_{\alpha\beta} = P^{(1)}_{\alpha\beta}, \tag{6.86}$$

$$q^{*(1)}_{\alpha} = 2u_\gamma P^{(1)}_{\alpha\gamma}. \tag{6.87}$$

By proceeding similarly as for the case $\mathrm{Pr} \leq 1$ the same equations are recovered, but with exchanged relaxation parameters for the viscosity and thermal conductivity respectively:

$$\mu = \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T_0, \tag{6.88}$$

$$\kappa = \frac{D}{2}\left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T_0. \tag{6.89}$$

6.3.4.1 Thermo-Hydrodynamic Equations

Summing up contributions of first and second order, the Fourier-Navier-Stokes equations for quasi-incompressible flows are recovered as follows:

$$\partial_t \rho + \partial_\alpha \left(\rho u_\alpha\right) = 0, \tag{6.90}$$

$$\partial_t u_\alpha + u_\beta \partial_\beta u_\alpha = -\frac{1}{\rho}\partial_\alpha \left(\rho T_0\right) - \frac{1}{\rho}\partial_\beta \Pi_{\alpha\beta}, \tag{6.91}$$

$$\partial_t T + u_\alpha \partial_\alpha T = -\frac{1}{\rho C_v}\rho T_0 \left(\partial_\alpha u_\alpha\right) - \frac{1}{\rho C_v}\left(\partial_\alpha u_\beta\right)\Pi_{\alpha\beta} + \frac{1}{\rho C_v}\partial_\alpha \left(\kappa \partial_\alpha T\right), \tag{6.92}$$

where, as before, the stress tensor is defined as,

$$\Pi_{\alpha\beta} = -\mu S_{\alpha\beta},$$

with the strain rate tensor,

$$S_{\alpha\beta} = \partial_\alpha u_\beta + \partial_\beta u_\alpha.$$

Expressions for dynamic viscosity and thermal conductivity are given by,

$$\mu = \begin{cases} \left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T_0, & \text{if } \mathrm{Pr} \leq 1, \\ \left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T_0, & \text{if } \mathrm{Pr} > 1, \end{cases} \tag{6.93}$$

$$\kappa = \begin{cases} C_p\left(\frac{1}{2\beta_2} - \frac{1}{2}\right)\rho T_0, & \text{if } \mathrm{Pr} \leq 1, \\ C_p\left(\frac{1}{2\beta_1} - \frac{1}{2}\right)\rho T_0, & \text{if } \mathrm{Pr} > 1, \end{cases} \tag{6.94}$$

and the expression for the Prandtl number becomes:

$$\mathrm{Pr} = \begin{cases} \frac{(1-\beta_1)\beta_2}{(1-\beta_2)\beta_1}, & \text{if } \mathrm{Pr} \leq 1, \\ \frac{(1-\beta_2)\beta_1}{(1-\beta_1)\beta_2}, & \text{if } \mathrm{Pr} > 1. \end{cases} \tag{6.95}$$

The heat capacities are fixed to the dimension of the problem as $C_v = C_p = \frac{D}{2}$.

6.3.5 Entropic multi-relaxation formulation

In order to simulate turbulent flows including under-resolved regimes, the entropic multi-relaxation LBM [29] formulation is here employed. The entropic multirelaxation formulation is designed to capture the sub-grid dynamics of the momentum field and it is thus applied only to the evolution of the f-lattice (6.29) in the fluid region. However, indirectly, since the evolution of the temperature depends on the velocity field, also the temperature field benefits from the entropic extension.
The kinetic equations (6.29) and (6.30) are rewritten in the following way:

$$f_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) = (1-\beta_1) f_i + \beta_1 f_i^{mirr} + 2[\beta_1 - \beta_2]\left(f_i^* - f_i^{eq}\right), \tag{6.96}$$

$$g_i(\boldsymbol{x} + \boldsymbol{v}_i, t+1) = g_i(\boldsymbol{x}, t) + 2\beta_1\left(g_i^{eq} - g_i^*\right) + 2\beta_2\left(g_i^* - g_i\right), \tag{6.97}$$

where f_i^{mirr} is the mirror state which will be specified below. Following [29], the populations f_i may be written in their moment representation:

$$f_i = k_i + s_i + h_i, \tag{6.98}$$

where k_i is the kinematic part which depends only on the locally conserved fields; s_i is the shear part, and depends on the stress tensor $\Pi = \sum_i \boldsymbol{v}_i \otimes \boldsymbol{v}_i f_i$ and optionally on other non-conserved fields; finally, h_i includes a linear combination of the remaining higher-order moments. With this representation, the mirror-state f_i^{mirr} can be written in the form:

$$f_i^{mirr} = k_i + \left[2s_i^{eq} - s_i\right] + \left[\gamma h_i^{eq} + (1-\gamma)h_i\right], \tag{6.99}$$

where γ is the entropic stabilizer, which is computed by minimizing the entropy H of the post-collision state. This reduces to solving the non-linear equation

$$\sum_{i=1}^{n_Q} \Delta h_i \ln \left[1 + \frac{(1-\beta_1\gamma)\,\Delta h_i - (2\beta_1 - 1)\Delta s_i + 2[\beta_1 - \beta_2]\,(f_i^* - f_i^{eq})}{f_i^{eq}} \right] = 0, \tag{6.100}$$

where $\Delta s_i = s_i - s_i^{eq}$ and $\Delta h_i = h_i - h_i^{eq}$. An estimate of the parameter γ is derived by expanding Eq. (6.100) to the first non-vanishing order in $\Delta s_i / f_i^{eq}$ and $\Delta h_i / f_i^{eq}$:

$$\gamma = \frac{1}{\beta_1} - \left(2 - \frac{1}{\beta_1}\right) \frac{\langle \Delta s | \Delta h \rangle}{\langle \Delta h | \Delta h \rangle} - \frac{1}{\beta_1} \frac{\langle 2[\beta_1 - \beta_2]\,(f_i^* - f_i^{eq})\,| \Delta h \rangle}{\langle \Delta h | \Delta h \rangle}, \tag{6.101}$$

where $\langle \cdot | \cdot \rangle$ is the entropic scalar product:

$$\langle X | Y \rangle = \sum_{i=1}^{n_Q} \frac{X_i Y_i}{f_i^{eq}}. \tag{6.102}$$

6.3.6 Boundary conditions

In this Section, the general treatment of boundary conditions, for both Neumann and Dirichlet cases, is described.
The treatment of the boundary conditions plays a fundamental role in the accuracy and stability of the simulations. Special care has to be taken when dealing with turbulent flows and complex geometries. Moreover, boundary conditions algorithms need to guarantee stability to the entire numerical scheme. For these reasons, the boundary conditions presented in [82], where populations are reconstructed by an approximation of the equilibrium plus a non-equilibrium part, are here extended to the thermal case. This reconstruction is performed based on the Grad's distribution, already employed previously for isothermal and multi-speed compressible models. This treatment is particularly useful if spatial gradients cannot be neglected or have to be explicitly enforced. Moreover, it enjoys higher accuracy and wider stability ranges compared to the conventional interpolated bounce-back scheme [82]. These virtues establish the Grad's approximation as a preferable method to model either boundary conditions or the interface between two media when flows with high Reynolds number are studied. The application of the Grad's boundary conditions for the computation of the mass and momentum populations has already been proposed in [82], therefore the focus here will be mainly on the energy conserving lattice.

In Fig. 6.1 a boundary node located at $\boldsymbol{x}_{\mathrm{BC}}$ is shown for the case of the D2Q9 lattice. A lattice node is defined *boundary node* if any of its populations have crossed the boundary. After each propagation step a boundary

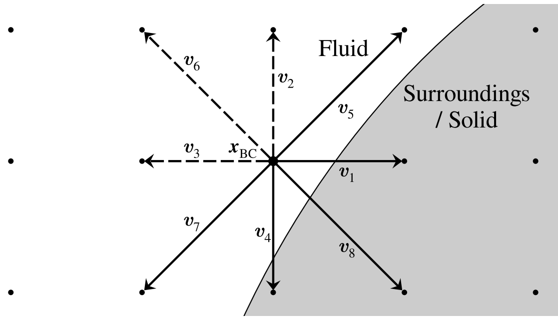


Figure 6.1: Schematic of the 8 convecting populations accommodated in $\boldsymbol{x}_{\mathrm{BC}}$. The components in the directions 2, 3 and 6 must be replaced as they have crossed the boundary during the last propagation step.

node accommodates also populations which have just crossed the border: for the case shown in Fig. 6.1 these distributions are marked with dashed arrows for populations moving from solid to fluid. Such populations are carrying information which are lacking or distorted and for this reason they will be denoted as *missing populations*.
In order to assign the appropriate information to the missing populations, f_i^{miss} and g_i^{miss}, the Grad's boundary condition employs the following approximations:

$$f_i^{\mathrm{miss}} = f_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}) + f_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}, \nabla \boldsymbol{u}_{tgt}), \tag{6.103}$$

$$g_i^{\mathrm{miss}} = g_i^{eq}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}) + g_i^{(1)}(\rho_{tgt}, \boldsymbol{u}_{tgt}, T_{tgt}, \nabla \boldsymbol{u}_{tgt}, \nabla T_{tgt}), \tag{6.104}$$

where ρ_{tgt}, $\boldsymbol{u}_{tgt}$, T_{tgt}, $\nabla \boldsymbol{u}_{tgt}$ and ∇T_{tgt} are the target density, velocity, temperature, velocity Jacobian and temperature gradient respectively, while $f_i^{(1)}$ and $g_i^{(1)}$ are the first-order terms of the multi-scale expansion of the distributions f_i and g_i. The equilibrium terms are computed as usual by plugging the moments of Table 6.1 in the Grad's distribution. Similarly, the first-order populations are derived by substituting the respective first order moments, specified below.

6.3.6.1 Non-equilibrium moments of f-lattice

For the f-populations the only first-order moment which needs to be included is the non-equilibrium pressure tensor [82]:

$$P_{\alpha\beta}^{(1)} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} f_i^{(1)} = -\frac{1}{2\beta_{1/2}} \rho T_0 S_{\alpha\beta}. \tag{6.105}$$

6.3.6.2 Non-equilibrium moments of g-lattice

Based on the derivation presented in Sec. 6.3.4, the first-order populations can be written as:

$$g_i^{(1)} = -\frac{1}{2\beta_{2/1}} \left(\partial_t^{(1)} + v_{i\alpha} \delta_{\alpha\beta} \right) g_i^{eq} + \left(1 - \frac{\beta_{1/2}}{\beta_{2/1}} \right) g_i^{*(1)}, \tag{6.106}$$

where subscript 1/2 depends on the Prandtl number. By projecting Eq. 6.106 on the desired moment space, expressions for the non-equilibrium moments are obtained:

$$q_{\alpha}^{(1)} = -\frac{1}{2\beta_{2/1}} \left(\partial_t^{(1)} q_{\alpha}^{eq} + \partial_\beta R_{\alpha\beta}^{eq} \right) + \left(1 - \frac{\beta_{1/2}}{\beta_{2/1}} \right) q_{\alpha}^{*(1)}, \tag{6.107}$$

$$R_{\alpha\beta}^{(1)} = -\frac{1}{2\beta_{2/1}} \left(\partial_t^{(1)} R_{\alpha\beta}^{eq} + \partial_\gamma S_{\alpha\beta\gamma}^{eq} \right) + \left(1 - \frac{\beta_{1/2}}{\beta_{2/1}} \right) R_{\alpha\beta}^{*(1)}. \tag{6.108}$$

The first term on the RHS of Eq. (6.107) is developed by replacing the equilibrium moments with their expressions (6.26) and (6.27), and second term by replacing respective expression for the quasi-equilibrium moments. The first-order time derivatives of the flow variables correspond to the Euler equations for thermal quasi-incompressible flows Eqs. (6.60), (6.61) and (6.63). Explicit computation of Eq. (6.107), including the pertinent total energy definition (6.20), leads to:

$$q_{\alpha}^{(1)} = -\frac{1}{2\beta_{2/1}} 2\rho C_v T_0 \partial_\alpha T + 2u_\beta P_{\alpha\beta}^{(1)}. \tag{6.109}$$

As for the derivation of $R_{\alpha\beta}^{(1),f}$, it is firstly observed that the contribution of the quasiequilibrium moment $R_{\alpha\beta}^{*(1)}$ is vanishing by construction. The derivatives left in brackets in Eq. (6.108) are then developed by means of

Eqs. (6.60)−(6.63) and all terms of the order of $O(u^3)$ are neglected in conformity with our closure relation; the result reads:

$$\partial_t^{(1)} R_{\alpha\beta}^{eq} + \partial_\gamma S_{\alpha\beta\gamma}^{eq} = 2\rho T_0 \left[S_{\alpha\beta} \left(E + 2T_0 \right) + u_\alpha \partial_\beta E + u_\beta \partial_\alpha E \right]. \quad (6.110)$$

By plugging in the definition of total energy and continuing to neglect the $O(u^3)$ terms, the expression for $R_{\alpha\beta}^{(1)}$ is finally derived:

$$R_{\alpha\beta}^{(1)} = -\frac{1}{2\beta_{2/1}} 2\rho T_0 \left[C_v \left(u_\alpha \partial_\beta T + u_\beta \partial_\alpha T \right) + S_{\alpha\beta} \left(2T_0 + C_v T \right) \right]. \quad (6.111)$$

6.3.6.3 Target quantities

So far, the missing populations were found in terms of flow density ρ, velocity $\boldsymbol{u}$, temperature T and their derivatives. Now, target quantities for fluid density ρ_{tgt}, velocity $\boldsymbol{u}_{tgt}$ and temperature T_{tgt} remain to be specified, in order to be used in the formulas of the previous Sections 6.3.6.1 and 6.3.6.2. As for the density, a bounce-back scheme is used to compute ρ_{tgt} that ensures no mass-flux condition:

$$\rho_{tgt} = \sum_{i \in \overline{D}} f_i^{bb} + \sum_{i \notin \overline{D}} f_i, \quad (6.112)$$

where $f_{v_i}^{bb} = f_{-v_i}$ for the population pointing from the solid to the fluid. The target velocity vector $\boldsymbol{u}_{tgt}$ is found by weighted linear interpolation between the velocity of the wall $\boldsymbol{U}_{\text{wall}}$ and the flow velocity at the neighboring fluid nodes $\boldsymbol{u}_{f,i}$, such that:

$$\boldsymbol{u}_{\text{tgt}} = \frac{1}{n_{\overline{D}}} \sum_{i \in \overline{D}} \frac{\delta_i \boldsymbol{u}_{f,i} + \boldsymbol{U}_{wall}}{1 + \delta_i}. \quad (6.113)$$

Here, δ_i are the normalized distances of $\boldsymbol{x}_{BC}$ from the wall along direction $\boldsymbol{v}_{-i} = -\boldsymbol{v}_i$, $\delta_i = \frac{\delta_i'}{\|\boldsymbol{v}_{-i}\|\delta t}$, where δ_i' is the distance from $\boldsymbol{x}_{BC}$ to the wall in direction $\boldsymbol{v}_{-i}$ measured in lattice units. This implies a range for δ_i of $0 < \delta_i < 1$. $n_{\overline{D}}$ is the number of missing populations.

In order to find the target temperature T_{tgt}, two different cases have to be distinguished: the case of fixed temperature of the wall (Dirichlet) and the case of an adiabatic wall (Neumann), satisfying

$$\left. \frac{\partial T}{\partial n} \right|_{wall} = 0. \quad (6.114)$$

For the case of a fixed wall temperature T_{wall}, linear interpolation is used again, since the temperature at the wall is known:

$$T_{\mathrm{tgt}} = \frac{1}{n_{\overline{D}}} \sum_{i \in \overline{D}} \frac{\delta_i T_{f,i} + T_{wall}}{1+\delta_i}. \tag{6.115}$$

As for the adiabatic wall, the target temperature is computed by means of the no flux condition (6.114). This requires, for a generic boundary point $\boldsymbol{x}_{BC}$, the computation of the temperature at the corresponding point $\boldsymbol{x}_f$ distant $l = \sqrt{D}$ by bilinear interpolation (in 2D) or trilinear interpolation (in 3D), as described in Sec. 3.9.2. For a given temperature T_f at point $\boldsymbol{x}_f$, one simply needs to set:

$$T_{\mathrm{tgt}} = T_f, \tag{6.116}$$

where the approximation

$$\frac{\partial T}{\partial n} \simeq \frac{T_f - T_{tgt}}{l - \delta_n} = 0, \tag{6.117}$$

has been employed. Above, δ_n is the distance from the boundary node $\boldsymbol{x}_{BC}$ to the wall in the normal direction, i.e. the closest distance, and is measured in lattice units. The range of δ_n is $0 < \delta_n < l$.

In addition to the above target quantities, the computation of the non-equilibrium populations $f_i^{(1)}$ and $g_i^{(1)}$ of Eqs. (6.103) and (6.104) requires the knowledge of spatial gradients for the velocity and temperature fields. Semi-second-order approximations are used to discretize these derivatives. In summary, Eqs. (6.112) – (6.116) provide all the quantities required to compute the equilibrium and first-order moments, which in turn allow to find the missing populations by means of the Grad's distribution (6.38).

6.4 Conclusion

Two thermal lattice Boltzmann models have been presented, each one with its pros and cons. The two-population model is simpler, since the mass and momentum are decoupled from the temperature equation, which is solved by another lattice. This configuration provides an efficient way of computing thermal flows, including advection-diffusion problems, and viscous heating. However, with this model it is not possible to simulate acoustics, since the

mass, momentum and energy equations are not fully coupled.
On the other side, the multi-speed thermal model has all the same physical features as the two-population model, and, additionally, can be used to compute acoustics, due to the correct recovery of the compressible Fourier-Navier-Stokes equations, in the low-Mach number limit. The main drawback of the multi-speed model is the computational cost arising by the employment of multi-speed lattices, which increases the computational cost of the advection operation and the communication in the case of parallel implementation in multi-CPU computers. Moreover, the boundary conditions of the multi-speed model are more complex than the two-population model, resulting in some cases in slightly lower accuracy.

In the next Chapter, numerical applications of the previously described models are exposed. Next, an extension of the two-population model for the simulation of conjugate heat transfer problems between solid-solid and solid-fluids is presented and validated in Chapter 8.

Chapter 7

Thermal flows simulations

7.1 Introduction

In this Chapter, the simulations employed to validate the multi-speed and the two-population thermal models are presented. In a first part, simple, two-dimensional simulations of thermal flows are presented for the multi-speed model: the correct viscous heating recovery is validated by the thermal Couette flow, in Sec. 7.2. Two-dimensional non-linear acoustics is validated in Sec. 7.3, and finally, simulations of the heated circular and square cylinder is reported in Sec. 7.4.
In the second part of the Chapter, three-dimensional simulations of the non-rotating and rotating Rayleigh-Bénard convection employing the two-population LBM are reported in Sec. 7.5; different regimes has been simulated, by varying the Rayleigh and Ekmann numbers.
In Section 7.6 the conclusions will be drawn.

7.2 Thermal couette

The thermal Couette flow problem is used to test the capability of the multi-speed thermal model in evaluating the correct viscous heating for different Prandtl numbers [138]. The bottom wall of the domain was kept at a temperature T_C and at rest, while the top wall had a temperature $T_H > T_C$ and a uniform wall velocity U_w. The analytical solution for the velocity

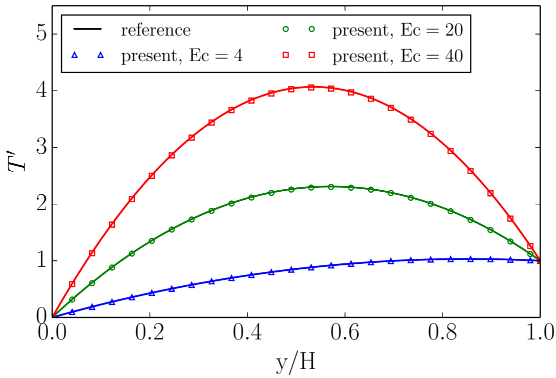


Figure 7.1: Temperature profile of the thermal Couette flow at various Eckert numbers for the fixed Prandtl number $\mathrm{Pr} = 0.71$.

field is a simple linear profile, while the solution for the temperature can be written using reduced temperature, $T' = (T - T_C)/(T_H - T_C)$, as:

$$T' = \frac{y}{H} + \frac{\mathrm{Pr} \cdot \mathrm{Ec}}{2} \frac{y}{H} \left(1 - \frac{y}{H}\right), \tag{7.1}$$

where y is the perpendicular distance from the bottom wall, H is the height of the channel, $\mathrm{Pr} = C_p \mu / \kappa$ is the Prandtl number, $\mathrm{Ec} = U_w^2 / \left(C_p \left(T_H - T_C\right)\right)$ is the Eckert number with $C_p = \frac{D+2}{2}$. These simulations were performed in a wide range of Pr and Ec numbers, with the following parameters: $U_w = 0.05$, $H = 100$, $T_C = 0.3675$, $\mu = 0.025$; other parameters can be defined using the Prandtl and Eckert numbers. Periodic boundary conditions were applied in the horizontal direction, while at the bottom and top walls, the Grad's boundary conditions were applied. Figure 7.1 represents the results of the simulation for different Eckert numbers, while the Prandtl number is fixed to $\mathrm{Pr} = 0.71$. The results, for fixed Eckert number $\mathrm{Ec} = 8.0$ and variable Prandtl numbers, are shown in Fig. 7.2. Both Figures 7.1 and 7.2 demonstrate that the present thermal lattice Boltzmann model recovers the analytical solution with a good accuracy.

7.3 Plane sound wave

The performance of the present multi-speed thermal lattice Boltzmann model was also tested for linear and non-linear acoustic cases, by means of simulation of acoustic waves with various amplitudes [138]. Initial conditions are

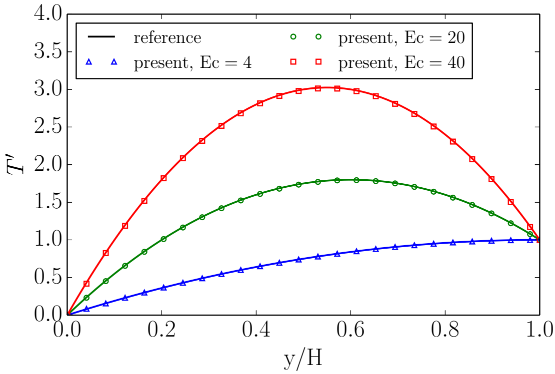


Figure 7.2: Temperature profile of the thermal Couette flow at various Prandtl numbers for the fixed Eckert number Ec = 8.0.

given as,

$$u_x(x, y) = \mathrm{Ma} \cdot c_s \cdot \sin\left(\frac{2\pi x}{L}\right), \tag{7.2}$$

$$u_y(x, y) = 0, \tag{7.3}$$

$$\rho(x, y) = \rho_0 \left(1.0 + \mathrm{Ma} \cdot \sin\left(\frac{2\pi x}{L}\right)\right), \tag{7.4}$$

where the Mach number Ma is varied in the range $[0.01, 0.2]$, $L = 256$ is the length of the domain, $c_s = \sqrt{2T}$ the speed of sound, and the viscosity is set to $\mu = 0.025$. Periodic boundary conditions were used in the $x-$ and $y-$direction. The simulations results of the multi-speed model are compared with a spectral element method [220] in Fig. 7.3, and shows how the present model is able to recover non-linear effects (so called 'N-shaped' waves) that appear with increasing Mach numbers.

7.4 Two-dimensional heated cylinders

In this Section, the thermal flow past a circular and a square cylinder, in two dimensions, in a forced convection regime, are studied [221]. The two-dimensional set-up is a rectangle of $L_x \times L_y = [21D \times 20D]$. The objects are placed at $L_u = 5D$ downstream from the inlet, where D is the characteristic size of the object, and symmetrically in the other direction. A characteristic size of the object of $D = 30$ lattice units have been used. With the

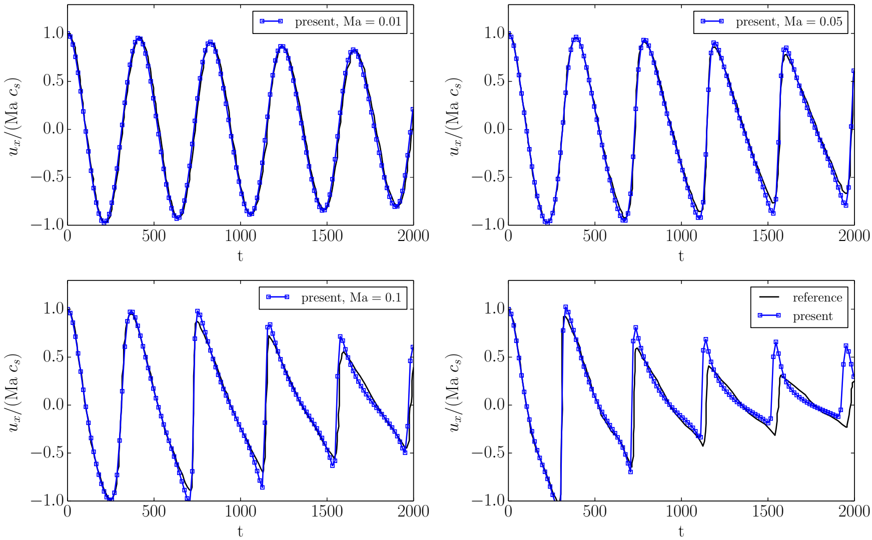


Figure 7.3: Velocity evolution of the sound wave measured at $N_x/4$ for different initial amplitudes proportional to the Mach number.

latter as well as with larger values, the simulations results are independent from the grid size for the range of Reynolds number considered here. The width of the domain is set to $H = 20D = 600$ in order to avoid influence of blockage of the flow. At the inlet all the populations are replaced with the equilibrium values that correspond to the inlet density ρ_0, velocity U_0, and temperature T_0. At the outlet, missing populations are replaced by equilibrium values evaluated with the local quantities at the previous time step. In order to avoid the influence of the outlet on the bulk flow the domain is taken large enough ($L = 21D$). On the top and bottom surfaces, the free-slip boundary condition is imposed for velocity, while the adiabatic wall condition is employed for the temperature.
The non-dimensional numbers characterizing the thermal flow are the Reynolds number,

$$\mathrm{Re_D} = \frac{\mathrm{U_0 D}}{\nu},$$

and the Prandtl number,

$$\mathrm{Pr} = \frac{\mathrm{c_p}\mu}{\kappa}.$$

In the present study the temperatures are set as $T_w = T_0$, $T_\infty = T_0 - \Delta T$, where $T_0 = 1 - \sqrt{2/5}$ is the reference temperature of the D2Q5^2 lattice and $\Delta T = 0.01$. The inlet velocity is set to $U_0 = 0.01$. Given different desired Reynolds numbers and Prandtl numbers, ν and κ are computed and used to set the relaxations parameters. The Prandtl number is set to $\mathrm{Pr} = 0.72$. For the two geometries, the Reynolds number is varied between $10 < \mathrm{Re} < 200$ and $5 < \mathrm{Re} < 200$ for the circle and the square respectively.

7.4.1 Numerical validation

The numerical validation of the set-up is structured as follows: in the first part, an overview of the flow field at different Reynolds numbers and regimes is provided by means of streamlines and isotherms. Next, the structure of the flow is analyzed from a pure hydrodynamic point of view, and compared with available simulations and experimental results, analyzing separately steady and unsteady regimes for the two different geometries, and for the specified range of Reynolds numbers. At the end of the section, the thermal characteristics of the flow are exposed, focusing on the mean Nusselt number measured for the two objects and compared with available reference data.

7.4.1.1 Flow and isotherms pattern

Streamlines The computational results for the circular cylinder are shown in Fig. 7.4 by means of streamlines at selected Reynolds numbers, $\mathrm{Re} = 10$, $\mathrm{Re} = 40$, $\mathrm{Re} = 60$, and $\mathrm{Re} = 200$, representing four different flow regimes. From a Reynolds of about $\mathrm{Re} = 5$, the flow field behind the circular cylinder detaches from the downstream part of the cylinder wall and forms a pair of symmetrical and stationary vortices, which reattaches downstream at the wake length L_r. Representative of this situation, are the streamlines depicted in Fig. 7.4 for the $\mathrm{Re} = 10$ case. By increasing the Reynolds number, the size of the recirculation zone increases and the flow field is steady up to a Reynolds number of around $\mathrm{Re} = 46$ [222]; this increase is clearly shown by the streamlines for the two Reynolds cases $\mathrm{Re} = 10$ and $\mathrm{Re} = 40$. From the critical $\mathrm{Re} = 46$, the flow undergoes the first instability which leads to an oscillatory, periodic, two-dimensional flow: the von Kàrmàn vortex street. The instantaneous streamlines at $\mathrm{Re} = 60$ in Fig. 7.4 represents this mode of the flow. From the critical Reynolds number $\mathrm{Re}_{3\mathrm{D}}$ on, the flow is supposed to become three dimensional, with different evolving modes of the vortex street. There are many suggestions from experimental, numerical

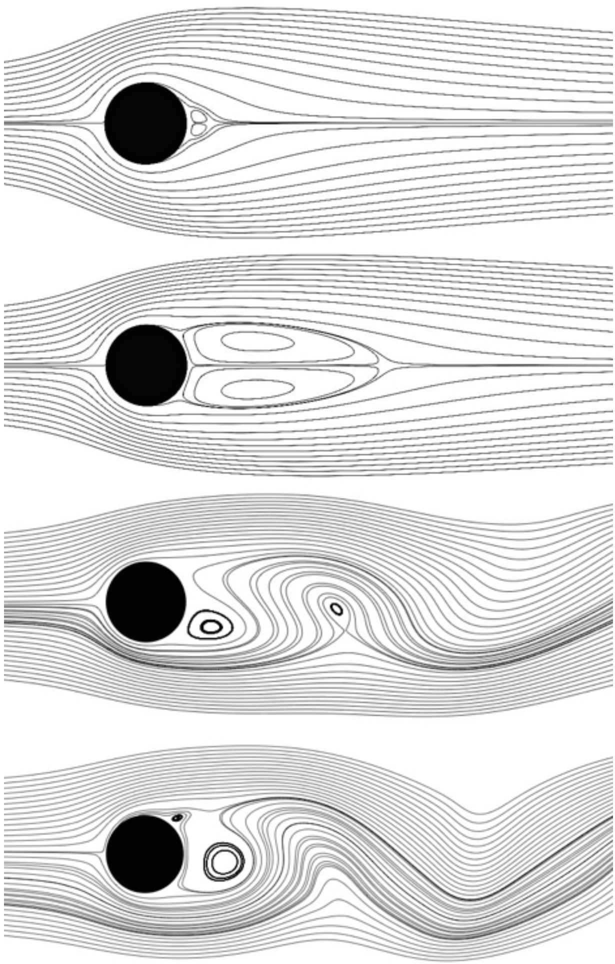


Figure 7.4: Streamlines around the circular cylinder for four different Reynolds numbers. In order from the top: Re = 10, Re = 40, Re = 60, and Re = 200.

and analytical works for the value of $\mathrm{Re}_{3\mathrm{D}}$, and it is found to be in the range between $140 < \mathrm{Re}_{3\mathrm{D}} < 190$ [222, 223]. For this reason, simulations for $\mathrm{Re} > 200$ are not performed in this work. However, thanks to the entropic formulation of the lattice Boltzmann method, simulations can be run stably for Reynolds numbers up to $\mathrm{Re} = 5000$ on the same grid ($D = 30$).

The streamlines for the situation of Re = 200 are shown in Fig. 7.4 for completeness.

Similarly to the circular case, streamlines of the square cylinder are shown in Fig. 7.5 at the same selected Reynolds numbers, Re = 10, Re = 40, Re = 60, and Re = 200, representing the same flow regimes described above. The steady regime with a separated flow is found to be in the range $5 < \mathrm{Re} < 40$ [224, 225] for a square two-dimensional cylinder, and it is represented in Fig. 7.5 by the two Reynolds Re = 10 and Re = 40. The unsteady two-dimensional regime ranges from a Reynolds of around Re = 40 to a critical Reynolds between $150 < \mathrm{Re}_{3D} < 175$ [224, 226], and the streamlines representatives of this situation are shown in Fig. 7.5 for Re = 60. Same as for the cylinder, the range for the Reynolds of the simulations has been considered only until Re = 200 due to transition to a three-dimensional flow.
To conclude this sub-section, it is stressed that the streamlines for both circular and square cylinder compare qualitatively very well with the literature results [223, 225, 227].

Isotherms Figures 7.6 and 7.7 show the instantaneous isotherms for circular and square cylinder respectively, at the same Reynolds numbers as for the streamlines of the previous section. For both geometries, the isotherms are symmetric with respect to the downstream direction in the steady flow regime, while in the unsteady regime the temperature is transported by the vortex street, forming hot spots downstream in the flow field. Moreover, it is possible to observe that isotherms becomes more dense near the bluff body walls as the Reynolds number increases, meaning that the gradient of temperature becomes steeper near the wall by increasing Reynolds. In general, the front surface for both geometries has the maximum clustering of isotherms indicating the maximum gradients of temperature are in this region, and therefore also the maximum local Nusselt number. Interesting to observe is that by increasing the Reynolds number the recirculation zone increases, for both geometries, in the steady zone, and it decreases in the unsteady zone, while the crowding of isotherms in the rear part of the geometry continuously increases for both range of Reynolds for both geometries. This is mainly due to the increases of recirculating flow in the steady regime, and due to the rolling of the von Kàrmàn vortices in the unsteady periodic regime [225].
The isotherms for both circular and square cylinder compare qualitatively very well with the literature results [225, 228].

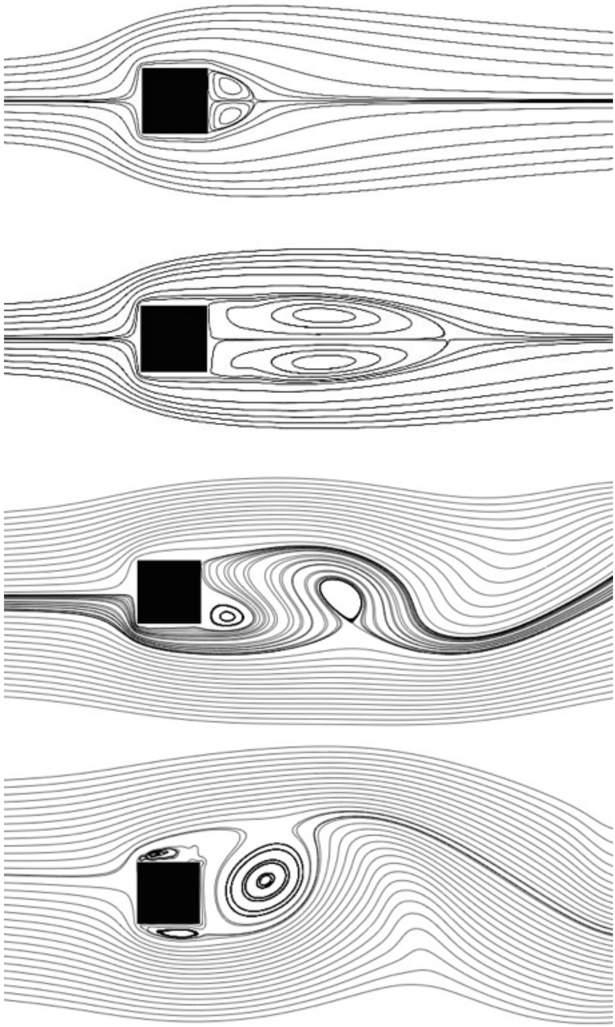


Figure 7.5: Streamlines around the square cylinder for four different Reynolds numbers. In order from the top: Re = 10, Re = 40, Re = 60, and Re = 200.

7.4.1.2 Flow structure

As anticipated above, in the range of Reynolds numbers considered for the model validation, the flow assumes mainly two regimes: a steady and an

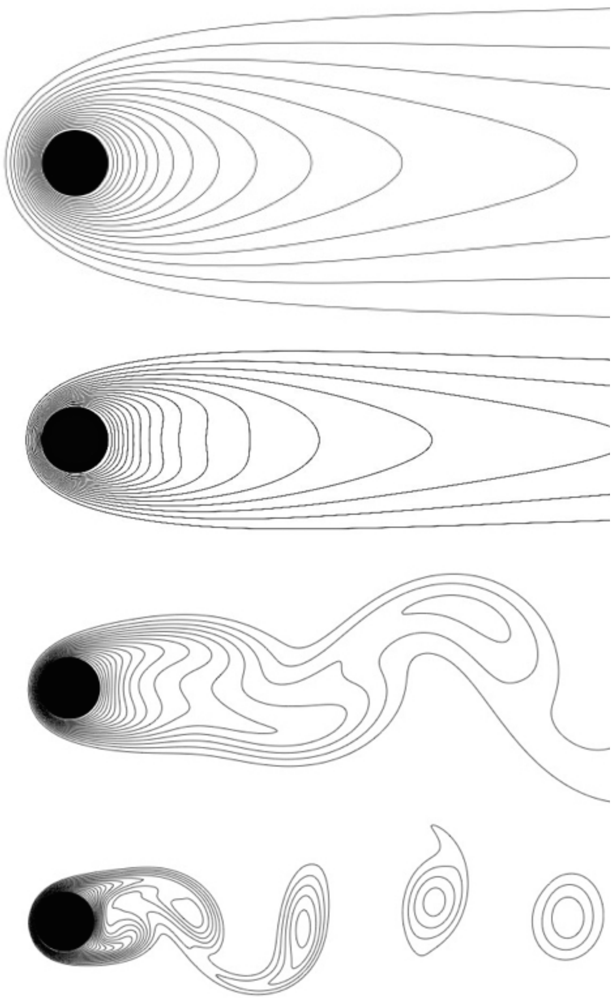


Figure 7.6: Isotherms around the circular cylinder for four different Reynolds numbers. In order from the top: Re = 10, Re = 40, Re = 60, and Re = 200. Step between each isotherms is $\delta T = 0.0005$.

unsteady regime. In Fig. 7.8, the wake-bubble length behind the circular and square cylinders is plotted as a function of the Reynolds number, in the steady flow regime. The wake length is the measure of the distance between the two stagnation points behind the bluff body: one attached to the solid geometry, the other in the free-stream. The present results are shown to-

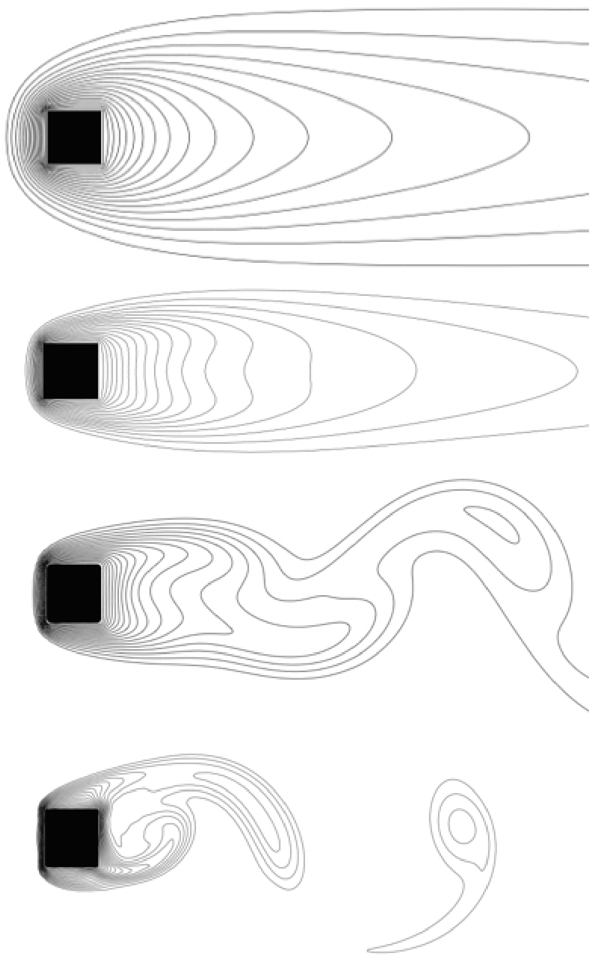


Figure 7.7: Isotherms around the square cylinder for four different Reynolds numbers. In order from the top: Re = 10, Re = 40, Re = 60, and Re = 200. Step between each isotherms is $\delta T = 0.0005$.

gether with the reference values of [229, 230] for the circular cylinder, and of [225, 231] for the square. In the unsteady regime, a characteristic length of the flow is the mean wake recirculation length, which consists in measuring the wake length similarly as for the steady case, but for the mean flow field averaged over many vortex shedding cycles. In Figure 7.9 the computed mean wake recirculation lengths are presented for the unsteady regime, for

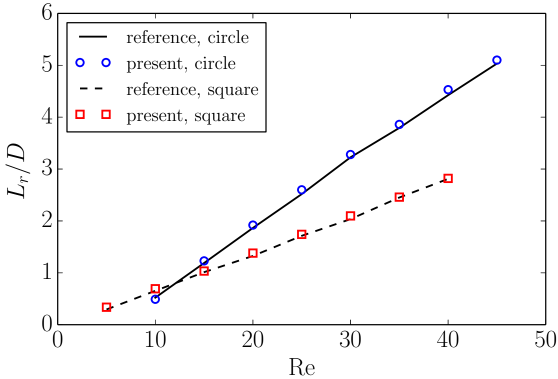


Figure 7.8: Wake recirculation length for square and circular cylinder. Reference data taken from [229, 230] for the circular cylinder, and from [225, 231] for the square.

both the circular and the square geometries, together with the reference values of [232] for the circle and of [225] for square. Both the results for steady and unsteady regimes for both the geometries of the wake length compare very well with the references, over the entire Reynolds number range considered in the simulations.

In order to characterize the unsteady but periodic nature of the shedding flow field induced by the first instability, important parameter is the vortex shedding frequency, usually measured in terms of the Strouhal number:

$$\mathrm{St} = \frac{f\ D}{U_\infty}. \tag{7.5}$$

In order to compute the Strouhal number, a fast Fourier transform of the time series for the y-velocity component, $v(t)$, measured $2.5D$ downstream the solid, was performed. The characteristic frequency of the flow is extracted then from the fast Fourier transform giving the corresponding peak. The results for the Strouhal number for both geometries are shown in Fig. 7.10. For both geometries the Strouhal number is near or in between the references provided, even if the range of Reynolds number is quite restricted, and the sensitivity in the measure of the Strouhal number is therefore highly affected. The results provided for the wake structure and the wake shedding in the unsteady regime suggests that the model and the boundary conditions presented in this work are able to reproduce the correct characteristics of the flow from the hydrodynamic point of view.

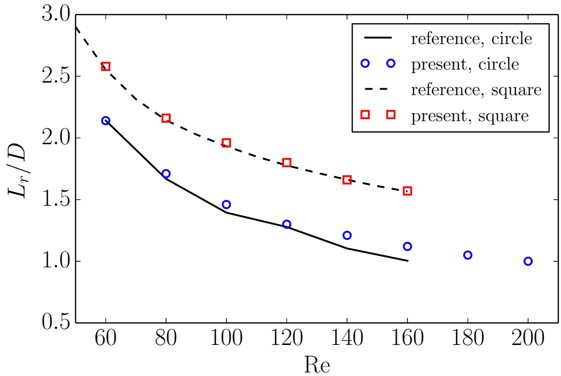


Figure 7.9: Mean wake recirculation length for square and circular cylinder. Reference data taken from [232] for the circular cylinder, and from [225] for the square.

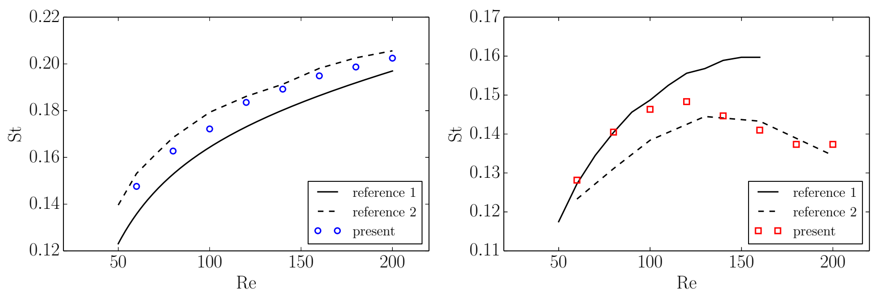


Figure 7.10: Strouhal number versus Reynolds number for circular cylinder (left) and the square cylinder (right). Reference data from [233, 234] for the circular cylinder, and from [225, 227] for the square cylinder.

7.4.2 Flow heat transfer

An estimation of the convective heat transfer between the solid and the fluid is provided by the Nusselt number. The local Nusselt number is evaluated by:

$$\mathrm{Nu} = -\frac{\partial \theta}{\partial n}, \tag{7.6}$$

where $\theta = \frac{T - T_\infty}{T_w - T_\infty}$ is the reduced temperature, and n is the coordinate normal with respect to the wall nondimensionalized by the characteristic size of the cylinder. The average Nusselt number over the entire geometry wall can then be evaluated by:

$$\mathrm{Nu_m} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{Nu_i}, \tag{7.7}$$

where $\mathrm{Nu_i}$ is the Nusselt number measured at the sample point i of N points distributed on the wall. Evaluation of the mean Nusselt number was performed on-line, in the in-house code, after advection and boundary conditions, and after locally conserved fields were computed.
The average Nusselt number for the circular and square cylinders is plotted as a function of the Reynolds number in Fig. 7.11, for the steady and unsteady flow regimes. The results are compared with reference [223] for the circular case and with [225] for the square.

From a Reynolds number of about $140 < \mathrm{Re} < 160$ the Nusselt number starts to deviate from reference values and for both geometries. This deviation could be attributed to the tendency of the flow to become three-dimensional, while the simulations are performed in two dimensions. Apart from this deviation, the results for the Nusselt number for both geometries are very promising, and suggest that the boundary conditions presented here are suitable for both thermo and hydrodynamic computations, and for complex geometries.

7.5 Rotating and non-rotating Rayleigh-Bénard convection

Thermal convective turbulence is ubiquitous phenomenon observed in both Nature and technological applications. In the geophysical and astrophysical context, convective motion driven by temperature stratification under

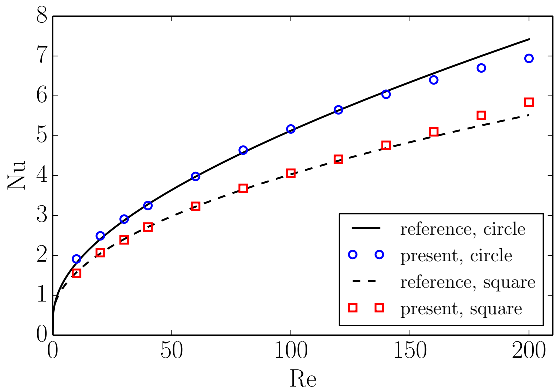


Figure 7.11: Nusselt number versus Reynolds number for circular and square cylinder. Reference data from [223] for the circle and from [225] for the square.

the effect of gravity takes place in stars and planets [214, 235–237], as well as in atmospheric and oceanic circulation [215, 216, 238, 239]. In Nature, buoyancy driven flows can also occur in biological systems in the so-called thermal-bio-convection phenomena [240, 241]. In engineering and technological applications, thermal convection plays important role in process technology [242], in heat exchangers and chip cooling [217], as well as in the quality of indoor circulation [218]. In the quasi-totality of the above examples, thermal convection is combined with other relevant physical phenomena, such as rotation of the frame of reference, radiative heat transfer, phase change, compressibility effects, magneto-hydrodynamic effects or chemical reactions [243].

A simplified paradigm for these situations is the Rayleigh-Bénard convection (RBC) [244, 245] when the reference frame is at rest, or the rotating Rayleigh-Bénard convection (RRBC) in the case of rotating reference frame [246, 247]. RBC plays a prominent role in the development of stability theory in hydrodynamics [248, 249] and is paradigmatic in pattern formation and in the study of spatial-temporal chaos [250, 251]. Special attention was devoted for long to the non-rotating RBC, because of its relevance to planetary systems, providing the limit behavior when convection overcomes rotational effects. In particular, convective heat transfer in RBC systems gives the upper bound on those of the rotating systems [247].

In RBC, an infinite fluid layer of height H is heated from below, cooled from above, and is under the influence of buoyancy forces. The non-dimensional parameters characterizing the system are the Rayleigh number Ra, representing the ratio between the thermally induced buoyancy forces and the viscous and thermal diffusive forces, the Prandtl number Pr, representing the ratio between viscous and thermal diffusion, and the aspect ratio A of the geometry:

$$\mathrm{Ra} = \frac{\mathrm{g}\lambda\Delta\mathrm{TH}^3}{\nu\alpha_{\mathrm{th}}}, \qquad \mathrm{Pr} = \frac{\nu}{\alpha_{\mathrm{th}}}, \qquad \mathrm{A} = \frac{\mathrm{H}}{\mathrm{L}}. \tag{7.8}$$

Here, g is the gravity acceleration, λ is the thermal expansion coefficient, ΔT is the temperature difference between the bottom and top boundaries of the fluid layer, H is the distance between the boundaries, ν is the kinematic viscosity, α_{th} is the thermal diffusivity and L the horizontal characteristic size of the domain. When the value of Ra exceeds the critical value $\mathrm{Ra_c}$, the system becomes unstable to convective motion. For the canonical Rayleigh-Bénard problem without the effect of rotation, with no-slip boundaries the critical Rayleigh number is $\mathrm{Ra_c} = 1708$ [252].
The key response of the system to the imposed Ra is the heat flux h from bottom to top. The dimensionless heat flux

$$\mathrm{Nu} = hH/(\alpha_{th}\Delta T), \tag{7.9}$$

is the Nusselt number representing the ratio between the actual heat flux and its pure conductive part. Employing the Oberbeck-Bussinesq approximation [253, 254], the Nusselt number for incompressible flows becomes

$$\mathrm{Nu} = \frac{\langle u_z T\rangle_A - \alpha\partial_z \langle T\rangle_A}{\alpha_{th}\Delta T/H}, \tag{7.10}$$

where $\langle\cdot\rangle_A$ denotes the average over any horizontal plane and over the time. The Nusselt number is a particularly important quantity in RBC since it provides a global characterization of the strength of convection. Much work has been done in order to estimate scaling theories for the turbulent heat transport $\mathrm{Nu} = \mathrm{Nu}(\mathrm{Ra}, \mathrm{Pr}, \mathrm{A})$. Two main theories were verified by experiments: the classical scaling law of Malkus [255] $\mathrm{Nu} \sim (\mathrm{Ra}/\mathrm{Ra_c})^{1/3}$, which was mainly verified for $\mathrm{Ra} \geq 10^{10}$ and an other one, more appropriate for moderate regimes ($\mathrm{Ra} \leq 10^{10}$), with $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$ [256, 257]. Other scaling laws were proposed for different Rayleigh and Prandtl number regimes [244, 258–261].

When rotational effects are included, new modes of convection may develop depending on the strength of the rotation. In order to characterize the strength of the rotation, another parameter, the Ekman number E is introduced. The Ekman number quantifies the importance of the viscous forces against Coriolis forces and is defined as:

$$\mathrm{E} = \frac{\nu}{2\Omega H^2}, \tag{7.11}$$

where Ω is the angular rotation rate. Rotation tends to suppress or delay the onset of convection [262]. As a consequence, the critical Rayleigh number for the onset of convection is not constant anymore, but instead grows with the rotation rate as:

$$\mathrm{Ra_c} = \mathrm{c_2 E}^{-4/3}, \tag{7.12}$$

where $c_2 \simeq 8.696$ as $\mathrm{E} \to 0$ [248]. Moreover, the Ekman number influences the turbulent heat transfer so that the parameter space becomes even wider with respect to RBC, i.e. $\mathrm{Nu} = \mathrm{Nu(Ra, E, Pr, A)}$. By assuming that Malkus marginal boundary layer arguments [248] hold also for rotating systems, King and Aurnou [263] predicted that rotating convective heat transfer scales steeply as $\mathrm{Nu} \sim (\mathrm{Ra/Ra_c})^3$, and that this steep, cubic scaling holds from the onset of convection up to the Rayleigh number $\mathrm{Ra_T}$ at which the thermal boundary layer is nested with the Ekman layer, i.e. $\mathrm{Ra_T} \sim \mathrm{E}^{-3/2}$ [247]. After $\mathrm{Ra_T}$, the scaling $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$ resumes, in agreement with non-rotating RBC.

RBC was subject of extensive research using both experiments and simulations. The goal is not only to explore the response of the system in a range of Ra and Pr, but rather to find an asymptotic regime in the high-Rayleigh-number convection in the Boussinesq approximation. With recent experiments, it was possible to reach $\mathrm{Ra} \simeq 10^{14} - 10^{17}$ [264–267] while direct numerical simulations (DNS) have just started to enter the transition regime where the boundary layer becomes turbulent, i.e. $\mathrm{Ra} \simeq 10^{12}$ [243, 268].
While large parameter range were explored extensively for non-rotating Rayleigh-Bńeard convection, the situation is different for the rotational RBC. Both experiments and DNS, face difficulties in entering the turbulent, but rotationally constrained regime [269]. Experiments can reach high levels of turbulence relatively easily, but encounter problems in ensuring Coriolis forces to be dominant in the force balance. On the other side, DNS suffer from the huge range of spatial and temporal scales that need to be resolved [269]. Recent parameters values that can be achieved by experiments for RRBC are $\mathrm{E} \geq 10^{-8}$ and $\mathrm{Ra} \leq 10^{13}$ while DNS are limited by $\mathrm{E} \geq 10^{-7}$ and $\mathrm{Ra} \leq 10^{12}$ [270].
Alternative modeling approaches can be employed for the simulation of both RBC and RRBC. For the case of the non-rotating Rayleigh-Bénard convection, Large Eddy Simulation (LES) are employed [271–273], and they have

been indicated as the only way to reach very high values of Rayleigh number. However, in order to capture accurately the near-wall stresses and heat transfer, grid resolution requirements become similar to DNS in the near-wall region. Even higher values of Rayleigh number can be achieved by the so-called transient RANS (T-RANS), which were applied up to $\mathrm{Ra} \simeq 2 \cdot 10^{16}$ [274, 275]. However, due to the empirical and tunable models employed in this case, the prediction capability is reduced [273].
Concerning RRBC, a system of reduced equations valid in the limit $(\mathrm{Ro_c}, \mathrm{E}) \rightarrow 0$, where $\mathrm{Ro_c} = \mathrm{E}\sqrt{\mathrm{Ra}/\mathrm{Pr}}$ is the convective Rossby number, has been developed in [276, 277] (NonHydrostatic Quasigeostrophic equations [278]). These equations allowed to discover a rich dynamical behavior in the low Ekman - high Rayleigh regimes, characterized by several distinct flow structures. However, they are limited to the stress-free boundary conditions case; thus, while the behavior described by this reduced set of equations in the bulk of the flow is well reproduced, the heat transfer in the near-wall regions may result incorrect with respect to the cases with no-slip boundary conditions [269], since in this case the dynamics is controlled by the viscous boundary layer.

Despite the ever growing interest in simulating turbulent flows with the LBM [6, 72, 102, 279], only a restricted number of works can be found in the literature for the simulation of non-rotating RBC and none can be found for the cases with rotation. The majority of the works concentrated on two-dimensional simulation of the classical laminar and turbulent Rayleigh-Bénard convection, the vertical natural convection or other kind of natural convection [280–285], while only few contribution reported three-dimensional simulations [10, 286], the latter including phase-changes.

In this context, the aim of this Section is to simulate three-dimensional RBC and RRBC for a wide range of Rayleigh numbers, and for different rotation rates, in the resolved as well as in the unresolved regime. This way, the outstanding capabilities of the two-population lattice Boltzmann model coupled with the entropic multiple-relaxation model for sub-grid simulations are demonstrated. Differently from LES simulations for RBC, coarse grid can also be employed near the walls, while still retaining the accuracy necessary for accurate comparison of first and second order statistics.

7.5.1 Exact difference method for forces

The forces are included in the LBM model via the exact difference method [287]. The post-collision populations are modified according to:

$$f_i' \to f_i' + \left[f_i^{eq}(\rho, \boldsymbol{u} + \delta\boldsymbol{u}) - f_i^{eq}(\rho, \boldsymbol{u})\right], \tag{7.13}$$

where $\delta\boldsymbol{u}$ is the velocity increment due to the force $\boldsymbol{F}$:

$$\delta\boldsymbol{u} = \frac{\boldsymbol{F}}{\rho}\delta t, \tag{7.14}$$

with the time step $\delta t = 1$.
Below, two different types of forces act on the system, depending on the case of interest. For the non-rotating Rayleigh-Bénard convection only buoyancy forces are included, for which the Oberbeck-Bussinesq approximation is employed, while otherwise, if the case with rotation is considered, also Coriolis forces need to be included. The force vector becomes thus:

$$\{F_x, F_y, F_z\} = \{-2\rho u_y\Omega, \quad 2\rho u_x\Omega, \quad \rho g\lambda\,(T - T_m)\}, \tag{7.15}$$

where $T_m = T_0 + 0.5\Delta T$ is the mean temperature between hot and cold walls, the gravity is oriented in the positive z-direction, and where $\Omega = 0$ is set for the non-rotating RBC.

7.5.2 Set-up

Simulations are performed in a three-dimensional rectangular domain, as shown in Fig. 7.12, of size $[L_x \times L_y \times L_z] = [H\mathrm{A} \times H\mathrm{A} \times H]$, where A is the aspect ratio of the domain. On the top and bottom plates, fixed cold (T_C) and hot (T_H) temperature boundary conditions are employed, while for the velocity no-slip boundary conditions are employed. In the x- and y-directions periodic boundary conditions are considered in all the simulations, in order to approximate an infinite horizontal extent ($\mathrm{A} \to \infty$). This approximation is valid when the largest characteristic horizontal length-scales appearing naturally in the solution are smaller than the horizontal extent $H\mathrm{A}$. For this reason, the aspect ratio A is adapted for each simulation, in order to keep this assumption valid, while maintaining the computational cost of the simulations reasonable.
Buoyancy forces are applied in the positive z-direction, while in the case of rotation ($\Omega > 0$), the rotational velocity Ω is applied in the negative

z-direction. In order to keep the simulations in the incompressible regime, the characteristic free-fall velocity,

$$U = \sqrt{g\lambda\Delta T H}, \tag{7.16}$$

is kept below $U < 0.06$.

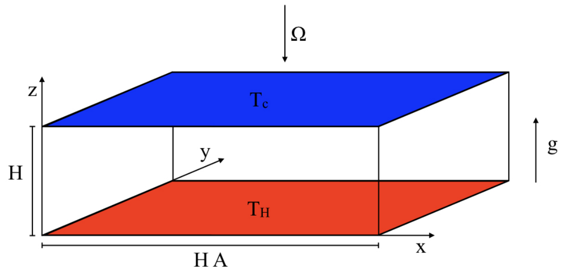


Figure 7.12: Geometry of simulations set-ups.

7.5.3 Non-rotating Rayleigh-Bénard convection

First, an overview over the flow field obtained for the different cases under consideration is provided. Next, the general Nusselt number behavior as a function of the Rayleigh number is presented and compared to available correlations. Finally detailed comparisons of first and second order statistics of different quantities are presented for the simulation at Rayleigh $\mathrm{Ra} = 10^7$. Since one of the focus of the present Section is on subgrid features of the thermal entropic LBM, here in the specific case of thermal convection, a discussion on the different length scales involved in the simulation has to be done. As a general rule, in DNS simulations the minimal grid size is bounded by two key scales, i.e. the Kolmogorov length scale $\eta = (\nu^3/\epsilon)^{1/4}$, where ϵ is the dissipation of turbulent kinetic energy [288, 289], and the viscous and thermal boundary layers thickness δ and δ_T respectively, which has to be resolved with at least six grid points [273]. For the specific case of RBC, Kerr [288] has estimated the Kolmogorov length scale as

$$\frac{\eta}{H} = \frac{\sqrt{Pr}}{[(\mathrm{Nu}-1)\mathrm{Ra}]^{1/4}}. \tag{7.17}$$

For LES simulations, the requirement on grid resolution is often expressed in terms of mean grid spacing, which is usually set to be at least half of the Taylor micro-scale λ [273]

$$\frac{\lambda}{H} = \frac{\sqrt{15}u^*}{[(\mathrm{Nu}-1)\mathrm{Ra}]^{1/2}}, \tag{7.18}$$

where u^* is also estimated, and is given by [273, 288]:

$$u^* = \sqrt{0.0027\mathrm{Ra}^{1.04} + 0.0312\mathrm{Ra}^{0.92}}. \tag{7.19}$$

Moreover, in the near wall region, LES simulations typically require resolution similar to that of DNS [273] in order to resolve the thermal boundary layer $\delta_T/H = 0.5/\mathrm{Nu}$.
In Table 7.1, an overview of the different cases simulated in this Section is provided, together with the parameters and grids employed in each simulation, as well as the different length scales estimates. The Prandtl number is kept fixed to $\mathrm{Pr} = 0.7$ in each simulation.

Case	N_x	A	$\frac{\Delta x}{H}$	$\frac{\eta}{H}$	$\frac{\lambda}{2H}$	$\frac{\delta_T}{H}$	$\frac{\Delta x}{\eta}$	$\frac{\Delta x}{\delta_T}$
$\mathrm{Ra} = 10^7$	736	8	0.0109	0.0076	0.0588	0.0321	1.43	0.34
$\mathrm{Ra} = 10^9$	768	2	0.0026	0.0017	0.0406	0.0079	1.52	0.33
$\mathrm{Ra} = 10^{11}$	384	0.5	0.0013	0.0004	0.0271	0.0021	3.25	0.62

Table 7.1: Simulations parameters and estimated length scales for Rayleigh-Bénard simulations. From left to right: Rayleigh number Ra, grid points in x- and y-directions $N_x = N_y$, aspect ratio A, grid spacing $\Delta x = \Delta y = \Delta z$, Kolmogorov length scale η, Taylor micro-scale λ, thermal boundary layer δ_T, relative grid spacing with respect to Kolmogorov length scale $\Delta x/\eta$ and relative grid spacing with respect to thermal boundary layer $\Delta x/\delta_T$.

7.5.3.1 Flow topology

In Figures 7.13-7.15, instantaneous volume renderings of temperature as well as of a slice of temperature and vorticity ω, near the bottom wall, at the half height of the thermal boundary layer $z = 0.5\delta_T$, are provided for the three simulations considered here without rotation. For renderings, red toward orange/yellow colors represent hot temperatures, while blue toward light blue colors represent cold temperatures. From renderings, for every case it is possible to clearly see the thermal plumes, which are localized

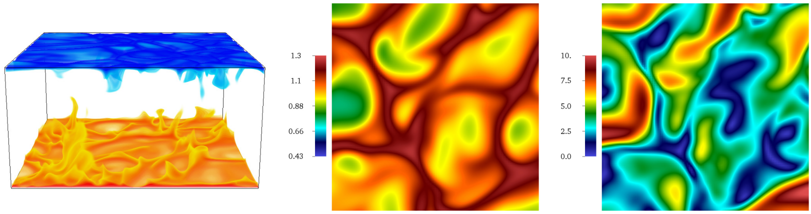


Figure 7.13: Visualization of instantaneous flow field at $\mathrm{Ra} = 10^7$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

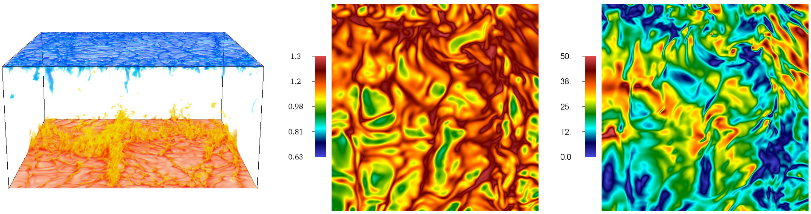


Figure 7.14: Visualization of instantaneous flow field at $\mathrm{Ra} = 10^9$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

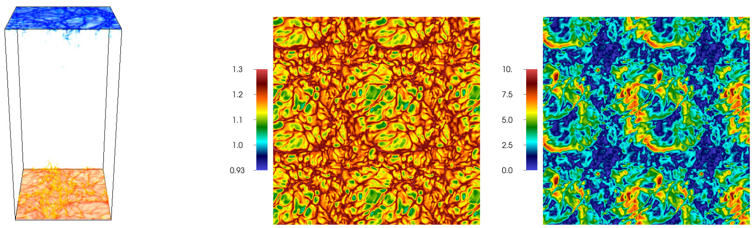


Figure 7.15: Visualization of instantaneous flow field at $\mathrm{Ra} = 10^{11}$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

portions of fluid which have a temperature contrast with the background [243].

As the Rayleigh is increased, thermal plumes become smaller and less coherent. This is also reflected in the temperature slices near the bottom wall, where thermal plumes can be clearly distinguished by the dark red color. A reduction of the turbulent scales associated with thermal plumes can also be observed in the slices of vorticity, which show a decrease of the turbulent structure by increasing the Rayleigh. Thermal plumes detaching from the bottom and top walls with hot and cold temperature, respectively, are responsible for the large amount of heat transport across the convection cell [290]. As can be seen clearly in Fig. 7.13, thermal plumes have a mushroom-like shape, while close to the walls they become sheet-like. With the help of the slices of temperature, one can see that sheet-like plumes create a thin network, surrounding thermal cells, while mushroom-like plumes are emitted from the intersection spots, as shown in the volume renderings.The progressively increasing Rayleigh number has a major effect on the formation of structures like cells and plumes. For lower Rayleigh the size of plumes are of the same order of the domain, and they are more or less homogenous over the domain. When the Rayleigh number is increased thermal plumes becomes smaller and less coherent, and at the same time they tend to organize at the border of the cells, which in contrast become more coherent and visible. This can be regarded as the effect of the large-scale circulation ("wind") developing progressively stronger with increasing Rayleigh number [291–293]. The same progressive decrease of the plume size can also be observed in the turbulent structure visible by the slices of vorticity.

7.5.3.2 Nusselt number scaling

The formation of smaller structures is accompanied with a progressively thinner thermal boundary layer. This is also reflected in an increase of the Nusselt number with increasing Rayleigh number, for which comparison with the scaling laws [247] is provided in Fig. 7.16 and shows good agreement. This is also in agreement with the 2/7 scaling for $\mathrm{Ra} < 10^{10}$ [256, 257, 294] and the 1/3 scaling law for $\mathrm{Ra} > 10^{10}$ [255]. Different ways of computing the Nusselt number exist; resorting to the conductive and convective heat transfer definitions, one can write, following [288]:

$$\mathrm{Nu} = 1 + \sqrt{\mathrm{RaPr}}\,\langle wT \rangle_V / (U \Delta T), \tag{7.20}$$

where $\langle \cdot \rangle_V$ represent the average over the entire volume. Another approach computes the Nusselt number directly from the heat conduction at the walls,

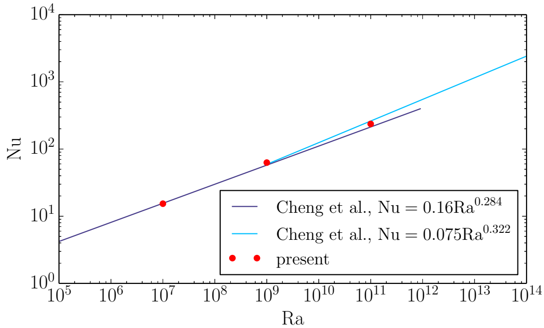


Figure 7.16: Mean Nusselt number for the different Rayleigh numbers considered in RBC.

i.e.:

$$\mathrm{Nu} = -\frac{\partial \overline{T}}{\partial z}\frac{H}{\Delta T}, \tag{7.21}$$

where the mean temperature gradient is evaluated by a semi-second-order finite difference scheme. Since our simulations are run in the under-resolved regime, the results reported in Fig. 7.16 employs Eq. (7.20). Clearly, not all the scales of the flow can be resolved, due to the under-resolution of the computational grid. Thus, even employment of Eq. (7.20) leads in an under estimation of the Nusselt number when the under resolution is severe, like for the case $\mathrm{Ra} = 10^{11}$. For this case, in fact, the computed Nusselt number is in between the two scaling laws [247], while it should be right on top of the scaling law with the highest Nusselt number, pertinent to the $\mathrm{Ra} > 10^{10}$ regime.

7.5.3.3 Flow field comparison

Particularly interesting for our simulations is the case $\mathrm{Ra} = 10^7$, since estimates of the length scales employed in both LES and DNS [273, 290] can be compared to our simulations. Compared to the DNS simulations [290], the grid employed in the LBM simulations is 2.35 times finer in the x- and y-directions. On the other side, in the z-direction the grid of present simulations is 1.31 to 84.12 times coarser transitioning from the bulk to the wall. The finer resolution in the x- and y-direction has to be attributed to the cartesian grid with constant spacing employed by the lattice Boltzmann method, and is not a resolution requirement of the simulations by

it self. For LES simulations similar considerations can be done: in the x- and y-directions the LES grid is 5.73 coarser than the LBM, while in the z-direction the resolution vary from 2.55 finer in the bulk to 5.19 coarser at the wall for the LBM. In general, while in the directions tangent to the walls the resolution can be kept coarser in the DNS and LES simulations with respect to LBM, at the wall the LBM method experience much coarser grid, also with respect to LES. In fact, while LES can be employed at coarser resolution with respect to DNS in the bulk, in order to achieve accurate results it requires DNS like resolutions at the wall [273].
Hereafter, the case $\mathrm{Ra} = 10^7$ is taken as a reference for which a more detailed study is performed, thanks to the data available from DNS [290], which can be employed as a comparison for first and second order statics of various quantities. In the left Figure 7.17, the profiles of mean and root mean square (rms) temperature are presented. The temperature profiles are plotted as a function of the normalized vertical coordinate (normal to the bottom wall) $z^* = z/(H/\mathrm{Nu})$. Good comparison can be achieved for both first and second order statistics, while the thermal boundary layer is resolved with only two points ($\delta_T/H = 1/(2Nu)$ i.e. $\delta_T^*/H = \delta_T Nu/H = 0.5$). The mean temperature $\overline{T}$ shows the typical behavior of Rayleigh-Bénard convection, where after the lowering of the mean temperature in the boundary layer, a state of perfect mixing with $\overline{T} = T_m$ is achieved and kept constant until the opposite wall is approached. The root mean square temperature T_{rms}, instead, presents the typical peak just in correspondence with the thermal boundary layer, i.e. where plumes emissions happens most frequently enhancing the mixing with the mixed bulk (mean) air. Good comparisons can

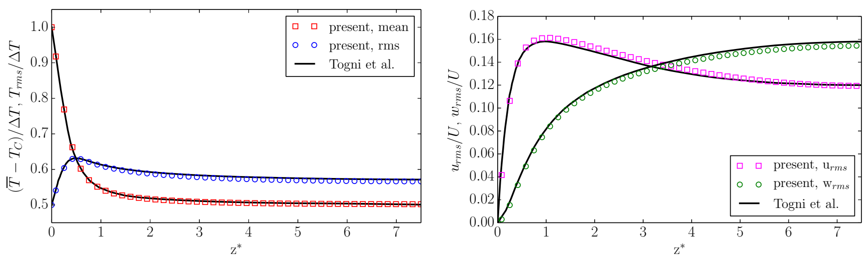


Figure 7.17: Temperature and velocity field statistics. Left: mean and root mean square temperature. Right: root mean square x- and z-components of velocity.

also be observed for the root mean square of x- and z-components of velocity, u_{rms} and w_{rms} respectively, depicted in the right Fig. 7.17. Vertical

(w_{rms}) and horizontal (u_{rms}) rms velocity components show two completely different behaviors. The vertical rms velocity, which is produced by the buoyancy acting on the fluid [290], reaches his maximum in the bulk, and it is transported toward the boundary layer by pressure mechanisms, where it is redistributed to the rms velocity components parallel to the walls by pressure-strain rate [290], where they reach their maximum. Finally, the rms velocity components parallel to the wall redistribute their energy to the bulk and in the boundary layer, where it is dissipated.
Compared in Figure 7.18, is a more sensitive quantity, i.e. the production of root mean square temperature, defined as $-2\overline{w'T'}\frac{d\overline{T}}{dz}$, which include the co-variance of vertical velocity and temperature $\overline{w'T'}$, which is also considered the production of turbulent kinetic energy, and the mean vertical temperature gradient $d\overline{T}/dz$ [290]. Despite a slight deviation in the peak value, the

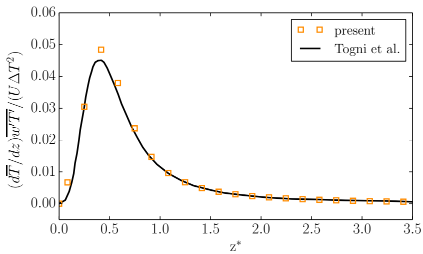


Figure 7.18: Production of root mean square temperature distribution along the vertical direction.

agreement is also in this case satisfactory.

7.5.3.4 Sub-grid features and grid requirements

Interesting to analyze here, is the grid dependence of the previous simulation in order to understand the minimal grid requirements for the thermal LBM to simulate Rayleigh-Bénard convection. For the case with Ra $= 10^7$ the results for three different grids are presented, i.e. ranging from a very coarse grid with $H = 32$ to a finer gird $H = 64$ and the grid employed in the previous section $H = 92$. In Figure 7.19, the mean and rms temperature as well as rms x- and z-velocity are reported for the different grid.

For a grid resolution of $H = 32$, the boundary layer in the mean temperature is over estimated, while the root mean square of temperature and velocity are in general underestimated. Despite the underestimation of rms

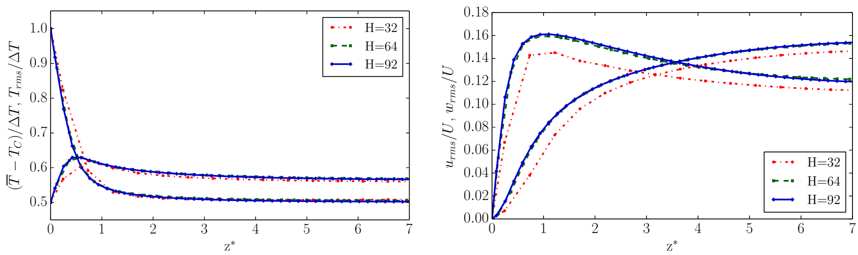


Figure 7.19: Temperature (left) and velocity field (right) statistics for the different girds $H = 32$, $H = 64$ and $H = 92$.

quantities, the main physical feature of the flows are anyway captured even for extremes low values of the grid resolution. When the grid is increased to $H = 64$ point, the statistics already does not show any significant change from the correct values reported for a grid of $H = 92$, even if also for this case the under resolution of the grid is quite important.
In Table 7.2, the Nusselt number evaluated with both Eq. (7.20) and Eq. (7.21) is reported for the different grids. The Nusselt number computed with the

H	32	64	92
Nu_{DNS}		15.59	
$\mathrm{Nu} = 1 + \sqrt{\mathrm{RaPr}}\,\langle wT \rangle_V / (U\Delta T)$	12.33	14.93	15.35
ϵ [%]	20.9	4.2	1.5
$\mathrm{Nu} = -\frac{\partial \overline{T}}{\partial z}\frac{H}{\Delta T}$	10.89	14.63	15.48
ϵ [%]	30.15	6.16	0.7

Table 7.2: Nusselt number evaluation for with two different methods, Eq. (7.20) and Eq. (7.21), and for three different grids $H = 32$, $H = 64$ and $H = 92$. Reported is also the relative error with respect to the reference value $\mathrm{Nu}_{\mathrm{DNS}}$ reported in [290].

grid $H = 32$ reflects the under resolution also observed by the over estimation of the thermal boundary layer and by the under estimation in the rms values, with both types of evaluation, with a minimum relative error of $\epsilon = 30.15\%$. With the grid $H = 64$, which reported already good results for statistics, the error is already below 5% by evaluating with the mean convective heat flux Eq. (7.20), while evaluation with direct computation of the gradient Eq. (7.21) leads to an higher error, since a slight over estimation of the thermal boundary layer is probably present due to the too low amount of grid points. In general, it can be stated that for a Rayleigh number of $\mathrm{Ra} = 10^7$ the minimum, constant grid spacing, resolution is $H = 64$. This is an exceptional feature of the present model, since for this case the Kolmogorov scale is underestimated by a factor $\Delta z/\eta = 2.06$ while the boundary layer is resolved with only one point since $\delta_T/\Delta z = 2.05$.

7.5.4 Rotating Rayleigh-Bénard convection

Here, presented are the simulations of rotating RBC, where the viscous fluid layer is under the effect of rotation. This is accounted for by the Ekman number E which decreases with increasing rotation rate. This section discuss the results for the cases in which the Ekaman number is kept in the weak to medium strong rotation, i.e. $E = 5.6 \cdot 10^{-5} - 4.3 \cdot 10^{-3}$, and for different Rayleigh number. The present results are compared with the DNS of [246], with the different cases and related simulations parameters reported in Table 7.3. The Prandtl number is kept fixed to $\mathrm{Pr} = 1$ and the convective Rossby number to $\mathrm{Ro_c} = 0.75$.

Case	E	Ra	$\mathrm{Ro_c}$	N_x	A	$\Delta x/H$
$\widetilde{\mathrm{Ra}} = 21.4$	$4.27 \cdot 10^{-3}$	$3.09 \cdot 10^{4}$	0.75	128	2	0.0156
$\widetilde{\mathrm{Ra}} = 28.1$	$2.83 \cdot 10^{-3}$	$7.03 \cdot 10^{4}$	0.75	128	2	0.0156
$\widetilde{\mathrm{Ra}} = 44.6$	$1.41 \cdot 10^{-3}$	$2.81 \cdot 10^{5}$	0.75	128	2	0.0156
$\widetilde{\mathrm{Ra}} = 57.2$	$9.76 \cdot 10^{-4}$	$5.91 \cdot 10^{5}$	0.75	128	2	0.0156
$\widetilde{\mathrm{Ra}} = 92.9$	$4.72 \cdot 10^{-4}$	$2.53 \cdot 10^{6}$	0.75	256	2	0.0078
$\widetilde{\mathrm{Ra}} = 138.7$	$2.58 \cdot 10^{-4}$	$8.44 \cdot 10^{6}$	0.75	256	2	0.0078
$\widetilde{\mathrm{Ra}} = 207.2$	$1.42 \cdot 10^{-4}$	$2.81 \cdot 10^{7}$	0.75	384	2	0.0052
$\widetilde{\mathrm{Ra}} = 326.5$	$7.15 \cdot 10^{-5}$	$1.10 \cdot 10^{8}$	0.75	384	2	0.0052

Table 7.3: Simulations parameters and grid sizes. From left to right: Rayleigh number Ra, Ekmann number E, compensated Rayleigh number $\widetilde{\mathrm{Ra}}$, convective Rossby number $\mathrm{Ro_c}$, number of grid points in x- and y-direction $N_x = N_y$, aspect ratio A and non-dimensional grid spacing $\Delta x/H = \Delta y/H = \Delta z/H$.

7.5.4.1 Flow topology

In order to visualize the main flow field features at different Rayleigh number for the rotating convection, four main representative flow situations are

presented in the following. Figures 7.20 to 7.23, provides instantaneous volume renderings of temperature as well as of a slice of temperature T and vertical vorticity ω_z, near the bottom wall, at the half height of the thermal boundary layer $z = 0.5\delta_T$.
In Figure 7.20 the flow field for the case $\mathrm{Ra} = 3.09 \cdot 10^4$ is shown. Both tem-

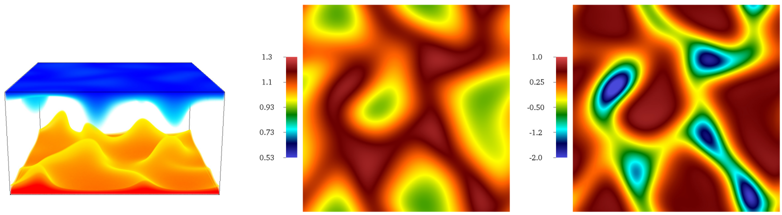


Figure 7.20: Visualization of instantaneous flow field at $\mathrm{Ra} = 3.09 \cdot 10^4$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

perature and vertical vorticity show a flow field characterized by aperiodic convection-rolls pattern associated to Küppers-Lortz instability [246]. As

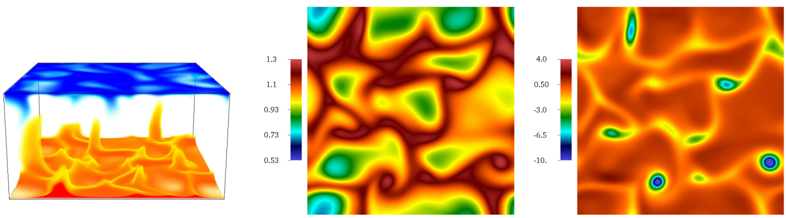


Figure 7.21: Visualization of instantaneous flow field at $\mathrm{Ra} = 5.91 \cdot 10^6$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

soon as the Rayleigh is increased, convection rolls give way to chaotic interactions between vertical vortices associated with convection cells [246]. This can be seen in Fig. 7.21. At this Rayleigh, thermal plumes can already be better distinguishable both in the volume rendering by the mushroom like plumes, but also in the temperature slice by the dark red color. This state persists until a Rayleigh number of $\mathrm{Ra} \simeq 6 \cdot 10^5$, above which, thin thermal plumes spontaneously appear in the flow field, as shown in Fig. 7.22. These

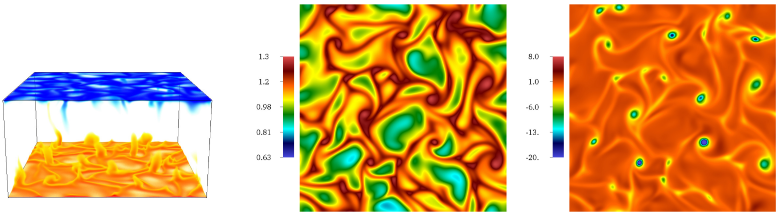


Figure 7.22: Visualization of instantaneous flow field at $\mathrm{Ra} = 8.44 \cdot 10^6$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

plumes are most prevalent near the junctions of convection cells, where cell boundaries merge with the thermal boundary layers attached to the hot and cold walls [246]. As the Rayleigh number is further increased, above

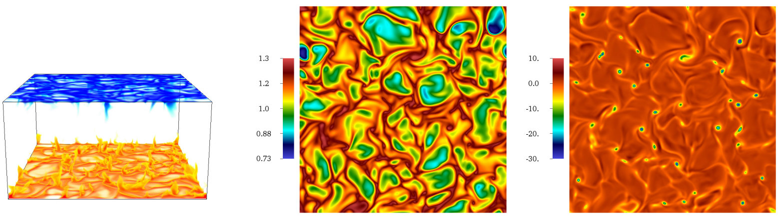


Figure 7.23: Visualization of instantaneous flow field at $\mathrm{Ra} = 1.1 \cdot 10^8$ by volume rendering of temperature T (left); slice at $z = 0.5\delta_T$ of instantaneous temperature T (center); slice at $z = 0.5\delta_T$ of instantaneous normalized vorticity ω (right).

$\mathrm{Ra} > 2 \cdot 10^6$ the flow field is dominated by coherent plumes, but the cellular convection pattern becomes more difficult to discern, as shown in Fig. 7.23. Important to notice, is that, unlike in non-rotating thermal convection, the vertical vorticity associated to the thermal plumes is always cyclonic (of the same sign of the externally applied rotation). This is reflected in the vertical vorticity slices of Figures 7.21-7.23, where vertical vorticity is negative (blue) in the zones associated with thermal plumes, also discernible as the intersection of the boundary thermal cells.

7.5.4.2 Flow field comparison

Quantitive comparisons for the flow field for the rotating Rayleigh-Bénard convection are provided in the following. In Figure 7.24, the mean temperature profiles for different Ra are compared with the DNS [246]. A part

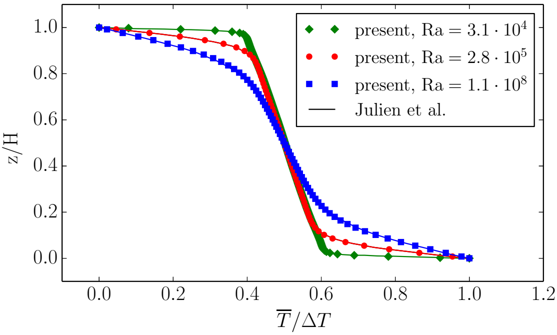


Figure 7.24: Mean temperature profiles comparison for the cases Ra = 30900, Ra = 281000 and Ra = 110000000. Symbols: present; solid lines: reference [246].

the good comparison between the present work and the reference, the plots show the important role played by rotation in RRBC, i.e. the finiteness of the mean temperature gradient $\left|\partial\overline{T}/\partial z\right|$ in the bulk of the fluid layer, which even by further increasing the Ra does not exceed a given value. This is in contrast to non-rotating RBC, where the mean temperature gradient in the bulk is always $\left|\partial\overline{T}/\partial z\right| = 0$ for sufficiently high Rayleigh.

In Figure 7.25, the magnitude of the mean temperature gradient is plotted as a function of Ra. The plot confirms, with both DNS and present simulations, a saturation of the mean temperature gradient at a value of about $\left|\partial\overline{T}/\partial z(H/\Delta T)\right| = 0.2$. An explanation for this saturation involves the specific dynamic of thermal plumes in RRBC, which is different from that of RBC, i.e. a vortical flow associated to the plumes. The vortical thermal plumes imply a second mechanism in the flow field, i.e. what is called "lateral mixing" [246]. The cyclonic vorticity associated to thermal plumes, in fact, has the consequence of attracting and merging vortices which are sufficiently close (since vertical vorticity associated with thermal plumes has the same sign, causing co-rotation). The main consequence of the co-rotation and merging is to mix the fluid along horizontal planes; this impedes the penetration of buoyant elements at greater distances from their original boundaries, causing the stratification of the temperature responsible of the finite mean temperature gradient.

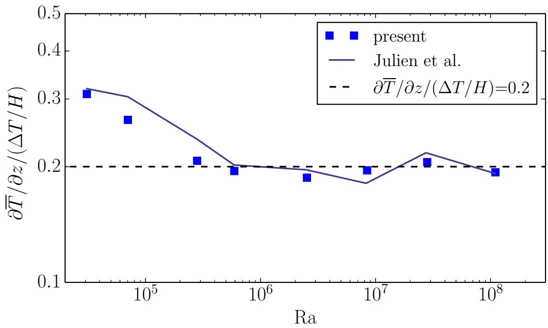


Figure 7.25: Mean negative temperature gradient at mid-layer as a function of Ra.

Second order statistics in the bulk are compared in Fig. 7.26 for both temperature and vertical velocity. Once again the comparison between the present simulations and the reference DNS are good over a wide range of Ra. Especially, the scaling found in [246] can be reproduced for both rms

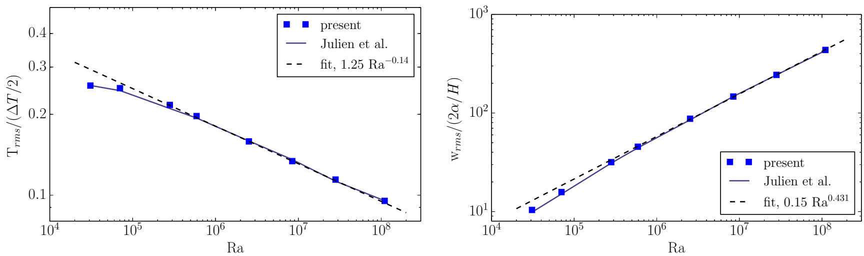


Figure 7.26: Comparison of non-dimensional root mean square values of temperature (left) and vertical velocity (right) in the mid plane layer as a function of Rayleigh number.

temperature and velocity and it corresponds to the free-falls behavior, also observed in the non-rotating RBC, and indicates that the Rayleigh number is sufficiently high such that the dynamic of the thermal plumes in the bulk is no more controlled by the boundary conditions [246]. Furthermore, it is possible to notice that the scaling exponent for rms temperature and rms vertical velocity as a function of Ra are consistent with $-0.14 \simeq -1/7$ and $0.431 \simeq 3/7$, which are scaling exponent typical of RBC. This does not imply that the turbulence dynamic in the bulk is independent on the rotation rate ($\mathrm{Ro_c}$), since these effect becomes evident in the pre factor of the Rayleigh scaling with varying $\mathrm{Ro_c}$. Also interesting to notice, is that in the high Rayleigh regime the Nusselt number can by estimated as $\mathrm{Nu} \sim T_{rms}/(\Delta_T/2) w_{rms}/(2\alpha/H)$ which in turn gives $\mathrm{Nu} \sim \mathrm{Ra}^{2/7}$, similar to the scaling observed at high Ra in non rotating RBC. This is further confirmed in Fig. 7.27, where the Nusselt number computed in the present simulations is compared with the DNS, as function of the different Ra.

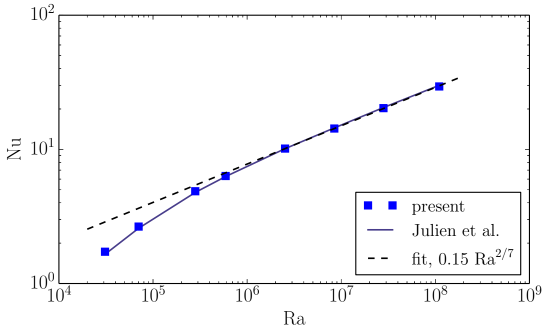


Figure 7.27: Comparison of Nusselt number Nu as a function of Rayleigh number.

Also for the Nusselt number, the present simulations are in good agreement with the reference, and the correct scaling as a function of Ra is reproduced.

7.6 Conclusion

Two entropic thermal lattice Boltzmann models were validated by means of two-and three-dimensional simulations. The multi-speed thermal model was validated for more fundamental problems, in order to prove the correct recovery of the Fourier-Navier-Stokes equations; these include the thermal Couette flow, the plane sound wave or the flow around heated bluff bodies, as done for the first time in [138, 221].
The two-population model has been throughly validated in [90], and thus

here it has been employed and validated for the simulations of more complex set-ups, like rotating and non-rotating turbulent Rayleigh-Bénard convection.
Each simulation showed satisfactory comparisons with analytical results and DNS simulations, demonstrating the capability of the models to reproduce the correct expected physics. Particularly interesting is the employment of the two-population model for the simulation of turbulent convecting flows, like the Rayleigh-Bénard convection: the model demonstrated in fact, that, in combination with the entropic multi-relaxation model, under-resolved simulations can be performed. Good results for first and second order statistics, as well as for the Nusselt number were obtained, thus demonstrating, among the others, the powerful subgrid features of the model. This, together with the simplicity of the two-population model, poses the lattice Boltzmann method as a powerful and efficient alternative to the simulation of thermal turbulent flows for practical applications.

7.7 Outlook

Two main application domains has been tackled, additionally to the simulations presented in this Chapter, with the two-population model. Preliminary simulations demonstrated promising results and thus further efforts may be devoted in those directions. In the particular case, in the following, the preliminary results of the rapidly rotating Rayleigh-Bénard convection and of a piston engine compression are presented, together with the difficulties related to the specific set-ups.

7.7.1 Rapidly rotating Rayleigh-Bénard convection

Simulations of rapidly rotating Rayleigh-Bénard convection (RRRBC) poses additional difficulties, in terms of grid requirements, with respect to the non-rotating and rotating RBC; additional small scales associated with the Ekman layer, in fact, appear in the flow and need to be resolved. In the range of Rayleigh and Ekman numbers that can be reached by state of the art direct numerical simulation, i.e. $\mathrm{E} \geq 10^{-7}$ and $\mathrm{Ra} \leq 10^{12}$ [270], the smallest scales are associated with turbulence and Ekman layer, and no more with thermal boundary layer, like in the case without rotation of the frame of reference. The Ekman layer is characterized by an $O(\mathrm{E}^{1/2}H)$ thickness, which in DNS is resolved with a minimum of ten grid points [269]. In the present simulations, with the employment of a Cartesian grid

with constant spacing, the computational const of such simulation would become prohibitive, in particular considering the large number of time-steps required to reach statistically likely results ($O(10^7)$ time steps); for this reason a grid of $H = 1024$ point is here employed. Such a large number of time-steps is required in order to ensure that the Coriolis force is sufficiently low, thus guaranteeing accuracy and stability of the scheme. The flow velocity related to Coriolis force can be estimated as $U_{Cor} = 2\Omega H$. Since it is required, in general, that the maximum velocity in the incompressible LBM is around $U_{max} \simeq 0.05$, this leads to a rotational velocity of $\Omega = 2.55 \cdot 10^{-5}$. In [295], the time required to run simulations of rapidly rotating RBC is around $t = 10^3 \tau_{Cor}$, where $\tau_{Cor} = 1/(2\Omega)$; in our case, the characteristic time related to Coriolis forces is around $\tau_{Cor} = 1.961 \cdot 10^4$, so that the estimate for the number of time steps follows.
Employment of such a coarse grid, $H = 1024$ points, imply that only one point is used to resolve the Ekman layer. In the following, the possibility to run simulations of RRRBC on such a coarse grid is reported. In the particular case, a simulation with the following parameters is exposed: $\mathrm{Ra} = 2.15 \cdot 10^9$, $\mathrm{E} = 3.16 \cdot 10^{-6}$ and $\mathrm{RaE}^{4/3} = 100$.
Figure 7.28 shows two snapshots, from two different views, of iso-surfaces of vorticity colored by temperature for the simulation of RRRBC considered here. First of all, the figure shows the formation of a large-scale vor-

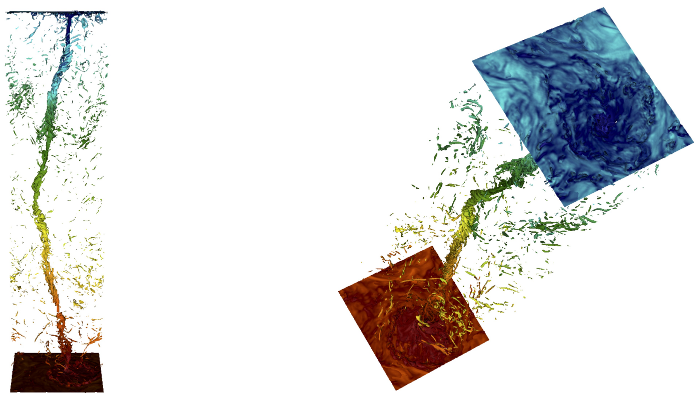


Figure 7.28: Lateral (left) and top-lateral (right) view of a snapshot of the iso-surface of vorticity colored with temperature for the RRRBC simulation. Red color represents warm temperature while blue color represents cold regions.

tex, which connect the bottom wall to the top wall. This is in qualitative

agreement with similar results reported in the literature for the parameters employed in the present simulation, see e.g. [295]. Important to notice, is that the simulation remains stable, despite its extreme under-resolution, in particular concerning Ekman layer. Despite the under-resolution, the Nusselt number varied from around Nu $\simeq$ 60 and Nu $\simeq$ 120, thus in a range which includes the results reported in [270].
Although the present simulation demonstrates feasibility of simulations of RRRBC in extremely under-resolved regimes, much work has to be conducted in this direction, providing a thorough validation over a wide range of parameters.

7.7.2 Engine piston compression

Simulation of internal combustion engines (ICE) is a topic of tremendous interest, and a lot of efforts are devoted in developing numerical methods for this specific field of fluid mechanics. Along this, employment of lattice Boltzmann models in this context would be very interesting. While first simulations of the intake process in ICE with LBM has recently been presented in [72] and compared to the DNS results of [296] for an engine-like geometry, the compression part still needs to be simulated. This is due to a lack of approaches to correctly reproduce the right thermodynamics of compression in a closed geometry.
Here, it is proposed to employ the thermal two-population model to simulate an engine-like compression. This kind of model is chosen since it enjoys a large temperature range which can be simulated, allowing the temperature to increase drastically during the compression. To that end, an heat source of the form

$$\frac{dT}{dt} = \frac{1}{\rho C_p}\frac{dp_0}{dt}, \tag{7.22}$$

is added in the temperature equation in order to account for the increase of the base pressure due to the compression. Here, $\frac{dp_0}{dt}$ is the temporal derivative of the mean pressure inside the cylinder of the engine, and it is computed by a second order finite difference scheme. The temperature variation is introduced in the g lattice by an exact difference scheme similar to Sec. 7.5.1 as

$$g_i' \to g_i' + \left[g_i^{eq}(\rho, \boldsymbol{u}, T+\delta T) - g_i^{eq}(\rho, \boldsymbol{u}, T)\right], \tag{7.23}$$

where δT is the temperature increase due to the heat surce $\frac{dT}{dt}$:

$$\delta T = \frac{1}{\rho C_p}\frac{dp_0}{dt}\delta t. \tag{7.24}$$

In order to validate the scheme, an adiabatic compression has been simulated, starting from an initial density of $\rho_i = 1.0$ and temperature $T_i = 100.0$, and with zero velocity. The compression was characterized by a compression ratio of $CR = 3$. Adiabatic wall boundary conditions were employed, while the boundary conditions for moving walls presented in Sec. 2.3.3.1 has been employed for the moving piston. The bore of the engine was resolved with $B = 150$ points, while the stroke with $S = 120$. An adiabatic exponent of $\gamma_{ad} = 1.4$ is chosen. Analytical expressions for the density and temperature evolutions are given as follow. The density evolution inside the combustion chamber is dictated by the displacement of the piston as

$$\rho(\varphi) = \rho_i \frac{2CR}{1 + CR + (CR - 1)\cos\varphi} \tag{7.25}$$

where φ is the crank angle of the piston. Given the density and assuming an adiabatic process, the temperature evolution becomes

$$T(\varphi) = T_i(\rho/\rho_i)^{\gamma_{ad}-1}. \tag{7.26}$$

Results for density and temperature evolution for the engine compression, i.e. for half cycle of the engine stroke, are shown in Fig. 7.29. Both den-

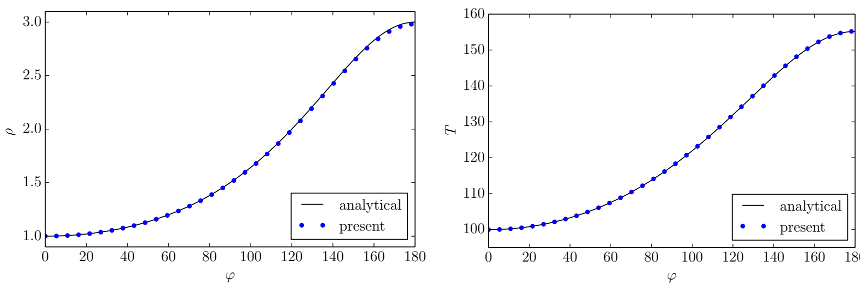


Figure 7.29: Density (left) and temperature (right) evolution during adiabatic compression as a function of φ.

sity and temperature evolutions compare well with the analytical evolution, demonstrating that the heat source term (7.22) may be used to simulate engine compression.

The next steps would be to extend the compression process to cases with realistic turbulent flow field and fixed wall temperatures, like that presented in the DNS [297] for an engine like geometry, similar to the one employed in the present simulation.

Chapter 8

Conjugate heat transfer model

8.1 Introduction

Conjugate heat transfer (CHT) arising from the coupled thermal dynamics between fluids and solids is found in a large variety of natural phenomena and technological applications, ranging from heat exchangers and blade cooling in jet engines to the heat transfer in chemical and nuclear reactors as well as in micro devices. The CHT condition at the interface between two or more materials, usually a fluid and a solid, is modeled by satisfying two physical requirements: (i) the temperature continuity and (ii) the normal heat flux continuity that ensures the energy conservation. Conventional CFD solvers employ a large variety of strategies to solve the conjugate heat transfer problem, but they differs mainly by the degree to which the different domains are coupled. "Strongly coupled" approaches are generally more robust, but they require a reformulation of the computational algorithm for the entire interface involving CHT. On the other hand, "weakly coupled" approaches generally re-use already existing physical codes which are then coupled at the interface [298]. In general, these solvers require extrapolations and iterative schemes to meet the correct conditions at the interface. Such features make conventional algorithms particularly hard to implement and computationally demanding for complex interface geometries [299].

First simulations of conjugate heat transfer with the LBM where limited to stationary cases and flat walls. In [300, 301], the fluid-solid interface was set at the mid-points between the solid and fluid nodes. In this case, populations are free to fly crossing the CHT interface, while only thermal conductivities are changed from fluid to solid state and vice-versa. Thanks to its simplicity, this approach was adopted in [302] to simulate CHT in a channel with wall-mounted obstacles, while in [303] the same model was employed to study a counterflow heat exchanger. In [304], the interface was placed in order to coincide with the lattice nodes. The populations crossing the interface were modified so that the ratio of the derivatives obtained by computing the heat-flux from populations in the two media is the same as for the temperature gradients, i.e. inversely proportional to the conductivity ratio.
More recently, variable heat capacities were enabled by different models. One of the first LBM approaches which modeled thermal inertia was the scheme of [305], where both conjugate conditions are directly enforced on the zero- and first-order moments of the distributions and explicit expressions for the missing populations were derived. In [306], different weight coefficients are assigned to the equilibrium distributions of the two domains and a rescaling factor is applied to the populations crossing the interface: all these coefficients are found by means of an asymptotic analysis. Alternatively, in [307], a source term was introduced to decouple the thermal diffusivity from the thermal conductivity in the temperature equation.
A further progress in the computation of CHT with the LBM was gained with models for curved geometries. In [308], a basic bounce-back approach was enhanced with spatial gradients from the surrounding populations in such a way as to satisfy the conjugate conditions at the interface and to maintain the second-order accuracy. This method brings along some limitations, namely complicated formulation already in two-dimensions, lack of a general algorithm to handle both flat and curved interfaces and unavailability of the normal heat flux, important for the Nusselt number computations. These limitations were overcome in [309] with a counter-extrapolation method where the surrounding temperature field and the conjugate conditions are explicitly used to derive the interface temperature. This information is converted to a Dirichlet boundary condition which is implemented by means of a mid-point bounce-back scheme. This last step somewhat weakens the approach, as it brings to drops in accuracy and stability, especially when simulating high Reynolds number flows with curved boundaries [82]. Finally, in [310], the idea of the source term at the interface has been revised. In order to avoid non-locality of the approach of [307], a staircase treatment of the fluid-solid interface has been employed. However, such an approach

suffers from a reduced accuracy due to staircase boundary approximation.

While the difficulties related to the different heat capacities between fluid and solid or simulating complex curved geometries have been addressed in some of the works cited above, the main concern here is to design a CHT interface treatment which guarantees both stability and accuracy at high Reynolds numbers. Here, an algorithm for high Reynolds numbers, variable Prandtl number and variable thermal diffusivity, for the treatment of CHT boundary conditions is enabled [219]. The possibility to handle different heat capacities between fluid and solid is retained, for both flat and curved boundaries treatment.

The CHT model is an extension of the two-population model of Sec. 6.3, then here only the solid domain treatment and the conjugate heat transfer conditions are presented. The CHT condition is included here similar to [309], with a counter extrapolation method. However, in this work the temperature computed at the interface is used to evaluate both target temperature and the temperature gradient, which in turn are needed to construct equilibrium and non-equilibrium moments employed in the Grad's approximations. This guarantees greater stability and accuracy at high Reynolds numbers, and enables easy extension to three-dimensional set-ups.

8.2 Solid domain treatment

In conjugate heat transfer problems, the domain is usually composed by a fluid domain and a solid domain. In the present approach, the fluid domain is treated exactly in the same way as in the two-population model Sec. 6.3, while the solid domain is treated similarly, but without considering f-populations for the evolution of the mass and momentum: the density is in fact considered to be fixed at $\rho_s = 1$, while the speed is always zero $\boldsymbol{u}_s = 0$. This way, the conservation law, the moments, kinetic equation and equilibrium can be adapted for the solid domain, as presented in the following.
The only conservation law required in the solid region is the conservation of total energy

$$2\rho_s E_s = 2\rho_s C_s T = \sum_{i=1}^{n_Q} g_i^s, \tag{8.1}$$

where C_s is the heat capacity of the solid, which is chosen based on the heat capacity ratio between fluid and solid.

The equilibrium moment for the g-populations in the solid are derived by the equilibrium moments in the fluid, by setting the density to ρ_s and the velocity to $u_\alpha = 0$, thus

$$q_{\alpha}^{eq,s} = \sum_{i=1}^{n_Q} v_{i\alpha} g_i^{eq,s} = 0, \tag{8.2}$$

$$R_{\alpha\beta}^{eq,s} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} g_i^{eq,s} = 2\rho_s T_0 \left(E_s + T_0\right) \delta_{\alpha\beta}, \tag{8.3}$$

$$S_{\alpha\beta\gamma}^{eq,s} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} v_{i\gamma} g_i^{eq,s} = 0. \tag{8.4}$$

In the solid domain the kinetic equations for f-populations are not solved, since the mass is constant and the solid is at rest. Moreover, the Prandtl number is immaterial inside the solid and there is no need to adopt the quasi-equilibrium model for the evolution of the g-lattice. The kinetic equation in the solid for the g-populations thus reduces to:

$$g_i^s \left(\boldsymbol{x} + \boldsymbol{v}_i, t+1\right) - g_i^s \left(\boldsymbol{x}, t\right) = 2\beta_{2,s} \left(g_i^{eq,s} - g_i^s\right), \tag{8.5}$$

where the relaxation parameter $\beta_{2,s}$ defines the thermal diffusivity in the solid:

$$\alpha_s = \frac{k_s}{\rho_s C_s} = \left(\frac{1}{2\beta_{2,s}} - \frac{1}{2}\right) T_0. \tag{8.6}$$

The equilibrium distribution function for the g-lattice in the solid domain, is found by plugging equilibrium moments (8.2)-(8.4) into the Grad's distribution Eq. (6.38), leading to

$$g_i^{eq,s} = W_i \left(2\rho_s E_s + \frac{\left(R_{\alpha\beta}^{eq,s} - 2\rho_s E_s T_0 \delta_{\alpha\beta}\right)\left(v_{i\alpha} v_{i\beta} - T_0 \delta_{\alpha\beta}\right)}{2T_0^2}\right). \tag{8.7}$$

Deriving the hydrodynamic limit of Eq. (8.5) similar as in Sec. 6.3.4, the temperature diffusion equation of the form

$$\partial_t T = \partial_\alpha \left(\alpha_s \partial_\alpha T\right) = \alpha_s \partial_{\alpha\alpha} T, \tag{8.8}$$

is obtained for the heat conduction in the solid.

8.3 Conjugate heat transfer boundary condition

In order to recover conjugate heat transfer boundary conditions, the same boundary conditions as in Sec. 6.3.6 are here considered, where, however, ad-hoc target quantities and derivatives need to be employed. To apply expression (6.38) to an interface between fluid and solid, the local temperature at the boundary node needs to be computed using the CHT interface conditions. Once the local temperature at the boundary node (both on the solid side and fluid side) are known, the temperature gradients are computed at this location using one-sided derivatives as (2D, see Fig. 8.1):

$$\partial_x T|_{\boldsymbol{x}_{\mathrm{BC}}} \approx \left.\frac{\delta T}{\delta x}\right|_{\boldsymbol{x}_{\mathrm{BC}}} = \frac{T_{\mathrm{tgt}} - T(\boldsymbol{x}_3)}{\Delta x}, \tag{8.9}$$

$$\partial_y T|_{\boldsymbol{x}_{\mathrm{BC}}} \approx \left.\frac{\delta T}{\delta y}\right|_{\boldsymbol{x}_{\mathrm{BC}}} = \frac{T(\boldsymbol{x}_2) - T_{\mathrm{tgt}}}{\Delta x}, \tag{8.10}$$

with $\Delta x = 1$. This procedure allows to employ the Grad's populations which have consistent values for the local temperature and local heat flux, thus resulting in a stable and accurate boundary condition for the fluid-solid interface. This local temperature at the $\boldsymbol{x}_{BC}$ node is found by applying the requirements for conjugate heat transfer at the interface, as explained below with the help of Fig. 8.1.

As a first step, the temperature at the actual interface location $\boldsymbol{x}_{\mathrm{I}}$, which is the projection of $\boldsymbol{x}_{\mathrm{BC}}$ on the CHT interface along its normal, has to be reconstructed. Such a point is subject to the following conjugate conditions:

- temperature continuity:

$$T(\boldsymbol{x}_{\mathrm{I}})|_{\mathrm{solid}} = T(\boldsymbol{x}_{\mathrm{I}})|_{\mathrm{fluid}} = T(\boldsymbol{x}_{\mathrm{I}}), \tag{8.11}$$

- normal heat flux continuity:

$$k_s \left.\frac{\delta T}{\delta n}\right|_{\boldsymbol{x}_{\mathrm{I}},\mathrm{solid}} = k_f \left.\frac{\delta T}{\delta n}\right|_{\boldsymbol{x}_{\mathrm{I}},\mathrm{fluid}}, \tag{8.12}$$

where k_f and k_s are thermal conductivities in the fluid and solid respectively. The last condition is discretized by means of a first-order approximation:

$$k_s \frac{T(\boldsymbol{x}_s) - T(\boldsymbol{x}_I)}{l} = k_f \frac{T(\boldsymbol{x}_I) - T(\boldsymbol{x}_f)}{l}\ , \tag{8.13}$$

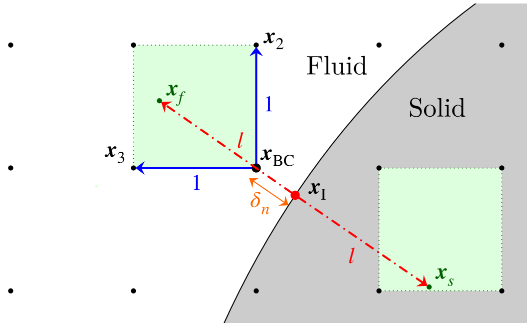


Figure 8.1: Schematic of the boundary treatment employed at the fluid-solid interface to reconstruct, for instance, the populations crossing the interface and coming to the node in $\boldsymbol{x}_{\mathrm{BC}}$.

where $\boldsymbol{x}_s$ and $\boldsymbol{x}_f$ are points on the direction normal to the interface. These inner points may not lie on the grid (similar to the case in Figure 8.1); hence, the temperature at these locations is computed using bilinear interpolation in two-dimensions, and trilinear interpolation in three-dimensions. The distance l in Fig. 8.1 of the inner points from the interface is set to $l = \sqrt{2}$ in 2D, while in 3D to $l = \sqrt{3}$, in lattice units.

Once the temperature at points $\boldsymbol{x}_s$, $\boldsymbol{x}_f$ is computed, the interface temperature $T(\boldsymbol{x}_{\mathrm{I}})$ is derived by rearranging Eq. (8.13):

$$T(\boldsymbol{x}_{\mathrm{I}}) = \frac{k_s T(\boldsymbol{x}_s) + k_f T(\boldsymbol{x}_f)}{k_s + k_f}. \tag{8.14}$$

The temperature in $\boldsymbol{x}_{\mathrm{I}}$, $\boldsymbol{x}_s$ and $\boldsymbol{x}_f$ is now known and its value at any point along the connecting line can be computed by linear interpolation. This approach is then used to derive T_{tgt} for the boundary node $\boldsymbol{x}_{\mathrm{BC}}$. With such a configuration, the target temperature for $\boldsymbol{x}_{\mathrm{BC}}$ is given by:

$$T_{\mathrm{tgt}} = T(\boldsymbol{x}_{\mathrm{I}}) + \frac{\delta_n}{l}\left(T(\boldsymbol{x}_f) - T(\boldsymbol{x}_{\mathrm{I}})\right). \tag{8.15}$$

After using Eqs. (8.9) and (8.10) to compute the temperature gradients, all the information needed to apply the Grad's boundary conditions at $\boldsymbol{x}_{\mathrm{BC}}$ are available and the missing populations of the g-lattice are thus obtained. The same procedures applies to the boundary nodes which lie on the solid side of the interface, where the major difference consists in replacing $T(\boldsymbol{x}_f)$ with $T(\boldsymbol{x}_s)$ in Eq. (8.15). It is evident from Figure 8.1 that this approach requires a minimum resolution of three to five grid points in the solid region, thus precluding simulations of sharp or thin objects without employing grid refinement.

8.4 Simulations

In order to systematically validate the above conjugate heat transfer model, four 2D benchmarks with increasing complexity are considered, followed by the simulation of a complex 3D system. First, a simple conduction between two solids with different thermal properties is simulated, followed by the simulation of conjugate heat transfer within a concentric annulus. A backward-facing step (BFS) flow is then proposed in order to validate the model for a flat fluid-solid interface with complex flow. Curved boundaries are later evaluated by means of a flow past an internally heated cylinder. Finally, turbulent heat transfer over a wall-mounted matrix of cubes is simulated to probe the coupling between the CHT model and the entropic LBM.

8.4.1 Conduction 2D

In the following, two purely conduction problems are considered: one with straight walls and different types of boundary conditions, and the other with curved walls.

In the first set-up, two solid materials with different thermal properties are vertically juxtaposed. The boundary conditions and a sample temperature solution obtained with the present model are illustrated in Fig. 8.2. Top, right and bottom borders are subject to Dirichlet boundary conditions: the left halves of the top and bottom boundaries reduced temperature are set to $T'_h = 1$ and $T'_c = 0$ respectively, the right border to $T'_m = 0.5$, where $T' = \frac{T-T_c}{T_h-T_c}$, and the right halves of the top and bottom boundaries undergo linear thermal ramps between the end temperatures. The left border is kept adiabatic. Temperature contours solution is plotted for the case $k_{s,2}/k_{s,1} = 10$. Simulations have been performed for different thermal conductivity ratios and the results at steady state are reported below. The ELBM solutions are obtained with 79×40 nodes, whereas the reference solutions are computed with the PDE toolbox provided by Matlab on a grid composed of more than 5 millions finite element cells in order to ensure convergence of the solution.
The temperature profiles along the horizontal interface and the vertical section placed in the middle of the domain are shown in Fig. 8.3. All the plots

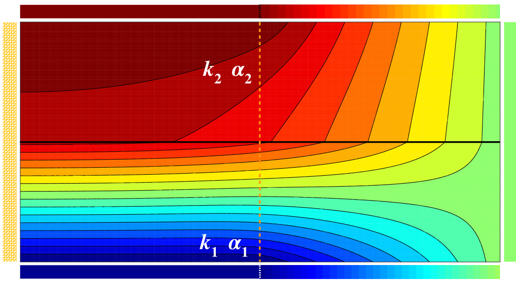


Figure 8.2: Setup and temperature solution for the pure conduction problem in a box. Temperature contours solution is plotted for the case $k_{s,2}/k_{s,1} = 10$.

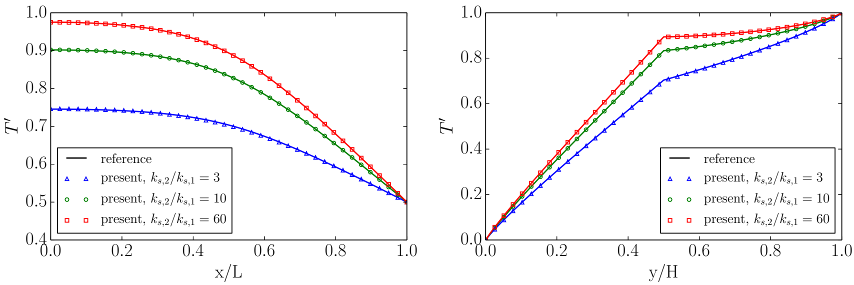


Figure 8.3: Temperature profiles along the interface (left) and along the vertical section (right) indicated in Fig. 8.2. The ratio for the volumetric specific heats has been fixed to $C_{s,2}/C_{s,1} = 10$.

exhibit an excellent agreement between the reference solution and the LBM results. Note that conductivity and thermal diffusivity ratios are different ($C_{s,2}/C_{s,1} = 10$), proving that our model is able to decouple the thermal diffusivity $\alpha_s = \frac{k_s}{\rho_s C_s}$ from the thermal conductivity k_s.

The complexity of the set-up is next increased, considering the problem of a concentric annulus made of two different solid materials. The set-up, boundary conditions and a sample temperature solution obtained with the present model are shown in Fig. 8.4. The outer border ($r = R_2$) is subject to Dirichlet boundary condition described by Eq. 8.16. The Dirichlet boundary condition for the temperature applied to the external wall is:

$$T(R_2, \theta) = T_c + \frac{T_h - T_c}{2}(1 + \cos[n\theta]), \quad n \in \mathbb{Z}, \tag{8.16}$$

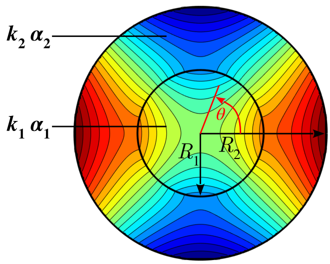


Figure 8.4: Setup and temperature solution for the concentric annulus problem. Temperature solution is plotted for the case $k_{s,2}/k_{s,1} = 100$.

with $n = 2$, while conjugate heat transfer condition is applied between the two annulus, at $r = R_1$. As for the previous case, simulations have been performed for different thermal conductivity ratios and the results at steady state are reported below. The ELBM solutions are obtained with a resolution of $R_1 = 40$ lattice nodes, whereas the reference analytical solutions were presented in [299]. In Fig. 8.5 the temperature profiles are exposed for three different thermal conductivity ratios $k_{s,2}/k_{s,1} = 0.1$, $k_{s,2}/k_{s,1} = 1$ and $k_{s,2}/k_{s,1} = 100$. The profiles are plotted along $\theta = 0$ and $\theta = 90°$ as a function of the radius r, and at the conjugate heat transfer interface (R_1) as a function of the angle θ. As for the previous cases, the LBM results are

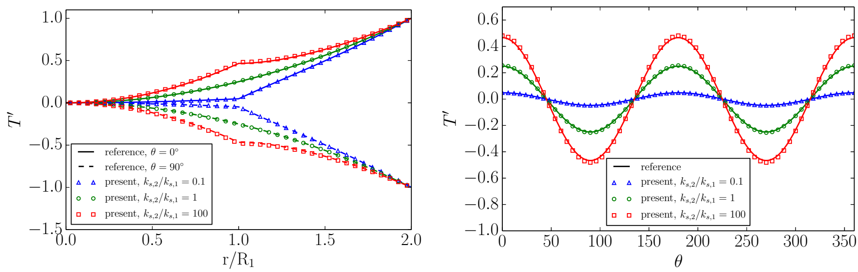


Figure 8.5: Temperature profiles along $\theta = 0$ and $\theta = 90°$ as a function of the radius r (left) and at the conjugate heat transfer interface (R_1) as a function of the angle θ (right).

in good agreement with the analytical derivations.
In order to analyze the order of convergence of the present model and boundary conditions for curved walls, the error of the solution is computed with respect to analytical solution for different grid sizes. The result is shown

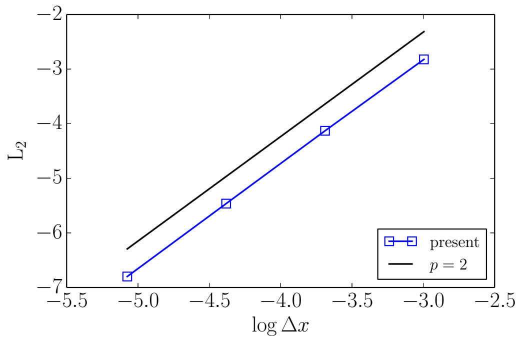


Figure 8.6: Grid convergence study for the case $k_{s,2}/k_{f,1} = 100$. The order of convergence is $p = 1.91$.

in Fig. 8.6. The error is computed based on the temperature field at the interface and it is defined as:

$$L_2 = \sqrt{\frac{\sum_i \left(T_{i,LBM} - T_{i,ref}\right)^2}{\sum_i T_{i,ref}^2}}, \tag{8.17}$$

where $T_{i,LBM}$ is the sample i of the temperature computed by the present LBM model and $T_{i,ref}$ is the sample i of the temperature computed analytically. The plot shows that the second order convergence of the lattice Boltzmann method is retained ($p = 1.91$) also in the case of curved walls and with conjugate heat transfer interface.

8.4.2 Backward-facing step flow

In order to validate the model for the case of a flat fluid-solid interface, the backward-facing step (BFS) flow is chosen as a benchmark due to the interplay between numerous features which characterize it: flow separation, reattachments and recirculation zones. The BFS problem has been investigated in the literature, and studies involving conjugate heat transfer have also been performed. In the present work the results from [311] will be used as a reference. The set-up, together with the boundary conditions is illustrated in Fig. 8.7, together with a sample temperature solution for the case $k_{s,2}/k_{f,1} = 10$. The inflow has a uniform temperature $T_c = 1/3$ and a parabolic u profile: $u(y) = 4U_{\max}\left(yh - y^2\right)/h^2$, $U_{\max} = 0.05$. The outflow is enforced to have fully-developed conditions. The bottom wall is kept at constant temperature $T_h = 1$, with all other walls set adiabatic. LBM results obtained with $h = 20$ lattice units. The domain is split into a top

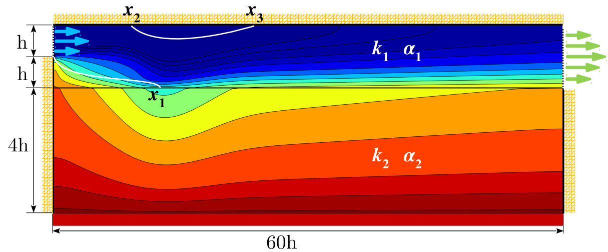


Figure 8.7: Setup and temperature solution for the BFS flow problem. Temperature solution plotted for the case $k_{s,2}/k_{f,1} = 10$. White lines indicate the points where the vorticity goes to 0.

region which is fluid and a bottom one which is solid. The fluid enters the channel through an inlet (of size h) which is half the height of the channel (of size $2h$), giving rise to the backward-facing step flow. After the step, the flow turns down due to the lower pressure above bottom wall, and reattaches at the bottom wall at x_1. At the same time, the flow separates on the top wall at x_2, creating an upper recirculation zone which ends at x_3.

The flow properties for this setup were chosen to be Pr $= 0.71$ and Re $= 800$. The Reynolds number is here defined as:

$$\mathrm{Re} = \frac{2hU_{\mathrm{mean}}}{\nu} , \tag{8.18}$$

where U_{mean} is the mean entrance velocity, which for a parabolic profile is equal to $U_{\mathrm{mean}} = \frac{2}{3}U_{\mathrm{max}}$.

Before testing the conjugate heat transfer model, a validation of the velocity field is required. This is accomplished in the first place by measuring the normalized recirculation lengths of the flow:

$$x_1/h = 12.10 , \quad x_2/h = 9.95 , \quad x_3/h = 20.65, \tag{8.19}$$

which match well with the data available in the literature [311–315]. As a further validation, in Fig. 8.8 the profiles of the horizontal velocity u along three vertical sections are presented, juxtaposed with the results from [312]. Good agreement between the profiles shows the validity of the ELBM

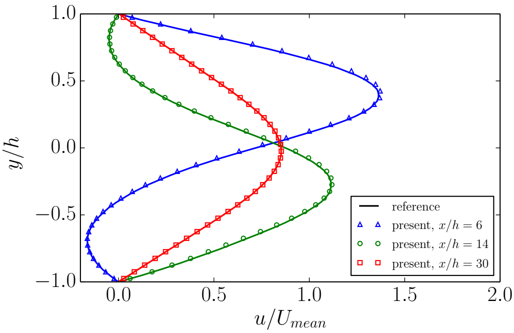


Figure 8.8: Profiles of the normalized horizontal velocity u/U_{mean} along different vertical sections. Reference solution from [312].

as flow solver. In order to validate the proposed conjugate heat transfer model, temperature profiles of the ELBM solution are provided in Fig. 8.9 and compared to the data of [311], for conductivity ratios $k_{s,2}/k_{f,1} = 1000$, $k_{s,2}/k_{f,1} = 100$, $k_{s,2}/k_{f,1} = 10$ and $k_{s,2}/k_{f,1} = 1$. In Figure 8.9 (left), temperature profiles along the vertical section located at $x/h = 6$ show that the temperature field assumes a linear profile in the solid region, which is mainly due to the horizontal stretching of the domain that softens any bi-dimensional effect. The same plots allow to observe the change of slope occurring at the interface as a result of the continuous heat flux and its dependence on the conductivity ratio. In addition, Figure 8.9 (right) shows that the lowest temperature at the interface is reached where the cold bulk flow reattaches at the bottom wall, i.e. around x_1 or slightly downstream. Conversely, the highest temperatures occur in the primary recirculation zone, for $x < x_1$, due to the flow stagnation. All the temperature profiles in Fig. 8.9 excellently match the solution from [311]. A grid convergence study for this problem was also performed and the outcome is summarized in Fig. 8.10 for the error on the temperature field. The error was defined as for the previous set-up (Eq. 8.17), with the reference temperature from the solution presented in [311]. Convergence rate to the solution turns out to have a slope of $p = 1.82$, which is considered a satisfactory result given the semi-second-order interpolations employed in the Grad's boundary conditions. Finally, the assessment of the Nusselt number along the fluid-solid interface is presented in Fig. 8.11 for different conductivity ratios: $k_{s,2}/k_{f,1} = 1000$, $k_{s,2}/k_{f,1} = 100$, $k_{s,2}/k_{f,1} = 10$ and $k_{s,2}/k_{f,1} = 1$. In

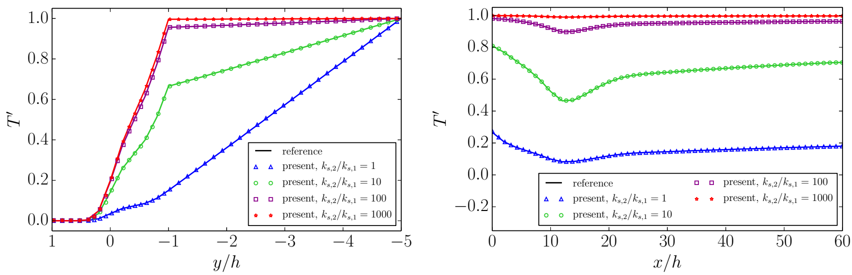


Figure 8.9: Temperature profiles along the vertical section located at $x/h = 6$ (left) and along the fluid-solid interface (right). Reference solutions from [311].

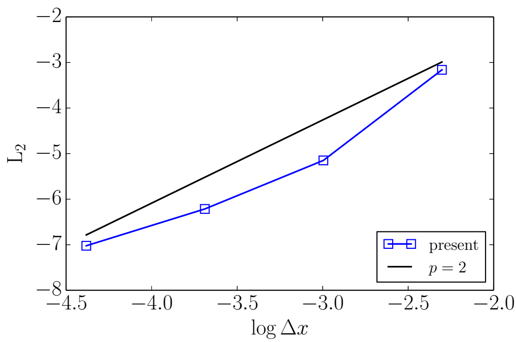


Figure 8.10: Grid convergence study for the case $k_{s,2}/k_{f,1} = 100$. The order of convergence is $p = 1.82$.

this system the Nusselt number is computed as:

$$\mathrm{Nu}(x) = -\frac{2h}{T_h - T_c} \frac{\partial T}{\partial y}\bigg|_{\mathrm{Int},f} . \tag{8.20}$$

The plot shows that the highest rate of heat transfer occurs at the lower reattachment point x_1, where the cold flow impacts the wall and efficiently removes heat from the solid. Conversely, downstream the fluid becomes warmer and it runs parallel to the wall, reducing the rate of heat transfer. Moreover, in the upstream stagnation region the fluid is not efficiently replenished and it heats up to almost the solid temperature, reducing the Nusselt number to its lowest values. The plot exhibits an excellent agreement between the present simulation and the reference, despite the sensitive nature of the Nusselt number.

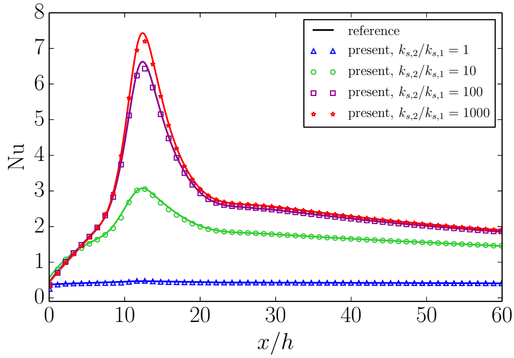


Figure 8.11: Nusselt number profiles along the fluid-solid interface for different conductivity ratios. Reference solutions from [311].

8.4.3 Flow past a heated cylinder

The next step of the validation process involves the simulation of a curved boundary. The benchmark chosen to perform such a verification is the system proposed in [316], where a cold flow passes over a circular cylinder, whose core is kept at a constant higher temperature. The hot core of the cylinder is also circular with half of its diameter. The complete setup is illustrated in Fig. 8.12 together with a sample temperature solution. A uniform inflow with velocity $U_\infty = 0.05$ and temperature $T_c = 1/3$ was considered, while the outflow is set to have normal gradients equal to zero. Periodic top and bottom boundaries are employed. The core of the cylinder is kept at constant temperature $T_h = 1$. The results reported here have been obtained with an external diameter of $D = 100$ lattice units. The

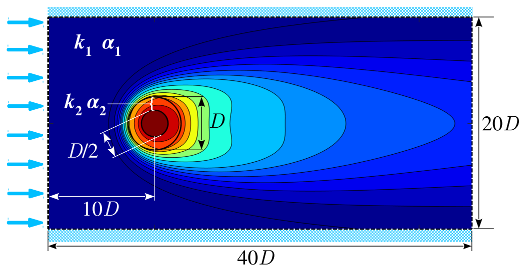


Figure 8.12: Setup and temperature solution for the flow past an internally heated cylinder. Temperature solution plotted for the case $k_{s,2}/k_{f,1} = 4$.

flow properties have been maintained constant for all simulations reported below, with Re = 40 andPr = 0.72. The Reynolds number is here defined as follows:

$$\mathrm{Re} = \frac{DU_\infty}{\nu}, \tag{8.21}$$

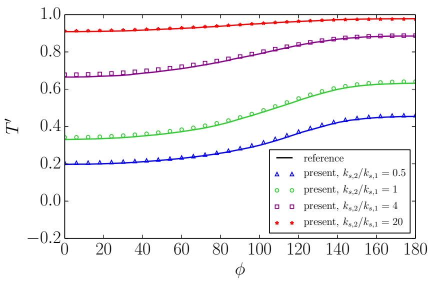
where U_∞ is the free-stream velocity upstream the cylinder.

Same as for the BFS flow, a validation of the flow field around the cylinder is first required. Specifically, the recirculation length of the wake region behind the cylinder is adopted as a validation parameter. The values obtained for the normalized recirculation length $\tilde{L} = \frac{2L}{D}$ are $\tilde{L} = 1.86$ and $\tilde{L} = 4.52$ for Re = 20 and Re = 40 respectively, which well agree with the results reported in the literature [78, 82, 230, 317, 318].

The validation of the conjugate heat transfer model is presented in Fig. 8.13, where the profiles for the temperature and the Nusselt number along the interface are compared to the results from [316] for the following conductivity ratios: $k_{s,2}/k_{f,1} = 20$, $k_{s,2}/k_{f,1} = 4$, $k_{s,2}/k_{f,1} = 1$ and $k_{s,2}/k_{f,1} = 0.5$. The Nusselt number is computed as:

$$\mathrm{Nu}(\phi) = -\frac{D}{T_h - T_c}\frac{\partial T}{\partial r}\bigg|_{\mathrm{Int},f}, \tag{8.22}$$

where r is the outward radial direction. The plots show a good agreement



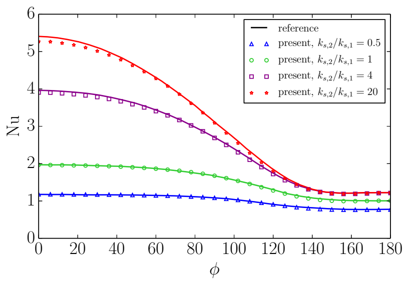


Figure 8.13: Profiles for the temperature (left) and the Nusselt number (right) along the fluid-solid interface of the cylinder. ϕ is the angle measured from the upstream stagnation point of the cylinder. Reference results from [316].

between the LBM results and the reference solutions. Small deviations in the Nusselt number for $\phi = 0$ can be detected, but they amount to at most 2.5% of the reference values; however, Ref. [316] does not report a validation procedure, thus undermining the overall accuracy of the results reported therein.

8.4.4 Turbulent flow over wall-mounted matrix of cubes

In order to demonstrate the capabilities of the described CHT model in the framework of the entropic thermal LBM, conjugate heat transfer with the flow in a turbulent regime was simulated. The references employed for this validation are the experiments described in [319, 320], and large eddy simulations (LES) from [321].
The experimental set-up considered a narrow flat channel with a regular matrix of cubes mounted on one of the walls. Convective heat transfer was performed by powering up a cube located downstream such that the flow, in this case air, could be considered fully developed. Similar to the LES study, this allows to numerically model only the sub-channel unit surrounding the heated cube, with the surrounding influence rendered through periodic boundaries for the velocity field. In fact, the experiment was carefully designed to serve as a benchmark for numerical method which typically prefer smaller simulation domains [320]. While the flow is modeled with periodic boundaries, only the thermal field is characterized by inlet-outlet boundary conditions, whose overview is provided in Fig. 8.14. The cubes has a char-

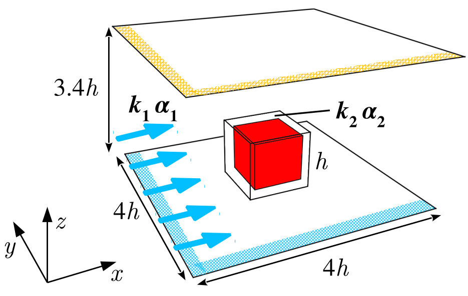


Figure 8.14: Numerical setup for the turbulent flow over a wall-mounted matrix of cubes.

acteristic size h corresponding to the edge length, the sub-channel unit has dimensions $4h \times 4h \times 3.4h$, with the latter being the channel height and $4h$ the distance between the centers of the matrix cubes. The flow enters the domain at uniform temperature $T_c = 20°\text{C}$, whereas at the outlet the normal gradient is set to zero. Both top and bottom walls are kept adiabatic with no-slip condition for the velocity. The cube comprises an inner cubic core of edge length $0.8h$ kept at constant temperature $T_h = 75°\mathrm{C}$ and an external epoxy layer with thickness $0.1h$. The simulations have been performed with h resolved by $h = 100$ lattice units. Conduction occurs within the epoxy layer (outer cube) enveloping the hot core and conjugate heat transfer takes place at the interface with the fluid medium. The detailed physical quantities describing the system are reported in [320, 321], with the characteristic thermal ratios being:

$$\frac{\alpha_{s,2}}{\alpha_{f,1}} = 5.599 \times 10^{-3}, \qquad \frac{k_{s,2}}{k_{f,1}} = 9.183. \tag{8.23}$$

In present simulations, the thermal diffusivity ratio is set to $\frac{\alpha_{s,2}}{\alpha_{f,1}} = 5.599 \times 10^{-1}$, leading to a heat capacity ratio of $\frac{c_{s,2}}{c_{v,1}} = 10$. This is due to the low grid resolution employed for the epoxy layer ($t = 10$ grid points), which in turn determines the total number of grid points in the domain, owing to the uniform grids used. However, this does not affect the quality of the comparisons with experiments and simulations, as demonstrated below, since only steady state profiles which are mildly depending on the heat capacity ratios, are compared.
The flow regime is defined by:

$$\mathrm{Pr} = 0.712\ , \qquad \mathrm{Re} = \frac{U_0 h}{\nu} = 3854, \tag{8.24}$$

where U_0 is the mean bulk velocity of the fluid before entering the matrix of cubes.

Similarly to the preceding 2D simulations, a validation of the velocity field is firstly presented. Figure 8.15 shows the time-averaged profiles for the horizontal velocity $\bar{u}$, upstream ($x = 1.2h$) and downstream ($x = 2.8h$) with respect to the cube, while for the same positions, Fig. 8.16 shows the velocity stresses $\overline{u'u'}$. Both mean profiles exhibit large velocities at higher values of z, where the flow is only marginally obstructed by the cube. Figure 8.15 shows that, at $x = 1.2h$, for low z values the fluid is slowed down by the presence of the cube, as it forces the flow to recirculate downwards and sideways or to shift upwards and climb over the leading edge. However, for very low z values a negative u velocity can be detected, as it belongs to those streamlines which have no other option but recirculating backwards against the main stream, forming a horse-shoe vortex. On the other hand, at position $x = 2.8h$, the mean streamwise velocity exhibits negative values for any $z < h$, as the fluid that has streamed sideways or above the cube recirculates back to fill the gap in the wake region. A good match with

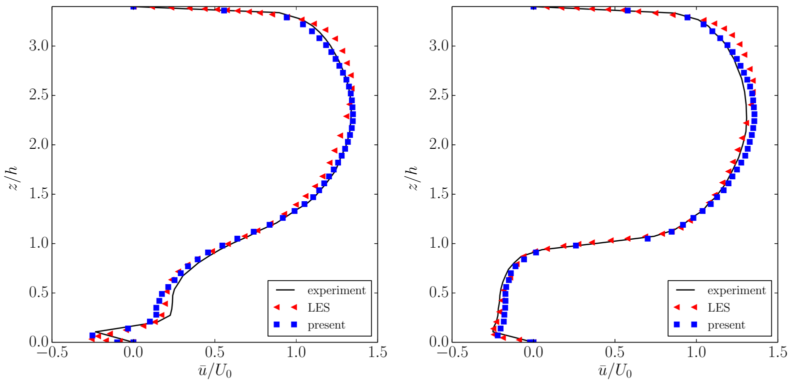


Figure 8.15: Profiles of the mean horizontal velocity $\bar{u}$ along two vertical lines located at $y = 2h$ (middle plane) and $x = 1.2h$ (left), $x = 2.8h$ (right). Experimental measurements from [320]; LES results from [321].

both experimental and numerical results can be observed. Correspondingly, stresses are more pronounced near the the top and bottom walls, as well as near the top of the cube, where the main recirculation zone starts to form in the wake behind the cube, as shown in Figure 8.16. Also for the stresses good agreement of the ELBM results with both the experiments and LES is observed.

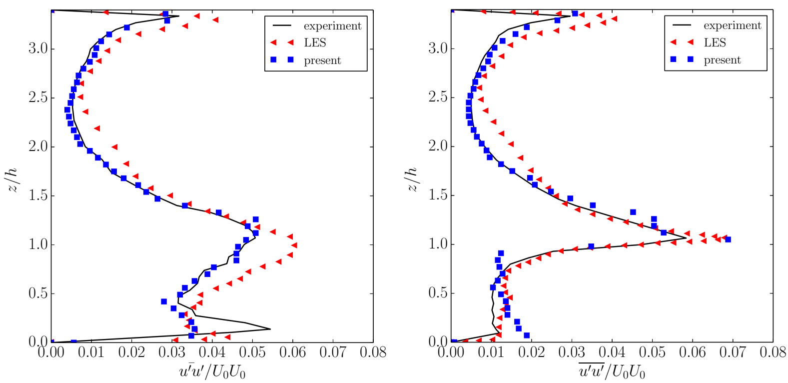


Figure 8.16: Profiles of the horizontal velocity stresses $\overline{u'u'}$ along two vertical lines located at $y = 2h$ (middle plane) and $x = 1.2h$ (left), $x = 2.8h$ (right). Experimental measurements from [320]; LES results from [321].

To provide an overview of the flow field, in Fig. 8.17 the isosurface of the mean vorticity colored with mean velocity magnitude is shown. One can notice the horse-shoe vortex formed in front of the cube, the mean recirculation bubbles near the sides of the cube and the mean wake vortex.

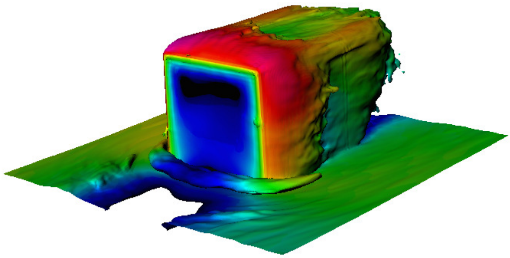


Figure 8.17: Isosurface of mean vorticity magnitude field colored by mean velocity magnitude. Isosurface of mean vorticity magnitude is fixed at 5% of the maximal mean vorticity, while range for the mean velocity magnitude is from 0 to U_0.

Regarding the temperature field, different characteristic profiles are here employed for comparison. In Figure 8.18 the temperature distributions along three normals to the surface are presented and compared to the results from the LES simulations of [321]. By resolving the temperature field also in the epoxy layer, it is possible to observe the effect of the step in conductivity on the temperature distributions, which at the fluid-solid interface has a jump in the gradient. Moreover, the steeper profile exhibited by the temperature in case (b) is an indication of the greater heat flux that affects the upstream face of the cube. Finally, the higher mean temperature observed away from the cube in case (c) indicates an increase of the mean flow temperature in the wake of the cube, due to the heat transfer from the heated cube. A satisfactory agreement is reached for all the profiles.

In Figures 8.19 and 8.20, the distribution of the cube surface temperature along the two main mid-planes is presented. As expected, the upwind face exhibits lower surface temperature than the downstream one, since the vortex in the wake region accumulates heat and does not convect it efficiently.

It is interesting to observe that the lateral faces of Fig. 8.20 demonstrate a slightly higher temperatures closer to the upwind edge, where the flow detaches and cannot fully cool the surface. Also in this case the present

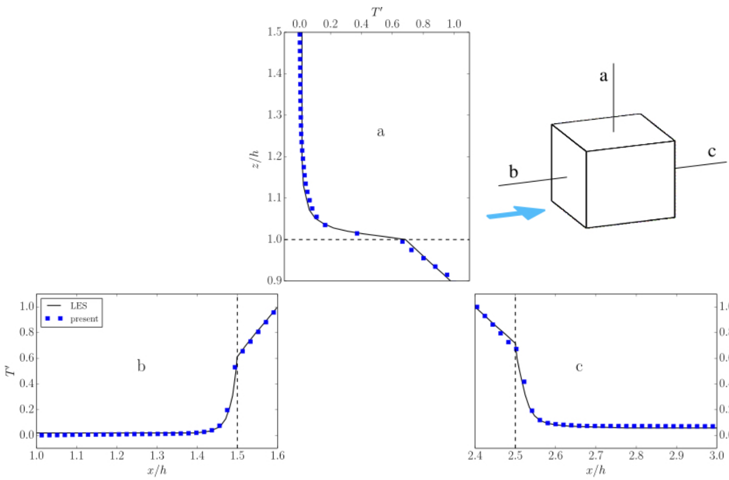


Figure 8.18: Temperature profiles along three normals to the surface of the cube. All profiles lie on the vertical mid-plane ($y = 2h$) and are located at $x = 2h$ (a) and $z = 0.5h$ (b, c). Also the temperature distribution within the epoxy layer is drawn. LES results from [321].

model produces results in agreements with the experimental and numerical references. However, one can notice an under-prediction of the temperature near points A and D of Figure 8.19, corresponding to the bottom wall of the domain. Similar findings were reported by the results from LES simulations [320], and are due to the fact that heat conduction in the bottom wall has not been resolved, and instead, adiabatic condition has been enforced.

Finally, in Figure 8.21 a snapshot of the temperature field is presented at mid-vertical plane corresponding to points ABCD of Fig. 8.20. One can observe the fixed hot temperature of the inner cube, which decreases progressively in the epoxy layer, and then drops dramatically in the fluid region, with some hot spots depending on the flow field. In general, the temperature in the wake of the cube is higher than upstream the cube, as expected, due to heat transfer from the solid cube to the fluid.

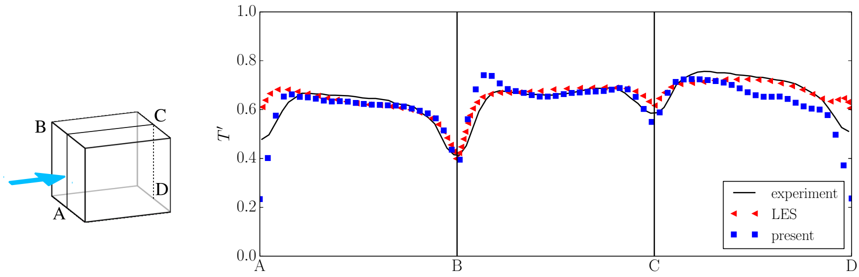


Figure 8.19: Surface temperature along the longitudinal mid-planes of the cube.

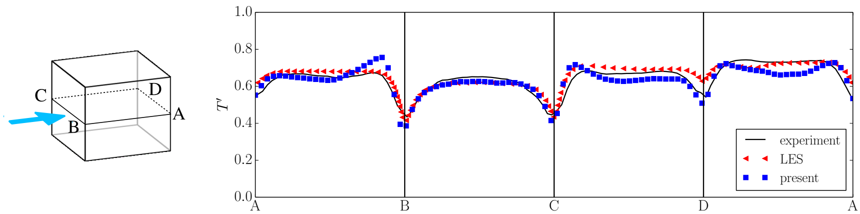


Figure 8.20: Surface temperature along the horizontal mid-planes of the cube.

8.5 Conclusion

An extension of the entropic two-population model for the simulation of flows with conjugate heat transfer was presented. The extension relies on two simple but fundamental features: consistent Grad's boundary conditions for the temperature dynamics and the conjugate heat transfer condition for complex geometries. The use of the Grad's boundary conditions together with the entropic multirelaxation LBM enable stable and accurate simulations even for turbulent flows. The conjugate heat transfer condition between two different media, guaranteed through counter-extrapolation method, extends the capabilities of the Grad's boundary conditions to further complex flows involving heat transfer. This is validated by means of three two-dimensional and one three-dimensional set-ups. In two-dimensions, the conjugate heat transfer between two-solid with varying boundary conditions at the borders has been successfully simulated and validated. Next, the well studied backward-facing step, together with a heated solid domain, has been simulated. Finally, for two-dimensional problems,

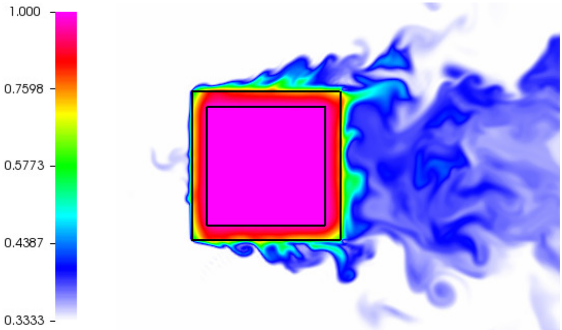


Figure 8.21: Snapshot of the temperature field (in lattice units and logarithmic scale) at mid-height plane of the cube. Inner black line represents inner hot cube, while outer black line represents interface cube between epoxy and fluid.

the complexity has been increased by simulating the flow around a circular cylinder heated by an internal smaller cylinder. Also in this case the results were promising for different thermal conductivities. Regarding the three-dimensional set-up, the turbulent flow field around a wall mounted, internally heated cube was simulated. Good comparison of velocity and temperature profiles with LES simulations and experiments highlights the accuracy and reliability of the proposed model.

Use of the present boundary conditions for imposing the CHT conditions at the interface requires at least three to five grid points to resolve the solid structures. This, when combined with the uniform grids employed here, precludes simulations of thin objects or sharp corners. However, block grid refinement techniques such as [322] can be combined with the present approach. Viable promising solution for the simulation of three-dimensional turbulent flows coupled with the heat transfer in solid were presented and successfully validated in a lattice Boltzmann framework.

Finally, it should be noted that the present approach could be easily extended to multi-speed lattices, thus enabling conjugate heat transfer computations also with compressible flows.

Chapter 9

Grid refinement for thermal and compressible entropic lattice Boltzmann models

9.1 Introduction

All simulations presented in this work were limited sofar to cartesian meshes with a constant grid spacing in the whole domain. However, in realistic applications, employment of such meshes would lead to unaffordable computational costs, due to the spatial grid required in order to solve entirely or partially the spatial and temporal flow field scales. Despite that, in general, small scale structures are often confined into a small portion of the simulation domain, with the possibility then, to optimize the mesh employing refinement block where needed. In general, two main approaches to block grid refinement can be found in the literature: a cell-centered volumetric approach [323, 324] and a node-centered approach [325–328]. In the cell-centered approach populations are continuous over the grid interface and exact conservation of mass and momentum across the grid border may be achieved. However, staggering effects on the fine level require additional filtering. On the other side, the node-centered description requires interpolation and rescaling of the population at the grid interface. Here, the node-centered approach is considered.

9.2 Grid refinement algorithm for the lattice Boltzmann method

In the following, the grid refinement algorithm [322] is briefly reminded. The main focus is to extend the approach to the thermal and compressible models presented in Chapters 6 and 3 respectively. For more details about the algorithm and implementation the reader is referred to [322, 328].

The present approach assumes that finer-level grids are nested into the coarse level grid, and only one level is present for each patch. It follows, that the most crucial part of the algorithm is the information transfer from coarse to fine level and vice-versa. This is realized through an overlapping region which extents the nominal domain of the two patches. In this region both fine and coarse grids are present. The minimal interface width for the proposed coupling mechanism is $2 \times \max\left(|v_i|,\ i \in n_Q\right)$ and $\max\left(|v_i|,\ i \in n_Q\right)$ for the fine- and coarse-level grid, respectively.
Two types of nodes are considered within the overlapping region: twin nodes, i.e. those nodes located in places where both coarse and fine level nodes are overlapping, and reconstruction nodes, i.e. those nodes that belong to the fine level interface and do not coincide with any coarse level nodes. Thus the information on those nodes is not readily available and information from the neighboring grid nodes need to be taken into account in order to complete the information exchange between the grids.

In standard LBM on regular meshes, non-dimensional quantities are evaluated in lattice units, which are specific to the chosen grid. Employment of multiple grids with different grid spacing, yields a set of lattice units which need to be rescaled appropriately in order to ensure identical non-dimensional quantities. The spatial scaling of the grid spacing is straightforwardly defined as

$$r = \frac{\delta x_c}{\delta x_f}, \tag{9.1}$$

where δx_c and δx_f are grid spacing in the coarse and fine grid respectively, and r is the rescaling factor. More possibilities exists for the time scale; while diffusive scaling relates the time and space scales as $\delta t \sim \delta x^2$, the convective scaling requires $\delta t \sim \delta x$. Here, the convective scaling is considered, due to its greater numerical efficiency; it follows

$$r = \frac{\delta t_c}{\delta t_f}. \tag{9.2}$$

As a result, the macroscopic quantities such as pressure, velocity and temperature are continuous across the grid border. However, for a continuous Reynolds number $\mathrm{Re} = UL/\nu$ the viscosity scales as

$$\nu_f = r\nu_c. \tag{9.3}$$

Since viscosity is related to a general relaxation parameter β by

$$\nu = \left(\frac{1}{2\beta_1} - \frac{1}{2}\right) T, \tag{9.4}$$

in the compressible model (with $T = T_0$ in the thermal model), a scaling of the relaxation parameter is required as

$$\beta_{1,f} = \frac{1}{1 + r(1/\beta_{1,c} - 1)}. \tag{9.5}$$

For the rescaling operation, populations are decomposed into $f_i = f_i^{eq}(M_{\mathcal{C}}) + f_i^{neq}(M_{\mathcal{C}}, \nabla M_{\mathcal{C}})$, where the equilibrium distribution is written as a function of the conserved moments $M_{\mathcal{C}}$ and the non-equilibrium part additionally depends on their gradients. Since the conserved fields are continuous across the interface, the equilibrium does not require any rescaling. However, the non-equilibrium part is proportional to the gradients and therefore needs rescaling. The rescaling procedure is discussed separately for each thermal and compressible lattice Boltzmann model in Sections 9.3 and 9.4 respectively.

The coupling from the coarse to the fine grid requires only the rescaling procedure as outlined above. The fine level grid however requires an additional reconstruction of the hanging or reconstruction nodes that do not correspond to a coarse level node. As no information from the coarse level is available at this location, the information of the neighboring nodes is required to be included through an interpolation scheme. Analogous to [328], a centered four point stencil is here used in each spatial dimension of the Lagrange polynomial. Note that in contrast to [328], biased interpolation at the corners is avoided to prevent the anisotropy from influencing the solution. This is particularly important for the thermal and compressible model, where a biased interpolation stencil triggers spurious artifacts at the grid interface. While stability issues caused by these artifacts for all entropic models presented here were not encountered, instabilities might be triggered for models without a stabilization mechanism, such as LBGK. This interpolation is used for the macroscopic quantities of the flow field needed to compute the equilibrium part of the populations as well as the non-equilibrium part of the populations using their values at the twin nodes

on the coarse level grid.
In the following, rescaling required for the grid refinement algorithm is described for the thermal and compressible entropic lattice Boltzmann models.

9.3 Populations rescaling for the thermal two-population model

The two-population model, is based on the incompressible lattice Boltzmann method for what concerns mass and momentum equations. In that case, the non-equilibrium part of the population can be approximated as

$$f_i^{neq} \approx \frac{W_i}{2T_0^2} P_{\alpha\beta}^{(1)} [v_{i\alpha} v_{i\beta} - T_0 \delta_{\alpha\beta}], \tag{9.6}$$

where non-equilibrium part of the pressure tensor is given as

$$P_{\alpha\beta}^{(1)} = -\frac{1}{2\beta_1} \rho T_0 \left(\partial_\alpha u_\beta + \partial_\beta u_\alpha \right). \tag{9.7}$$

Note that strain rate tensor is given in fine and coarse lattice units respectively, and therefore requires rescaling to assure continuity of the non-equilibrium fields. Straightforward substitution yields the relation between coarse and fine level non-equilibrium as

$$f_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} f_{i,c}^{neq}. \tag{9.8}$$

For the thermal two-population model the rescaling of the f- and g-populations cannot be applied in the same way as for the f-populations, Eq. (9.8), because two independent relaxation parameters appear in the kinetic equations. Moreover, a distinction between the two cases of $\mathrm{Pr} \leq 1$ and $\mathrm{Pr} > 1$ has to be taken into account.

For $\mathrm{Pr} \leq 1$ there is no contribution of the quasi-equilibrium state in the kinetic equation for f, so that the same rescaling as Eq. (9.8) can be applied for the f-populations. For the g-populations, however, this is no more valid, since the non-equilibrium part depends on two independent relaxation parameters. This can be verified by deriving an analytical expression for the non-equilibrium g-populations through Chapman-Enskog expansion, as in [219]. The non-equilibrium part of the g-population depends on the higher-order, non-conserved moments as

$$g_i^{neq} = g_i^{neq} \left[q_\alpha^{(1)}(\beta_1, \beta_2), R_{\alpha\beta}^{(1)}(\beta_2) \right], \tag{9.9}$$

where $q_\alpha^{(1)}$ and $R_{\alpha\beta}^{(1)}$ are the first and second order non-equilibrium moments of the g-populations. The $q_\alpha^{(1)}$ moment includes a contribution of β_1, different from the moment $R_{\alpha\beta}^{(1)}(\beta_2)$ and the higher-order moments, which need to be excluded before rescaling of the non-equilibrium part. The $q_\alpha^{(1)}$ moment can be written analytically as

$$q_\alpha^{(1)}(\beta,\beta_1) = -\frac{1}{2\beta_2}\rho D T_0 \partial_\alpha T + 2u_\beta P_{\alpha\beta}^{(1)}(\beta_1), \tag{9.10}$$

so that the contribution in β_1 can be separated from the rest of the contributions to the non-equilibrium part of the g-populations. This contribution can be rescaled separately according to

$$P_{\alpha\beta,f}^{(1)} = \frac{\beta_{1,c}}{r\beta_{1,f}} P_{\alpha\beta,c}^{(1)}. \tag{9.11}$$

After subtraction of the term dependent on β_1 the rescaling is performed as usual. The final result reads

$$g_{i,f}^{neq} = \frac{\beta_{2,c}}{r\beta_{2,f}}\left[g_{i,c}^{neq} - W_i \frac{2u_\beta P_{\alpha\beta,c}^{(1)} v_{i\alpha}}{T_0}\right] + W_i \frac{2u_\beta P_{\alpha\beta,f}^{(1)} v_{i\alpha}}{T_0}. \tag{9.12}$$

Note that the non-equilibrium pressure tensor $P_{\alpha\beta}^{(1)}$ can be computed directly from f-populations by

$$P_{\alpha\beta}^{(1)} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta}(f_i - f_i^{eq}). \tag{9.13}$$

For the case $\mathrm{Pr} > 1$, similar considerations can be applied, and the final results are, for the f-populations,

$$f_{i,f}^{neq} = \frac{\beta_{2,c}}{r\beta_{2,f}} f_{i,c}^{neq}, \tag{9.14}$$

while for the g-populations,

$$g_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}}\left[g_{i,c}^{neq} - W_i \frac{2u_\beta P_{\alpha\beta,c}^{(1)} v_{i\alpha}}{T_0}\right] + W_i \frac{2u_\beta P_{\alpha\beta,f}^{(1)} v_{i\alpha}}{T_0}, \tag{9.15}$$

where the non-equilibrium pressure tensor is rescaled as

$$P_{\alpha\beta,f}^{(1)} = \frac{\beta_{2,c}}{r\beta_{2,f}} P_{\alpha\beta,c}^{(1)}. \tag{9.16}$$

9.4 Populations rescaling for compressible models

As for the thermal model, also in the compressible model the populations can not be rescaled directly, since the non-equilibrium part depends on two relaxation parameters. For both f- and g-populations the non-equilibrium parts can be derived analytically as a function of the higher-order, non-conserved moments [130]. Here, only the case $\mathrm{Pr} \leq 1$ is considered; for the case $Pr > 1$ the relaxation parameters β_1 and β_2 needs simply to be interchanged.
For the f-populations, the non-equilibrium part depends on the higher-order, non-conserved moments as

$$f_i^{(1)} = f_i^{(1)} \left[P_{\alpha\beta}^{(1)}(\beta_1), Q_{\alpha\beta\gamma}^{(1)}(\beta_1, \beta_2), R_{\alpha\beta\gamma\mu}^{(1)}(\beta_1)\right], \tag{9.17}$$

where $P_{\alpha\beta}^{(1)}$ is the non-equilibrium pressure tensor, and $Q_{\alpha\beta\gamma}^{(1)}$ and $R_{\alpha\beta\gamma\mu}^{(1)}$ are the third- and fourth-order non-equilibrium moments. In this case, the different relaxation shows up only in the $Q_{\alpha\beta\gamma}^{(1)}$ tensor, which can be written analytically as

$$\begin{aligned} Q_{\alpha\beta\gamma}^{(1)}(\beta_1, \beta_2) = &-\frac{1}{2\beta_2}\rho T \left[\partial_\alpha T \delta_{\beta\gamma} + \partial_\beta T \delta_{\alpha\gamma} + \partial_\gamma T \delta_{\alpha\beta}\right] \\ &+ u_\alpha P_{\beta\gamma}^{(1)}(\beta_1) + u_\beta P_{\alpha\gamma}^{(1)}(\beta_1) + u_\gamma P_{\alpha\beta}^{(1)}(\beta_1). \end{aligned} \tag{9.18}$$

The contribution related to β_3 can be subtracted from the rest of the non-equilibrium part and rescaled separately according to the proper relaxation. The non-equilibrium part without the contribution related to β_3 can be written as

$$\begin{aligned} \overline{f}_i^{neq} = &f_i^{neq} \\ &+ \frac{W_i Y_{i\alpha\beta\gamma}}{6T^3}\Big[u_\alpha P_{\beta\gamma}^{(1)}(\beta_1) + u_\beta P_{\alpha\gamma}^{(1)}(\beta_1) + u_\gamma P_{\alpha\beta}^{(1)}(\beta_1) \\ &- Q_{\alpha\beta\gamma}^{(1)}(\beta_1, \beta_2)\Big], \end{aligned} \tag{9.19}$$

where

$$Y_{i\alpha\beta\gamma} = v_{i\alpha} v_{i\beta} v_{i\gamma} - 3 v_{i\gamma} T \delta_{\alpha\beta}. \tag{9.20}$$

At this point the reduced non-equilibrium part can be rescaled according to

$$\overline{f}_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} \overline{f}_{i,c}^{neq}, \tag{9.21}$$

and the final non-equilibrium populations become

$$
\begin{aligned}
f_{i,f}^{neq} &= \overline{f}_{i,f}^{neq} \\
&- \frac{\beta_{2,c}}{r\beta_{2,f}} \frac{W_i Y_{i\alpha\beta\gamma}}{6T^3} \left[u_\alpha P_{\beta\gamma,c}^{(1)}(\beta_1) + u_\beta P_{\alpha\gamma,c}^{(1)}(\beta_1) + u_\gamma P_{\alpha\beta,c}^{(1)}(\beta_1) \right. \\
&\left. - Q_{\alpha\beta\gamma,c}^{(1)}(\beta_1, \beta_2) \right],
\end{aligned}
\tag{9.22}
$$

where $P_{\alpha\beta}^{(1)}$ and $Q_{\alpha\beta\gamma}^{(1)}$ can be computed as

$$
P_{\alpha\beta}^{(1)} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} (f_i - f_i^{eq}),
\tag{9.23}
$$

and

$$
Q_{\alpha\beta\gamma}^{(1)} = \sum_{i=1}^{n_Q} v_{i\alpha} v_{i\beta} v_{i\gamma} (f_i - f_i^{eq}).
\tag{9.24}
$$

For g-populations a similar procedure is applied. The non-equilibrium part can be expressed as

$$
g_i^{(1)} = g_i^{(1)} \left[K^{(1)}(\beta_1), q_\alpha^{(1)}(\beta_1, \beta_2), R_{\alpha\beta}^{(1)}(\beta_1) \right],
\tag{9.25}
$$

where $K^{(1)}$, $q_\alpha^{(1)}$ and $R_{\alpha\beta}^{(1)}$ are the zeroth-, first- and second-order non-equilibrium moments of g-populations. Similar to the f-populations, the different contribution on the relaxation, β_2, shows up only in the $q_\alpha^{(1)}$ tensor; this can be expressed analytically as [130]

$$
q_\alpha^{(1)} = -\frac{1}{2\beta_2} \rho T \left(2C_v - D\right) \partial_\alpha T + u_\alpha K^{(1)}(\beta_1).
\tag{9.26}
$$

Also for this case then, the contribution related to β_2 can be subtracted from the rest of the non-equilibrium g-populations and thus can be rescaled separately according the the proper relaxation. The non-equilibrium part without the contribution related to β_2 can be written as

$$
\overline{g}_i^{neq} = g_i^{neq} + W_i \frac{\left[u_\alpha K^{(1)}(\beta_1) - q_\alpha^{(1)}(\beta_1, \beta_2) \right] v_{i\alpha}}{T}.
\tag{9.27}
$$

The reduced non-equilibrium part can be rescaled as

$$
\overline{g}_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} \overline{g}_{i,c}^{neq},
\tag{9.28}
$$

thus the final non-equilibrium populations become

$$g_{i,f}^{neq} = \overline{g}_{i,f}^{neq} + \frac{\beta_{2,c}}{r\beta_{2,f}} W_i \frac{\left[u_\alpha K^{(1)}(\beta_1) - q_\alpha^{(1)}(\beta_1, \beta_2)\right] v_{i\alpha}}{T}, \tag{9.29}$$

where $K^{(1)}$ and $q_\alpha^{(1)}$ can be directly computed from populations as

$$K^{(1)} = \sum_{i=1}^{n_Q} (g_i - g_i^{eq}), \tag{9.30}$$

and

$$q_\alpha^{(1)} = \sum_{i=1}^{n_Q} v_{i\alpha}(g_i - g_i^{eq}). \tag{9.31}$$

9.5 Simulations

Stability and accuracy of the proposed grid refinement algorithm, when applied to thermal and compressible flows, are investigated below. The two-population model for thermal flows and grid refinement algorithm is here validated by means of the Rayleigh-Bénard convection, in Sec. 9.5.1, and by the flow past a heated sphere, in Sec. 9.5.2. The compressible model and grid refinement algorithm are validated by the viscous supersonic flow field around the NACA0012 airfoil, in Sec. 9.5.3.

9.5.1 Rayleigh-Bénard convection

The Rayleigh-Bénard set-up of Section 7.5 is here considered. Presented are the results of the thermal two-population model coupled with the grid refinement algorithm for thermal flows and their comparison with the DNS simulation of [290]. The Rayleigh and Prandtl number are $\mathrm{Ra} = 10^7$ and $\mathrm{Pr} = 0.7$. The computational domain is a box with periodicity in the x- and y-directions, where a fixed temperature is imposed at the bottom (hot) and top (cold) walls. The size of the domain is $L_x \times L_y \times L_z = 8H \times 8H \times H$, where $H = 64$ is the resolution in the coarse grid level. On top of the coarse grid, one more refinement patch is added at both hot and cold walls in order to increase the resolution in the boundary layers. To trigger convection an initial random perturbation is imposed on a linear temperature profile. The buoyancy force is computed according to the Boussinesq assumption, and

it is implemented as reported in Sec. 7.5.1.
An instantaneous volume rendering of the temperature in the domain is shown in Fig. 9.1. One can notice the cold temperature plumes in the up-

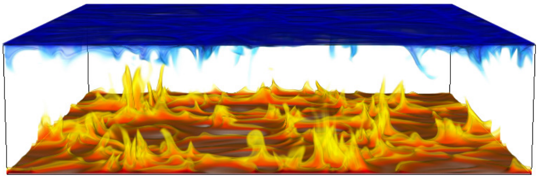


Figure 9.1: Volume rendering of temperature for the Rayleigh-Bénard convection at $\mathrm{Ra} = 10^7$.

per part of the box and the hot temperature plumes developing from the bottom of the box. Quantitatively, in Fig. 9.2, on the left, the mean and the rms temperature profiles are compared with the recent DNS data [290] and the agreement is good. All statistics are collected after the initial transient at $40t_L$ and sampled every $2.5t_L$ for a time period of $100t_L$, where $t_L = 2H/U$ denotes the large eddy turnover time with the free fall velocity $U = \sqrt{g\lambda\Delta TH}$. The temperature profiles are plotted as a function of the normalized vertical coordinate (normal to the bottom wall) $z^* = z/(H/\overline{\mathrm{Nu}})$, where $\overline{\mathrm{Nu}}$ is the mean Nusselt number, defined as $\overline{\mathrm{Nu}} = \frac{d\overline{T}/dn}{\Delta T/H}$, where $d\overline{T}/dn$ is the mean temperature gradient at top and bottom walls. In Figure 9.2, on the right, the rms velocity profiles are studied for the u and w component and compared to the reference data. Analogous to the temperature profiles, a good agreement is also observed for the velocity fluctuations. To complete the validation for the Rayleigh-Bénard convection, the resulting Nusselt number is compared in the following; while in present simulation the Nusselt number at the wall is evaluated as $\mathrm{Nu}_{ELBM} = 15.67$, the DNS [290] reports $\mathrm{Nu}_{DNS} = 15.59$. This amounts to a relative discrepancy of $\epsilon_{\mathrm{Nu}} = 5.13 \cdot 10^{-3}$.

It is important to notice that the transition at the grid interface is smooth for all quantities presented here, from the mean and rms temperature profiles to the rms x- and z-velocity profiles. This is attributed to the entropic stabilizer γ for which a snapshot through the domain is shown in Fig. 9.3. It is evident that larger deviations of γ from the LBGK value $\gamma = 2$ appear in the bulk rather than near the walls where the resolution is higher. Also in this case larger deviations in the γ arise in the immediate neighborhood of the level interface, which seems to be detected and compensated by the KBC model. Thus demonstrating the self-adaptive nature of the entropic

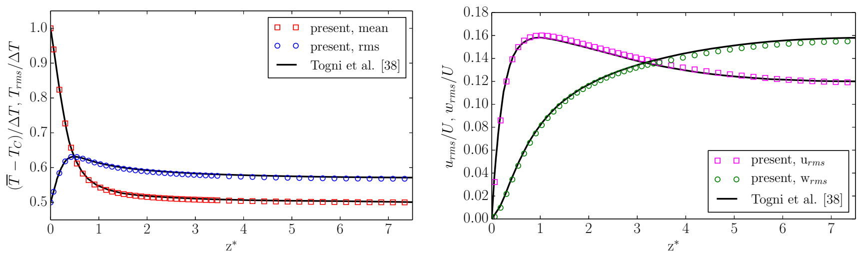


Figure 9.2: Comparison of mean and rms temperature profiles (left) and of rms u- and w-velocity components profiles for the Rayleigh-Bénard convection at $\mathrm{Ra} = 10^7$. The rms temperature profile is shifted by 0.5 in order to ease visualization of the plot.

stabilizer γ for all flow situations including grid refinement and complex wall boundary conditions. This self-adaptive nature of γ makes the simulations parameter-free.

9.5.2 Flow past a heated sphere

Complexity with respect to the thermal two-population scheme with grid refinement for flows involving flat walls is here increased, by considering curved walls. For this purpose, a heated sphere with the surface at constant temperature is simulated. The flow is considered for a Reynolds number $\mathrm{Re} = 3700$ and Prandtl number $\mathrm{Pr} = 0.7$. The computational domain is chosen as $[-7D, 23D] \times [-10D, 10D] \times [-10D, 10D]$ with the sphere centered at the origin. Four refinement patches are located closely around the sphere, where the finest patch resolves the sphere with $D = 240$. A snapshot of the temperature distribution around and in the wake of the sphere for this simulation along with the refinement patches is shown in Fig. 9.4. The figure shows the hot temperature streams in the shear layer and the back of the sphere, where the flow recirculates. Further downstream, hot fluid mixes with cold fluid and the temperature becomes diluted. In order to quantitatively validate the heated sphere simulation, the mean reduced Nusselt number distribution $\overline{\mathrm{Nu}}/\sqrt{\mathrm{Re}} = \frac{d\overline{T}/dn}{\Delta T/D}/\sqrt{\mathrm{Re}}$ around the sphere is shown in comparison to experimental results [329] in Fig. 9.5, where n is the normal coordinate with respect to the sphere surface and ΔT the temperature difference between fluid and sphere surface. The plot shows a good comparison with the experiment.

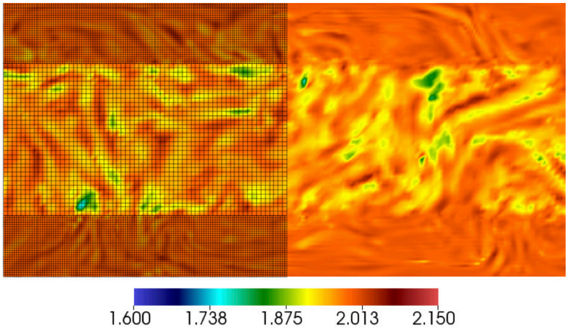


Figure 9.3: Slice through the Rayleigh-Bénard convection at Ra $= 10^7$ showing the spatial distribution of the stabilizer γ for the KBC model and the refinement patch (only left half shown here).

The numerical validation of the grid refinement algorithm is concluded by entering the compressible regime for which the simulation of the two-dimensional viscous supersonic flow around a NACA0012 airfoil is taken as an example.

9.5.3 Supersonic NACA0012 airfoil

The set-up consists of a two-dimensional simulation of the viscous supersonic flow field around a NACA0012 airfoil, at zero angle of attack $\mathcal{A} = 0°$. The free-stream Mach number is set to $\mathrm{Ma}_\infty = U_\infty/a_\infty = 1.5$, where $a_\infty = \sqrt{\gamma_{ad} T_\infty}$ is the speed of sound with $\gamma_{ad} = 1.4$ and $T_\infty = 0.8$, while the Reynolds number, based on the chord C of the airfoil, is $\mathrm{Re} = \mathrm{CU}_\infty/\nu = 10000$. The computational domain is prescribed as $[-7.5C, 17.5C] \times [-7C, 7C]$ with the airfoil centered at the origin. Two refinement patches are placed closely around the airfoil, where the finest patch resolves the airfoil chord with $C = 1200$ grid points (see Fig. 9.6).
Due to the high Mach number in this simulation, shifted lattices as presented in Sec. 4.2 are employed. In particular, the D2Q49 lattice with a shift in the freestream direction of $U_x = 1$ is used.

In Figure 9.6, a snapshot of the temperature distribution along with the two refinement patches around the airfoil, indicated by the shadowed regions in the lower half of the domain, is shown. The main features of the viscous supersonic flow field are evident from the temperature distribution: in front of the airfoil, the formation of a bow shock may be observed, yielding a jump

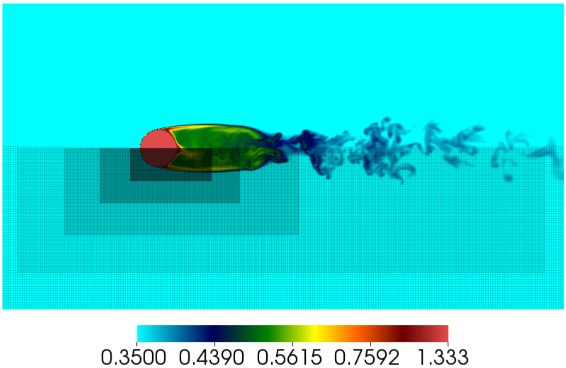


Figure 9.4: Snapshot of the instantaneous temperature distribution around and in the wake of the heated sphere at Re = 3700 along with the refinement patches (only lower half shown here).

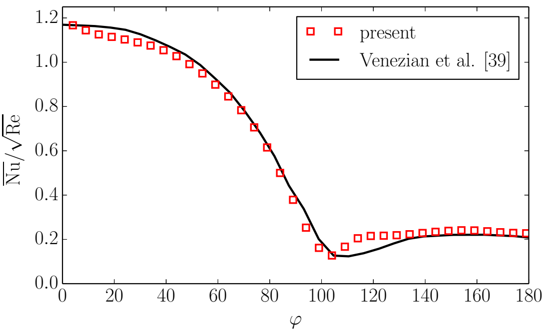


Figure 9.5: Average Nusselt number distribution around the heated sphere at Re = 3700.

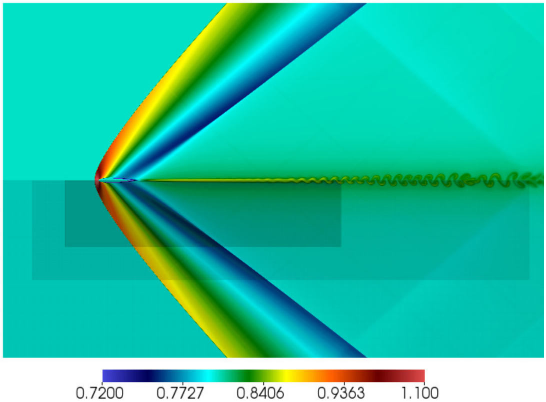


Figure 9.6: Snapshot of temperature distribution around the NACA0012 airfoil at $\mathcal{A} = 0°$, Ma $= 1.5$ and Re $= 10000$. Grid refinement patches are shown by the shadowed regions around the airfoil in the lower half of the domain.

and drastic increase in temperature. At the trailing edge of the airfoil, an oblique shock wave develops from the shear layer as a lambda shock. Further downstream, in the shear layer, vortex shedding is initiated. Important to notice is that the shock waves penetrate through various refinement levels, where special care usually needs to be taken to avoid artificial reflections at the interface. It is apparent that for the proposed refinement algorithm, no reflections or similar artifacts are observed and thus making it suitable also for compressible flows and the related shock dynamics.

For a quantitative comparison, the mean pressure coefficient distribution upstream the airfoil ($x/C \in [-1, 0]$), on the airfoil surface ($x/C \in [0, 1]$) and downstream the airfoil ($x/C \in [1, 1.5]$), is plotted in Fig. 9.7 along with the simulation results reported in [185]. Statistics have been collected after $50t^* = 50C/U_\infty$ flow times, and at every time step in the coarse level. Evidently, an excellent comparison can be reported.

Before concluding the numerical validation, a snapshot of the entropic estimate distribution around the airfoil in Fig. 9.8 is presented. From the picture, two main observations can be made. First, it is apparent that the entropic estimate adapts to the main physical features of the flow. In particular, large deviations from the resolved limit value $\alpha = 2$ may be

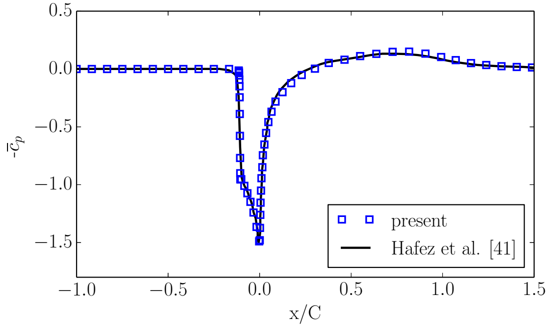


Figure 9.7: Comparison of the pressure coefficient distribution upstream the airfoil ($x/C \in [-1, 0]$), on the airfoil surface ($x/C \in [0, 1]$) and downstream the airfoil ($x/C \in [1, 1.5]$) for the supersonic simulation of the flow around a NACA0012 airfoil at $\mathcal{A} = 0°$, Ma $= 1.5$ and Re $= 10000$.

observed near the bow shock, through the expansion wave preceding the oblique shock at the trailing edge, and in the oblique shock itself. A second key observation concerns the interplay of the entropic estimate with the physical feature of the flow and the grid refinement patches. For example, it can be seen that the entropic estimate exhibits larger deviations from the fine to the coarse grid when a shock wave crosses the interface (see interface position in Fig. 9.6). These deviations in the entropic estimate arises naturally from the entropy condition Eq. (3.101) and plays a central role in sustaining the flow field, damping the spurious oscillations near the shock regions or damping the spurious noise, which would otherwise appear near the different grid interfaces.

9.6 Conclusion

An extension of the grid refinement algorithm to thermal and compressible models is achieved by an appropriate rescaling of the populations and thus widening the range of applicability of grid refinement algorithms. Accuracy and robustness is established through three set-ups in the thermal and compressible flow regimes, for which local grid refinement is crucial in order to obtain accurate results at a reasonable computational cost. The implicit subgrid features of entropy-based lattice Boltzmann models render the solutions stable and allow for significant under-resolution while retaining

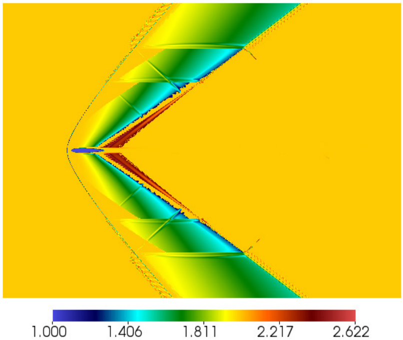


Figure 9.8: Snapshot of the distribution of the entropic estimate α around the NACA0012 airfoil.

accuracy. The entropic stabilizer adapts to the flow features and refinement patches, which enables multi-scale simulations where the fine-to-coarse level projection and vice versa is implicit to the model. These features are particularly important in the simulation of the supersonic airfoil, where the shock waves cross the refinement patches without being reflected and destabilizing the flow.

Chapter 10

Conclusions and future work

In this thesis, novel lattice Boltzmann models for the Fourier-Navier-Stokes equations are presented.
In particular, a new model for compressible flows is developed in order to simulate laminar and turbulent flows in the sub-, trans- and supersonic regimes. The model thus distinguishes itself from conventional solvers, which require identification of the flow field regime and then employing the appropriate numerical approach for it. The model relies on multi-speed lattices, accurate entropic equilibrium and entropic relaxation, thus avoiding any tuning parameter, turbulence model, or tracking of the shock front. Moreover, detailed boundary conditions were presented in order to incorporate the presence of solid walls.
The simplicity and robustness of the new approach are highlighted by the fact that the construction (adherence to the second law of thermodynamics) and boundary conditions presented here are independent of the choice of the lattice (standard and multi-speed lattices) and simulation dimensions. This, when viewed in combination with the simplicity of LB methods, could make ELBM a competitive and viable option for compressible flow simulations.
Additionally, optimization strategies for the compressible model have been proposed; shifted lattices can be employed in order to shift the operating domain in terms of Mach number, while pruning together with entropic guided equilibrium can be employed to reduce the computational cost and memory overhead and, at the same time, to increase the accuracy of the model.

The model has been validated for many two- and three dimensional setups, including shock dynamics, trans- and super-sonic aerodynamics, vortex-shock interaction, compressible decaying turbulence and turbulence-shock interaction, among others.
Considering the lack of compressible lattice Boltzmann solvers in the literature, much progress has been made in this thesis by employing the second principle of thermodynamics at the core of the new model. However, much room for improvements exists. Some of the limitations of the model presented here include restriction of the operable Mach number, which needs to be in the range zero to three for predominantly unidirectional flows and below two for flows with opposing or changing flow directions. Further benchmarking is needed to accurately establish the presence or absence of numerical diffusion for the shock-turbulence interaction problems. Grid refinement can help alleviate the issue to accuracy and can lead to DNS quality results. Also, the temperature range allowed (ratio of minimum to maximum temperature observed in the domain) is rather restricted for applications such as combustion where the highest temperature could be more than eight times the lowest temperature observed in the domain. This also restricts the application of this model only to flows with moderate compressibility. Possible directions for further improving the application domain of the models are also outlined in the thesis.

Additionally, two new thermal models are also presented; the first is derived directly from the compressible model by employing smaller lattices and an expansion of the accurate entropic equilibrium for the low Mach number regime. The model can be used to simulate convection-diffusion processes, viscous heating and acoustics.
The second model relies on a two-population approach, in which the total energy is conserved on a second lattice. This model is also able to reproduce convection-diffusion and viscous heating processes correctly; however, acoustics can not be simulated. The main advantage of the two-populations model over the thermal multi-speed model is the reduced computational overhead. Moreover, the two-population model has also been extended for the simulation of conjugate heat transfer problems, by defining a new interface boundary condition and solid region treatment.
Both models are validated by a large number of simulations, demonstrating the great capabilities to recover correctly the Fourier-Navier-Stokes equations in the proper operating domain.

The ensemble of models presented in this thesis provides a number of instruments which can be readily used to simulate thermal and compressible

flows in many situations. Apart from simplicity in the choice of grid and implementation, the present kinetic methods may significantly reduce the computational costs with respect to conventional fluid dynamics solvers, especially for complex flows set-ups like those presented in this work. In this sense, future works may be devoted in applying the present models to a large number of important applications involving thermal and compressible flows: from compressible aerodynamics all the way to geophysical thermal convecting flows.
Extensions may be pursued in the direction of magneto-hydrodynamic or multi-phase flows, where thermal and compressibility phenomena also plays an important role.

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